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48

E. Robert Fernholz

Stochastic Portfolio Theory



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To Luisa

Preface

This monograph introduces *stochastic portfolio theory*, a novel mathematical framework for analyzing portfolio behavior and equity market structure, and is intended for investment professionals and students of mathematical finance. Stochastic portfolio theory is descriptive as opposed to normative, and is consistent with the observable characteristics of actual portfolios and real markets; it is a theoretical tool for practical applications.

As a theoretical tool, stochastic portfolio theory provides insight into questions of market equilibrium and arbitrage, and it can be used to construct portfolios with controlled behavior. In practice, stochastic portfolio theory can be applied to portfolio optimization and performance analysis, and has been the basis for successful investment strategies employed for over a decade by the institutional equity manager INTECH, where I have served as chief investment officer.

Stochastic portfolio theory is descended from the classical portfolio theory of Harry Markowitz (1952), as is the rest of mathematical finance. Nevertheless, stochastic portfolio theory represents a significant departure from the current theory of dynamic asset pricing. Dynamic asset pricing theory is a normative theory that started with the general equilibrium model for financial markets due to Arrow (1953), evolved through the capital asset pricing models of Sharpe (1964) and Merton (1969), and currently consists in a theory of market equilibrium postulated on the absence of arbitrage and the existence of an equivalent martingale measure, proposed by Harrison and Kreps (1979).

Stochastic portfolio theory is a descriptive theory that is applicable under the wide range of assumptions and conditions that may hold in actual

equity markets. Unlike dynamic asset pricing theory, stochastic portfolio theory is consistent with either equilibrium or disequilibrium, with either arbitrage or no-arbitrage, and it remains valid regardless of the existence of an equivalent martingale measure.

This volume is organized as follows: Chapter 1 is an introduction to the basic structures of stochastic portfolio theory, including stocks, portfolios, and the market portfolio. Here it is shown that the logarithmic *growth rate* of a portfolio determines the portfolio's long-term behavior. The concept of the *excess growth rate* of a portfolio is introduced, and its role in portfolio behavior is explored.

The diversity of the distribution of capital in the market is introduced in Chapter 2, and we consider the case of a market in which the capital distribution of the market is essentially "stable" over time, in which capital will not concentrate into a single stock. We determine conditions that will be compatible with market stability, and the nature of the consequences that stability engenders.

Chapter 3 introduces one of the central concepts of stochastic portfolio theory, *portfolio generating functions*. These functions can be used to construct a wide variety of portfolios, and they allow the returns on these portfolios to be decomposed into components with specific characteristics. Conditions implying the existence of arbitrage are discussed in this chapter, and it appears that these conditions may be present in actual markets.

Chapter 4 extends the generating function methodology of the previous chapter to stocks identified by rank rather than by name. Since the capital distribution of the market is determined by the ranked weights of the stocks, this produces portfolios that depend in some manner on the capital distribution. This chapter serves two purposes: It brings together the two basic concepts of capital distribution and portfolio generation introduced in the previous two chapters, and it also provides the structure needed in order to apply portfolio generating functions to subsets of an equity universe. Since most stock indices are subsets of some larger universe, the results obtained here are required for practical applications.

Chapter 5 uses the ranked market weights of the previous chapter to construct stable models for the distribution of capital in an equity market. Such models give a more complete characterization of the structure of a stable equity market than we were able to obtain with the results of Chapter 2. As an application, a stable model is constructed for the U.S. equity market over the 10-year period from 1990 to 1999.

To understand the behavior of functionally generated portfolios, it is necessary to observe some examples of simulated portfolios. In Chapter 6 we present a number of simulations, with charts showing the decomposition of the portfolio return. The results of these simulations raise questions regarding market equilibrium models and the absence of arbitrage, so further investigation into these matters may be warranted.

Chapter 7 is devoted to practical applications of stochastic portfolio the-

ory. The first example is a diversity-weighted index, a functionally generated passive alternative to traditional capitalization-weighted indices. Also considered in this chapter is an analysis of the effects that changes in market diversity and the distribution of capital have on portfolio return.

Each chapter includes a number of problems of varying levels of difficulty. The unmarked problems should be fairly routine exercises; those marked (!) are likely to be significantly more challenging, and may actually be in the class of those marked (!!), which are believed to be open research problems as of this writing.

At the end of each chapter there is a section of notes containing references to related work, along with a brief summary of the principal results of the chapter, without proofs. In the summaries, the equations are numbered with the same numbering as where they originally appeared in the chapter, in order to allow convenient referencing. At the end of the book there is an appendix outlining a practical method for calculating local times associated with simulated functionally generated portfolios.

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E. Robert Fernholz
Princeton, New Jersey
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1

Stochastic Portfolio Theory

In this chapter we introduce the basic definitions for stocks and portfolios, and prove preliminary results that are used throughout the later chapters. The mathematical definitions and notation that we use can be found in Karatzas and Shreve (1991), and the model for continuous stock prices can be found in, e.g., Karatzas (1997), Karatzas and Shreve (1998), and Duffie (1992). The definitions, notation, and stock price model are all fairly standard in current mathematical finance, and we make a number of fairly standard assumptions to simplify the presentation.

We shall assume that the number of companies in the market is finite and fixed, that the total number of shares of each company is constant, and that companies do not merge or break up. Trading is continuous in time, with no transaction costs, taxes, or problems with the indivisibility of shares. Since shares are infinitely divisible, there is no loss of generality in assuming that each company has a single share outstanding, and we shall do so. Dividends will be paid continuously rather than discretely, and when dividends are not essential to the discussion, we shall assume that there are no dividends.

The stock prices and portfolio values we consider follow random processes defined on a probability space $\{\Omega, \mathcal{F}, P\}$, and a.s., “almost surely,” indicates that an event has probability one as measured by P . The term “random” indicates measurable dependence on $\omega \in \Omega$, although such dependence is usually implicit. The source of randomness in our model is a standard n -dimensional Brownian motion for some positive integer n ,

$$W = \{W(t) = (W_1(t), \dots, W_n(t)), \mathcal{F}_t, t \in [0, \infty)\},$$

defined on $\{\Omega, \mathcal{F}, P\}$, where $\{\mathcal{F}_t\}$ is the augmentation under P of the natural filtration $\{\mathcal{F}_t^W = \sigma(W(s); 0 \leq s \leq t)\}$. The dimension n is chosen to be large enough to avoid unnecessary dependencies among the stocks we define.

A process $\{X(t), \mathcal{F}_t, t \in [0, \infty)\}$ is *adapted* if $X(t)$ is $\mathcal{F}_t$ -measurable for $t \in [0, \infty)$. The filtration $\{\mathcal{F}_t, t \in [0, \infty)\}$ is the only filtration we consider, so all adapted processes, martingales, etc., are defined with respect to this filtration. For continuous, square-integrable local martingales M and N we can define the *cross-variation process* $\langle M, N \rangle$. The cross-variation process is adapted, continuous, and of bounded variation, and the operation $\langle \cdot, \cdot \rangle$ is bilinear on the real vector space of continuous, square-integrable local martingales. We use the notation $\langle M \rangle = \langle M, M \rangle$, where $\langle M \rangle$ is called the *quadratic variation process* of M , and has continuous, nondecreasing sample paths. The Brownian motion process defined above is a continuous, square-integrable martingale, and it is characterized by its cross-variation processes

$$\langle W_i, W_j \rangle_t = \delta_{ij}t, \quad t \in [0, \infty),$$

where $\delta_{ij} = 1$ if $i = j$, and 0 otherwise.

A *continuous semimartingale* X is a measurable, adapted process that has the decomposition

$$X(t) = X(0) + M_X(t) + V_X(t), \quad t \in [0, \infty), \quad \text{a.s.}, \quad (1.0.1)$$

where M_X is a continuous, square-integrable local martingale and V_X is a continuous, adapted process of bounded variation. It can be shown that this decomposition is a.s. unique (see Karatzas and Shreve (1991)), so we can define the cross-variation process for continuous semimartingales X and Y by

$$\langle X, Y \rangle = \langle M_X, M_Y \rangle,$$

where M_X and M_Y are the local martingale parts of X and Y , respectively. The quadratic variation process for X is similarly defined by $\langle X \rangle = \langle M_X \rangle$.

1.1 Stocks and Portfolios

We shall use a *logarithmic model* for continuous-time stock price processes. The logarithmic model is advantageous for the type of analysis that we do, and our use of this model does not imply a preference for logarithmic utility. In fact, utility functions play at most a minor role in stochastic portfolio theory.

1.1.1 Definition. Let n be a positive integer. A *stock price process* X is a process that satisfies the stochastic differential equation

$$d \log X(t) = \gamma(t) dt + \sum_{\nu=1}^n \xi_\nu(t) dW_\nu(t), \quad t \in [0, \infty), \quad (1.1.1)$$

where $(W_1, \dots, W_n)$ is a Brownian motion, γ is measurable, adapted, and satisfies $\int_0^T |\gamma(t)| dt < \infty$, for all $T \in [0, \infty)$, a.s., and the ξ_ν , $\nu = 1, \dots, n$, are measurable, adapted, and satisfy

- (i) $\int_0^T (\xi_1^2(t) + \dots + \xi_n^2(t)) dt < \infty$, $T \in [0, \infty)$, a.s.;
- (ii) $\lim_{t \rightarrow \infty} t^{-1} (\xi_1^2(t) + \dots + \xi_n^2(t)) \log \log t = 0$, a.s.;
- (iii) $\xi_1^2(t) + \dots + \xi_n^2(t) > 0$, $t \in [0, \infty)$, a.s.

Equation (1.1.1) can be integrated directly, with

$$\log X(t) = \log X_0 + \int_0^t \gamma(s) ds + \int_0^t \sum_{\nu=1}^n \xi_\nu(s) dW_\nu(s), \quad t \in [0, \infty),$$

where the positive constant X_0 represents the initial value of the stock. In exponential form, the stock price process can be expressed as

$$X(t) = X_0 \exp\left(\int_0^t \gamma(s) ds + \int_0^t \sum_{\nu=1}^n \xi_\nu(s) dW_\nu(s)\right), \quad t \in [0, \infty). \quad (1.1.2)$$

Frequently a stock price process will be referred to simply as a *stock*. In Definition 1.1.1, $X(t)$ represents the price of the stock at time $t \geq 0$, and it follows from (1.1.2) that $X(t) > 0$ for all $t \in [0, \infty)$. Since we assume that each company has a single share of stock outstanding, $X(t)$ also represents the total capitalization of the company at time t . In (1.1.1), the process γ is called the *growth rate* process of X , and the ξ_ν are called the *volatility* processes of X . The process ξ_ν represents the sensitivity of X to the ν th source of uncertainty, W_ν .

We shall see that condition (ii) of Definition 1.1.1 ensures that the volatility of the stock does not rise so quickly as to render meaningless the stock's growth rate. The growth rate is terminology from the logarithmic model: In classical portfolio theory, the standard model uses the rate of return, which we introduce in (1.1.4) below. Although the rate of return is used in classical portfolio theory, the growth rate is a better indicator of long-term stock behavior than the rate of return. We prove this in Section 1.3, where we also show that the growth rate of a portfolio determines the long-term behavior of the portfolio value. Since it is the growth rate rather than the rate of return that should be of interest to long-term investors, it is the growth rate that will be of interest to us here.

Remark. The finite time domain $[0, T]$, $T > 0$, is commonly used in mathematical finance due to the need for Girsanov's theorem (see Duffie (1992) or Karatzas and Shreve (1998)). Since we do not depend on this theorem, all our results hold for $[0, \infty)$. In this chapter and the next we consider asymptotic events, so the infinite time domain is necessary. However, in nonasymptotic settings we restrict our consideration to the finite domain $[0, T]$ in order to comply with convention. $\square$

It is clear from Definition 1.1.1 that $\log X(t)$ is a continuous semimartingale with bounded variation component

$$\gamma(t) dt, \quad t \in [0, \infty),$$

and local martingale component

$$\sum_{\nu=1}^n \xi_{\nu}(t) dW_{\nu}(t), \quad t \in [0, \infty).$$

By Itô's rule (see Itô (1951) or Karatzas and Shreve (1991), Section 3.3) applied to $X(t) = \exp(\log X(t))$,

$$dX(t) = X(t) d\log X(t) + \frac{1}{2} X(t) d\langle \log X \rangle_t, \quad t \in [0, \infty), \quad \text{a.s.},$$

where

$$d\langle \log X \rangle_t = \sum_{\nu=1}^n \xi_{\nu}^2(t) dt, \quad t \in [0, \infty), \quad \text{a.s.},$$

so X is a continuous semimartingale that satisfies

$$dX(t) = \left(\gamma(t) + \frac{1}{2} \sum_{\nu=1}^n \xi_{\nu}^2(t) \right) X(t) dt + X(t) \sum_{\nu=1}^n \xi_{\nu}(t) dW_{\nu}(t), \quad (1.1.3)$$

$t \in [0, \infty)$, a.s. If we define the *rate of return* process α by

$$\alpha(t) = \gamma(t) + \frac{1}{2} \sum_{\nu=1}^n \xi_{\nu}^2(t), \quad t \in [0, \infty), \quad (1.1.4)$$

then (1.1.3) becomes

$$dX(t) = \alpha(t) X(t) dt + X(t) \sum_{\nu=1}^n \xi_{\nu}(t) dW_{\nu}(t), \quad t \in [0, \infty), \quad \text{a.s.},$$

and we have the standard model for a stock price process. This can also be written in the form

$$\frac{dX(t)}{X(t)} = \alpha(t) dt + \sum_{\nu=1}^n \xi_{\nu}(t) dW_{\nu}(t), \quad t \in [0, \infty), \quad \text{a.s.}, \quad (1.1.5)$$

where $dX(t)/X(t)$ can be interpreted as the “instantaneous” return on X . In the same sense, $d\log X(t)$ can be interpreted as the instantaneous logarithmic, or geometric, return on X .

Suppose that we have a family of stocks X_i , $i = 1, \dots, n$, defined by

$$d\log X_i(t) = \gamma_i(t) dt + \sum_{\nu=1}^n \xi_{i\nu}(t) dW_{\nu}(t), \quad t \in [0, \infty), \quad (1.1.6)$$

in differential form, or, equivalently, by

$$X_i(t) = X_0^i \exp\left(\int_0^t \gamma_i(s) ds + \int_0^t \sum_{\nu=1}^n \xi_{i\nu}(s) dW_\nu(s)\right), \quad (1.1.7)$$

in exponential form. Consider the matrix-valued process ξ defined by $\xi(t) = (\xi_{i\nu}(t))_{1 \leq i, \nu \leq n}$ and define the *covariance* process σ , where $\sigma(t) = \xi(t)\xi^T(t)$. For any $x \in \mathbb{R}^n$ and $t \in [0, \infty)$,

$$x\sigma(t)x^T = x\xi(t)\xi^T(t)x^T = x\xi(t)(x\xi(t))^T \geq 0, \quad (1.1.8)$$

so $\sigma(t)$ is positive semidefinite for all $t \in [0, \infty)$.

The cross-variation processes for $\log X_i$ and $\log X_j$ are related to σ by

$$\sigma_{ij}(t) dt = d\langle \log X_i, \log X_j \rangle_t = \sum_{\nu=1}^n \xi_{i\nu}(t)\xi_{j\nu}(t) dt, \quad (1.1.9)$$

$t \in [0, \infty)$, a.s. Since the processes $\xi_{i\nu}$ are assumed to be locally square-integrable in Definition 1.1.1, it follows that for all i and j ,

$$\int_0^t |\sigma_{ij}(s)| ds < \infty, \quad t \in [0, \infty), \quad \text{a.s.}$$

For $i = 1, \dots, n$, the process σ_{ii} is called the *covariance* process of X_i .

1.1.2 Definition. A *market* is a family $\mathcal{M} = \{X_1, \dots, X_n\}$ of stocks, each defined as in (1.1.7), such that $\sigma(t)$ is nonsingular for all $t \in [0, \infty)$, a.s. The market $\mathcal{M}$ is *nondegenerate* if there is a number $\varepsilon > 0$ such that

$$x\sigma(t)x^T \geq \varepsilon \|x\|^2, \quad x \in \mathbb{R}^n, \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.1.10)$$

The market $\mathcal{M}$ has *bounded variance* if there exists a number $M > 0$ such that

$$x\sigma(t)x^T \leq M \|x\|^2, \quad x \in \mathbb{R}^n, \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.1.11)$$

Note that in (1.1.7) the number of stocks is equal to the dimension of the Brownian motion process W , and that σ is assumed to be nonsingular in Definition 1.1.2. Nonsingularity of σ is not always necessary for our purposes, but the slight generality gained by removing it is not especially relevant.

1.1.3 Lemma. *For the market covariance process σ , $\sigma(t)$ is positive definite for all $t \in [0, \infty)$, a.s.*

Proof. We saw in (1.1.8) that for all $t \in [0, \infty)$, $\sigma(t)$ is positive semidefinite. Since Definition 1.1.2 states that $\sigma(t)$ is nonsingular, it follows that $\sigma(t)$ is positive definite for all $t \in [0, \infty)$, a.s. $\square$

1.1.4 Definition. A *portfolio* in the market $\mathcal{M}$ is a measurable, adapted vector-valued process π , $\pi(t) = (\pi_1(t), \dots, \pi_n(t))$, for $t \in [0, \infty)$, such that π is a.s. bounded on $[0, \infty)$ and

$$\pi_1(t) + \dots + \pi_n(t) = 1, \quad t \in [0, \infty), \quad \text{a.s.}$$

Remark. We have included no *riskless asset* in either the market or in portfolios. Our purpose here is to study the behavior of stock portfolios, and the existence of a riskless asset is irrelevant. $\square$

The component processes π_i of a portfolio represent the *proportions*, or *weights*, of the corresponding stocks in the portfolio. Two portfolios are *equal* if their weights are equal for all $t \in [0, \infty)$, a.s. We shall say that a stock is *held* in a portfolio if the corresponding weight is positive: If the portfolio holds no shares of a given stock, then the weight of that stock is zero. A negative value for $\pi_i(t)$ indicates a short sale in the i th stock, so the weights of portfolio with no short sales are all nonnegative.

Suppose we have a portfolio π and that $Z_\pi(t) > 0$ represents the value of an investment in π at time t . Then the amount invested in the i th stock X_i is

$$\pi_i(t)Z_\pi(t),$$

so, heuristically speaking, if the price of X_i changes by $dX_i(t)$, the induced change in the portfolio value is

$$\pi_i(t)Z_\pi(t) \frac{dX_i(t)}{X_i(t)}.$$

Hence the total change in the portfolio value at time t is

$$dZ_\pi(t) = \sum_{i=1}^n \pi_i(t)Z_\pi(t) \frac{dX_i(t)}{X_i(t)},$$

or, equivalently,

$$\frac{dZ_\pi(t)}{Z_\pi(t)} = \sum_{i=1}^n \pi_i(t) \frac{dX_i(t)}{X_i(t)}. \quad (1.1.12)$$

This equation shows that $dZ_\pi(t)/Z_\pi(t)$, the instantaneous return of the portfolio, is the weighted average of the instantaneous returns of the component stocks, $dX_i(t)/X_i(t)$. Here we wish to study the nature of solutions to (1.1.12) in the context of our logarithmic model. For background regarding solutions to stochastic differential equations, see Karatzas and Shreve (1991). The following proposition expresses Z_π in differential form.

1.1.5 Proposition. *Let π be a portfolio in $\mathcal{M}$. Then the process Z_π satisfies*

$$d \log Z_\pi(t) = \gamma_\pi(t) dt + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t), \quad (1.1.13)$$

for $t \in [0, \infty)$, a.s., where

$$\gamma_\pi(t) = \sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \left(\sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) \right). \quad (1.1.14)$$

The properties of γ_i , π_i , and $\xi_{i\nu}$ ensure that $\log Z_\pi$ is a continuous semimartingale. For any initial value $Z_\pi(0) > 0$, (1.1.13) can be integrated directly to obtain

$$Z_\pi(t) = Z_\pi(0) \exp\left(\int_0^t \gamma_\pi(s) ds + \int_0^t \sum_{i,\nu=1}^n \pi_i(s) \xi_{i\nu}(s) dW_\nu(s)\right), \quad (1.1.15)$$

as a strong solution of (1.1.12). From this it is clear that $Z_\pi(t) > 0$, for all $t \in [0, \infty)$, a.s.

Proof of Proposition 1.1.5. The process Z_π defined by (1.1.15) is clearly adapted; we must check that $Z_\pi(t)$ satisfies (1.1.12).

By Itô's rule applied to $Z_\pi(t) = \exp(\log Z_\pi(t))$, (1.1.13) implies that, a.s., for $t \in [0, \infty)$,

$$dZ_\pi(t) = Z_\pi(t) d\log Z_\pi(t) + \frac{1}{2} Z_\pi(t) d\langle \log Z_\pi \rangle_t, \quad (1.1.16)$$

so, a.s., for all $t \in [0, \infty)$,

$$\frac{dZ_\pi(t)}{Z_\pi(t)} = \gamma_\pi(t) dt + \frac{1}{2} d\langle \log Z_\pi \rangle_t + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t).$$

Now, a.s., for $t \in [0, \infty)$,

$$\begin{aligned} d\langle \log Z_\pi \rangle_t &= \sum_{i,j=1}^n \pi_i(t) \pi_j(t) d\langle \log X_i, \log X_j \rangle_t \\ &= \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) dt, \end{aligned} \quad (1.1.17)$$

by (1.1.9). Since by definition

$$\gamma_\pi(t) = \sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \left(\sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) \right),$$

it follows that a.s., for $t \in [0, \infty)$,

$$\frac{dZ_\pi(t)}{Z_\pi(t)} = \sum_{i=1}^n \pi_i(t) \gamma_i(t) dt + \frac{1}{2} \sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) dt + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t).$$

By (1.1.9), $\sigma_{ii}(t) = \sum_{\nu=1}^n \xi_{i\nu}^2(t)$, so (1.1.3) implies that for $i = 1, \dots, n$,

$$dX_i(t) = \left(\gamma_i(t) + \frac{1}{2} \sigma_{ii}(t) \right) X_i(t) dt + X_i(t) \sum_{\nu=1}^n \xi_{i\nu}(t) dW_\nu(t),$$

for $t \in [0, \infty)$, a.s. Therefore,

$$\frac{dZ_\pi(t)}{Z_\pi(t)} = \sum_{i=1}^n \pi_i(t) \frac{dX_i(t)}{X_i(t)},$$

for $t \in [0, \infty)$, a.s. □

The process Z_π is called the *portfolio value process* for π , and γ_π in (1.1.14) is called the *portfolio growth rate process* for π . The process $\sigma_{\pi\pi}$ defined by

$$\sigma_{\pi\pi}(t) = \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t), \quad t \in [0, \infty), \quad (1.1.18)$$

is called the *portfolio variance process*, and (1.1.17) implies that

$$\langle \log Z_\pi \rangle_t = \int_0^t \sigma_{\pi\pi}(s) ds, \quad t \in [0, \infty), \quad \text{a.s.},$$

so (1.1.16) can be written

$$\frac{dZ_\pi(t)}{Z_\pi(t)} = d \log Z_\pi(t) + \frac{1}{2} \sigma_{\pi\pi}(t) dt, \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.1.19)$$

If the market $\mathcal{M}$ has bounded variance, then the portfolio variance process $\sigma_{\pi\pi}$ is a.s. bounded on $[0, \infty)$ for any portfolio π .

The process γ_π^* defined by

$$\gamma_\pi^*(t) = \frac{1}{2} \left(\sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) \right), \quad t \in [0, \infty), \quad (1.1.20)$$

is called the *excess growth rate process* of the portfolio. With this notation, we have

$$\gamma_\pi(t) = \sum_{i=1}^n \pi_i(t) \gamma_i(t) + \gamma_\pi^*(t), \quad t \in [0, \infty), \quad \text{a.s.}, \quad (1.1.21)$$

and (1.1.20) is equivalent to

$$\gamma_\pi^*(t) = \frac{1}{2} \left(\sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \sigma_{\pi\pi}(t) \right), \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.1.22)$$

1.1.6 Corollary. *Let π be a portfolio and let Z_π be its value process. Then*

$$d \log Z_\pi(t) = \sum_{i=1}^n \pi_i(t) d \log X_i(t) + \gamma_\pi^*(t) dt, \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.1.23)$$

Proof. By (1.1.1), a.s., for $t \in [0, \infty)$,

$$\sum_{i=1}^n \pi_i(t) d \log X_i(t) = \sum_{i=1}^n \pi_i(t) \gamma_i(t) dt + \sum_{i,\nu=1}^n \xi_{i\nu}(t) dW_\nu(t). \quad (1.1.24)$$

From (1.1.13), (1.1.14), and (1.1.20), we have, a.s., for $t \in [0, \infty)$,

$$d \log Z_\pi(t) = \sum_{i=1}^n \pi_i(t) \gamma_i(t) dt + \gamma_\pi^*(t) dt + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t),$$

and (1.1.23) follows from this and (1.1.24). $\square$

From (1.1.23) we see that the instantaneous logarithmic return of the portfolio, $d \log Z_\pi(t)$, is the weighted average of the instantaneous logarithmic returns of the component stocks, $d \log X_i(t)$, plus the excess growth rate. It follows from (1.1.22) that $\gamma_\pi^*(t)$ is half the difference between the weighted-average variance of the individual stocks and the portfolio variance. Heuristically, $\gamma_\pi^*(t)$ can be regarded as a measure of the efficacy of portfolio diversification in reducing the volatility of Z_π compared to that of its component stocks. That diversification can lower portfolio volatility is a well-known result of classical portfolio theory, but it may be less universally recognized that diversification also influences the portfolio growth rate. In Section 1.2 we show that $\gamma_\pi^*(t)$ is positive for portfolios that hold more than a single stock and no short sales. This means that for any such portfolio, the weighted-average variance of the individual stocks in the portfolio is greater than the portfolio variance. This will no longer be true in general if the portfolio has short sales.

Let us define the *portfolio rate of return* process for π to be

$$\alpha_\pi(t) = \sum_{i=1}^n \pi_i(t) \alpha_i(t), \quad t \in [0, \infty), \quad (1.1.25)$$

where α_i , $i = 1, \dots, n$, is the rate of return of X_i defined as in (1.1.4). This is the classical equation for the portfolio rate of return, and it expresses this rate of return as the weighted average of the rates of return of the stocks in the portfolio. In contrast to the classical case, the portfolio growth rate exceeds the weighted average of the growth rates of the component stocks, and the amount by which the portfolio growth rate exceeds this weighted average is precisely the excess growth rate.

From (1.1.5), for $i = 1, \dots, n$,

$$\frac{dX_i(t)}{X_i(t)} = \alpha_i(t) dt + \sum_{\nu=1}^n \xi_{i\nu}(t) dW_\nu(t), \quad t \in [0, \infty),$$

so (1.1.12) becomes

$$\frac{dZ_\pi(t)}{Z_\pi(t)} = \sum_{i=1}^n \pi_i(t) \frac{dX_i(t)}{X_i(t)} = \alpha_\pi(t) dt + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t), \quad (1.1.26)$$

for $t \in [0, \infty)$, a.s. From (1.1.19) and (1.1.13) we have, a.s., for $t \in [0, \infty)$,

$$\frac{dZ_\pi(t)}{Z_\pi(t)} = \gamma_\pi(t) dt + \frac{1}{2} \sigma_{\pi\pi}(t) dt + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t),$$

so

$$\alpha_\pi(t) = \gamma_\pi(t) + \frac{1}{2} \sigma_{\pi\pi}(t), \quad t \in [0, \infty), \quad \text{a.s.},$$

just as in (1.1.4).

1.1.7 Example. (Portfolio optimization I) Classical Markowitz (1952) portfolio optimization is accomplished by minimizing the portfolio variance

$$\sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t)$$

under the linear constraints

$$\sum_{i=1}^n \pi_i(t) \alpha_i(t) \geq \alpha_0 \quad (1.1.27)$$

and

$$\pi_1(t) + \cdots + \pi_n(t) = 1 \quad \text{with} \quad \pi_1(t), \dots, \pi_n(t) \geq 0.$$

Hence, the optimization finds the portfolio with no short sales that has the minimum variance for a given portfolio rate of return. Such an optimization can be carried out using conventional quadratic programming techniques (see Wolfe (1959)).

If we wish to minimize the portfolio variance under a constraint on the portfolio growth rate rather than on the portfolio rate of return, then (1.1.27) is replaced by

$$\sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \left(\sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) \right) \geq \gamma_0. \quad (1.1.28)$$

This constraint is nonlinear, and conventional quadratic programs cannot be used in this case. However, (1.1.28) is equivalent to

$$\sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) \geq \gamma_0 + \frac{1}{2} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t), \quad (1.1.29)$$

and we shall see later that in certain cases the quadratic term in (1.1.29) can be controlled, and hence quadratic programming can be used. $\square$

1.1.8 Example. (Logarithmic utility) Suppose that we wish to find the portfolio π that maximizes the expected value of $\log Z_\pi(t)$ for $t \in [0, \infty)$. Since for $t \in [0, \infty)$,

$$d \log Z_\pi(t) = \gamma_\pi(t) dt + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t),$$

and since the expected value of the last term, the martingale component of $d \log Z_\pi(t)$, is zero, maximizing the expected value of $\log Z_\pi(t)$ amounts to maximizing the portfolio growth rate, $\gamma_\pi(t)$. Now, for $t \in [0, \infty)$,

$$\gamma_\pi(t) = \sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \frac{1}{2} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t), \quad (1.1.30)$$

by (1.1.14). Conventional quadratic programming can be used to maximize (1.1.30) under the constraints

$$\pi_1(t) + \cdots + \pi_n(t) = 1 \quad \text{with} \quad \pi_1(t), \dots, \pi_n(t) \geq 0.$$

It can be shown that maximizing the expected value of $\log Z_\pi(t)$ will produce the portfolio with the greatest asymptotic value, a.s., but such portfolios carry too high a level of risk for most investors. $\square$

Dividends are used by companies to distribute earnings back to their stockholders. In our context, dividend payments allow a stock to have returns without affecting the capitalization or the market weight of the stock. Although it is frequently unnecessary to include dividends in our discussion, sometimes they play an important role, and accordingly, we introduce them now. We shall assume that dividends are paid continuously.

1.1.9 Definition. A *dividend rate* process is a measurable, adapted process δ that satisfies

$$\int_0^t |\delta(s)| ds < \infty, \quad t \in [0, \infty), \quad \text{a.s.}$$

Usually, dividend rates are assumed to be nonnegative, but this assumption is not necessary. For a stock X with dividends, i.e., with an associated dividend rate process δ , we define the *total return* process $\hat{X}$ by

$$\hat{X}(t) = X(t) \exp\left(\int_0^t \delta(s) ds\right), \quad t \in [0, \infty). \quad (1.1.31)$$

The total return process $\hat{X}$ represents the value of an investment in the stock X with all dividends continuously reinvested. If $\delta = 0$, then $\hat{X} = X$. It follows from (1.1.31) that $\hat{X}(0) = X(0)$ and that

$$d \log \hat{X}(t) = d \log X(t) + \delta(t) dt, \quad t \in [0, \infty).$$

It is convenient to define the process

$$\rho(t) = \gamma(t) + \delta(t), \quad t \in [0, \infty),$$

which we call the *augmented growth rate*.

Let $\delta_1, \dots, \delta_n$ be the respective dividend rates of the stocks $X_1, \dots, X_n$ in the market $\mathcal{M}$. For any portfolio π , we define the *dividend rate process* δ_π for the portfolio by

$$\delta_\pi(t) = \sum_{i=1}^n \pi_i(t) \delta_i(t), \quad t \in [0, \infty),$$

and the *total return process* $\widehat{Z}_\pi$ of π by

$$\widehat{Z}_\pi(t) = Z_\pi(t) \exp\left(\int_0^t \delta_\pi(s) ds\right), \quad t \in [0, \infty). \quad (1.1.32)$$

As with individual stocks,

$$d \log \widehat{Z}_\pi(t) = d \log Z_\pi(t) + \delta_\pi(t) dt, \quad t \in [0, \infty).$$

The process $\widehat{Z}_\pi$ represents the value of a portfolio with the same weights as π , but in which all dividends are reinvested proportionally across the entire portfolio according to the weight of each stock. Hence the reinvestment of the dividends modifies the value of $\widehat{Z}_\pi$ while preserving the weights of the portfolio π . We also define the *augmented growth rate* for π by

$$\rho_\pi(t) = \gamma_\pi(t) + \delta_\pi(t), \quad t \in [0, \infty).$$

For a market $\mathcal{M}$ without dividends, $Z_\pi = \widehat{Z}_\pi$ for all portfolios π .

1.1.10 Problem. (!) Develop a portfolio model for zero-coupon bonds, where the present value of each of the bonds depends on a random, time-dependent yield curve.

1.1.11 Problem. For $p > 0$, characterize the portfolio π that maximizes the expected value of $Z_\pi^p(t)$ for $t \in [0, \infty)$.

1.1.12 Problem. Generalize the definition of the dividend process as follows: For $i = 1, \dots, n$, let Δ_i be a continuous, adapted process of locally bounded variation such that $\Delta_i(0) = 0$, and interpret $\Delta_i(t)$ to be the total proportional dividends paid by the i th stock up to time $t > 0$. Then

$$d \log \widehat{X}_i(t) = d \log X_i(t) + d\Delta_i(t), \quad t \in [0, \infty).$$

For a portfolio π , define $\widehat{Z}_\pi$ similarly to (1.1.32).

1.2 Relative Return and the Market Portfolio

It is frequently of interest to measure the performance of stocks or portfolios relative to a given benchmark portfolio or index. A natural benchmark is the market portfolio, consisting of all the shares of all the stocks in the market. We first define the relative return of a stock versus a portfolio.

1.2.1 Definition. For a stock X_i , $1 \leq i \leq n$, and portfolio η , the process

$$\log(X_i(t)/Z_\eta(t)), \quad t \in [0, \infty), \quad (1.2.1)$$

is called the *relative return* process of X_i versus η .

The relative return process (1.2.1) is a continuous semimartingale with

$$\begin{aligned} \langle \log(X_i/Z_\eta), \log(X_j/Z_\eta) \rangle_t &= \langle \log X_i, \log X_j \rangle_t - \langle \log X_i, \log Z_\eta \rangle_t \\ &\quad - \langle \log X_j, \log Z_\eta \rangle_t + \langle \log Z_\eta \rangle_t, \end{aligned} \quad (1.2.2)$$

for $t \in [0, \infty)$, a.s. If we define the process $\sigma_{i\eta}$ by

$$\sigma_{i\eta}(t) = \sum_{j=1}^n \eta_j(t) \sigma_{ij}(t), \quad t \in [0, \infty),$$

for $i = 1, \dots, n$, then

$$d\langle \log X_i, \log Z_\eta \rangle_t = \sigma_{i\eta}(t) dt, \quad t \in [0, \infty) \quad \text{a.s.}$$

The *relative covariance* process τ^η is the matrix-valued process

$$\tau^\eta(t) = (\tau_{ij}^\eta(t))_{1 \leq i, j \leq n}, \quad t \in [0, \infty),$$

where

$$\tau_{ij}^\eta(t) = \sigma_{ij}(t) - \sigma_{i\eta}(t) - \sigma_{j\eta}(t) + \sigma_{\eta\eta}(t), \quad t \in [0, \infty), \quad (1.2.3)$$

for $i, j = 1, \dots, n$, with $\sigma_{\eta\eta}(t) = \eta(t)\sigma(t)\eta^T(t)$. Then for all i and j ,

$$d\langle \log(X_i/Z_\eta), \log(X_j/Z_\eta) \rangle_t = \tau_{ij}^\eta(t) dt, \quad t \in [0, \infty), \quad \text{a.s.}, \quad (1.2.4)$$

and

$$d\langle \log(X_i/Z_\eta) \rangle_t = \tau_{ii}^\eta(t) dt, \quad t \in [0, \infty), \quad \text{a.s.}$$

Since $\langle \log(X_i/Z_\eta) \rangle_t$ is a.s. nondecreasing,

$$\tau_{ii}^\eta(t) \geq 0, \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.2.5)$$

1.2.2 Lemma. For a portfolio η , $\tau^\eta(t)$ is positive semidefinite with rank $n - 1$, for $t \in [0, \infty)$, a.s., and the null space of $\tau^\eta(t)$ is spanned by $\eta(t)$.

Proof. Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, $x \neq 0$, and let $t \in [0, \infty)$. Then it follows from (1.2.3) that

$$x\tau^\eta(t)x^T = \sigma(t)x^T - 2x\sigma(t)\eta^T(t) \sum_{i=1}^n x_i + \eta(t)\sigma(t)\eta^T(t) \left(\sum_{i=1}^n x_i \right)^2. \quad (1.2.6)$$

There are two cases we shall consider. First suppose that $\sum_{i=1}^n x_i = 0$. Then it follows from (1.2.6) that

$$x\tau^\eta(t)x^T = x\sigma(t)x^T > 0,$$

since $\sigma(t)$ is a.s. positive definite by Lemma 1.1.3, and $x \neq 0$. Now suppose that

$$\sum_{i=1}^n x_i = a \neq 0.$$

Let $y = a^{-1}x$. Since

$$x\tau^\eta(t)x^T = a^2 y\tau^\eta(t)y^T,$$

it suffices to consider $y\tau^\eta(t)y^T$. Since $\sum_{i=1}^n y_i = 1$, (1.2.6) now becomes

$$\begin{aligned} y\tau^\eta(t)y^T &= y\sigma(t)y^T - 2y\sigma(t)\eta^T(t) + \eta(t)\sigma(t)\eta^T(t) \\ &= (y - \eta(t))\sigma(t)(y - \eta(t))^T \\ &= 0 \end{aligned} \quad (1.2.7)$$

if and only if $y = \eta(t)$, since $\sigma(t)$ is a.s. positive definite by Lemma 1.1.3. Hence, $x = a\eta(t)$, so $\eta(t)$ spans the null space of $\tau^\eta(t)$, a.s., and the rank of $\tau^\eta(t)$ is $n - 1$. This holds for any $t \in [0, \infty)$, a.s. $\square$

The *relative variance* process of π versus η is defined for $t \in [0, \infty)$ by

$$\tau_{\pi\pi}^\eta(t) = \pi(t)\tau^\eta(t)\pi^T(t).$$

We can use the same calculation as in (1.2.7) to show that, a.s., for $t \in [0, \infty)$,

$$\pi(t)\tau^\eta(t)\pi^T(t) = (\pi(t) - \eta(t))\sigma(t)(\pi(t) - \eta(t))^T = \eta(t)\tau^\pi(t)\eta^T(t), \quad (1.2.8)$$

so

$$\tau_{\pi\pi}^\eta = \tau_{\eta\eta}^\pi.$$

Lemma 1.2.2 implies that the relative variance of two portfolios is zero if and only if the two portfolios are equal. This would not be the case if $\sigma(t)$ were singular.

The portfolio we now introduce is perhaps the most important portfolio we shall consider.

1.2.3 Definition. The portfolio μ with weights $\mu_1, \dots, \mu_n$ defined by

$$\mu_i(t) = \frac{X_i(t)}{X_1(t) + \cdots + X_n(t)}, \quad t \in [0, \infty), \quad (1.2.9)$$

for $i = 1, \dots, n$, is called the *market portfolio*, and the weights μ_i are called the *market weights*.

For the market portfolio, $0 < \mu_i(t) < 1$, for $t \in [0, \infty)$ and $i = 1, \dots, n$, so it has positive weights for all the stocks. The importance of the market portfolio is derived from its status as the canonical benchmark for equity portfolio performance. It can easily be verified that μ satisfies the requirements of Definition 1.1.4. If we let

$$Z_\mu(t) = X_1(t) + \cdots + X_n(t), \quad t \in [0, \infty), \quad (1.2.10)$$

then $Z_\mu(t)$ satisfies (1.1.12) with proportions $\mu_i(t)$ given by (1.2.9). Hence, the value of the market portfolio represents the combined capitalization of all the stocks in the market.

In recognition of the special status of the market portfolio, we shall reserve the notation μ to represent this portfolio, and $Z_\mu(t)$ in (1.2.10) to represent its value. We shall also use the notation τ to represent τ^μ , τ_{ij} to represent τ_{ij}^μ , and for a portfolio π , we shall use $\tau_{\pi\pi}$ to represent $\tau_{\pi\pi}^\mu$.

Definition 1.2.3 and (1.2.10) imply that the market weight processes μ_i are quotient processes

$$\mu_i(t) = X_i(t)/Z_\mu(t), \quad t \in [0, \infty),$$

for $i = 1, \dots, n$. By Definition 1.2.1, the process

$$\log \mu_i(t) = \log(X_i(t)/Z_\mu(t)), \quad t \in [0, \infty), \quad (1.2.11)$$

represents the relative return of the i th stock versus the market portfolio. From this it follows that

$$d\langle \log \mu_i, \log \mu_j \rangle_t = d\langle \log(X_i/Z_\mu), \log(X_j/Z_\mu) \rangle_t = \tau_{ij}(t) dt, \quad (1.2.12)$$

by (1.2.4), and Itô's rule applied to $\mu_i(t) = \exp(\log \mu_i(t))$ implies that, a.s., for $t \in [0, \infty)$,

$$\begin{aligned} d\mu_i(t) &= \mu_i(t) d\log \mu_i(t) + \frac{1}{2} \mu_i(t) d\langle \log \mu_i \rangle_t \\ &= \mu_i(t) d\log \mu_i(t) + \frac{1}{2} \mu_i(t) \tau_{ii}(t) dt. \end{aligned} \quad (1.2.13)$$

From this we have

$$d\langle \mu_i, \mu_j \rangle_t = \mu_i(t) \mu_j(t) d\langle \log \mu_i, \log \mu_j \rangle_t = \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \quad (1.2.14)$$

for $t \in [0, \infty)$, a.s. Note that relations of the form (1.2.12) and (1.2.14) are unique to the market weights, and similar results cannot be expected to hold for arbitrary portfolio weights.

1.2.4 Definition. For portfolios π and η , the *relative return* process of π versus η is defined by

$$\log(Z_\pi(t)/Z_\eta(t)), \quad t \in [0, \infty).$$

The *relative total return* process is defined by

$$\log(\widehat{Z}_\pi(t)/\widehat{Z}_\eta(t)), \quad t \in [0, \infty).$$

Let us consider the structure of the relative return process. Suppose that π and η are portfolios. Then, a.s., for $t \in [0, \infty)$,

$$d \log Z_\pi(t) = \sum_{i=1}^n \pi_i(t) d \log X_i(t) + \gamma_\pi^*(t) dt,$$

by Corollary 1.1.6, so, a.s., for $t \in [0, \infty)$,

$$d \log(Z_\pi(t)/Z_\eta(t)) = \sum_{i=1}^n \pi_i(t) d \log(X_i(t)/Z_\eta(t)) + \gamma_\pi^*(t) dt. \quad (1.2.15)$$

When $\eta = \mu$, the market portfolio, this equation can be expressed in a particularly useful form.

1.2.5 Proposition. *Let π be a portfolio in the market $\mathcal{M}$. Then*

$$d \log(Z_\pi(t)/Z_\mu(t)) = \sum_{i=1}^n \pi_i(t) d \log \mu_i(t) + \gamma_\pi^*(t) dt, \quad (1.2.16)$$

a.s., for $t \in [0, \infty)$.

Proof. By (1.2.11), for $i = 1, \dots, n$,

$$\log \mu_i(t) = \log(X_i(t)/Z_\mu(t)), \quad t \in [0, \infty),$$

and if we combine this with (1.2.15) for $\eta = \mu$, (1.2.16) follows. $\square$

This proposition shows that we can represent the relative return of a portfolio versus the market portfolio in terms of the changes in the market weights. This relationship is central to the development of the theory of portfolio generating functions in Chapters 3 and 4.

1.3 Portfolio Behavior and Optimization

Traditionally, portfolio theory has emphasized the expected rate of return and variance of a portfolio of stocks. In this section we show that the growth rate rather than the rate of return determines the long-term behavior of a portfolio of stocks, so for long-term investment, it would seem reasonable to consider growth rates rather than rates of return. Let us now show that the portfolio growth rate determines long-term portfolio behavior.

1.3.1 Proposition. *For any portfolio π in $\mathcal{M}$,*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log Z_\pi(T) - \int_0^T \gamma_\pi(t) dt \right) = 0, \quad \text{a.s.} \quad (1.3.1)$$

Proof. By Proposition 1.1.5, a.s., for $t \in [0, \infty)$,

$$\log(Z_\pi(T)/Z_\pi(0)) = \int_0^T \gamma_\pi(t) dt + \int_0^T \sum_{i, \nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t).$$

For $t \in [0, \infty)$, let

$$V(T) = \log(Z_\pi(T)/Z_\pi(0)) - \int_0^T \gamma_\pi(t) dt = \int_0^T \sum_{i, \nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t).$$

Then V is a continuous martingale with

$$\langle V \rangle_t = \int_0^t \sigma_{\pi\pi}(s) ds, \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.3.2)$$

Since the proportions in π are bounded, condition (ii) of Definition 1.1.1 implies that

$$\lim_{t \rightarrow \infty} t^{-1} \sigma_{\pi\pi}(t) \log \log t = 0, \quad \text{a.s.}$$

This and (1.3.2), along with some elementary calculus, imply that Lemma 1.3.2 below can be applied to (1.3.1). The conclusion follows. $\square$

1.3.2 Lemma. *Let M be a continuous local martingale such that*

$$\lim_{t \rightarrow \infty} t^{-2} \langle M \rangle_t \log \log t = 0, \quad \text{a.s.} \quad (1.3.3)$$

Then

$$\lim_{t \rightarrow \infty} t^{-1} M(t) = 0, \quad \text{a.s.}$$

Proof. By extending the measure space Ω if necessary, we can construct a one-dimensional Brownian motion W_0 independent of M , and then we can define

$$M_0(t) = M(t) + W_0(t), \quad t \in [0, \infty).$$

Then M_0 is a continuous local martingale with

$$\langle M_0 \rangle_t = \langle M \rangle_t + t, \quad t \in [0, \infty), \quad \text{a.s.}, \quad (1.3.4)$$

and (1.3.3) and (1.3.4) imply that

$$\lim_{t \rightarrow \infty} t^{-2} \langle M_0 \rangle_t \log \log t = 0, \quad \text{a.s.} \quad (1.3.5)$$

From (1.3.4) we see that

$$\lim_{t \rightarrow \infty} \langle M_0 \rangle_t = \infty, \quad \text{a.s.}, \quad (1.3.6)$$

so the time change theorem for local martingales (Karatzas and Shreve (1991), Theorem 3.4.6) can be applied to show that there exists a Brownian motion B such that

$$B(\langle M_0 \rangle_t) = M_0(t), \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.3.7)$$

Due to (1.3.6), we can apply the law of the iterated logarithm for Brownian motion (Karatzas and Shreve (1991), Theorem 2.9.23), which, along with (1.3.7), implies that

$$\limsup_{t \rightarrow \infty} \frac{|M_0(t)|}{\sqrt{2\langle M_0 \rangle_t \log \log \langle M_0 \rangle_t}} = 1, \quad \text{a.s.} \quad (1.3.8)$$

From (1.3.5) it follows that $\langle M_0 \rangle_t$ grows more slowly than t^2 , so we can replace $\log t$ by $\log \langle M_0 \rangle_t$ in (1.3.5), and we have

$$\lim_{t \rightarrow \infty} t^{-2} \langle M_0 \rangle_t \log \log \langle M_0 \rangle_t = 0, \quad \text{a.s.}$$

Hence,

$$\lim_{t \rightarrow \infty} t^{-1} \sqrt{\langle M_0 \rangle_t \log \log \langle M_0 \rangle_t} = 0, \quad \text{a.s.},$$

and this and (1.3.8) imply that

$$\lim_{t \rightarrow \infty} t^{-1} M_0(t) = 0, \quad \text{a.s.}$$

Since the strong law of large numbers for Brownian motion (Karatzas and Shreve (1991), Problem 2.9.3) implies that

$$\lim_{t \rightarrow \infty} t^{-1} W_0(t) = 0, \quad \text{a.s.},$$

the proposition follows. $\square$

Since single stocks can be considered portfolios, Proposition 1.3.1 also applies to single stocks.

1.3.3 Corollary. *Let X be a stock with growth rate γ . Then*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log X(T) - \int_0^T \gamma(t) dt \right) = 0, \quad \text{a.s.} \quad (1.3.9)$$

Proof. Apply Proposition 1.3.1 to a portfolio in which the weight corresponding to X is 1 and all the other weights are 0. $\square$

Proposition 1.3.1 shows that the portfolio growth rate is an important determinant of portfolio performance, especially over the long term. As we know from Proposition 1.1.5, the portfolio growth rate is the weighted average of the growth rates of the component stocks, plus the excess growth rate of the portfolio. Since the excess growth rate is an essential part of the portfolio growth rate, we need to develop tools to assist in its calculation. The following lemma can be interpreted to imply that the excess growth rate γ_π^* is “numeraire invariant,” and is of particular interest when the numeraire is the market portfolio.

1.3.4 Lemma. *Let π and η be portfolios. Then a.s., for $t \in [0, \infty)$,*

$$\gamma_\pi^*(t) = \frac{1}{2} \left(\sum_{i=1}^n \pi_i(t) \tau_{ii}^\eta(t) - \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}^\eta(t) \right). \quad (1.3.10)$$

Proof. By (1.2.3), a.s., for $t \in [0, \infty)$,

$$\sum_{i=1}^n \pi_i(t) \tau_{ii}^\eta(t) = \sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - 2 \sum_{i=1}^n \pi_i(t) \sigma_{i\eta}(t) + \sigma_{\eta\eta}(t),$$

and

$$\begin{aligned} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}^\eta(t) &= \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) - \sum_{i=1}^n \pi_i(t) \sigma_{i\eta}(t) \\ &\quad - \sum_{j=1}^n \pi_j(t) \sigma_{j\eta}(t) + \sigma_{\eta\eta}(t), \end{aligned}$$

and (1.3.10) follows. $\square$

Let us consider an example to see how this lemma can be used in portfolio optimization.

1.3.5 Example. (Portfolio optimization II) Certain stock portfolios, variously called “risk-controlled” portfolios or “enhanced index” portfolios, are constructed to maintain a low relative variance with a particular index or benchmark portfolio. The “tracking error” of such a portfolio is the square root of the relative variance, and is typically held to about 2% a year. Hence, optimization in this case should minimize the tracking error, or equivalently, the relative variance of the portfolio versus the benchmark.

Suppose that η is a benchmark index, and we use (1.3.10) to represent γ_π^* , and then we optimize the portfolio by minimizing the variance of the portfolio relative to the benchmark,

$$\sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}^\eta(t), \quad (1.3.11)$$

under the constraints

$$\sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \sum_{i=1}^n \pi_i(t) \tau_{ii}^\eta(t) \geq \gamma_0 + \frac{1}{2} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}^\eta(t) \quad (1.3.12)$$

and

$$\pi_1(t) + \cdots + \pi_n(t) = 1 \quad \text{with} \quad \pi_1(t), \dots, \pi_n(t) \geq 0.$$

Lemma 1.3.4 implies that (1.3.12) is equivalent to (1.1.29) in Example 1.1.7,

$$\sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) \geq \gamma_0 + \frac{1}{2} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t).$$

The last term in (1.3.12) is half the square of the tracking error, so this term will be about 0.02% a year if the tracking error is held to about 2%. In practice, the inaccuracy induced by ignoring 0.02% a year in the estimated portfolio growth rate is likely to be inconsequential. Hence, we can delete the last term in (1.3.12), and we have

$$\sum_{i=1}^n \pi_i(t) \gamma_i(t) + \frac{1}{2} \sum_{i=1}^n \pi_i(t) \tau_{ii}^\eta(t) \geq \gamma_0. \quad (1.3.13)$$

Since this is linear in π , conventional quadratic programming can be used for optimization with this constraint. $\square$

The next lemma simplifies the expression for the excess growth rate of a portfolio.

1.3.6 Lemma. *Let π be a portfolio. Then a.s., for $t \in [0, \infty)$,*

$$\gamma_\pi^*(t) = \frac{1}{2} \sum_{i=1}^n \pi_i(t) \tau_{ii}^\pi(t). \quad (1.3.14)$$

Proof. Let $\eta = \pi$ in Lemma 1.3.4. From (1.2.8) we have

$$\pi(t) \tau^\pi(t) \pi^T(t) = (\pi(t) - \pi(t)) \sigma(t) (\pi(t) - \pi(t))^T = 0,$$

so (1.3.10) reduces to (1.3.14). $\square$

This lemma allows us to draw some conclusions about the positivity of the excess growth rate.

1.3.7 Proposition. *Let π be a portfolio with nonnegative weights. Then*

$$\gamma_\pi^*(t) \geq 0, \quad t \in [0, \infty), \quad \text{a.s.} \quad (1.3.15)$$

If for all i , $0 \leq \pi_i(t) < 1$, for all $t \in [0, \infty)$, then

$$\gamma_\pi^*(t) > 0, \quad t \in [0, \infty), \quad \text{a.s.}$$

Proof. Since by (1.2.5), $\tau_{ii}^\pi(t) \geq 0, t \in [0, \infty)$, a.s., for all i , all of the terms on the right-hand side of (1.3.14) are nonnegative, a.s. Therefore, (1.3.15) holds.

The condition $0 \leq \pi_i(t) < 1$ for all $t \in [0, \infty)$ implies that at least two of the weights $\pi_1(t), \dots, \pi_i(t)$ are positive. Lemma 1.2.2 implies that for $t \in [0, \infty)$, $\tau^\pi(t)$ is positive semidefinite with rank $n - 1$, a.s. For $i = 1, \dots, n$ and $t \in [0, \infty)$,

$$\tau_{ii}^\pi(t) = (0, \dots, 1, \dots, 0) \tau^\pi(t) (0, \dots, 1, \dots, 0)^T \geq 0, \quad (1.3.16)$$

where the i th coordinate of $(0, \dots, 1, \dots, 0) \in \mathbb{R}^n$ is 1 and the other coordinates are all 0. Since $\tau^\pi(t)$ has rank $n - 1$, equality can hold in (1.3.16) for at most one value of i , and since at least two of $\pi_1(t), \dots, \pi_n(t)$ are positive for any $t \in [0, \infty)$, the right-hand side of (1.3.14) is positive. $\square$

This proposition implies that for multistock portfolios with no short sales, the portfolio growth rate always exceeds the weighted average of the growth rates of the component stocks. In a market where all the stock growth rates are equal, all multistock portfolios without short sales have a higher growth rate than the common growth rate of the component stocks.

1.3.8 Example. (Portfolio optimization III) In practice, it is quite difficult, if not impossible, to accurately forecast the growth rates $\gamma_i(t)$. Moreover, it may not be unreasonable to assume that over the long term most stocks will have about the same average growth rate, at least for the stocks in a large-stock index η . If all the stocks share a common growth rate $\gamma(t)$, then Lemma 1.3.6 implies that the growth rate for the benchmark index η is

$$\gamma_\eta(t) = \gamma(t) + \frac{1}{2} \sum_{i=1}^n \eta_i(t) \tau_{ii}^\eta(t), \quad t \in [0, \infty), \quad \text{a.s.}$$

If the tracking error of π relative to η can be assumed to be small enough, as it was in Example 1.3.5, then the same reasoning as in that example implies that minimizing

$$\sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}^\eta(t)$$

under the constraints

$$\frac{1}{2} \sum_{i=1}^n \pi_i(t) \tau_{ii}^\eta(t) \geq \gamma_0 + \frac{1}{2} \sum_{i=1}^n \eta_i(t) \tau_{ii}^\eta(t), \quad (1.3.17)$$

where $\gamma_0 > 0$ is constant, and

$$\pi_1(t) + \dots + \pi_n(t) = 1 \quad \text{with} \quad \pi_1(t), \dots, \pi_n(t) \geq 0,$$

produces a portfolio with growth rate about γ_0 greater than the benchmark's growth rate. The constraint in (1.3.17) is linear, so conventional quadratic programming can be used. This optimization involves only the covariance process τ^η , so it is not necessary to predict any future growth rates or rates of return. $\square$

1.4 Notes and Summary

The continuous-time model for a stock price process has evolved over the years, beginning with Bachelier (1900) and Samuelson (1965). The logarithmic model we use was first presented in Fernholz and Shay (1982), and is equivalent to the standard model of Merton (1969) (see also Merton (1990) and Karatzas and Shreve (1998)). The logarithmic model is advantageous for analyzing long-term or asymptotic events, because the log-price processes resemble ordinary linear random walks rather than the exponential random walks of the standard representation.

The conditions in Definition 1.1.2 may be reminiscent of conditions that imply that $\mathcal{M}$ is *complete*, but are actually not sufficient to imply market completeness (see Karatzas and Shreve (1998)). The nondegeneracy condition (1.1.10) is somewhat stronger than nonsingularity, and is also fairly common in the literature. It can be found, for example, in Karatzas and Shreve (1991), Karatzas and Kou (1996), and as *uniform ellipticity* in Duffie (1992).

Definition 1.1.4 differs from the traditional definition of portfolio, since the traditional definition gives the value of the investment in each stock rather than the weights (see, e.g., Karatzas and Shreve (1998)). Because of this difference, we must separately calculate the value of the portfolio.

Equation (1.1.12) is a classical equation that dates back to the original work of Markowitz (1952) in discrete time and Merton (1969) in continuous time. Equation (1.1.25) is the classical representation for the portfolio rate of return, also from Markowitz (1952) and Merton (1969).

With regard to Example 1.1.8, Breiman (1961) showed that maximizing the expected value of $\log Z_\pi(t)$ will a.s. produce the portfolio with the greatest asymptotic value. Merton and Samuelson (1974) and Samuelson (1979) argued that such portfolios carry too high a level of risk for most investors. A discussion of logarithmic utility can be found in Karatzas and Shreve (1998).

The market portfolio was first presented in our current setting, Definition 1.2.3, in Fernholz (1999a). In the classical theory of capital markets, the importance of the market portfolio was derived from the renowned capital asset pricing model (CAPM) of Sharpe (1964), which endowed this portfolio with remarkable return characteristics (see also Karatzas and Shreve (1998)).

Chapter Summary

Section 1.1: With the logarithmic model, a family of stocks, i.e., stock price processes, X_i , $i = 1, \dots, n$, is defined by

$$d \log X_i(t) = \gamma_i(t) dt + \sum_{\nu=1}^n \xi_{i\nu}(t) dW_\nu(t), \quad t \in [0, \infty), \quad (1.1.6)$$

where the coefficients satisfy the conditions specified in Definition 1.1.1. We assume that each company has a single share of stock outstanding, so X_i represents the total capitalization of the company. The *growth rate* process γ_i for the i th stock is related to the *rate of return* process α_i in the standard representation by

$$\alpha_i(t) = \gamma_i(t) + \frac{\sigma_{ii}(t)}{2}, \quad t \in [0, \infty), \quad (1.1.4)$$

where $\sigma_{ii}(t) = \xi_{i1}^2(t) + \dots + \xi_{in}^2(t)$ is the *variance process* of the stock.

A *market* $\mathcal{M}$ is a family of stocks of the form (1.1.6) that satisfies certain regularity conditions. A portfolio π is represented by its weights in each stock, $\pi_i(t), \dots, \pi_n(t)$, at time t , and the weights are bounded and sum to 1. The expression for portfolio growth rate is somewhat more complicated under the logarithmic representation than under the standard representation, where the portfolio rate of return is a simple weighted average of the rates of return of the stocks. Under the logarithmic representation, if we let $Z_\pi(t)$ represent the value of π at time t , then

$$d \log Z_\pi(t) = \gamma_\pi(t) dt + \sum_{i,\nu=1}^n \pi_i(t) \xi_{i\nu}(t) dW_\nu(t), \quad (1.1.13)$$

a.s., for $t \in [0, \infty)$. In (1.1.13)

$$\gamma_\pi(t) = \sum_{i=1}^n \pi_i(t) \gamma_i(t) + \gamma_\pi^*(t) \quad (1.1.21)$$

is the *portfolio growth rate*, and

$$\gamma_\pi^*(t) = \frac{1}{2} \left(\sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) \right),$$

with $\sigma_{ij}(t) = \xi_{i1}(t) \xi_{j1}(t) + \dots + \xi_{in}(t) \xi_{jn}(t)$, is called the *excess growth rate* of the portfolio. The excess growth rate would seem to be an unwelcome complication in the expression for portfolio return. However, it is precisely this complication that provides insight into aspects of portfolio behavior that remain obscure under the standard representation.

Section 1.2: The *market portfolio* μ is defined by the *market weights* $\mu_1, \dots, \mu_n$ defined by

$$\mu_i(t) = \frac{X_i(t)}{X_1(t) + \dots + X_n(t)}, \quad t \in [0, \infty), \quad (1.2.9)$$

for $i = 1, \dots, n$. The market portfolio is the canonical performance benchmark for all other portfolios. The value Z_μ of the market portfolio satisfies

$$Z_\mu(t) = X_1(t) + \dots + X_n(t), \quad t \in [0, \infty), \quad \text{a.s.}, \quad (1.2.10)$$

so the market weights are quotient processes,

$$\mu_i(t) = X_i(t)/Z_\mu(t), \quad t \in [0, \infty), \quad \text{a.s.},$$

for $i = 1, \dots, n$.

For any portfolio π , it is important to consider the performance of the portfolio relative to the market, and this satisfies

$$d \log(Z_\pi(t)/Z_\mu(t)) = \sum_{i=1}^n \pi_i(t) d \log \mu_i(t) + \gamma_\pi^*(t) dt, \quad (1.2.16)$$

a.s., for $t \in [0, \infty)$. Hence, we can analyze the relative portfolio performance in terms of the changing market weights and the excess growth rate, so it will be important for us to develop an understanding of the behavior of both the market weights and the excess growth rate.

Section 1.3: The growth rate of a portfolio determines its long-term behavior, in the sense that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log Z_\pi(T) - \int_0^T \gamma_\pi(t) dt \right) = 0, \quad \text{a.s.} \quad (1.3.1)$$

Since a stock can be considered a portfolio holding a single stock, a relation of the form (1.3.1) also holds for stocks.

Since the growth rate of a portfolio determines its long-term behavior, the growth rate is an important parameter in optimization. In portfolios of large stocks, it may be reasonable to assume that all the stocks have about the same growth rate. In this case, the first term in the expression for the growth rate in (1.1.21) will equal this common growth rate, and the portfolio growth rate will depend only on the excess growth rate. Therefore, for large-stock portfolios, the excess growth rate is a critical optimization parameter.

2

Stock Market Behavior and Diversity

In this chapter we study the diversity of the distribution of capital in an equity market. Heuristically speaking, a market is “diverse” if the capital is spread among a reasonably large number of stocks. We show that the excess growth rate of the market is related to the diversity of the capital distribution, and we use this relationship to study the long-term behavior of market diversity under the hypothesis that all the stocks have the same growth rate. It might seem that in such a market, diversity would naturally be maintained, but we shall see that this is not so, and in fact, such markets have a tendency to concentrate capital into single stocks. Dividend payments are a natural means to maintain market diversity, and we investigate the structure of this mechanism. Finally, we propose market *entropy* as a measure of market diversity, and study a derived portfolio called the *entropy-weighted* portfolio.

To analyze the long-term behavior of stocks, portfolios, or the market itself, it is appropriate that we consider the time-average values rather than the expected values of the processes under consideration. In practice, we are able to observe the time-average value, whereas the expected value is merely a theoretical construct. Hence, for the growth rate γ_i of a stock X_i , we shall consider

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \gamma_i(t) dt$$

rather than $E\gamma_i(t)$. Likewise, for a market weight μ_i , we shall study

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log \mu_i(t) dt$$

rather than $E \log \mu_i(t)$.

2.1 The Long-Term Behavior of the Market

In this section we shall investigate the long-term relative performance of the stocks in the market. This will also allow us to characterize the long-term behavior of certain simple portfolios. For some of the results here, we need to impose a structural condition on the market.

2.1.1 Definition. The market $\mathcal{M}$ is *coherent* if for $i = 1, \dots, n$,

$$\lim_{t \rightarrow \infty} t^{-1} \log \mu_i(t) = 0, \quad \text{a.s.} \quad (2.1.1)$$

Since $\log \mu_i(t) < 0$, condition (2.1.1) holds if none of the stocks declines too rapidly. Note that since $\mu_i(t) = X_i(t)/Z_\mu(t)$, (2.1.1) is equivalent to

$$\lim_{t \rightarrow \infty} t^{-1} (\log X_i(t) - \log Z_\mu(t)) = 0, \quad \text{a.s.} \quad (2.1.2)$$

2.1.2 Proposition. Let $\mathcal{M}$ denote the market with stocks $X_1, \dots, X_n$. Then the following statements are equivalent:

(i) $\mathcal{M}$ is coherent;

(ii) for $i = 1, \dots, n$, $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\gamma_i(t) - \gamma_\mu(t)) dt = 0, \quad \text{a.s.};$

(iii) for $i, j = 1, \dots, n$, $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\gamma_i(t) - \gamma_j(t)) dt = 0, \quad \text{a.s.}$

Proof. We shall prove that (i) implies (ii) implies (iii) implies (i).

Suppose $\mathcal{M}$ is coherent. Then (2.1.2) states that for $i = 1, \dots, n$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} (\log X_i(T) - \log Z_\mu(T)) = 0, \quad \text{a.s.}$$

By Proposition 1.3.1,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log Z_\mu(T) - \int_0^T \gamma_\mu(t) dt \right) = 0, \quad \text{a.s.},$$

and by Corollary 1.3.3,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log X_i(T) - \int_0^T \gamma_i(t) dt \right) = 0, \quad \text{a.s.}$$

These three equations imply condition (ii).

Condition (iii) follows immediately from condition (ii).

Now, suppose that condition (iii) holds. It is convenient here to explicitly show the dependence of all random variables and processes on $\omega \in \Omega$.

Corollary 1.3.3 and condition (iii) imply that there is a subset $\Omega' \subset \Omega$ with $P(\Omega') = 1$ such that for $\omega \in \Omega'$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log X_i(T, \omega) - \int_0^T \gamma_i(t, \omega) dt \right) = 0, \quad (2.1.3)$$

for $i = 1, \dots, n$, and

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\gamma_i(t, \omega) - \gamma_j(t, \omega)) dt = 0, \quad (2.1.4)$$

for $i, j = 1, \dots, n$.

Let us choose $\omega \in \Omega'$. Then (2.1.3) and (2.1.4) with $j = 1$ imply that for $i = 1, \dots, n$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log X_i(T, \omega) - \int_0^T \gamma_1(t, \omega) dt \right) = 0. \quad (2.1.5)$$

Hence, for $i = 1, \dots, n$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\max_{1 \leq i \leq n} (\log X_i(T, \omega)) - \int_0^T \gamma_1(t, \omega) dt \right) = 0,$$

which is equivalent to

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log \left(\max_{1 \leq i \leq n} X_i(T, \omega) \right) - \int_0^T \gamma_1(t, \omega) dt \right) = 0, \quad (2.1.6)$$

for $i = 1, \dots, n$. Now, for $t \in [0, \infty)$,

$$X_1(t, \omega) \leq X_1(t, \omega) + \dots + X_n(t, \omega) \leq n \max_{1 \leq i \leq n} X_i(t, \omega),$$

so, for $t \in [0, \infty)$,

$$\log X_1(t, \omega) \leq \log Z_\mu(t, \omega) \leq \log n + \log \left(\max_{1 \leq i \leq n} X_i(t, \omega) \right). \quad (2.1.7)$$

Since

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log n = 0,$$

it follows from (2.1.5), (2.1.6), and (2.1.7) that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log Z_\mu(T, \omega) - \int_0^T \gamma_1(t, \omega) dt \right) = 0. \quad (2.1.8)$$

From (2.1.5) and (2.1.8), we have

$$\lim_{T \rightarrow \infty} \frac{1}{T} (\log X_i(T, \omega) - \log Z_\mu(T, \omega)) = 0,$$

and since this holds for any $\omega \in \Omega'$, $\mathcal{M}$ is coherent by (2.1.2). $\square$

This proposition means that in a coherent market, the time-average difference between the growth rates of any two stocks will be zero. Note that this pertains only to the differences; the time average of the growth rate of an individual stock may not exist. An example of a coherent market is one in which all the stocks have the same growth rate process.

2.1.3 Corollary. *Suppose that all the stocks in the market $\mathcal{M}$ have the same growth rate process. Then $\mathcal{M}$ is coherent.*

Proof. If all the stocks have the same growth rate process, then (iii) of Proposition 2.1.2 holds. Hence $\mathcal{M}$ is coherent. $\square$

For the case that the growth rates of the stocks are constant, the converse of this corollary also holds.

2.1.4 Corollary. *Suppose that all the stocks in the market $\mathcal{M}$ have constant growth rates. Then $\mathcal{M}$ is coherent if and only if the growth rates are all equal.*

Proof. If the growth rates are all equal, $\mathcal{M}$ is coherent by Corollary 2.1.3. If X_i and X_j have different constant growth rates, then (iii) of Proposition 2.1.2 will fail, so $\mathcal{M}$ is not coherent. $\square$

To proceed from here, we need to prove several lemmas that relate the size of the weights of a portfolio to its excess growth rate. The first one establishes a lower bound on the relative variances defined in (1.2.3).

2.1.5 Lemma. *Let π be a portfolio in a nondegenerate market. Then there exists an $\varepsilon > 0$ such that for $i = 1, \dots, n$,*

$$\tau_{ii}^\pi(t) \geq \varepsilon(1 - \pi_i(t))^2, \quad t \in [0, \infty), \quad \text{a.s.} \quad (2.1.9)$$

Proof. Let $\varepsilon > 0$ be chosen as in (1.1.10) so that

$$x\sigma(t)x^T \geq \varepsilon \|x\|^2, \quad x \in \mathbb{R}^n, t \in [0, \infty), \quad \text{a.s.} \quad (2.1.10)$$

For $1 \leq i \leq n$ and $t \in [0, \infty)$, let $x(t) = (\pi_1(t), \dots, \pi_i(t) - 1, \dots, \pi_n(t))$. Then, a.s., for $t \in [0, \infty)$,

$$\tau_{ii}^\pi(t) = \sigma_{ii}(t) - 2\sigma_{i\pi}(t) + \sigma_{\pi\pi}(t) = x(t)\sigma(t)x^T(t) \geq \varepsilon \|x(t)\|^2,$$

by (2.1.10). Since,

$$\|x(t)\|^2 \geq (1 - \pi_i(t))^2, \quad t \in [0, \infty), \quad \text{a.s.,}$$

the lemma follows. $\square$

For a portfolio π , it is convenient to introduce the notation

$$\pi_{\max}(t) = \max_{1 \leq i \leq n} \pi_i(t), \quad t \in [0, \infty). \quad (2.1.11)$$

With this notation, we can restate Lemma 2.1.5 in its most useful form.

2.1.6 Lemma. *Let π be a portfolio in a nondegenerate market. Then there exists an $\varepsilon > 0$ such that for $i = 1, \dots, n$,*

$$\tau_{ii}^\pi(t) \geq \varepsilon(1 - \pi_{\max}(t))^2, \quad t \in [0, \infty), \quad \text{a.s.} \quad (2.1.12)$$

Proof. This follows immediately from Lemma 2.1.5. $\square$

The next lemma strengthens Proposition 1.3.7 for nondegenerate markets.

2.1.7 Lemma. *Let π be a portfolio with nonnegative weights in a nondegenerate market. Then there exists an $\varepsilon > 0$ such that a.s., for $t \in [0, \infty)$,*

$$\gamma_\pi^*(t) \geq \varepsilon(1 - \pi_{\max}(t))^2. \quad (2.1.13)$$

Proof. By Lemma 1.3.6, a.s., for $t \in [0, \infty)$,

$$\gamma_\pi^*(t) = \frac{1}{2} \sum_{i=1}^n \pi_i(t) \tau_{ii}^\pi(t) \geq \frac{\varepsilon}{2} (1 - \pi_{\max}(t))^2,$$

where ε is chosen as in Lemma 2.1.6, since the $\pi_i(t)$ are nonnegative. $\square$

Lemma 2.1.7 shows that in a nondegenerate market, if $\pi_{\max}(t)$ is bounded away from 1, then $\gamma_\pi^*(t)$ is bounded away from 0. The next lemma shows that in a market with bounded variance, if $\gamma_\pi^*(t)$ is bounded away from 0, then $\pi_{\max}(t)$ is bounded away from 1.

2.1.8 Lemma. *Let π be a portfolio in a market with bounded variance such that for $i = 1, \dots, n$, $0 \leq \pi_i(t) < 1$, for all $t \in [0, \infty)$, a.s. Then there exists a number $\varepsilon > 0$ such that*

$$\pi_{\max}(t) \leq 1 - \varepsilon \gamma_\pi^*(t), \quad t \in [0, \infty), \quad \text{a.s.} \quad (2.1.14)$$

Proof. Since the market is assumed to have bounded variance, we can choose M as in (1.1.11) so that

$$x\sigma(t)x^T \leq M \|x\|^2, \quad x \in \mathbb{R}^n, t \in [0, \infty), \quad \text{a.s.} \quad (2.1.15)$$

Hence, for $1 \leq i \leq n$,

$$\sigma_{ii}(t) \leq M, \quad t \in [0, \infty), \quad \text{a.s.} \quad (2.1.16)$$

For any integer k , $1 \leq k \leq n$, $\pi_k(t) < 1$, so we can define

$$\eta_i(t) = \begin{cases} \pi_i(t)/(1 - \pi_k(t)) & \text{if } i \neq k, \\ 0 & \text{if } i = k, \end{cases} \quad (2.1.17)$$

for $t \in [0, \infty)$, $i = 1, \dots, n$. Then $(\eta_1(t), \dots, \eta_n(t))$ defines a portfolio η with nonnegative weights, and (2.1.16) implies that, a.s., for $t \in [0, \infty)$,

$$\sum_{i=1}^n \eta_i(t) \sigma_{ii}(t) - \sigma_{\eta\eta}(t) \leq \sum_{i=1}^n \eta_i(t) \sigma_{ii}(t) \leq M. \quad (2.1.18)$$

Let

$$x = (\eta_1(t), \dots, \eta_{k-1}(t), -1, \eta_{k+1}(t), \dots, \eta_n(t)). \quad (2.1.19)$$

Then $\|x\|^2 \leq 2$, so for $k = 1, \dots, n$,

$$\sigma_{kk}(t) - 2\sigma_{k\eta}(t) + \sigma_{\eta\eta}(t) = x\sigma(t)x^T \leq 2M, \quad (2.1.20)$$

$t \in [0, \infty)$, a.s., by (2.1.15). By (1.1.20), a.s., for $t \in [0, \infty)$,

$$\begin{aligned} 2\gamma_\pi^*(t) &= \sum_{i=1}^n \pi_i(t) \sigma_{ii}(t) - \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \sigma_{ij}(t) \\ &= \pi_k(t) \sigma_{kk}(t) + (1 - \pi_k(t)) \sum_{i=1}^n \eta_i(t) \sigma_{ii}(t) \\ &\quad - \pi_k^2(t) \sigma_{kk}(t) - 2\pi_k(t) (1 - \pi_k(t)) \sum_{i=1}^n \eta_i(t) \sigma_{ik}(t) \end{aligned} \quad (2.1.21)$$

$$\begin{aligned} &\quad - (1 - \pi_k(t))^2 \sum_{i,j=1}^n \eta_i(t) \eta_j(t) \sigma_{ij}(t) \\ &= (\pi_k(t) - \pi_k^2(t)) (\sigma_{kk}(t) - 2\sigma_{k\eta}(t) + \sigma_{\eta\eta}(t)) \\ &\quad + (1 - \pi_k(t)) \left(\sum_{i=1}^n \eta_i(t) \sigma_{ii}(t) - \sigma_{\eta\eta}(t) \right) \\ &\leq (1 - \pi_k(t)) (2M + M), \end{aligned} \quad (2.1.22)$$

where (2.1.21) follows from (2.1.17), and (2.1.22) is implied by (2.1.18) and (2.1.20). Since (2.1.22) holds for all k , $1 \leq k \leq n$, (2.1.14) follows with $\varepsilon = 2/(3M)$. $\square$

A portfolio π is *constant-weighted* if the weight processes π_i are all constant in t . The next proposition gives some insight into the behavior of constant-weighted portfolios.

2.1.9 Proposition. *Suppose that the market $\mathcal{M}$ is nondegenerate and coherent, and that π is a constant-weighted portfolio with at least two positive weights and no negative weights. Then*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \log(Z_\pi(T)/Z_\mu(T)) > 0, \quad \text{a.s.}$$

Proof. Suppose that π is constant-weighted with

$$\pi_i(t) = p_i, \quad t \in [0, \infty),$$

for $i = 1, \dots, n$, where the p_i are nonnegative constants that sum to one with $1 > p = \max_{1 \leq i \leq n} p_i$. Since $\mathcal{M}$ is nondegenerate, Lemma 2.1.7 implies that there exists an $\varepsilon > 0$ such that

$$\gamma_\pi^*(t) \geq \varepsilon(1-p)^2, \quad t \in [0, \infty), \quad \text{a.s.},$$

so,

$$\frac{1}{T} \int_0^T \gamma_\pi^*(t) dt \geq \varepsilon(1-p)^2, \quad T \in [0, \infty), \quad \text{a.s.} \quad (2.1.23)$$

By (1.2.16),

$$d \log(Z_\pi(t)/Z_\mu(t)) = \sum_{i=1}^n p_i d \log \mu_i(t) + \gamma_\pi^*(t) dt,$$

a.s., for $t \in [0, \infty)$. Therefore, a.s.,

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T} \left(\log(Z_\pi(T)/Z_\mu(T)) - \int_0^T \gamma_\pi^*(t) dt \right) &= \lim_{T \rightarrow \infty} \frac{1}{T} \left(\sum_{i=1}^n p_i \log \mu_i(T) \right) \\ &= \sum_{i=1}^n p_i \left(\lim_{T \rightarrow \infty} \frac{1}{T} \log \mu_i(T) \right) \\ &= 0, \end{aligned}$$

by 2.1.1, since $\mathcal{M}$ is coherent. This and (2.1.23) imply that

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \log(Z_\pi(T)/Z_\mu(T)) \geq \varepsilon(1-p)^2, \quad \text{a.s.}$$

□

2.2 Stock Market Diversity

In this section we give a formal definition of market diversity, and we show that diversity can be characterized in terms of the excess growth rate of the market. We use this relationship to determine market conditions that are compatible with market diversity.

All economically developed nations have some form of antitrust legislation to prevent the excessive concentration of capital and economic power in a few giant corporations. Here we are not concerned with the economic rationale for antitrust legislation, but rather with the effect such legislation may have on the distribution of capital in the equity market. Any credible

antitrust law should prevent prolonged concentration of practically all the market capital into a single company, and from a realistic point of view, in an economy such as that of the U.S., it is unlikely that a single company could account for even half of the total market capitalization. The condition we impose in the following definition is a weak consequence of actual antitrust laws, and any market model bearing even a remote resemblance to the U.S. equity market can safely be assumed to satisfy it.

Recall that $\mu_{\max}$ represents the value of the largest of the market weights at a given time, as in (2.1.11).

2.2.1 Definition. The market $\mathcal{M}$ is *diverse* if there exists a number $\delta > 0$ such that

$$\mu_{\max}(t) \leq 1 - \delta, \quad t \in [0, \infty), \quad \text{a.s.}$$

$\mathcal{M}$ is *weakly diverse* on $[0, T]$ if there exists a number $\delta > 0$ such that

$$\frac{1}{T} \int_0^T \mu_{\max}(t) dt \leq 1 - \delta, \quad \text{a.s.} \quad (2.2.1)$$

By this definition, a market is diverse if at no time a single stock accounts for almost the entire market capitalization, and is weakly diverse if this holds on average over history. These are fairly weak empirical requirements, and it is clear that actual equity markets of any importance satisfy both of these conditions. Nevertheless, we shall see that market diversity has strong mathematical consequences.

The lemmas in Section 2.1 allow us to characterize diversity in terms of the excess growth rate of the market portfolio.

2.2.2 Proposition. *If the market $\mathcal{M}$ is nondegenerate and diverse, then there is a $\delta > 0$ such that*

$$\gamma_{\mu}^*(t) \geq \delta, \quad t \in [0, \infty), \quad \text{a.s.} \quad (2.2.2)$$

Conversely, if $\mathcal{M}$ has bounded variance and there exists a $\delta > 0$ such that (2.2.2) holds, then $\mathcal{M}$ is diverse.

Proof. Suppose $\mathcal{M}$ is nondegenerate and diverse, so there is a $\delta > 0$ such that

$$\mu_{\max}(t) \leq 1 - \delta, \quad t \in [0, \infty), \quad \text{a.s.}$$

Since $\mathcal{M}$ is nondegenerate, Lemma 2.1.7 implies that we can choose $\varepsilon > 0$ such that

$$\gamma_{\mu}^*(t) \geq \varepsilon(1 - \mu_{\max}(t))^2, \quad t \in [0, \infty), \quad \text{a.s.}$$

Therefore,

$$\gamma_{\mu}^*(t) \geq \varepsilon\delta^2, \quad t \in [0, \infty), \quad \text{a.s.},$$

and (2.2.2) follows.

Now suppose that $\mathcal{M}$ has bounded variance and there exists $\delta > 0$ such that (2.2.2) holds. Since $\mathcal{M}$ has bounded variance, Lemma 2.1.8 implies that we can choose $\varepsilon > 0$ such that, a.s., for $t \in [0, \infty)$,

$$\mu_{\max}(t) \leq 1 - \varepsilon \gamma_{\mu}^*(t) \leq 1 - \varepsilon \delta,$$

and hence $\mathcal{M}$ is diverse. $\square$

Proposition 1.3.1 and Corollary 1.3.3 show that the long-term behavior both of portfolios and of stocks is determined by their growth rates. If all the stocks in the market have the same growth rate, then (1.1.21) implies that the growth rate of the market portfolio is

$$\gamma_{\mu}(t) = \gamma(t) + \gamma_{\mu}^*(t), \quad t \in [0, \infty), \quad \text{a.s.} \quad (2.2.3)$$

If the stocks in the market all have the same growth rate, Corollary 2.1.3 states that the market will be coherent, and Proposition 2.1.2(ii) implies that, asymptotically, its growth rate will be the same as the common growth rate of the stocks. It follows that, over the long term, the contribution of $\gamma_{\mu}^*(t)$ to $\gamma_{\mu}(t)$ in (2.2.3) must be minimal.

2.2.3 Proposition. *Suppose that all the stocks in the market $\mathcal{M}$ have the same growth rate. Then*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \gamma_{\mu}^*(t) dt = 0, \quad \text{a.s.} \quad (2.2.4)$$

Proof. Since all the growth rates of the stocks are equal, Corollary 2.1.3 states that $\mathcal{M}$ is coherent. By Proposition 2.1.2(ii),

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\gamma(t) - \gamma_{\mu}(t)) dt = 0, \quad \text{a.s.},$$

and this and (2.2.3) imply (2.2.4). $\square$

Proposition 2.2.3 shows that over the long term, the average excess growth rate of this market is asymptotically negligible. The following corollary shows that this has implications regarding the diversity of the market.

2.2.4 Corollary. *Suppose that the market $\mathcal{M}$ is nondegenerate. If all the stocks in $\mathcal{M}$ have the same growth rate, then $\mathcal{M}$ is not diverse.*

Proof. If $\mathcal{M}$ is diverse, Proposition 2.2.2 implies that there exists a $\delta > 0$ such that $\gamma_{\mu}^*(t) \geq \delta$, a.s., for $t \in [0, \infty)$. In this case

$$\frac{1}{T} \int_0^T \gamma_{\mu}^*(t) dt \geq \delta, \quad T \in [0, \infty), \quad \text{a.s.}$$

But this contradicts Proposition 2.2.3. $\square$

2.2.5 Problem. (!!) For a nondegenerate market in which all the stocks have the same growth rate, calculate $E(\mu_{\max}(t))$ as a function of time.

2.2.6 Problem. (!!) What can be said about the stochastics of changes in leadership in a nondegenerate market in which all the stocks have the same growth rate?

2.2.7 Corollary. *Suppose that the market $\mathcal{M}$ is nondegenerate. If all the stocks in $\mathcal{M}$ have constant growth rates, then $\mathcal{M}$ is not diverse.*

Proof. Corollary 1.3.3 implies that all stocks except those that share the highest growth rate will represent a negligible part of the market value in the long term. But then the (sub)market composed of the stocks that share the highest growth rate satisfies the hypotheses of the previous corollary, and hence is not diverse. $\square$

This corollary implies that, in some sense, an equity market with constant growth rates and covariances is unstable and has a tendency to concentrate essentially into a single stock. Although these Corollaries 2.2.4 and 2.2.7 show that a common growth rate among the stocks in a market is not sufficient to maintain market diversity, we should be able to maintain diversity if we allow companies to redistribute capital in some manner. Dividend payments are a means of redistributing capital, and any such redistribution can be considered to be a dividend in a generalized sense. Accordingly, let us consider a market in which there are nonnegative dividend rates and all the stocks have the same augmented growth rate. This models a situation in which all the companies have the same potential for capital growth, but some of the companies elect to distribute part of their capital in the form of dividends rather than reinvesting in themselves.

2.2.8 Proposition. *Suppose that all the stocks in the market $\mathcal{M}$ have nonnegative dividend rates and the same augmented growth rate. Then*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\gamma_\mu^*(t) - \delta_\mu(t)) dt \leq 0. \quad \text{a.s.} \quad (2.2.5)$$

Proof. Let ρ be the common augmented growth rate process. Define the process W by

$$W(t) = \hat{X}_1(t) + \cdots + \hat{X}_n(t), \quad t \in [0, \infty).$$

The process W represents the value of a portfolio with $W(0) = Z_\mu(0)$ in which the dividends of each stock are reinvested in the same stock. Since all the $\hat{X}_i$ have the same augmented growth rate ρ , the same steps as in the proof of Proposition 1.3.1 establish that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log W(T) - \int_0^T \rho(t) dt \right) = 0, \quad \text{a.s.}$$

Since for all i , $X_i(t) \leq \widehat{X}_i(t)$, a.s., for $t \in [0, \infty)$, it follows that

$$Z_\mu(t) \leq W(t), \quad t \in [0, \infty), \quad \text{a.s.},$$

and hence,

$$\limsup_{t \rightarrow \infty} \frac{1}{T} \left(\log Z_\mu(T) - \int_0^T \rho(t) dt \right) \leq 0, \quad \text{a.s.} \quad (2.2.6)$$

By Proposition 1.3.1,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log Z_\mu(T) - \int_0^T \gamma_\mu(t) dt \right) = 0, \quad \text{a.s.} \quad (2.2.7)$$

Equations (2.2.6) and (2.2.7) imply that

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\gamma_\mu(t) - \rho(t)) dt \leq 0, \quad \text{a.s.} \quad (2.2.8)$$

Now,

$$\rho_\mu(t) = \rho(t) + \gamma_\mu^*(t), \quad t \in [0, \infty), \quad \text{a.s.},$$

and also

$$\rho_\mu(t) = \gamma_\mu(t) + \delta_\mu(t), \quad t \in [0, \infty), \quad \text{a.s.},$$

so

$$\rho(t) - \gamma_\mu(t) = \delta_\mu(t) - \gamma_\mu^*(t), \quad t \in [0, \infty), \quad \text{a.s.}$$

Therefore, (2.2.8) is equivalent to (2.2.5), and the proposition is proved. $\square$

The relation (2.2.8), that the growth rate of the market cannot exceed the common augmented growth rate of the stocks, may be worth restating as a corollary.

2.2.9 Corollary. *Suppose that all the stocks in the market $\mathcal{M}$ have non-negative dividend rates and the same augmented growth rate ρ . Then*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\gamma_\mu(t) - \rho(t)) dt \leq 0. \quad \text{a.s.}$$

Proof. See (2.2.8). $\square$

Proposition 2.2.8 shows that in a market in which the stocks have the same augmented growth rate, the average dividend rate must at least equal the average excess growth rate of the market over the long term. Since the excess growth rate is related to market diversity by Proposition 2.2.2, in order for a market of this type to remain diverse, at least some dividends must be paid.

2.2.10 Corollary. *Suppose that the market $\mathcal{M}$ is nondegenerate, and that all the stocks in $\mathcal{M}$ have nonnegative dividend rates and the same augmented growth rate. If $\mathcal{M}$ is diverse, then there exists a $\delta > 0$ such that*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \delta_\mu(t) dt \geq \delta, \quad \text{a.s.} \quad (2.2.9)$$

Proof. If $\mathcal{M}$ is diverse, then by Proposition 2.2.2 there is a $\delta > 0$ such that $\gamma_\mu^*(t) \geq \delta$, for all $t \in [0, \infty)$, a.s. By Proposition 2.2.8, a.s.,

$$\begin{aligned} 0 &\leq \liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\delta_\mu(t) - \gamma_\mu^*(t)) dt \leq \liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\delta_\mu(t) - \delta) dt \\ &= \liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \delta_\mu(t) dt - \delta, \end{aligned}$$

and the corollary follows. $\square$

2.2.11 Problem. (!) Show that under the hypotheses of Corollary 2.2.10,

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \delta_i(t) dt > 0, \quad \text{a.s.},$$

for $i = 1, \dots, n$. Can this inequality be strengthened? $\square$

Although it is not explicit in Corollary 2.2.10, presumably the dividend rates of the larger stocks must be greater than those of the smaller stocks in order for diversity to be maintained. We study this in more detail in the next section, where we relax the requirement that all the stocks have the same growth rate or augmented growth rate.

2.3 Entropy as a Measure of Market Diversity

Entropy is a measure of the uniformity of a probability distribution that has been used in statistical mechanics and information theory, and here we shall use it as a measure of market diversity. Since in this section we do not consider asymptotic events, we shall follow convention and restrict our time domain to $[0, T]$.

Let us recall that the *entropy* function $\mathbf{S}$ is defined by

$$\mathbf{S}(x) = - \sum_{i=1}^n x_i \log x_i \quad (2.3.1)$$

for all x in the set

$$\Delta^n = \{x \in \mathbb{R}^n : x_1 + \dots + x_n = 1; 0 < x_i < 1, i = 1, \dots, n\}. \quad (2.3.2)$$

2.3.1 Definition. Let μ be the market portfolio. Then the *market entropy process* $\mathbf{S}(\mu)$ is defined by

$$\mathbf{S}(\mu(t)) = - \sum_{i=1}^n \mu_i(t) \log \mu_i(t), \quad t \in [0, T].$$

It follows from this definition that $\mathbf{S}(\mu)$ is a continuous semimartingale, and that $0 < \mathbf{S}(\mu(t)) \leq \log n$, for all $t \in [0, T]$, a.s. Definition 2.2.1 provided a criterion for determining whether or not a market is diverse; entropy provides a measure of the degree of the diversity in the market. Let us see how these two concepts are related.

2.3.2 Proposition. *The market $\mathcal{M}$ is diverse if and only if there is an $\varepsilon > 0$ such that*

$$\mathbf{S}(\mu(t)) \geq \varepsilon, \quad t \in [0, T], \quad \text{a.s.} \quad (2.3.3)$$

Proof. The proof depends on the continuity of the entropy function (2.3.1) when extended to the closure of Δ^n ,

$$\overline{\Delta}^n = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 1; \quad 0 \leq x_i \leq 1, \quad i = 1, \dots, n\}.$$

The function $\mathbf{S}$ is nonnegative on the compact set $\overline{\Delta}^n$, and $\mathbf{S}(x) = 0$ only on the vertices. If a neighborhood of the vertices in $\overline{\Delta}^n$ is deleted, then $\mathbf{S}$ is bounded away from 0 on the rest of $\overline{\Delta}^n$. $\square$

We can define a portfolio associated with the market entropy process $\mathbf{S}(\mu)$.

2.3.3 Definition. Let μ be the market portfolio. The portfolio π with weights defined by

$$\pi_i(t) = \frac{-\mu_i(t) \log \mu_i(t)}{\mathbf{S}(\mu(t))}, \quad t \in [0, T], \quad (2.3.4)$$

for $i = 1, \dots, n$, is called the *entropy-weighted* portfolio.

It can be verified that π satisfies the conditions of Definition 1.1.4. The ratio

$$\frac{\pi_i(t)}{\mu_i(t)} = - \frac{\log \mu_i(t)}{\mathbf{S}(\mu(t))}, \quad t \in [0, T],$$

decreases with increasing $\mu_i(t)$. Hence π is less concentrated than μ in those stocks with the highest market weights.

2.3.4 Theorem. *Let μ be the market portfolio and π be the entropy-weighted portfolio, and let Z_μ and Z_π be their portfolio value processes, respectively. Then, a.s., for $t \in [0, T]$,*

$$d \log \mathbf{S}(\mu(t)) = d \log (Z_\pi(t)/Z_\mu(t)) - \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} dt. \quad (2.3.5)$$

Proof. From (1.2.14) we have

$$d\langle \mu_i, \mu_j \rangle_t = \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \quad t \in [0, \infty), \quad \text{a.s.,}$$

and this and Itô's rule imply that, a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log \mathbf{S}(\mu(t)) &= \sum_{i=1}^n D_i \log \mathbf{S}(\mu(t)) d\mu_i(t) \\ &\quad + \frac{1}{2} \sum_{i,j=1}^n D_{ij} \log \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt. \end{aligned} \quad (2.3.6)$$

In dealing with these computations, as well as some in the following chapters, it is worth recalling that for a positive, twice continuously differentiable function S ,

$$\begin{aligned} D_{ij} \log S(x) &= D_j \left(\frac{D_i S(x)}{S(x)} \right) = \frac{D_{ij} S(x)}{S(x)} - \frac{D_i S(x) D_j S(x)}{S^2(x)} \\ &= \frac{D_{ij} S(x)}{S(x)} - D_i \log S(x) D_j \log S(x). \end{aligned} \quad (2.3.7)$$

Since $D_i \mathbf{S}(x) = -\log x_i - 1$, (2.3.6) and (2.3.7) imply that, a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log \mathbf{S}(\mu(t)) &= - \sum_{i=1}^n \frac{\log(\mu_i(t))}{\mathbf{S}(\mu(t))} d\mu_i(t) - \frac{1}{\mathbf{S}(\mu(t))} \sum_{i=1}^n d\mu_i(t) \\ &\quad - \sum_{i,j=1}^n \frac{(\log \mu_i(t) + 1)(\log \mu_j(t) + 1)}{2\mathbf{S}^2(\mu(t))} \mu_i(t) \mu_j(t) \tau_{ij}(t) dt \\ &\quad - \frac{1}{2\mathbf{S}(\mu(t))} \sum_{i=1}^n \mu_i(t) \tau_{ii}(t) dt. \end{aligned} \quad (2.3.8)$$

Now, a.s., for $t \in [0, T]$,

$$\begin{aligned} &\sum_{i,j=1}^n (\log \mu_i(t) + 1)(\log \mu_j(t) + 1) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt \\ &= \sum_{i,j=1}^n \log \mu_i(t) \log \mu_j(t) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt \\ &\quad + 2 \sum_{i,j=1}^n \log \mu_i(t) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt \\ &\quad + \sum_{i,j=1}^n \mu_i(t) \mu_j(t) \tau_{ij}(t) dt \\ &= \sum_{i,j=1}^n \log \mu_i(t) \log \mu_j(t) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \end{aligned}$$

since $\mu(t)$ is in the null space of $\tau(t)$ by Lemma 1.2.2. This and the fact that $\sum_{i=1}^n d\mu_i(t) = 0$ imply that (2.3.8) is equivalent to

$$\begin{aligned} d \log \mathbf{S}(\mu(t)) &= - \sum_{i=1}^n \frac{\log(\mu_i(t))}{\mathbf{S}(\mu(t))} d\mu_i(t) \\ &\quad - \sum_{i,j=1}^n \frac{\log \mu_i(t) \log \mu_j(t)}{2\mathbf{S}^2(\mu(t))} \mu_i(t) \mu_j(t) \tau_{ij}(t) dt \\ &\quad - \frac{1}{2\mathbf{S}(\mu(t))} \sum_{i=1}^n \mu_i(t) \tau_{ii}(t) dt, \end{aligned} \quad (2.3.9)$$

a.s., for $t \in [0, T]$.

By Proposition 1.2.5 we have, a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\pi(t)/Z_\mu(t)) &= \sum_{i=1}^n \pi_i(t) d \log \mu_i(t) + \gamma_\pi^*(t) dt \\ &= \sum_{i=1}^n \frac{\pi_i(t)}{\mu_i(t)} d\mu_i(t) - \frac{1}{2} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}(t) dt \\ &= - \sum_{i=1}^n \frac{\log \mu_i(t)}{\mathbf{S}(\mu(t))} d\mu_i(t) \\ &\quad - \sum_{i,j=1}^n \frac{\log \mu_i(t) \log \mu_j(t)}{2\mathbf{S}^2(\mu(t))} \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \end{aligned} \quad (2.3.10)$$

where (2.3.10) is implied by (1.2.13) and Lemma 1.3.4. Equation (2.3.5) now follows from (2.3.9), (2.3.11), and Lemma 1.3.6. $\square$

Theorem 2.3.4 implies that the semimartingale decomposition of the logarithm of the market entropy satisfies

$$\begin{aligned} d \log \mathbf{S}(\mu(t)) &= \left(\gamma_\pi(t) - \gamma_\mu(t) - \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} \right) dt \\ &\quad + \sum_{i,\nu=1}^n (\pi_i(t) - \mu_i(t)) \xi_{i\nu}(t) dW_\nu(t), \end{aligned} \quad (2.3.12)$$

for $t \in [0, \infty)$, a.s. If the market is “stable” over the long term, then it is reasonable to require that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log \mathbf{S}(\mu(T)) = 0, \quad \text{a.s.}$$

From (2.3.12) we see that, a.s.,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log \mathbf{S}(\mu(T)) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left(\gamma_\pi(t) - \gamma_\mu(t) - \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} \right) dt,$$

since the last term in (2.3.12) vanishes in the limit by Lemma 1.3.2. Hence, we have

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left(\gamma_\pi(t) - \gamma_\mu(t) - \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} \right) dt = 0, \quad \text{a.s.}$$

Therefore, in order to have long-term stability, it is necessary that, on average, $\gamma_\pi(t)$ be greater than $\gamma_\mu(t)$. So, over the long term, we can expect the entropy-weighted portfolio to outperform the market portfolio. In the case of a diverse market, this result can be strengthened.

2.3.5 Corollary. *Let μ be the market portfolio and π be the entropy-weighted portfolio, and suppose that the market $\mathcal{M}$ is nondegenerate and diverse. Then for a sufficiently large number T ,*

$$Z_\pi(T)/Z_\pi(0) > Z_\mu(T)/Z_\mu(0), \quad \text{a.s.} \quad (2.3.13)$$

Proof. Theorem 2.3.4 implies that, a.s.,

$$\begin{aligned} \log(Z_\pi(T)/Z_\pi(0)) &= \log(Z_\mu(T)/Z_\mu(0)) \\ &\quad + \log(\mathbf{S}(\mu(T))/\mathbf{S}(\mu(0))) + \int_0^T \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} dt. \end{aligned}$$

Now, $\mathbf{S}(\mu(0)) \leq \log n$, and Proposition 2.3.2 implies that, for a diverse market, there is a $\delta_1 > 0$ such that $\mathbf{S}(\mu(t)) > \delta_1$ for all $t \in [0, T]$, a.s. Also, Proposition 2.2.2 implies that, for a nondegenerate, diverse market, there is $\delta_2 > 0$ such that $\gamma_\mu^*(t) > \delta_2$ for all $t \in [0, T]$, a.s. Hence, a.s.,

$$\log(Z_\pi(T)/Z_\pi(0)) > \log(Z_\mu(T)/Z_\mu(0)) + \log \delta_1 - \log \log n + \frac{\delta_2 T}{\log n},$$

and (2.3.13) follows for $T > \delta_2^{-1} \log n (\log \log n - \log \delta_1)$. $\square$

The following corollary brings dividend rates into the equation.

2.3.6 Corollary. *Let μ be the market portfolio and π be the entropy-weighted portfolio, and let $\hat{Z}_\mu$ and $\hat{Z}_\pi$ be their total return processes, respectively. Then, a.s., for all $t \in [0, T]$,*

$$d \log(\hat{Z}_\pi(t)/\hat{Z}_\mu(t)) = d \log \mathbf{S}(\mu(t)) + \left(\delta_\pi(t) - \delta_\mu(t) + \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} \right) dt. \quad (2.3.14)$$

Proof. The proof follows immediately from Theorem 2.3.4 and the definition of the total return processes $\hat{Z}_\mu$ and $\hat{Z}_\pi$. $\square$

If the market is nondegenerate and diverse, then $\log \mathbf{S}(\mu(t))$ in (2.3.14) is bounded and hence contributes only negligibly to the long-term variation of $\log(\hat{Z}_\pi(t)/\hat{Z}_\mu(t))$. In this case, the long-term risk level of π is the same

as that of μ , and therefore it could be argued that the return of π should not be greater than that of μ . In this case, if T is large enough, we should expect that

$$\int_0^T (\delta_\mu(t) - \delta_\pi(t)) dt \geq \int_0^T \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} dt.$$

The right-hand side of this inequality is positive, and for the left-hand side to be positive, the dividends of μ must be greater than the dividends of π , at least on average over $[0, T]$. Since μ is more concentrated than π in the larger stocks in the market, this means that, on average, the larger stocks must have higher dividend rates than the smaller stocks. We could then conclude that for diversity to be maintained, the larger stocks must redistribute capital by paying dividends.

2.3.7 Problem. (!!) Find a dividend rate structure for a nondegenerate, diverse market that is compatible with the capital asset pricing model (see Sharpe (1964), Karatzas and Shreve (1998), Remark 4.6.7, and Fernholz (1999a)).

2.4 Notes and Summary

Market diversity was first defined in Fernholz (1999a), and could be considered a consequence of antitrust law. Antitrust law is universal in modern economies, because at least since Smith (1776), it has been generally accepted that excessive concentration of either production or capital is likely to interfere with competition, and be detrimental to a national economy.

Since the classical continuous-time model used by Merton (1969) has constant growth rates and covariances, Corollary 2.2.7 implies that the market under the classical model would have deteriorated essentially into a single stock. Similar inconsistencies in the classical continuous-time model were observed by Rosenberg and Ohlson (1976).

The entropy function in (2.3.1) was introduced by Shannon (1948) as a measure of randomness in probability and information theory.

Chapter Summary

Section 2.1: Coherence is a minimal stability condition for a stock market. Simply stated, a market is coherent if none of the stocks descends into nothingness too quickly. Formally, a market is *coherent* if, for $i = 1, \dots, n$,

$$\lim_{t \rightarrow \infty} t^{-1} \log \mu_i(t) = 0, \quad \text{a.s.} \quad (2.1.1)$$

In a nondegenerate, coherent market, constant-weighted portfolios of more than a single stock will asymptotically outperform the market, a.s.

Section 2.2: Diversity is the opposite of concentration, and a diverse market is one that avoids extreme concentration of capital into single stocks. Formally, a market is *diverse* if there exists a number $\delta > 0$ such that

$$\mu_{\max}(t) = \max_{1 \leq i \leq n} \mu_i(t) \leq 1 - \delta, \quad t \in [0, \infty), \quad \text{a.s.},$$

and is *weakly diverse* on $[0, T]$ if there exists a number $\delta > 0$ such that

$$\frac{1}{T} \int_0^T \mu_{\max}(t) dt \leq 1 - \delta, \quad \text{a.s.} \quad (2.2.1)$$

Market diversity can be characterized in terms of the excess growth rate of the market. If the market $\mathcal{M}$ is nondegenerate and diverse, then there is a $\delta > 0$ such that

$$\gamma_{\mu}^*(t) \geq \delta, \quad t \in [0, \infty), \quad \text{a.s.} \quad (2.2.2)$$

Conversely, if $\mathcal{M}$ has bounded variance and there exists a $\delta > 0$ such that (2.2.2) holds, then $\mathcal{M}$ is diverse.

Diversity is a subtle property. It would seem that equal growth rates for all stocks would ensure market diversity. However, the opposite is true: In a nondegenerate market, if all the stocks have the same growth rate, then the market is not diverse.

Section 2.3: Entropy is a measure of diversity that was introduced by Shannon (1948) in his mathematical formulation of information theory. The *market entropy* process $\mathbf{S}(\mu)$ is defined by

$$\mathbf{S}(\mu(t)) = - \sum_{i=1}^n \mu_i(t) \log \mu_i(t), \quad t \in [0, T].$$

Zero entropy occurs if all capital is concentrated in a single stock. In a diverse market, market entropy is bounded away from zero.

The portfolio π with weights defined by

$$\pi_i(t) = \frac{-\mu_i(t) \log \mu_i(t)}{\mathbf{S}(\mu(t))}, \quad t \in [0, T], \quad (2.3.4)$$

for $i = 1, \dots, n$, is called the *entropy-weighted portfolio*. The relative return of the entropy portfolio is related to the variation of the market entropy. In fact, a.s., for $t \in [0, T]$,

$$d \log \mathbf{S}(\mu(t)) = d \log(Z_{\pi}(t)/Z_{\mu}(t)) - \frac{\gamma_{\mu}^*(t)}{\mathbf{S}(\mu(t))} dt. \quad (2.3.5)$$

With this equation, it can be shown that in a nondegenerate and diverse market, for a sufficiently large number T ,

$$Z_{\pi}(T)/Z_{\pi}(0) > Z_{\mu}(T)/Z_{\mu}(0), \quad \text{a.s.} \quad (2.3.13)$$

This signifies that, with probability one, the entropy-weighted portfolio has return above that of the market portfolio, and any condition of this nature must be avoided in normative theories of equilibrium. Dividend payments by the larger stocks could correct this condition.

3

Functionally Generated Portfolios

Functionally generated portfolios are a generalization of the entropy-weighted portfolio defined in Section 2.3. In this chapter we show that a broad range of functions can be used to generate portfolios, and for functionally generated portfolios, a decomposition of the relative return analogous to that of the entropy-weighted portfolio in Theorem 2.3.4 remains valid. Theorem 2.3.4 is a prototype for the general results we present here.

The value of functionally generated portfolios derives from the fact the return decomposition is a probability-one phenomenon. With appropriate generating functions, this allows us to obtain probability-one constraints on the relative return of functionally generated portfolios over a fixed, finite time horizon. Hence, portfolio generating functions are a powerful tool for the creation of portfolios with well-defined return characteristics, and are an essential component of stochastic portfolio theory.

3.1 Portfolio Generating Functions

In this section we shall introduce the concept of functionally generated portfolios. The basic idea is that certain real-valued functions defined on Δ^n can be used to generate portfolios, and the behavior of these functions can provide insight into the behavior of the portfolios they generate. Let us start with a formal definition.

3.1.1 Definition. Let $\mathbf{S}$ be a positive continuous function defined on Δ^n , and let π be a portfolio. Then $\mathbf{S}$ *generates* π if there exists a measurable

process of bounded variation Θ such that

$$\log(Z_\pi(t)/Z_\mu(t)) = \log \mathbf{S}(\mu(t)) + \Theta(t), \quad t \in [0, T], \quad \text{a.s.} \quad (3.1.1)$$

The process Θ is called the *drift* process corresponding to $\mathbf{S}$.

If $\mathbf{S}$ generates π , then $\mathbf{S}$ is called the *generating function* of π , and π is said to be *functionally generated*.

Since $\log(Z_\pi/Z_\mu)$ and $\log \mathbf{S}(\mu)$ in (3.1.1) are both continuous and adapted, it follows that Θ is also continuous and adapted. Since Θ is of bounded variation, $\log \mathbf{S}(\mu)$ is a continuous semimartingale, and hence we can express (3.1.1) in differential form as

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{S}(\mu(t)) + d\Theta(t), \quad t \in [0, T], \quad \text{a.s.} \quad (3.1.2)$$

It is in this differential form that we shall usually consider generating functions. Note that the existence of the Itô differential $d \log \mathbf{S}(\mu(t))$ does not imply that $\mathbf{S}$ is necessarily differentiable, and we shall see in Chapter 4 that nondifferentiable generating functions play an important role in stochastic portfolio theory.

Equation (3.1.2) provides a decomposition of the relative return of π . As in the case of the entropy function in the previous chapter, in specific examples the generating-function component of the relative return often dominates the short-term behavior of the relative return, and the drift process often dominates the long-term behavior, especially if $\log \mathbf{S}$ is bounded on Δ^n .

3.1.2 Example. The entropy function

$$\mathbf{S}(x) = - \sum_{i=1}^n x_i \log x_i$$

of Section 2.3 is the prototype for portfolio generating functions. Theorem 2.3.4 shows that the weights of the portfolio it generates are

$$\pi_i(t) = - \frac{\mu_i(t) \log \mu_i(t)}{\mathbf{S}(\mu(t))},$$

and the drift process satisfies

$$d\Theta(t) = \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} dt.$$

□

The next proposition shows how the growth rates of a functionally generated portfolio π and of the market portfolio μ are related to the drift process Θ .

3.1.3 Proposition. *Let $\mathbf{S}$ generate the portfolio π with drift process Θ , and suppose that*

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbf{S}(\mu(t)) = 0, \quad \text{a.s.} \quad (3.1.3)$$

Then,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\int_0^T \gamma_\pi(t) dt - \int_0^T \gamma_\mu(t) dt - \Theta(T) \right) = 0, \quad \text{a.s.} \quad (3.1.4)$$

Proof. From (3.1.1), we have, a.s., for $t \in [0, T]$,

$$\begin{aligned} & \log \mathbf{S}(\mu(T)) + \Theta(T) \\ &= \log Z_\pi(T) - \log Z_\mu(T) \\ &= \int_0^T (\gamma_\pi(t) - \gamma_\mu(t)) dt + \int_0^T \sum_{i, \nu=1}^n (\pi_i(t) - \mu_i(t)) \xi_{i\nu}(t) dW_\nu(t), \end{aligned} \quad (3.1.5)$$

by (1.1.13). Apply $\lim_{T \rightarrow \infty} T^{-1}$ to both sides of (3.1.5). Then $\log \mathbf{S}(\mu(T))$ vanishes by (3.1.3), and the second integral vanishes by Lemma 1.3.2. What remains is (3.1.4). $\square$

Condition (3.1.3) of this proposition will be satisfied, for example, if $\log \mathbf{S}$ is bounded on Δ^n , and this is the condition that pertains in most applications.

The next proposition considers the total return process of a functionally generated portfolio. Although no new ideas are involved, it is worth stating for the sake of completeness.

3.1.4 Proposition. *Suppose that $\mathbf{S}$ generates the portfolio π with drift process Θ . Then a.s., for $t \in [0, T]$,*

$$\log(\widehat{Z}_\pi(t)/\widehat{Z}_\mu(t)) = \log \mathbf{S}(\mu(t)) + \int_0^t (\delta_\pi(s) - \delta_\mu(s)) ds + \Theta(t). \quad (3.1.6)$$

Proof. This follows immediately from (3.1.2) and the definition of $\widehat{Z}_\pi$. $\square$

We shall find that many functions defined on Δ^n generate portfolios. However, the dimension of Δ^n is $n - 1$, while there are n market weights, so there is no natural coordinate system on Δ^n that treats all the market weights in the same manner. Hence, to analyze these generating functions, it is convenient to use the standard coordinate system in $\mathbb{R}^n$, even though it is not a coordinate system on Δ^n . For this reason we consider functions that are defined on an open neighborhood $U \subset \mathbb{R}^n$ of Δ^n . Functions defined on Δ^n can always be extended to an open neighborhood by making the extension constant along lines parallel to one of the axes of $\mathbb{R}^n$. In this chapter we consider the case where the generating function is of class C^2 . (Recall that a real-valued function defined on an open subset of $\mathbb{R}^n$ is of

class C^2 if it is twice continuously differentiable in all n variables.) We use the notation D_i for the partial derivative with respect to the i th variable, and D_{ij} for the second partial derivative with respect to the i th and j th variables.

3.1.5 Theorem. *Let $\mathbf{S}$ be a positive C^2 function defined on a neighborhood U of Δ^n such that for all i , $x_i D_i \log \mathbf{S}(x)$ is bounded on Δ^n . Then $\mathbf{S}$ generates the portfolio π with weights*

$$\pi_i(t) = \left(D_i \log \mathbf{S}(\mu(t)) + 1 - \sum_{j=1}^n \mu_j(t) D_j \log \mathbf{S}(\mu(t)) \right) \mu_i(t), \quad (3.1.7)$$

for $t \in [0, T]$ and $i = 1, \dots, n$, and with a drift process Θ such that a.s., for $t \in [0, T]$,

$$d\Theta(t) = \frac{-1}{2\mathbf{S}(\mu(t))} \sum_{i,j=1}^n D_{ij} \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt. \quad (3.1.8)$$

Remark. Note that the weights $\pi_1, \dots, \pi_n$ depend only on the market weights, not on the covariance structure of the market. The covariance structure enters only in the relative covariance term, $\tau_{ij}(t)$, in the expression for $d\Theta(t)$. $\square$

Proof of Theorem 3.1.5. The weights $\pi_i(t)$ sum to 1, and the conditions on $\mathbf{S}$ ensure that π is a portfolio. It is clear from (3.1.8) that Θ is of bounded variation.

Equation (1.2.14) states that

$$d\langle \mu_i, \mu_j \rangle_t = \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \quad t \in [0, T], \quad \text{a.s.}$$

This and Itô's rule imply that a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log \mathbf{S}(\mu(t)) &= \sum_{i=1}^n D_i \log \mathbf{S}(\mu(t)) d\mu_i(t) \\ &\quad + \frac{1}{2} \sum_{i,j=1}^n D_{ij} \log \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt. \end{aligned} \quad (3.1.9)$$

From (2.3.7) we recall that for $x \in \Delta^n$,

$$D_{ij} \log \mathbf{S}(x) = \frac{D_{ij} \mathbf{S}(x)}{\mathbf{S}(x)} - D_i \log \mathbf{S}(x) D_j \log \mathbf{S}(x).$$

With this, we see that (3.1.9) is equivalent to

$$\begin{aligned} d \log \mathbf{S}(\mu(t)) &= \sum_{i=1}^n D_i \log \mathbf{S}(\mu(t)) d\mu_i(t) \\ &+ \frac{1}{2 \mathbf{S}(\mu(t))} \sum_{i,j=1}^n D_{ij} \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt \\ &- \frac{1}{2} \sum_{i,j=1}^n D_i \log \mathbf{S}(\mu(t)) D_j \log \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \end{aligned} \quad (3.1.10)$$

a.s., for $t \in [0, T]$.

In order for (3.1.2) to hold, the local martingale parts of $\log \mathbf{S}(\mu(t))$ and $\log(Z_\pi(t)/Z_\mu(t))$ must be equal. By Proposition 1.2.5 we have a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\pi(t)/Z_\mu(t)) &= \sum_{i=1}^n \pi_i(t) d \log \mu_i(t) + \gamma_\pi^*(t) dt \\ &= \sum_{i=1}^n \frac{\pi_i(t)}{\mu_i(t)} d\mu_i(t) - \frac{1}{2} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}(t) dt, \end{aligned} \quad (3.1.11)$$

by (1.2.13) and Lemma 1.3.4. Suppose that

$$\varphi(t) = 1 - \sum_{j=1}^n \mu_j(t) D_j \log \mathbf{S}(\mu(t)), \quad t \in [0, T], \quad (3.1.12)$$

and

$$\pi_i(t) = (D_i \log \mathbf{S}(\mu(t)) + \varphi(t)) \mu_i(t), \quad t \in [0, T]. \quad (3.1.13)$$

Then

$$\sum_{i=1}^n \pi_i(t) = \sum_{i=1}^n \mu_i(t) D_i \log \mathbf{S}(\mu(t)) + \varphi(t) = 1,$$

and (3.1.7) is satisfied. With φ defined as in (3.1.12), we have, a.s., for $t \in [0, T]$,

$$\begin{aligned} \sum_{i=1}^n \frac{\pi_i(t)}{\mu_i(t)} d\mu_i(t) &= \sum_{i=1}^n D_i \log \mathbf{S}(\mu(t)) d\mu_i(t) + \varphi(t) \sum_{i=1}^n d\mu_i(t) \\ &= \sum_{i=1}^n D_i \log \mathbf{S}(\mu(t)) d\mu_i(t), \end{aligned}$$

since $\sum_{i=1}^n d\mu_i(t) = 0$. Hence, the local martingale parts of $\log \mathbf{S}(\mu(t))$ and $\log(Z_\pi(t)/Z_\mu(t))$ are equal.

With $\pi_i(t)$ defined as in (3.1.13), we have, a.s., for $t \in [0, T]$,

$$\begin{aligned} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}(t) &= \sum_{i,j=1}^n D_i \log \mathbf{S}(\mu(t)) D_j \log \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) \\ &\quad + 2\varphi(t) \sum_{i,j=1}^n D_i \log \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) \\ &\quad + \varphi^2(t) \sum_{i,j=1}^n \mu_i(t) \mu_j(t) \tau_{ij}(t) \\ &= \sum_{i,j=1}^n D_i \log \mathbf{S}(\mu(t)) D_j \log \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t), \end{aligned}$$

since $\mu(t)$ is in the null space of $\tau(t)$ by Lemma 1.2.2. Hence, a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\pi(t)/Z_\mu(t)) &= \sum_{i=1}^n D_i \log \mathbf{S}(\mu(t)) d\mu_i(t) \\ &\quad - \frac{1}{2} \sum_{i,j=1}^n D_i \log \mathbf{S}(\mu(t)) D_j \log \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt. \end{aligned}$$

This equation and (3.1.10) imply that a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\pi(t)/Z_\mu(t)) \\ = d \log \mathbf{S}(\mu(t)) - \frac{1}{2 \mathbf{S}(\mu(t))} \sum_{i,j=1}^n D_{ij} \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \end{aligned}$$

so (3.1.8) is proved. $\square$

3.1.6 Example. Here are a few examples of simple generating functions and the portfolios they generate.

1. $\mathbf{S}(x) = 1$ generates the market portfolio μ with $\Theta(t) = 0$.
2. $\mathbf{S}(x) = w_1 x_1 + \cdots + w_n x_n$, where $w_1, \dots, w_n$ are nonnegative constants at least one of which is positive, generates the buy-and-hold portfolio that holds w_i shares of the i th stock. Here $\Theta(t) = 0$. This type of portfolio is commonly held by investors, at least in a piecewise manner.
3. $\mathbf{S}(x) = (x_1 \cdots x_n)^{1/n}$ generates the equal-weighted portfolio with $d\Theta(t) = \gamma_\pi^*(t) dt$. The Value Line Index is such a portfolio.
4. $\mathbf{S}(x) = x_1^{p_1} \cdots x_n^{p_n}$, where $p_1, \dots, p_n$ are constant and $p_1 + \cdots + p_n = 1$, generates the constant-weighted portfolio with weights $\pi_i(t) = p_i$ and $d\Theta(t) = \gamma_\pi^*(t) dt$.

5. (A single stock, with leverage) The function $\mathbf{S}(x) = x_1^2$ generates the portfolio π with weights $\pi_1(t) = 2 - \mu_1(t)$, and $\pi_i(t) = -\mu_i(t)$, for $i = 2, \dots, n$. This corresponds to a continuously rebalanced portfolio in which for each \$2 invested in stock X_1 , $-\$1$ is invested in the market portfolio. In this case, $d\Theta(t) = -\tau_{11}(t) dt$, so the drift process is decreasing. This shows that although investment in a single stock may be reasonable, it may not be wise to leverage this investment by shorting the market, at least over the long term. $\square$

3.1.7 Problem. (!!) Develop a theory of optimization for functionally generated portfolios.

3.1.8 Problem. (!!) Develop a theory of optimization that is dependent only on observable parameters such as market weights and market value.

3.1.9 Example. (Weighted-average capitalization) The weighted-average capitalization of the market is used sometimes as a measure of the concentration of capital in the market. The value of this weighted average would be

$$\sum_{i=1}^n \mu_i(t) X_i(t), \quad t \in [0, T].$$

The weighted average of the capitalization weights,

$$\sum_{i=1}^n \mu_i^2(t), \quad t \in [0, T],$$

is proportional to the weighted-average capitalization, and the square root of this weighted average,

$$\mathbf{S}(x) = \left(\sum_{i=1}^n x_i^2 \right)^{1/2},$$

generates the portfolio π with weights

$$\pi_i(t) = \frac{\mu_i^2(t)}{\mu_1^2(t) + \dots + \mu_n^2(t)}, \quad t \in [0, T],$$

for $i = 1, \dots, n$. This means that relative to the market, π is overweighted in the larger stocks and underweighted in the smaller stocks. The drift process for this portfolio satisfies

$$d\Theta(t) = -\gamma_\pi^*(t) dt, \quad t \in [0, T].$$

Since

$$\log \mathbf{S}(\mu(t)) = \frac{1}{2} \log \left(\sum_{i=1}^n \mu_i^2(t) \right), \quad t \in [0, T],$$

(3.1.2) implies that if the weighted average of the capitalization weights is the same at the beginning and end of a given period of time, then π will have a lower return than the market over that period. $\square$

3.1.10 Example. (Price-to-book ratio) Suppose that $b_i > 0$ represents the book value of the i th company, and suppose that b_i is constant. Then the *price-to-book ratio* at time t for this company is $X_i(t)/b_i$. This ratio is frequently used to distinguish *growth* stocks, those with higher price-to-book ratios, from *value* stocks, those with lower ratios. For our purposes, it is convenient to consider the ratio $\mu_i(t)/b_i$, but let us continue to call it the price-to-book ratio, since the two ratios have similar properties. We shall assume that $b_i > 0$ is constant, for $i = 1, \dots, n$.

The weighted-average price-to-book ratio of the market is given by

$$\sum_{i=1}^n \frac{\mu_i^2(t)}{b_i}, \quad t \in [0, T].$$

The function

$$\mathbf{S}(x) = \left(\sum_{i=1}^n \frac{x_i^2}{b_i} \right)^{1/2}$$

generates the portfolio π with weights

$$\pi_i(t) = \frac{\mu_i^2(t)}{b_i \mathbf{S}^2(\mu(t))}, \quad t \in [0, T],$$

for $i = 1, \dots, n$. Relative to the market, this portfolio is overweighted in growth stocks, and underweighted in value stocks. The drift process in this case is

$$d\Theta(t) = -\gamma_\pi^*(t), \quad t \in [0, T].$$

Reasoning similar to that of Example 3.1.9 reveals that over a period of time in which the weighted-average price-to-book ratio of the market remains fixed, the portfolio π will have lower return than the market. $\square$

Sometimes several generating functions can be combined to generate portfolios with hybrid characteristics. For example, (3.1.7) of Theorem 3.1.5 implies that if $\mathbf{S}_1, \dots, \mathbf{S}_k$ generate portfolios $\pi_1, \dots, \pi_k$, respectively, then for constants $p_1, \dots, p_k$ such that $p_1 + \dots + p_k = 1$, the function

$$\mathbf{S} = \mathbf{S}_1^{p_1} \mathbf{S}_2^{p_2} \dots \mathbf{S}_k^{p_k}$$

generates a portfolio with weights

$$\pi_i = p_1 \pi_{1i} + \dots + p_k \pi_{ki},$$

for $i = 1, \dots, n$, where π_{ji} is the i th weight of π_j , for $j = 1, \dots, k$. It also follows from Theorem 3.1.5 that if $\mathbf{S}_\pi$ generates π and $\mathbf{S}_\eta$ generates η , with drift processes Θ_π and Θ_η , respectively, then a.s., for $t \in [0, T]$,

$$\log(Z_\pi(t)/Z_\eta(t)) = \log(\mathbf{S}_\pi(\mu(t))/\mathbf{S}_\eta(\mu(t))) + \Theta_\pi(t) - \Theta_\eta(t). \quad (3.1.14)$$

Not all portfolios are functionally generated; let us now characterize those that are. Recall that a differential is *exact* if it is of the form $\sum_i D_i G(x) dx_i$ for some differentiable function G (see Spivak (1965)).

3.1.11 Proposition. *Suppose that $f_1, \dots, f_n$ are continuously differentiable real-valued functions defined on a neighborhood of Δ^n such that $\sum_{i=1}^n f_i(x) = 1$ for all $x \in \Delta^n$. Then the portfolio π defined by $\pi_i(t) = f_i(\mu(t))$ for $t \in [0, T]$ and $i = 1, \dots, n$ is functionally generated if and only if there exists a continuously differentiable real-valued function F defined on a neighborhood of Δ^n such that*

$$\sum_{i=1}^n \left(\frac{f_i(x)}{x_i} + F(x) \right) dx_i \quad (3.1.15)$$

is an exact differential.

Proof. Suppose that such functions $F, f_1, \dots, f_n$ exist. Since (3.1.15) is exact, there exists a function G such that for $x \in \Delta^n$,

$$D_i G(x) = \frac{f_i(x)}{x_i} + F(x),$$

for $i = 1, \dots, n$. Then the function $\mathbf{S} = e^G$ generates the portfolio π with $\pi_i(t) = f_i(\mu(t))$, $i = 1, \dots, n$.

Now suppose that π has a generating function $\mathbf{S}$ defined on a neighborhood U of Δ^n . For $x \in U$, let

$$f_i(x) = \left(D_i \log \mathbf{S}(x) + 1 - \sum_{j=1}^n x_j D_j \log \mathbf{S}(x) \right) x_i,$$

for $i = 1, \dots, n$, and let

$$F(x) = -1 + \sum_{j=1}^n x_j D_j \log \mathbf{S}(x).$$

By Theorem 3.1.5 the weights π_i satisfy $\pi_i(t) = f_i(\mu(t))$, for $i = 1, \dots, n$, for $t \in [0, T]$, a.s. For $x \in U$, the differential

$$\sum_{i=1}^n \left(\frac{f_i(x)}{x_i} + F(x) \right) dx_i = \sum_{i=1}^n D_i \log \mathbf{S}(x) dx_i = d \log \mathbf{S}(x)$$

is exact, so the proposition is proved. $\square$

3.1.12 Example. Here is an example of a portfolio π whose weights depend differentiably on the market portfolio weights but is not functionally generated. For $x \in \mathbb{R}^n$, let

$$\begin{aligned} f_1(x) &= x_1, \\ f_2(x) &= x_2 + \dots + x_n, \\ f_i(x) &= 0, \quad \text{for } i = 3, \dots, n, \end{aligned}$$

and let π be the portfolio defined by $\pi_i(t) = f_i(\mu(t))$, for $i = 1, \dots, n$. It can be shown (see Spivak (1965)) that if the differential in (3.1.15) is exact, then for all i and j ,

$$D_j \left(\frac{f_i(x)}{x_i} + F(x) \right) = D_i \left(\frac{f_j(x)}{x_j} + F(x) \right).$$

No function F satisfies these equations for all i and j , so π has no generating function. $\square$

Theorem 3.1.5 shows that a positive C^2 function $\mathbf{S}$ defined in a neighborhood of Δ^n generates a portfolio. It would seem reasonable that the values of $\mathbf{S}$ on Δ^n should uniquely determine the portfolio, and that the values of $\mathbf{S}$ in the complement of Δ^n should be irrelevant. We shall prove that this is, indeed, the case, but first we need a preliminary result.

3.1.13 Lemma. *Let f be a continuously differentiable real-valued function defined on a neighborhood U of Δ^n . Then f is constant on Δ^n if and only if for all $x \in \Delta^n$, $D_i f(x) = D_j f(x)$ for all i and j .*

Proof. Parameterize Δ^n by positive real numbers $t_1, \dots, t_{n-1}$ with $t_1 + \dots + t_{n-1} < 1$, such that $x_i = t_i$, for $i = 1, \dots, n-1$ and $x_n = 1 - t_1 - \dots - t_{n-1}$. Then for $i = 1, \dots, n-1$,

$$\frac{\partial f(x)}{\partial t_i} = D_i f(x) - D_n f(x)$$

for all $x \in \Delta^n$. If f is constant on Δ^n , then all its partial derivatives with respect to the parameters t_i are zero, which implies that $D_i f(x) = D_n f(x)$ for all i . Likewise, if $D_i f(x) = D_j f(x)$ for all i and j , then the partial derivatives with respect to all the t_i are zero, so f is constant on Δ^n . $\square$

We consider generating functions that generate the same portfolio for all values of the market weights $\mu_1(t), \dots, \mu_n(t)$, for $t \in [0, T]$. Since the initial value of the market portfolio can take on any value in Δ^n , (3.1.7) of Theorem 3.1.5 implies that C^2 functions $\mathbf{S}_1$ and $\mathbf{S}_2$ generate the same portfolio for all realizations of the market portfolio if and only if

$$\begin{aligned} D_i \log \mathbf{S}_1(x) + 1 - \sum_{j=1}^n x_j D_j \log \mathbf{S}_1(x) \\ = D_i \log \mathbf{S}_2(x) + 1 - \sum_{j=1}^n x_j D_j \log \mathbf{S}_2(x) \end{aligned} \quad (3.1.16)$$

for $i = 1, \dots, n$ and for all $x \in \Delta^n$.

3.1.14 Proposition. *Let $\mathbf{S}_1$ and $\mathbf{S}_2$ be positive C^2 functions defined on an open neighborhood of Δ^n . Then (3.1.16) holds for $i = 1, \dots, n$ and for all $x \in \Delta^n$ if and only if $\mathbf{S}_1/\mathbf{S}_2$ is constant on Δ^n .*

Proof. Suppose $\mathbf{S}_1$ and $\mathbf{S}_2$ are defined on the open neighborhood U of Δ^n , and that $\mathbf{S}_1/\mathbf{S}_2$ is constant on Δ^n . Define the function f for $x \in U$ by

$$f(x) = \log \mathbf{S}_1(x) - \log \mathbf{S}_2(x).$$

Then f is constant on Δ^n , so Lemma 3.1.13 implies that $D_i f(x) = D_j f(x)$ for all i and j , for all $x \in \Delta^n$. Therefore, for $x \in \Delta^n$,

$$D_i \log \mathbf{S}_1(x) - D_j \log \mathbf{S}_1(x) = D_i \log \mathbf{S}_2(x) - D_j \log \mathbf{S}_2(x),$$

for all i and j . Hence, the difference $D_i \log \mathbf{S}_1(x) - D_i \log \mathbf{S}_2(x)$ is the same for all i , and (3.1.7) of Theorem 3.1.5 implies that the weights generated by $\mathbf{S}_1$ and $\mathbf{S}_2$ are the same. It follows that $\mathbf{S}_1$ and $\mathbf{S}_2$ generate the same portfolio.

If (3.1.16) holds for $i = 1, \dots, n$ and for all $x \in \Delta^n$, then for all i and k ,

$$D_i \log(\mathbf{S}_1(x)/\mathbf{S}_2(x)) = D_k \log(\mathbf{S}_1(x)/\mathbf{S}_2(x)),$$

for all $x \in \Delta^n$. It follows by Lemma 3.1.13 that $\log(\mathbf{S}_1/\mathbf{S}_2)$ is constant on Δ^n . $\square$

The next proposition characterizes those functions that generate portfolios with increasing drift processes. A functionally generated portfolio with an increasing drift process will outperform the market portfolio over periods in which the generating remains constant or increases. For this reason, such portfolios are of particular interest. The entropy-weighted portfolio we considered in Section 2.3 had an increasing drift process, and we shall consider a more general class of such portfolios in Section 3.4.

3.1.15 Proposition. *Let $\mathbf{S}$ be a generating function such that for all $x \in \Delta^n$, the matrix $(D_{ij}\mathbf{S}(x))$ has at most one positive eigenvalue, and if there is a positive eigenvalue, it corresponds to an eigenvector orthogonal to Δ^n . Let π be the portfolio generated by $\mathbf{S}$. Then $\pi_i(t) \geq 0$, for $i = 1, \dots, n$, and Θ is nondecreasing, a.s. If for all $x \in \Delta^n$, $\text{rank}(D_{ij}\mathbf{S}(x)) > 1$, then Θ is strictly increasing, a.s.*

Proof. Suppose that $\mathbf{S}$ is a generating function such that for all $x \in \Delta^n$, the matrix $(D_{ij}\mathbf{S}(x))$ has at most one positive eigenvalue, and if there is a positive eigenvalue it corresponds to an eigenvector orthogonal to Δ^n . For any $x \in \Delta^n$, define $x(u) \in \Delta^n$ by

$$x(u) = uv_k + (1 - u)x$$

for $0 \leq u < 1$, where $v_k = (0, \dots, 1, \dots, 0)$ with 1 in the k th position and 0 elsewhere. Let

$$f(u) = \mathbf{S}(x(u)),$$

so

$$f'(u) = D_k \mathbf{S}(x(u)) - \sum_{i=1}^n x_i D_i \mathbf{S}(x(u)) \quad (3.1.17)$$

and

$$f''(u) = (v_k - x)(D_{ij}\mathbf{S}(x(u)))(v_k - x)^T \leq 0,$$

since $v_k - x$ is parallel to Δ^n and hence is composed of eigenvectors of $(D_{ij}\mathbf{S}(x(u)))$ that have nonpositive eigenvalues. This implies that f is convex on $[0, 1)$, so

$$f(u) \leq f(0) + uf'(0), \quad 0 \leq u < 1,$$

and therefore

$$0 < f(0) + uf'(0), \quad 0 \leq u < 1.$$

It follows from this and (3.1.17) that

$$0 \leq \mathbf{S}(x) + D_k\mathbf{S}(x) - \sum_{i=1}^n x_i D_i\mathbf{S}(x).$$

Hence, by (3.1.7) of Theorem 3.1.5, $\pi_k(t) \geq 0$, $k = 1, \dots, n$, for $t \in [0, T]$, a.s.

Suppose that $t \in [0, T]$, and that $\lambda_1, \dots, \lambda_n$ are the eigenvalues of $(D_{ij}\mathbf{S}(\mu(t)))$. Let $e_k = (e_{k1}, \dots, e_{kn})^T$ be a normalized eigenvector corresponding to the eigenvalue λ_k , for $k = 1, \dots, n$. In this case, for $i, j = 1, \dots, n$,

$$D_{ij}\mathbf{S}(\mu(t)) = \sum_{k=1}^n \lambda_k e_{ki} e_{kj},$$

so

$$\sum_{i,j=1}^n D_{ij}\mathbf{S}(\mu(t))\mu_i(t)\mu_j(t)\tau_{ij}(t) = \sum_{k=1}^n \lambda_k \sum_{i,j=1}^n \mu_i(t)\mu_j(t)e_{ki}e_{kj}\tau_{ij}(t). \quad (3.1.18)$$

If one of the λ_i is positive, it is orthogonal to Δ^n , and we can assume without loss of generality that it is λ_1 with $e_1 = \pm(n^{-1/2}, \dots, n^{-1/2})$. Suppose that $(\tau_{ij}(t))$ is positive semidefinite with null space generated by $\mu(t)$. Then

$$\sum_{i,j=1}^n \mu_i(t)\mu_j(t)e_{1i}e_{1j}\tau_{ij}(t) = 0,$$

and

$$\sum_{i,j=1}^n \mu_i(t)\mu_j(t)e_{ki}e_{kj}\tau_{ij}(t) > 0 \quad (3.1.19)$$

for $k = 2, \dots, n$. By hypothesis, none of the eigenvalues $\lambda_2, \dots, \lambda_n$ is positive, so (3.1.18) implies that

$$\sum_{i,j=1}^n D_{ij}\mathbf{S}(\mu(t))\mu_i(t)\mu_j(t)\tau_{ij}(t) \leq 0.$$

Since Lemma 1.2.2 implies that a.s., for all $t \in [0, T]$, $(\tau_{ij}(t))$ is positive semidefinite with null space generated by $\mu(t)$, it follows from (3.1.8) of Theorem 3.1.5 that Θ is nondecreasing, a.s.

If $\text{rank}(D_{ij}\mathbf{S}(x)) > 1$ for all $x \in \Delta^n$, then at least one of the eigenvalues $\lambda_2, \dots, \lambda_n$ is negative. Hence (3.1.18) and (3.1.19) imply that

$$\sum_{i,j=1}^n D_{ij}\mathbf{S}(\mu(t))\mu_i(t)\mu_j(t)\tau_{ij}(t) < 0,$$

and (3.1.8) of Theorem 3.1.5 implies that Θ is strictly increasing, a.s. $\square$

3.2 Time-Dependent Generating Functions

In this section we generalize the definition of generating functions to include time-dependent generating functions. There is a connection between time-dependent generating functions and the Black/Scholes option pricing model that allows us to view generating functions in the setting of option pricing theory. We shall not develop these ideas here, but they may be of interest for future research. For this reason, it is likely that the most interesting material in this section can be found in the problems at the end.

3.2.1 Definition. Let $\mathbf{S}$ be a positive continuous function defined on $\Delta^n \times [0, T]$, and let π be a portfolio. Then $\mathbf{S}$ *generates* π if there exists a measurable, adapted process of bounded variation Θ such that

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{S}(\mu(t), t) - D_t \log \mathbf{S}(\mu(t), t) dt + d\Theta(t), \quad (3.2.1)$$

for $t \in [0, T]$, a.s. The process Θ is called the *drift process* corresponding to $\mathbf{S}$.

Theorem 3.1.5 can be extended to time-dependent generating functions. A real-valued function defined on $U \times [0, T]$, where $U \subset \mathbb{R}^n$ is an open set, is of class $C^{2,1}$ if it is twice continuously differentiable in the first n variables, and continuously differentiable in the last (time) variable. For such a function, let D_t represent the partial derivative with respect to the last variable.

3.2.2 Theorem. Let $\mathbf{S}$ be a positive $C^{2,1}$ function defined on $U \times [0, T]$, where U is a neighborhood of Δ^n , and suppose that for all i , $x_i D_i \log \mathbf{S}(x, t)$ is bounded on $\Delta^n \times [0, T]$. Then $\mathbf{S}$ generates the portfolio π with weights

$$\pi_i(t) = \left(D_i \log \mathbf{S}(\mu(t), t) + 1 - \sum_{j=1}^n \mu_j(t) D_j \log \mathbf{S}(\mu(t), t) \right) \mu_i(t), \quad (3.2.2)$$

$t \in [0, T]$, for $k = 1, \dots, n$, and with a drift process Θ that satisfies

$$\begin{aligned} d\Theta(t) = & \frac{-1}{2\mathbf{S}(\mu(t), t)} \sum_{i,j=1}^n D_{ij}\mathbf{S}(\mu(t), t)\mu_i(t)\mu_j(t)\tau_{ij}(t) dt \\ & - D_t \log \mathbf{S}(\mu(t), t) dt, \quad t \in [0, T], \quad \text{a.s.} \end{aligned} \quad (3.2.3)$$

Proof. The fact that the weights $\pi_i(t)$ sum to 1 and the conditions on $\mathbf{S}$ ensure that π is a portfolio. It is clear from (3.2.3) that Θ is of bounded variation.

Itô's rule and (1.2.14) imply that a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log \mathbf{S}(\mu(t), t) = & \sum_{i=1}^n D_i \log \mathbf{S}(\mu(t), t) d\mu_i(t) + D_t \log \mathbf{S}(\mu(t), t) dt \\ & + \frac{1}{2} \sum_{i,j=1}^n D_{ij} \log \mathbf{S}(\mu(t), t) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt. \end{aligned}$$

The rest of the proof follows the proof of Theorem 3.1.2. $\square$

3.2.3 Example. Let $\mathbf{S}(x, t) = w_1(t)x_1 + \dots + w_n(t)x_n$, where the $w_i(t)$ are nonnegative C^1 functions defined on $[0, T]$, at least one of which is positive. Then $\mathbf{S}$ generates the portfolio that holds $w_i(t)$ shares of the i th stock, and $\Theta(t) = 0$. This portfolio is similar to conventional portfolios held by many investors. $\square$

Remark. (Option pricing) If a $C^{2,1}$ generating function $\mathbf{S}$ satisfies

$$D_t \mathbf{S}(\mu(t), t) = -\frac{1}{2} \sum_{i,j=1}^n D_{ij} \mathbf{S}(\mu(t), t) \mu_i(t) \mu_j(t) \tau_{ij}(t), \quad t \in [0, T], \quad (3.2.4)$$

then (3.2.3) implies that

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{S}(\mu(t), t), \quad t \in [0, T],$$

and (3.2.1) becomes a variation of the Black/Scholes option pricing model. Here $\log \mathbf{S}(\mu(t), t)$ corresponds to the option price, and $\log(Z_\pi(t)/Z_\mu(t))$ corresponds to the value of the hedging portfolio. $\square$

3.2.4 Problem. (!) Is there a nontrivial example of a time-dependent generating function that satisfies (3.2.4)?

3.2.5 Problem. (!!) Generalize the Black/Scholes option pricing model to include a drift process analogous to Θ in (3.2.1). Can this drift process be interpreted as the effect of variation in the volatility and interest rate processes in the model?

3.2.6 Problem. (!) In Example 3.1.10, suppose that the book values of the stocks are C^1 functions of time, $b_i(t) > 0$, $t \in [0, T]$, for $i = 1, \dots, n$. Let π be the portfolio generated by

$$S(x, t) = \left(\sum_{i=1}^n \frac{x_i^2}{b_i(t)} \right)^{1/2}, \quad t \in [0, T].$$

Can these variable book values be constructed in such a way that for a period of time in which the weighted-average price-to-book ratio of the market remains fixed, π will have the same, or greater, return than the market?

3.3 The No-Arbitrage Hypothesis

An *arbitrage opportunity* is a combination of investments in portfolios such that the sum of the initial values of the investments is zero and such that at some given nonrandom future time T , the sum of the values will be nonnegative with probability one and positive with positive probability. The no-arbitrage hypothesis states that there exist no arbitrage opportunities, at least when the portfolios comprising the arbitrage opportunities satisfy certain regularity conditions. No-arbitrage is a common hypothesis in current financial theory, and although there are examples of markets with arbitrage, these examples appear to be mathematical oddities that do not resemble “real” equity markets. In this chapter we show that arbitrage exists in weakly diverse markets in which the stocks do not pay dividends. Under those hypotheses, we construct a portfolio that dominates the market portfolio, and another portfolio that is dominated by the market portfolio is discussed in Problem 3.3.4.

The no-arbitrage hypothesis appears to be quite reasonable—we can think of at least two good reasons for the central position that no-arbitrage occupies in mathematical finance. First, over the short term, no-arbitrage appears to be an accurate representation of actual equity markets. Second, arbitrage opportunities are probability-one events, and outside mathematics there are no probability-one events (except for death and taxes, of course), so one could argue that no-arbitrage holds by default. However, the example we present shows that arbitrage opportunities exist in markets that appear to be indistinguishable from actual markets.

The definition of a portfolio, Definition 1.1.4, is quite general and allows for the existence of portfolios with somewhat unrealistic properties. For example, it is possible for the ratio of portfolio weights to market weights to be unbounded, so a portfolio might hold more than the total capitalization of some company. To avoid these unrealistic constructions, we must restrict somewhat the class of the portfolios we consider for testing the no-arbitrage hypothesis.

3.3.1 Definition. A portfolio π is *admissible* if:

- (i) for $i = 1, \dots, n$, $\pi_i(t) \geq 0$, $t \in [0, T]$;
- (ii) there exists a constant $c > 0$ such that

$$\widehat{Z}_\pi(t)/\widehat{Z}_\pi(0) \geq c\widehat{Z}_\mu(t)/\widehat{Z}_\mu(0), \quad t \in [0, T], \quad \text{a.s.};$$

- (iii) there exists a constant M such that for $i = 1, \dots, n$,

$$\pi_i(t)/\mu_i(t) \leq M, \quad t \in [0, T], \quad \text{a.s.}$$

Admissibility conditions vary in the literature, and a portfolio that satisfies Definition (3.3.1) may not be “admissible” in other settings. Condition (i) is imposed here because we are interested in portfolios without short sales. Condition (ii) implies limited negative performance relative to the market as numeraire. The market is a natural numeraire for equity managers whose performance is measured versus the market as benchmark. Condition (iii) prevents arbitrarily high overweighting of any particular stock relative to the market weighting. We are interested in arbitrage opportunities composed of admissible portfolios.

3.3.2 Definition. Let η and ξ be portfolios. Then η *dominates* ξ in $[0, T]$ if

$$\widehat{Z}_\eta(T)/\widehat{Z}_\eta(0) \geq \widehat{Z}_\xi(T)/\widehat{Z}_\xi(0), \quad \text{a.s.},$$

and

$$\widehat{Z}_\eta(T)/\widehat{Z}_\eta(0) > \widehat{Z}_\xi(T)/\widehat{Z}_\xi(0), \quad \text{a.s.}$$

If

$$\widehat{Z}_\eta(T)/\widehat{Z}_\eta(0) > \widehat{Z}_\xi(T)/\widehat{Z}_\xi(0), \quad \text{a.s.}, \quad (3.3.1)$$

then η *strictly dominates* ξ in $[0, T]$.

It is clear from this definition that if η strictly dominates ξ in $[0, T]$, then η dominates ξ in $[0, T]$.

Suppose that η and ξ are admissible portfolios such that η dominates ξ . Proposition 1.1.5 implies that the value of an investment in a portfolio is scalable by setting its initial value, so we can buy one dollar’s worth of η at time 0, and finance this purchase by selling one dollar’s worth of ξ short at the same time. Therefore, the total initial value of our portfolio holdings is zero. At time T , the dollar value of our holdings in η will be

$$\widehat{Z}_\eta(T)/\widehat{Z}_\eta(0), \quad (3.3.2)$$

and the dollar value we owe on the short sale of ξ will be

$$\widehat{Z}_\xi(T)/\widehat{Z}_\xi(0). \quad (3.3.3)$$

Definition 3.3.2 implies that, with probability one, (3.3.2) is not less than (3.3.3), and will be greater than (3.3.3) with positive probability. It follows that the total value of our holdings at time T will be nonnegative with probability one, and positive with positive probability. Hence, this combination of investments is an arbitrage opportunity. Therefore, proof of the existence of a pair of admissible portfolios, one of which dominates the other, refutes the no-arbitrage hypothesis.

3.3.3 Example. (An admissible, market-dominating portfolio) Let $\mathcal{M}$ be a market without dividends, and suppose that $\mathcal{M}$ is nondegenerate and weakly diverse in $[0, T]$. Consider the function $\mathbf{S}$ defined by

$$\mathbf{S}(x) = 1 - \frac{1}{2} \sum_{i=1}^n x_i^2. \quad (3.3.4)$$

Then $\mathbf{S}$ generates a portfolio π with weights

$$\pi_i(t) = \left(\frac{2 - \mu_i(t)}{\mathbf{S}(\mu(t))} - 1 \right) \mu_i(t), \quad (3.3.5)$$

for $i = 1, \dots, n$, and a drift process that satisfies

$$d\Theta(t) = \frac{1}{2\mathbf{S}(\mu(t))} \sum_{i=1}^n \mu_i^2(t) \tau_{ii}(t) dt.$$

We shall show that π is admissible and strictly dominates the market portfolio if T is sufficiently large.

Let us assume for now that $T > 0$, and we shall determine later how large it must be. We first show that π is admissible. From (3.3.4) it is clear that

$$\frac{1}{2} < \mathbf{S}(\mu(t)) < 1, \quad t \in [0, T], \quad \text{a.s.} \quad (3.3.6)$$

This and (3.3.5) imply that for $i = 1, \dots, n$,

$$0 < \pi_i(t) < 3\mu_i(t), \quad t \in [0, T], \quad \text{a.s.},$$

so conditions (i) and (iii) of Definition 3.3.1 are satisfied. Since for all i , $\tau_{ii}(t) \geq 0$, the drift process Θ is nondecreasing, so

$$\log(Z_\pi(T)/Z_\pi(0)) - \log(Z_\mu(T)/Z_\mu(0)) \geq \log(\mathbf{S}(\mu(T))/\mathbf{S}(\mu(0))), \quad \text{a.s.}$$

From (3.3.6) it follows that $\mathbf{S}(\mu(T))/\mathbf{S}(\mu(0)) \geq \frac{1}{2}$, so

$$Z_\pi(T)/Z_\pi(0) \geq \frac{1}{2} Z_\mu(T)/Z_\mu(0), \quad \text{a.s.},$$

and hence condition (ii) of Definition 3.3.1 is satisfied, so π is admissible.

Now we must show that π strictly dominates μ . Since $\mathcal{M}$ is nondegenerate, Lemma 2.1.6 implies that there is an $\varepsilon > 0$ such that for $i = 1, \dots, n$,

$$\tau_{ii}(t) \geq \varepsilon(1 - \mu_{\max}(t))^2, \quad t \in [0, T], \quad \text{a.s.}$$

This inequality and (3.3.6) imply that a.s.,

$$\begin{aligned} \Theta(T) &= \frac{1}{2} \int_0^T \frac{1}{\mathbf{S}(\mu(t))} \sum_{i=1}^n \mu_i^2(t) \tau_{ii}(t) dt \\ &\geq \frac{\varepsilon}{2} \int_0^T \sum_{i=1}^n \mu_i^2(t) (1 - \mu_{\max}(t))^2 dt \\ &\geq \frac{\varepsilon}{2n} \int_0^T (1 - \mu_{\max}(t))^2 dt, \end{aligned}$$

since $\sum_{i=1}^n \mu_i^2(t) \geq 1/n$. Since $\mathcal{M}$ is weakly diverse in $[0, T]$, there is a $\delta > 0$ such that

$$\frac{1}{T} \int_0^T (1 - \mu_{\max}(t)) dt > \delta, \quad \text{a.s.},$$

so by Schwarz's inequality,

$$\frac{1}{T} \int_0^T (1 - \mu_{\max}(t))^2 dt > \delta^2, \quad \text{a.s.}$$

Therefore,

$$\Theta(T) \geq \frac{\varepsilon \delta^2 T}{2n}, \quad \text{a.s.},$$

and this combined with (3.3.6) implies that

$$\log(Z_\pi(T)/Z_\pi(0)) - \log(Z_\mu(T)/Z_\mu(0)) \geq \frac{\varepsilon \delta^2 T}{2n} - \log 2, \quad \text{a.s.}$$

Hence, if

$$T > \frac{2n \log 2}{\varepsilon \delta^2}, \tag{3.3.7}$$

then π strictly dominates the market portfolio in $[0, T]$. $\square$

Remark. The arbitrage based on Example 3.3.3 would require a long time horizon, since the expression for T in (3.3.7) has the presumably small numbers ε and δ in the denominator. Perhaps short-term arbitrage can be excluded even in a nondegenerate, weakly diverse market. $\square$

It is difficult, if not impossible, to test the validity of the no-arbitrage hypothesis empirically. In the literature, no-arbitrage frequently follows from the assumed existence of an equivalent martingale measure, and the

existence of such a measure is not amenable to statistical verification. While statistical tests of various versions of the *efficient market hypothesis* have appeared, none of these constitute a test of no-arbitrage.

Example 3.3.3 shows that arbitrage is possible in a market that seems eminently well behaved. Consider the market conditions that lead to the existence of arbitrage. The assumption that the market is nondegenerate is completely compatible with the no-arbitrage hypothesis. Indeed, this assumption is sometimes used to help establish no-arbitrage. The lack of dividend payments in Example 3.3.3 could be viewed with suspicion, but this simplifying assumption is quite common in the literature, where no-arbitrage enjoys virtually complete hegemony. That leaves us with weak diversity.

From a normative point of view, weak diversity seems like an innocuous enough assumption, and it would surely be imposed upon an actual equity market by any credible antitrust regulation. Compare this mild assumption to the all-encompassing existence of an equivalent martingale measure. The former implies arbitrage, the latter no-arbitrage. The former is suggested by common sense, the latter is a complicated mathematical construct. The choice between these two assumptions determines whether or not arbitrage exists in an equity market.

Weak diversity may conform with common sense, but it, too, probably cannot be established empirically. Since the $\delta > 0$ in the definition of weak diversity is arbitrary, a statistical test for weak diversity will depend on detecting an event of arbitrarily small probability. Hence, weak diversity proscribes an event for which the probability of occurrence is so vanishingly small that it will never be observed. Since empirical verification depends on observation, it is unlikely that an empirical test can be devised for weak diversity.

In light of this discussion, it would seem that the no-arbitrage hypothesis must be relegated to the class of “empirically undecidable” statements, along with the older problem of determining the number of angels that can dance on the head of a pin.

3.3.4 Problem. Suppose that the market satisfies the hypotheses of Example 3.3.3. Show that the portfolio generated by $\mathbf{S}(x) = x_1^2$ in Example 3.1.6 (5) is strictly dominated by the market portfolio. Show that, in this case, weak diversity need only pertain to μ_1 , rather than $\mu_{\max}$.

3.3.5 Problem. (!) Construct an example of a weakly diverse market. Note that if the growth rate and volatility processes for the market model in (1.1.6) are all a.s. bounded, then there exists an equivalent martingale measure (see Karatzas and Shreve (1991), Section 5.8 A, for a proof), in which case Example 3.3.3 implies that the market is not weakly diverse.

3.3.6 Problem. (!) Find a dividend payment structure that can ensure no-arbitrage in a nondegenerate, weakly diverse market.

3.3.7 Problem. (!!) How much can the hypotheses that the market is nondegenerate and weakly diverse be weakened in Example 3.3.3 and still imply the existence of arbitrage?

3.3.8 Problem. (!!) In reference to the remark on page 60, what is the minimal time horizon for arbitrage in a nondegenerate, weakly diverse market with no dividends?

3.3.9 Problem. (!!) Construct a model that forbids “short-term arbitrage,” but allows “long-term arbitrage.” How does this model affect the pricing of long-term warrants?

3.4 Measures of Diversity

In Section 2.3 we used the entropy function as a measure of market diversity; in this section we shall present a general definition of such measures. We are interested in measures of diversity for two reasons. The first is that market diversity is an observable characteristic of equity markets that is amenable to stochastic analysis. Hence, it is useful to consider more general measures of diversity than the entropy function. The second reason is that measures of diversity can be used to construct portfolios with desirable investment characteristics, as we shall see in Chapters 6 and 7.

Recall that a real-valued function F defined on a subset of $\mathbb{R}^n$ is *symmetric* if it is invariant under permutations of the variables $x_i, i = 1, \dots, n$, and *concave* if for $0 < p < 1$ and $x, y \in \mathbb{R}^n$, $F(px + (1 - p)y) > pF(x) + (1 - p)F(y)$.

3.4.1 Definition. A positive C^2 function defined on an open neighborhood of Δ^n is a *measure of diversity* if it is symmetric and concave. A portfolio generated by a measure of diversity is called a *diversity-weighted* portfolio, and its proportions are called *diversity weights*.

In this definition, symmetry ensures that all stocks are treated in the same manner, and concavity implies that transferring capital from a larger company to a smaller one increases the value of the measure. The results of Section 3.1 imply that measures of diversity can be used to generate portfolios.

3.4.2 Proposition. Suppose that $\mathbf{S}$ is a measure of diversity that generates a portfolio π with drift process Θ . Then Θ is a.s. nondecreasing, and $\mu_i(t) \geq \mu_j(t)$ implies that $\pi_j(t)/\mu_j(t) \geq \pi_i(t)/\mu_i(t)$ for all $t \in [0, T]$, a.s.

Proof. If $\mathbf{S}$ is a measure of diversity, then by definition it is concave and C^2 . It is well known that for a concave C^2 function, the matrix $(D_{ij}\mathbf{S}(x))$ is negative semidefinite. A negative semidefinite matrix has no positive eigenvalues, so Proposition 3.1.15 implies that Θ is nondecreasing, a.s.

Now suppose that $x = (x_1, \dots, x_n) \in \Delta^n$ with $x_i \leq x_j$ for some $i < j$. Define

$$x(u) = (x_1, \dots, x_{i-1}, (1-u)x_i + ux_j, x_{i+1}, \dots, \\ \dots, x_{j-1}, ux_i + (1-u)x_j, x_{j+1}, \dots, x_n),$$

so $x(0) = x$ and $x(1)$ is x with the i th and j th coordinates reversed. Define $f(u) = \mathbf{S}(x(u))$, so f is C^2 and concave, and since $\mathbf{S}$ is symmetric, $f(0) = f(1)$. Now,

$$f'(u) = (x_j - x_i)(D_i \mathbf{S}(x(u)) - D_j \mathbf{S}(x(u))),$$

and the concavity of f implies that $f'(0) \geq 0$. Since $x_i \leq x_j$, it follows that $D_i \mathbf{S}(x) \geq D_j \mathbf{S}(x)$. Then (3.1.7) of Theorem 3.1.5 implies that for $\mu_i(t) \geq \mu_j(t)$, we have $\pi_j(t)/\mu_j(t) \geq \pi_i(t)/\mu_i(t)$. $\square$

This proposition shows that the weight ratios $\pi_i(t)/\mu_i(t)$ decrease with increasing market weight. Hence, if a stock's market weight increases, i.e., the stock goes up relative to the market, then the portfolio π sells some (fractional) shares of that stock.

Let us now consider some examples of measures of diversity.

3.4.3 Example. The entropy function

$$\mathbf{S}(x) = - \sum_{i=1}^n x_i \log x_i$$

of Section 2.3 is the archetypal measure of diversity, and its properties as a portfolio generating function were considered in Example 3.1.2. $\square$

3.4.4 Example. For $0 < p < 1$, let

$$\mathbf{D}_p(x) = \left(\sum_{i=1}^n x_i^p \right)^{1/p}. \quad (3.4.1)$$

This is a measure of diversity, and has in fact been used to construct an institutional equity investment product (see Section 7.2). The portfolio generated by $\mathbf{D}_p$ has weights

$$\pi_i(t) = \frac{\mu_i^p(t)}{(\mathbf{D}_p(\mu(t)))^p}, \quad t \in [0, T],$$

for $i = 1, \dots, n$. Note that as $p \rightarrow 1$, π approaches the market portfolio. The drift process, which is increasing, satisfies

$$d\Theta(t) = (1-p)\gamma_\pi^*(t) dt, \quad t \in [0, T].$$

The function $\mathbf{D}_p$ has advantages over the entropy function both for the purpose of generating portfolios and as a measure of diversity. When $\mathbf{D}_p$

is used to generate portfolios, the parameter p can be varied to adjust the risk and return characteristics of the portfolio that is generated. Moreover, $\mathbf{D}_p$ is scale invariant in the sense that if $x_1, \dots, x_n$ are positive numbers that do not necessarily add up to 1, then

$$\frac{\mathbf{D}_p(x_1, \dots, x_n)}{x_1 + \dots + x_n} = \mathbf{D}_p\left(\frac{x_1}{x_1 + \dots + x_n}, \dots, \frac{x_n}{x_1 + \dots + x_n}\right). \quad (3.4.2)$$

A normalized version of $\mathbf{D}_p$,

$$\tilde{\mathbf{D}}_p(x) = \left(n^{p-1} \sum_{i=1}^n x_i^p\right)^{1/p}, \quad (3.4.3)$$

that attains its maximum value of 1 when all the x_i are equal, can provide a valid comparison of the diversity of different-sized markets. For example, if each of the companies represented in the market were to break up into k equally capitalized new companies, resulting in a new market of nk stocks, the value of $\tilde{\mathbf{D}}_p$ would not change. Proposition 3.1.14 implies that $\mathbf{D}_p$ and $\tilde{\mathbf{D}}_p$ both generate the same portfolio. $\square$

3.4.5 Example. The geometric mean in Example 3.1.6 is a measure of diversity that generates a portfolio with all weights equal to n^{-1} and drift process such that $d\Theta(t) = \gamma_\pi^*(t) dt$. $\square$

3.4.6 Example. The generating function used in Example 3.3.3,

$$\mathbf{S}(x) = 1 - \frac{1}{2} \sum_{i=1}^n x_i^2,$$

is a measure of diversity. $\square$

3.4.7 Example. The *Gini coefficient* is frequently used by economists to measure the diversity of the distribution of wealth. It is usually defined as

$$G(x) = \frac{1}{2} \sum_{i=1}^n |x_i - n^{-1}|.$$

If we modify it to

$$\mathbf{G}(x) = 1 - \frac{1}{2} \sum_{i=1}^n |x_i - n^{-1}|,$$

this comes closer to Definition 3.4.1, but fails to be C^2 . An analysis of $\mathbf{G}$ will be given in Example 4.2.2; here we shall settle for a quadratic version of it. Let

$$\mathbf{S}(x) = 1 - \frac{1}{2} \sum_{i=1}^n (x_i - n^{-1})^2.$$

This measure of diversity is similar to that of Example 3.4.6 and generates a portfolio with weights

$$\pi_i(t) = \left(\frac{n^{-1} - \mu_i(t)}{\mathbf{S}(\mu(t))} + 1 - \sum_{j=1}^n \frac{\mu_j(t)(n^{-1} - \mu_j(t))}{\mathbf{S}(\mu(t))} \right) \mu_i(t), \quad t \in [0, T],$$

for $i = 1, \dots, n$, and an increasing drift process that satisfies

$$d\Theta(t) = \frac{1}{2\mathbf{S}(\mu(t))} \sum_{i=1}^n \mu_i^2(t) \tau_{ii}(t) dt, \quad t \in [0, T].$$

□

3.4.8 Example. The *Rényi entropy* is a generalization of the entropy function defined by

$$\mathbf{S}_p(x) = \frac{1}{1-p} \log \sum_{i=1}^n x_i^p,$$

for $p \neq 1$. As $p \rightarrow 1$, $\mathbf{S}_p$ tends to the usual entropy function. It can be shown that for $p < 1$, $\mathbf{S}_p$ is a measure of diversity, but for $p > 1$, $\mathbf{S}_p$ is not concave. For $p > 1$, the weights of the portfolio $\mathbf{S}_p$ generates may be negative, and the corresponding drift process may be locally decreasing. □

3.5 Notes and Summary

Functionally generated portfolios first appeared in Fernholz (1999b), and constitute one of the basic tools of stochastic portfolio theory. Functionally generated portfolios are a natural generalization of the entropy-weighted portfolio introduced in the previous chapter.

The importance of the price-to-book ratio presented in Example 3.1.10 may be due to Sharpe (1988) and Fama and French (1995), who claimed that this ratio is an essential factor in portfolio return.

The Black/Scholes option pricing model discussed in Section 3.2 was developed by Black and Scholes (1973), and appears in more general form in, e.g., Karatzas and Shreve (1998).

The no-arbitrage hypothesis discussed in Section 3.3 is central to modern mathematical finance—Karatzas (1997) characterizes it as “a basic tenet of the reality available to most of us” (see also Duffie (1992)). Examples of arbitrage in the literature (see, e.g., Karatzas (1997), Section 0.2) do not seem to resemble actual equity markets, so perhaps it was thought that arbitrage could occur only in bizarre circumstances. The use of an equivalent martingale measure to ensure no-arbitrage was originally proposed in Harrison and Kreps (1979) (see also Harrison and Pliska (1981) and Dybvig and Huang (1988)). Statistical tests of the efficient market hypothesis

are discussed in, e.g., Taylor (1986) and Malkiel (1990), but these tests are irrelevant for the no-arbitrage hypothesis. The nondegeneracy condition in Example 3.3.3 was used in Karatzas and Shreve (1991), Section 5.8, to help establish the existence of an equivalent martingale measure, and hence the validity of the no-arbitrage hypothesis.

Definition 3.4.1 is an adaptation of measures of diversity to equity markets. Measures of diversity have been used in probability and information theory (see Shannon (1948)), and in mathematical ecology (see Simpson (1949), Good (1953), and Hill (1973)). The measure of diversity $\mathbf{D}_p$ appeared in our context in Fernholz et al. (1998) and Fernholz (1999b). The Rényi entropy function was proposed by Rényi (1960).

Chapter Summary

Sections 3.1 and 3.2: The market portfolio μ defines a point on the set

$$\Delta^n = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 1; 0 < x_i < 1, i = 1, \dots, n\}.$$

A positive continuous function $\mathbf{S}$ defined on Δ^n *generates* a portfolio π if there exists a measurable process of bounded variation Θ such that

$$\log(Z_\pi(t)/Z_\mu(t)) = \log \mathbf{S}(\mu(t)) + \Theta(t), \quad t \in [0, T], \quad \text{a.s.} \quad (3.1.1)$$

The process Θ is called the *drift* process corresponding to $\mathbf{S}$. We say that π is a *functionally generated* portfolio, and the essential property of such a portfolio is that its return relative to the market is connected, through (3.1.1), to the behavior of the function that generates it.

Many functions generate portfolios. If $\mathbf{S}$ is a positive C^2 function defined on Δ^n such that for all i , $x_i D_i \log \mathbf{S}(x)$ is bounded on Δ^n , then $\mathbf{S}$ generates the portfolio π with weights

$$\pi_i(t) = \left(D_i \log \mathbf{S}(\mu(t)) + 1 - \sum_{j=1}^n \mu_j(t) D_j \log \mathbf{S}(\mu(t)) \right) \mu_i(t), \quad (3.1.7)$$

for $t \in [0, T]$ and $i = 1, \dots, n$, and a drift process that satisfies

$$d\Theta(t) = \frac{-1}{2\mathbf{S}(\mu(t))} \sum_{i,j=1}^n D_{ij} \mathbf{S}(\mu(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \quad (3.1.8)$$

a.s., for $t \in [0, T]$, where $\tau_{ij}(t) dt = d\langle \mu_i, \mu_j \rangle_t$.

Time-dependent generating functions can be defined in a similar manner, and there is a connection between these time-dependent generating functions and option pricing theory.

Section 3.3: In a nondegenerate, weakly diverse market without dividends, it is possible to generate a portfolio that will a.s. have higher return than the market portfolio over a fixed time period. This violates the no-arbitrage hypothesis of mathematical finance.

Consider the function $\mathbf{S}$ defined by

$$\mathbf{S}(x) = 1 - \frac{1}{2} \sum_{i=1}^n x_i^2. \quad (3.3.4)$$

This function generates a portfolio π with weights

$$\pi_i(t) = \left(\frac{2 - \mu_i(t)}{\mathbf{S}(\mu(t))} - 1 \right) \mu_i(t), \quad (3.3.5)$$

for $i = 1, \dots, n$, and a drift process that satisfies

$$d\Theta(t) = \frac{1}{2\mathbf{S}(\mu(t))} \sum_{i=1}^n \mu_i^2(t) \tau_{ii}(t) dt.$$

For this portfolio π , there exists a fixed time $T > 0$ such that

$$Z_\pi(T)/Z_\pi(0) > Z_\mu(T)/Z_\mu(0), \quad \text{a.s.}$$

Section 3.4: A positive C^2 function defined on an open neighborhood of Δ^n is a *measure of diversity* if it is symmetric and concave. A portfolio generated by a measure of diversity is called a *diversity-weighted* portfolio, and its proportions are called *diversity weights*.

Besides entropy, which was introduced in Chapter 2, the most important measure of diversity for our purposes is $\mathbf{D}_p$, defined by

$$\mathbf{D}_p(x) = \left(\sum_{i=1}^n x_i^p \right)^{1/p}, \quad (3.4.1)$$

for $0 < p < 1$. This is a measure of diversity, and in fact has been used to construct an institutional equity investment product (see Fernholz et al. (1998)). The portfolio generated by $\mathbf{D}_p$ has weights

$$\pi_i(t) = \frac{\mu_i^p(t)}{(\mathbf{D}_p(\mu(t)))^p}, \quad t \in [0, T],$$

for $i = 1, \dots, n$. The drift process, which is increasing, satisfies

$$d\Theta(t) = (1 - p)\gamma_\pi^*(t) dt, \quad t \in [0, T].$$

As $p \rightarrow 1$, the $\mathbf{D}_p$ -weighted portfolio tends to the market portfolio.

4

Portfolios of Stocks Selected by Rank

The distribution of capital is of fundamental importance in stochastic portfolio theory, as are functionally generated portfolios. In this chapter we shall combine these two concepts.

Usually we identify stocks by their *names*, i.e., their subscripts, X_1, X_2, X_3 , etc. However, with regard to the distribution of capital, it is advantageous to identify the stocks by their *ranks* rather than their names. If we wish to work with ranked stocks, we must have a precise definition of rank that allows us to use the available mathematical tools. Suppose $x_1, \dots, x_n \in \mathbb{R}$ is a set of numbers, and define

$$x_{(k)} = \max_{1 \leq i_1 < \dots < i_k \leq n} \min(x_{i_1}, \dots, x_{i_k}), \quad (4.0.1)$$

where $1 \leq k \leq n$. Then it is not difficult to see that $x_{(k)} \in \{x_1, \dots, x_n\}$, and

$$\max(x_1, \dots, x_n) = x_{(1)} \geq x_{(2)} \geq \dots \geq x_{(n)} = \min(x_1, \dots, x_n).$$

The value of (4.0.1) is that it expresses the ranked numbers in terms of maxima and minima, and the maximum and minimum functions are mathematically tractable. From (4.0.1) we derive the following definition.

4.0.1 Definition. Let $X_1, \dots, X_n$ be processes. Then for $k = 1, \dots, n$, the *kth rank process* of $\{X_1, \dots, X_n\}$ is defined by

$$X_{(k)}(t) = \max_{1 \leq i_1 < \dots < i_k \leq n} \min(X_{i_1}(t), \dots, X_{i_k}(t)), \quad t \in [0, T]. \quad (4.0.2)$$

We are particularly interested in the ranked market weights,

$$\mu_{(1)}(t) \geq \mu_{(2)}(t) \geq \cdots \geq \mu_{(n)}(t), \quad t \in [0, T],$$

where the $\mu_{(k)}$ are the rank processes associated with the market weights $\mu_1, \dots, \mu_n$. The ranked family of market weights $\{\mu_{(1)}(t), \dots, \mu_{(n)}(t)\}$ will be called the *capital distribution* of the market at time t .

The capital distribution for a subset of the market can also be defined, and this is of particular interest when the subset is an index such as the S&P 500 Index, which is composed of 500 of the largest stocks traded on U.S. stock exchanges. The two curves in Figure 4.1 show the capital distribution of the S&P 500 Index for December 30, 1997 (solid line), and for December 29, 1999 (broken line). The points on the curves in Figure 4.1 represent the weights of the stocks in the S&P 500 Index, expressed as percentages. For example, the leftmost point on the solid line represents General Electric, the company with the largest capitalization on December 30, 1997. As can be seen, its weight in the S&P 500 Index on that date was slightly over 3%. The leftmost point on the broken line represents Microsoft, the company with the largest capitalization on December 29, 1999. Although the S&P 500 Index is not the market, it usually holds about 70% of the total market capitalization including all of the largest stocks, so these curves give a reasonable idea of the capital distribution of the whole market. From the two curves, we can see that there was more concentration of

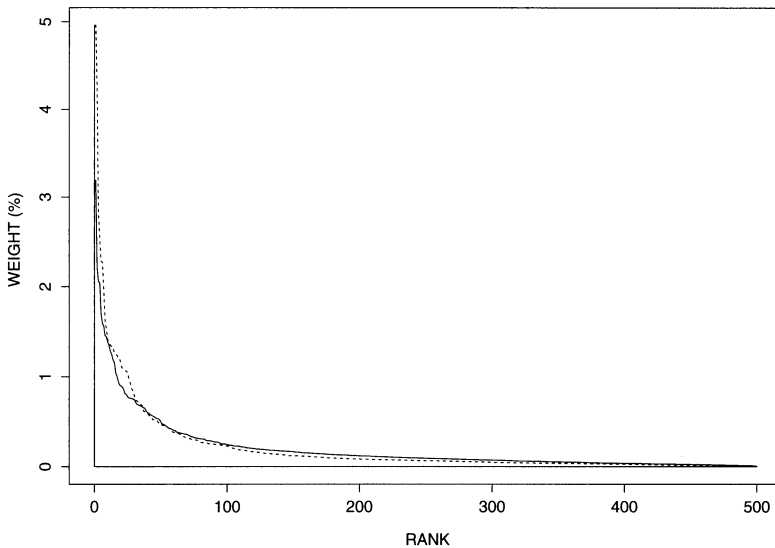


FIGURE 4.1. Capital distribution for the S&P 500 Index.
December 30, 1997 (solid line), and December 29, 1999 (broken line).

capital in the larger stocks at the end of 1999 than at the end of 1997. This means that, in some sense, there was a flow of capital from the smaller stocks into the larger stocks during 1998 and 1999.

The movement of capital that took place from 1997 to 1999 is likely to have had some effect on portfolios of stocks. The flow of capital from the smaller stocks to the larger stocks would improve the performance of the larger stocks at the expense of the smaller ones. Hence, a portfolio with weights more concentrated in the larger stocks would probably have had a performance advantage over one with less such concentration. The methodology we introduce in this chapter enables us to measure the effect that such movements of capital have on portfolio return.

The main result of this chapter is a version of Theorem 3.1.5 that holds for generating functions of the form

$$\mathbf{S}(\mu_1(t), \dots, \mu_n(t)) = S(\mu_{(1)}(t), \dots, \mu_{(n)}(t)), \quad (4.0.3)$$

where S is positive and C^2 in a neighborhood of Δ^n . This situation is common in applications because portfolios are frequently chosen exclusively from either a large-stock index or a small-stock index, and the selection of these indices depends on the ranked market weights of the stocks. This case is more complicated mathematically than Theorem 3.1.5, because the generating function $\mathbf{S}$ is not differentiable, and hence Itô's rule cannot be applied directly. Instead, semimartingale local times must be introduced to account for the changes in rank that occur randomly among the stock price processes. Consequently, the drift process for a portfolio generated by a function of the form (4.0.3) has two distinct parts: a smooth component similar to the drift function in (3.1.8) of Theorem 3.1.5, and a component involving local times for changes in rank among the market weights.

We first present some introductory material regarding semimartingale local times, and prove results we need for rank processes derived from continuous semimartingales. Then we prove the main theorem, and finally we provide a few examples of applications of the theory.

4.1 Rank Processes and Local Times

Our goal in this section is to derive a representation for the rank processes corresponding to the market weights $\mu_1, \dots, \mu_n$. In order to accomplish this, we must introduce the concept of semimartingale local times. Let us begin with a definition for a semimartingale local time, a measure of the amount of time a process spends near the origin.

4.1.1 Definition. Let X be a continuous semimartingale. Then the *local time* (at 0) for X is the process Λ_X defined for $t \in [0, T]$ by

$$\Lambda_X(t) = \frac{1}{2} \left(|X(t)| - |X(0)| - \int_0^t \operatorname{sgn}(X(s)) dX(s) \right), \quad (4.1.1)$$

where $\text{sgn}(x) = 2I_{(0,\infty)}(x) - 1$, with $I_{(0,\infty)}$ the indicator function of $(0, \infty)$.

For general background on local times, see Karatzas and Shreve (1991). Note that the asymmetry in sgn induces an asymmetry in the local time: In general, Λ_X differs from Λ_{-X} . It can be shown that $\Lambda_X(t)$ is almost surely nondecreasing in t , and satisfies

$$I_{\{0\}}(X(t)) d\Lambda_X(t) = d\Lambda_X(t), \quad t \in [0, T], \quad \text{a.s.} \quad (4.1.2)$$

(see Karatzas and Shreve (1991), Theorem 3.7.1). This implies, for example, that for one-dimensional Brownian motion B , Λ_B is a nonnegative random measure on $[0, T]$ that almost surely has support contained in the set $\{t : B(t) = 0\}$, and hence is singular with respect to Lebesgue measure. In order to effectively use local times, we shall assume that the stock price processes we consider exhibit a certain level of nondegeneracy.

4.1.2 Definition. The processes $X_1, \dots, X_n$ are *pathwise mutually nondegenerate* if:

- (i) for all $i \neq j$, $\{t : X_i(t) = X_j(t)\}$ has Lebesgue measure zero, a.s.;
- (ii) for all $i < j < k$, $\{t : X_i(t) = X_j(t) = X_k(t)\} = \emptyset$, a.s.

4.1.3 Problem. Show that the components of multidimensional Brownian motion $(W_1, \dots, W_n)$ are pathwise mutually nondegenerate, at least for $t > 0$. Condition (i) follows from the fact that, with probability one, the zero set of one-dimensional Brownian motion has Lebesgue measure 0, and condition (ii) follows from the fact that, with probability one, 2-dimensional Brownian motion never returns to the origin (see Karatzas and Shreve (1991)).

4.1.4 Problem. (!!) Does Definition 1.1.2, nondegenerate, imply Definition 4.1.2, pathwise mutually nondegenerate? If not, what additional conditions are necessary?

Stock price processes have certain properties that facilitate the application of local times. We show that the following definition describes one such property.

4.1.5 Definition. Let X be a continuous semimartingale with decomposition

$$X(t) = X(0) + M_X(t) + V_X(t), \quad t \in [0, T], \quad \text{a.s.}, \quad (4.1.3)$$

where M_X is a continuous local martingale and V_X is a continuous process with locally bounded variation. Then X is *absolutely continuous* if the random signed measures dV_X and $d\langle X \rangle$ on $[0, T]$ are both almost surely absolutely continuous with respect to Lebesgue measure.

Note that the sample paths of absolutely continuous semimartingales are usually not absolutely continuous functions of t : Consider, e.g., Brownian motion.

The next two lemmas show that C^2 functions of absolutely continuous semimartingales are themselves absolutely continuous semimartingales. This implies that stock price processes, the market value process, and market weight processes, are all absolutely continuous semimartingales.

4.1.6 Lemma. *Suppose X and Y are absolutely continuous semimartingales. Then the random signed measure $d\langle X, Y \rangle$ is almost surely absolutely continuous with respect to Lebesgue measure on $[0, T]$.*

Proof. The processes $\langle X + Y \rangle$ and $\langle X - Y \rangle$ are almost surely increasing on $[0, T]$, with

$$\langle X \pm Y \rangle_t = \langle X \rangle_t \pm 2\langle X, Y \rangle_t + \langle Y \rangle_t, \quad t \in [0, T], \quad \text{a.s.}$$

If A is a Lebesgue measurable subset of $[0, T]$, then, a.s.,

$$\int_A d\langle X \rangle_t \pm 2 \int_A d\langle X, Y \rangle_t + \int_A d\langle Y \rangle_t = \int_A d\langle X \pm Y \rangle_t \geq 0.$$

Hence, a.s.,

$$2 \left| \int_A d\langle X, Y \rangle_t \right| \leq \int_A d\langle X \rangle_t + \int_A d\langle Y \rangle_t. \quad (4.1.4)$$

Since X and Y are absolutely continuous, if A has Lebesgue measure zero, then the right-hand side of (4.1.4) is zero, so a.s.,

$$\int_A d\langle X, Y \rangle_t = 0. \quad (4.1.5)$$

Since this holds for any set A of Lebesgue measure zero, $d\langle X, Y \rangle$ is a.s. absolutely continuous. $\square$

The next lemma shows that when Itô's rule is applied to absolutely continuous semimartingales, the result is an absolutely continuous semimartingale.

4.1.7 Lemma. *Suppose $X_1, \dots, X_n$ are absolutely continuous semimartingales, and f is a real-valued C^2 function defined on the range of $(X_1, \dots, X_n)$ in $\mathbb{R}^n$. Then $f(X_1, \dots, X_n)$ is an absolutely continuous semimartingale.*

Proof. Itô's rule implies that the process Y defined by

$$Y(t) = f(X_1(t), \dots, X_n(t))$$

is a continuous semimartingale such that, a.s., for $t \in [0, T]$,

$$\begin{aligned} dY(t) &= \sum_{i=1}^n D_i f(X_1(t), \dots, X_n(t)) dX_i(t) \\ &\quad + \frac{1}{2} \sum_{i,j=1}^n D_{ij} f(X_1(t), \dots, X_n(t)) d\langle X_i, X_j \rangle_t. \end{aligned}$$

Let $dY(t) = dM_Y(t) + dV_Y(t)$ be the decomposition (4.1.3) for Y . Then

$$\begin{aligned} dV_Y(t) &= \sum_{i=1}^n D_i f(X_1(t), \dots, X_n(t)) dV_{X_i}(t) \\ &\quad + \frac{1}{2} \sum_{i,j=1}^n D_{ij} f(X_1(t), \dots, X_n(t)) d\langle X_i, X_j \rangle_t, \end{aligned} \tag{4.1.6}$$

$t \in [0, T]$, a.s. The first term on the right-hand side of (4.1.6) is absolutely continuous, since $D_i f(X_1(t), \dots, X_n(t))$ is almost surely continuous in t for $i = 1, \dots, n$, and $dV_{X_i}(t)$ is absolutely continuous by hypothesis. The second term on the right-hand side of (4.1.6) is absolutely continuous, since $D_{ij} f(X_1(t), \dots, X_n(t))$ is almost surely continuous in t and $d\langle X_i, X_j \rangle_t$ is absolutely continuous by Lemma 4.1.6.

Similar reasoning shows that the process $\langle Y \rangle$, which satisfies

$$d\langle Y \rangle_t = \sum_{i,j=1}^n D_i f(X_1(t), \dots, X_n(t)) D_j f(X_1(t), \dots, X_n(t)) d\langle X_i, X_j \rangle_t,$$

for all $t \in [0, T]$, a.s., is also absolutely continuous. $\square$

The next two lemmas show the relationship between the local times $\Lambda_{|X|}$ and Λ_X for an absolutely continuous semimartingale X . This relationship will be used in this chapter and the next.

4.1.8 Lemma. *Let X and Y be continuous semimartingales with X absolutely continuous, and suppose that the set $\{t : Y(t) = 0\}$ has Lebesgue measure 0, a.s. Then*

$$\int_0^t I_{\{0\}}(Y(s)) dX(s) = 0, \quad t \in [0, T], \quad \text{a.s.}$$

Proof. We have

$$dX(t) = dM_X(t) + dV_X(t), \quad t \in [0, T],$$

where M_X and V_X are defined as in the decomposition (4.1.3). By hypothesis, dV_X is almost surely absolutely continuous with respect to Lebesgue measure, and the Lebesgue measure of $\{t : Y(t) = 0\}$ is 0, a.s., so

$$\int_0^t I_{\{0\}}(Y(s)) dV_X(s) = 0, \quad t \in [0, T], \quad \text{a.s.}$$

Therefore,

$$\int_0^t I_{\{0\}}(Y(s)) dX(s) = \int_0^t I_{\{0\}}(Y(s)) dM_X(s), \quad t \in [0, T], \quad \text{a.s.}$$

The process U defined by

$$U(t) = \int_0^t I_{\{0\}}(Y(s)) dM_X(s), \quad t \in [0, T],$$

is a square-integrable continuous local martingale. Since $\{t : Y(t) = 0\}$ almost surely has Lebesgue measure 0 and $d\langle M_X \rangle$ is almost surely absolutely continuous with respect to Lebesgue measure,

$$\langle U \rangle_t = \int_0^t I_{\{0\}}(Y(s)) d\langle M_X \rangle_s = 0, \quad t \in [0, T], \quad \text{a.s.}$$

It follows that $U(t) = 0$ for $t \in [0, T]$, a.s. (see Karatzas and Shreve (1991), Problem 1.5.12 and Exercise 1.5.21). $\square$

4.1.9 Lemma. *Let X be an absolutely continuous semimartingale such that the set $\{t : X(t) = 0\}$ has Lebesgue measure 0, almost surely. Then*

$$\Lambda_{|X|}(t) = 2\Lambda_X(t), \quad t \in [0, T], \quad \text{a.s.}$$

Proof. By definition,

$$2\Lambda_X(t) = |X(t)| - |X(0)| - \int_0^t \operatorname{sgn}(X(s)) dX(s), \quad t \in [0, T], \quad (4.1.7)$$

so

$$d|X(t)| = 2d\Lambda_X(t) + \operatorname{sgn}(X(t)) dX(t), \quad t \in [0, T], \quad \text{a.s.} \quad (4.1.8)$$

Again by definition, a.s., for $t \in [0, T]$,

$$\begin{aligned} 2\Lambda_{|X|}(t) &= |X(t)| - |X(0)| - \int_0^t \operatorname{sgn}|X(s)| d|X(s)| \\ &= 2\Lambda_X(t) + \int_0^t \operatorname{sgn}(X(s)) dX(s) - 2 \int_0^t \operatorname{sgn}|X(s)| d\Lambda_X(s) \\ &\quad - \int_0^t \operatorname{sgn}|X(s)| \operatorname{sgn}(X(s)) dX(s) \end{aligned} \quad (4.1.9)$$

$$= 4\Lambda_X(t) - 2 \int_0^t I_{\{0\}}(X(s)) dX(s) \quad (4.1.10)$$

$$= 4\Lambda_X(t), \quad (4.1.11)$$

where (4.1.9) follows from (4.1.7) and (4.1.8), (4.1.10) from (4.1.2) and the fact that $\operatorname{sgn}(0) = -1$, and (4.1.11) from Lemma 4.1.8, since X is absolutely continuous and $X(s) = 0$ only on a set of Lebesgue measure 0, a.s. $\square$

Definition 4.0.1 constructs rank processes by means of maximum and minimum operations, so we shall consider these processes first.

4.1.10 Lemma. *Let X and Y be pathwise mutually nondegenerate absolutely continuous semimartingales. Then, a.s., for $t \in [0, T]$,*

$$\begin{aligned} d \max(X(t), Y(t)) &= I_{(0, \infty)}(X(t) - Y(t)) dX(t) \\ &\quad + I_{(0, \infty)}(Y(t) - X(t)) dY(t) + d\Lambda_{X-Y}(t). \end{aligned} \quad (4.1.12)$$

Proof. We shall do the calculation:

$$\begin{aligned} d \max(X(t), Y(t)) &= \frac{dX(t) + dY(t)}{2} + \frac{d|X(t) - Y(t)|}{2} \\ &= \frac{dX(t) + dY(t)}{2} + \frac{\operatorname{sgn}(X(t) - Y(t)) d(X(t) - Y(t))}{2} \\ &\quad + d\Lambda_{X-Y}(t) \end{aligned} \quad (4.1.13)$$

$$\begin{aligned} &= I_{(0, \infty)}(X(t) - Y(t)) dX(t) \\ &\quad + I_{(0, \infty)}(Y(t) - X(t)) dY(t) + d\Lambda_{X-Y}(t) \\ &= I_{(0, \infty)}(X(t) - Y(t)) dX(t) \\ &\quad + I_{(0, \infty)}(Y(t) - X(t)) dY(t) + d\Lambda_{X-Y}(t), \end{aligned} \quad (4.1.14)$$

where (4.1.13) follows from Definition 4.1.1, and (4.1.14) is implied by Lemma 4.1.8. $\square$

A similar result holds for the minimum function, since $\min(x, y) = -\max(-x, -y)$.

The following proposition is the result we need in order to prove the main theorem in the next section. The proposition shows that rank processes derived from pathwise mutually nondegenerate absolutely continuous semimartingales can be expressed in terms of the original processes, adjusted by local times.

4.1.11 Proposition. *Let $X_1, \dots, X_n$ be pathwise mutually nondegenerate absolutely continuous semimartingales, and for $t \in [0, T]$, let p_t be the random permutation of $\{1, \dots, n\}$ such that for $k = 1, \dots, n$,*

$$\begin{aligned} X_{p_t(k)}(t) &= X_{(k)}(t), \\ p_t(k) &< p_t(k+1) \quad \text{if} \quad X_{(k)}(t) = X_{(k+1)}(t). \end{aligned}$$

Then the rank processes $X_{(k)}$, $k = 1, \dots, n$, are continuous semimartingales such that a.s., for $t \in [0, T]$,

$$\begin{aligned} dX_{(k)}(t) &= \sum_{i=1}^n I_{\{i\}}(p_t(k)) dX_i(t) \\ &\quad + \frac{1}{2} d\Lambda_{X_{(k)} - X_{(k+1)}}(t) - \frac{1}{2} d\Lambda_{X_{(k-1)} - X_{(k)}}(t). \end{aligned} \quad (4.1.15)$$

Proof. Since this proof is somewhat technical, it is convenient to explicitly show the dependence of all random variables and processes on $\omega \in \Omega$. By hypothesis and Lemma 4.1.6, we can choose a subset $\Omega' \subset \Omega$ with $P(\Omega') = 1$ such that for $\omega \in \Omega'$, and $i, j, k \in \{1, \dots, n\}$, the following conditions apply:

- (1) $X_i(t, \omega)$ is continuous in t ;
- (2) for $i \neq j$, the set $\{t : X_i(t, \omega) = X_j(t, \omega)\}$ has Lebesgue measure 0;
- (3) for $i < j < k$, the set $\{t : X_i(t, \omega) = X_j(t, \omega) = X_k(t, \omega)\} = \emptyset$;
- (4) for $i \neq j$ and $t \in [0, T]$,

$$d \max(X_i(t, \omega), X_j(t, \omega)) = I_{(0, \infty)}(X_i(t, \omega) - X_j(t, \omega)) dX_i(t, \omega) \\ + I_{(0, \infty)}(X_j(t, \omega) - X_i(t, \omega)) dX_j(t, \omega) + d\Lambda_{X_i - X_j}(t, \omega);$$

- (5) for $i \neq j$ and $t \in [0, T]$,

$$d\Lambda_{X_i - X_j}(t, \omega) = \frac{1}{2} d\Lambda_{|X_i - X_j|}(t, \omega).$$

- (6) for $k = 1, \dots, n-1$ and $t \in [0, T]$,

$$d\Lambda_{X_{(k)} - X_{(k+1)}}(t, \omega) \\ = I_{\{0\}}(X_{(k)}(t, \omega) - X_{(k+1)}(t, \omega)) d\Lambda_{X_{(k)} - X_{(k+1)}}(t, \omega).$$

Conditions 1, 2, and 3 follow directly from Definitions 4.1.5 and 4.1.2; we can impose condition 4 because of Lemma 4.1.10; condition 5 follows from Lemma 4.1.9; and condition 6 follows from (4.1.2).

Suppose that $\omega \in \Omega'$, $t_0 \in [0, T]$, and $k \in \{1, \dots, n\}$, and let $m(\omega) = p_{t_0}(k, \omega)$. Then

$$X_{(k)}(t_0, \omega) = X_{m(\omega)}(t_0, \omega).$$

There are two cases we must consider. In the first case, for all $j \neq m(\omega)$,

$$X_j(t_0, \omega) \neq X_{m(\omega)}(t_0, \omega), \quad (4.1.16)$$

and in the second case, there is an $r(\omega) \neq m(\omega)$, $1 \leq r(\omega) \leq n$, such that

$$X_{r(\omega)}(t_0, \omega) = X_{m(\omega)}(t_0, \omega). \quad (4.1.17)$$

Let us consider the first case. Condition 1 implies that (4.1.16) continues to hold for all t in some neighborhood U of t_0 . Hence, for all $t \in U$,

$$X_{(k-1)}(t, \omega) > X_{(k)}(t, \omega) > X_{(k+1)}(t, \omega),$$

so for all $t \in U$, $p_t(k, \omega) = m(\omega)$, and condition 6 implies that (4.1.15) reduces to

$$dX_{(k)}(t, \omega) = dX_{m(\omega)}(t, \omega).$$

Now consider the second case, so (4.1.17) holds. Then conditions 1 and 3 imply that there is a neighborhood U of t_0 such that for all $t \in U$, either

$$X_{(k-1)}(t, \omega) > X_{(k)}(t, \omega) \geq X_{(k+1)}(t, \omega) > X_{(k+2)}(t, \omega), \quad (4.1.18)$$

in which case for $t \in U$,

$$X_{(k)}(t, \omega) = \max(X_{m(\omega)}(t, \omega), X_{r(\omega)}(t, \omega)); \quad (4.1.19)$$

or

$$X_{(k-2)}(t, \omega) > X_{(k-1)}(t, \omega) \geq X_{(k)}(t, \omega) > X_{(k+1)}(t, \omega), \quad (4.1.20)$$

in which case for $t \in U$,

$$X_{(k)}(t, \omega) = \min(X_{m(\omega)}(t, \omega), X_{r(\omega)}(t, \omega)). \quad (4.1.21)$$

Suppose that (4.1.18) and (4.1.19) hold. Then for $t \in U$,

$$\begin{aligned} dX_{(k)}(t, \omega) &= d \max(X_{m(\omega)}(t, \omega), X_{r(\omega)}(t, \omega)) \\ &= I_{(0, \infty)}(X_{m(\omega)}(t, \omega) - X_{r(\omega)}(t, \omega)) dX_{m(\omega)}(t, \omega) \\ &\quad + I_{(0, \infty)}(X_{r(\omega)}(t, \omega) - X_{m(\omega)}(t, \omega)) dX_{r(\omega)}(t, \omega) \\ &\quad + d\Lambda_{X_{r(\omega)} - X_{m(\omega)}}(t, \omega) \\ &= I_{\{m(\omega)\}}(p_t(k, \omega)) dX_{m(\omega)}(t, \omega) \\ &\quad + I_{\{r(\omega)\}}(p_t(k, \omega)) dX_{r(\omega)}(t, \omega) + \frac{1}{2} d\Lambda_{|X_{r(\omega)} - X_{m(\omega)}|}(t, \omega) \end{aligned} \quad (4.1.22)$$

$$\begin{aligned} &= I_{\{m(\omega)\}}(p_t(k, \omega)) dX_{m(\omega)}(t, \omega) \\ &\quad + I_{\{r(\omega)\}}(p_t(k, \omega)) dX_{r(\omega)}(t, \omega) + \frac{1}{2} d\Lambda_{X_{(k)} - X_{(k+1)}}(t, \omega) \\ &= \sum_{i=1}^n I_{\{i\}}(p_t(k, \omega)) dX_i(t, \omega) \\ &\quad + \frac{1}{2} d\Lambda_{X_{(k)} - X_{(k+1)}}(t, \omega) - \frac{1}{2} d\Lambda_{X_{(k-1)} - X_{(k)}}(t, \omega), \end{aligned} \quad (4.1.24)$$

where condition 4 implies (4.1.22), (4.1.23) follows from (4.1.19) and condition 5, and (4.1.24) follows from condition 6. Hence, if (4.1.18) and (4.1.19) hold, (4.1.15) is valid.

The proof is similar if (4.1.20) and (4.1.21) hold, so (4.1.15) is valid for all $\omega \in \Omega'$. Since $P(\Omega') = 1$, the proposition is proved. $\square$

We shall apply this proposition to the market weight processes.

4.1.12 Corollary. *Let $\mathcal{M}$ be a market of stocks $X_1, \dots, X_n$ that are path-wise mutually nondegenerate. Then the market weight processes $\mu_1, \dots, \mu_n$*

satisfy

$$\begin{aligned} d \log \mu_{(k)}(t) &= \sum_{i=1}^n I_{\{i\}}(p_t(k)) d \log \mu_i(t) \\ &\quad + \frac{1}{2} d \Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) - \frac{1}{2} d \Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}(t), \end{aligned} \quad (4.1.25)$$

a.s., for $t \in [0, T]$, where p_t is the random permutation of $\{1, \dots, n\}$ such that for $k = 1, \dots, n$,

$$\begin{aligned} \mu_{p_t(k)}(t) &= \mu_{(k)}(t), \\ p_t(k) < p_t(k+1) \quad &\text{if} \quad \mu_{(k)}(t) = \mu_{(k+1)}(t). \end{aligned} \quad (4.1.26)$$

Proof. Note that the permutation p_t is uniquely defined by (4.1.26), and associates each rank process with one of the original market weights that has the same value at time t .

That the market weights are absolutely continuous semimartingales follows from Definition 1.1.2 and Lemma 4.1.7. Hence, Proposition 4.1.11 can be applied, and the corollary follows. $\square$

4.1.13 Problem. (!) Extend Proposition 4.1.11 to include processes that are not pathwise mutually nondegenerate. For example, consider the case in which there exists $\Omega' \subset \Omega$, $P(\Omega') = 1$, such that for $\omega \in \Omega'$, the set

$$\{t : X_i(t, \omega) = X_j(t, \omega) = X_k(t, \omega)\}$$

is finite for all $i < j < k$.

4.2 Portfolios Generated by Functions of Ranked Market Weights

In the previous section we saw that the market weight processes are absolutely continuous processes, so if the stocks $X_1, \dots, X_n$ in the market $\mathcal{M}$ are pathwise mutually nondegenerate, Corollary 4.1.12 allows us to represent the ranked market weights $\mu_{(1)}, \dots, \mu_{(n)}$ in terms of $\mu_1, \dots, \mu_n$. In this section we use this representation to extend Theorem 3.1.5 to functions of the form (4.0.3).

It will be convenient to use the notation

$$\mu_{(\cdot)}(t) = (\mu_{(1)}(t), \dots, \mu_{(n)}(t)), \quad t \in [0, T]. \quad (4.2.1)$$

For $i, j = 1, \dots, n$, we define the *relative rank covariance processes* $\tau_{(ij)}$ by

$$\tau_{(ij)}(t) = \tau_{p_t(i)p_t(j)}(t), \quad t \in [0, T], \quad (4.2.2)$$

where p_t is the permutation defined in (4.1.26). Since for all i and j , τ_{ij} is a.s. an L^1 function of t , the same is true for $\tau_{(ij)}$.

4.2.1 Theorem. *Let $\mathcal{M}$ be a market of stocks $X_1, \dots, X_n$ that are path-wise mutually nondegenerate, let p_t be the random permutation defined by (4.1.26), and let $\mathbf{S}$ be a function defined on a neighborhood U of Δ^n . Suppose that there exists a positive C^2 function S defined on U such that for $(x_1, \dots, x_n) \in U$,*

$$\mathbf{S}(x_1, \dots, x_n) = S(x_{(1)}, \dots, x_{(n)}), \quad (4.2.3)$$

and for $i = 1, \dots, n$, $x_i D_i \log S(x)$ is bounded for $x \in \Delta^n$. Then $\mathbf{S}$ generates the portfolio π such that for $k = 1, \dots, n$,

$$\pi_{p_t(k)}(t) = \left(D_k \log S(\mu_{(\cdot)}(t)) + 1 - \sum_{j=1}^n \mu_{(j)}(t) D_j \log S(\mu_{(\cdot)}(t)) \right) \mu_{(k)}(t), \quad (4.2.4)$$

for all $t \in [0, T]$, a.s., with a drift process Θ that satisfies

$$\begin{aligned} d\Theta(t) = & \frac{-1}{2\mathbf{S}(\mu(t))} \sum_{i,j=1}^n D_{ij} S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) dt \\ & + \frac{1}{2} \sum_{k=1}^{n-1} (\pi_{p_t(k+1)}(t) - \pi_{p_t(k)}(t)) d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t), \end{aligned} \quad (4.2.5)$$

for all $t \in [0, T]$, a.s.

Remark. For a generating function of the form (4.2.3), $\mathbf{S}(\mu(t))$ measures the effect that changes in the capital distribution of the market have on the portfolio π . This is somewhat different from Theorem 3.1.5, where the generating function measures the dependence of the portfolio on individual stocks by name (i.e., index). $\square$

We shall refer to the first term on the right-hand side of (4.2.5) as the *smooth* component of the drift process, and the second term as the *local time* component.

Proof of Theorem 4.2.1. First we must verify that π defined by (4.2.4) is a portfolio, and that Θ defined by (4.2.5) is of bounded variation. If π satisfies (4.2.4), then $\sum_{i=1}^n \pi_i(t) = 1$, and the conditions on S imply that the processes π_i are adapted and bounded on $[0, T] \times \Omega$. Hence, π is a portfolio process. Regarding Θ , let us consider the two expressions on the right-hand side of (4.2.5) separately. The process represented by the first expression is almost surely of bounded variation, because $\tau_{(ij)}$ is an L^1 function of t and the rest of the terms are continuous in t . The second expression is a sum of local times multiplied by bounded functions, and hence is also of bounded variation. Therefore, Θ is almost surely of bounded variation.

We must show that the portfolio π defined by (4.2.4) and the drift process Θ defined by (4.2.5) satisfy (3.1.2). To accomplish this, we shall analyze

the generating function term $\log \mathbf{S}(\mu(t))$ in (3.1.2) and the relative return process $\log(Z_\pi(t)/Z_\mu(t))$, and show that the difference of these two terms satisfies (4.2.5). We first need some preliminary results.

Corollary 4.1.12 states that the ranked weight processes $\mu_{(k)}$, for $k = 1, \dots, n$, satisfy

$$\begin{aligned} d \log \mu_{(k)}(t) &= \sum_{i=1}^n I_{\{i\}}(p_t(k)) d \log \mu_i(t) \\ &\quad + \frac{1}{2} d \Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) - \frac{1}{2} d \Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}(t), \end{aligned} \quad (4.2.6)$$

for $t \in [0, T]$, a.s. This and (1.2.12) imply that, for $i, j = 1, \dots, n$,

$$d \langle \log \mu_{(i)}, \log \mu_{(j)} \rangle_t = \tau_{(ij)}(t) dt, \quad t \in [0, T], \quad \text{a.s.}$$

By Itô's rule applied to $\mu_{(i)}(t) = \exp(\log \mu_{(i)}(t))$, a.s., for $t \in [0, T]$,

$$d \mu_{(i)}(t) = \mu_{(i)}(t) d \log \mu_{(i)}(t) + \frac{1}{2} \mu_{(i)}(t) \tau_{(ii)}(t) dt, \quad (4.2.7)$$

and from this we have

$$d \langle \mu_{(i)}, \mu_{(j)} \rangle_t = \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) dt, \quad t \in [0, T], \quad \text{a.s.} \quad (4.2.8)$$

Let us also note that for all $t \in [0, T]$, $\sum_{i=1}^n \mu_{(i)}(t) = 1$, and hence

$$\sum_{i=1}^n d \mu_{(i)}(t) = 0.$$

Consider now the generating function component of the relative return, $\log \mathbf{S}(\mu(t))$. Itô's rule, along with (4.2.8), implies that a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log \mathbf{S}(\mu(t)) &= d \log S(\mu_{(\cdot)}(t)) \\ &= \sum_{i=1}^n D_i \log S(\mu_{(\cdot)}(t)) d \mu_{(i)}(t) \\ &\quad + \frac{1}{2 \mathbf{S}(\mu(t))} \sum_{i,j=1}^n D_{ij} S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) dt \\ &\quad - \frac{1}{2} \sum_{i,j=1}^n D_i \log S(\mu_{(\cdot)}(t)) D_j \log S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) dt. \end{aligned} \quad (4.2.9)$$

Now let us consider the relative return process $\log(Z_\pi(t)/Z_\mu(t))$. From (1.1.23) we have a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\pi(t)/Z_\mu(t)) &= \sum_{i=1}^n \pi_i(t) d \log \mu_i(t) + \gamma_\pi^*(t) dt \\ &= \sum_{i=1}^n \sum_{k=1}^n I_{\{i\}}(p_t(k)) \pi_{p_t(k)}(t) d \log \mu_i(t) + \gamma_\pi^*(t) dt \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=1}^n \pi_{p_t(k)}(t) \sum_{i=1}^n I_{\{i\}}(p_t(k)) d \log \mu_i(t) + \gamma_\pi^*(t) dt \\
&= \sum_{k=1}^n \pi_{p_t(k)}(t) d \log \mu_{(k)}(t)
\end{aligned} \tag{4.2.10}$$

$$\begin{aligned}
&+ \frac{1}{2} \sum_{k=1}^{n-1} (\pi_{p_t(k)}(t) - \pi_{p_t(k+1)}(t)) d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) \\
&+ \frac{1}{2} \left(\sum_{i=1}^n \pi_{p_t(i)}(t) \tau_{(ii)}(t) - \sum_{i,j=1}^n \pi_{p_t(i)}(t) \pi_{p_t(j)}(t) \tau_{(ij)}(t) \right) dt \\
&= \sum_{k=1}^n \frac{\pi_{p_t(k)}(t)}{\mu_{(k)}(t)} d\mu_{(k)}(t) \\
&+ \frac{1}{2} \sum_{k=1}^{n-1} (\pi_{p_t(k)}(t) - \pi_{p_t(k+1)}(t)) d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) \\
&- \frac{1}{2} \sum_{i,j=1}^n \pi_{p_t(i)}(t) \pi_{p_t(j)}(t) \tau_{(ij)}(t) dt,
\end{aligned} \tag{4.2.11}$$

where (4.2.10) follows from (4.2.6) and Lemma 1.3.4, and (4.2.11) follows from (4.2.7).

Let us simplify the first summation on the right-hand side of (4.2.11). If the weights π_i , $i = 1, \dots, n$, satisfy (4.2.4), then

$$\pi_{p_t(k)}(t) = (D_k \log S(\mu_{(\cdot)}(t)) + \varphi(t)) \mu_{(k)}(t), \quad t \in [0, T], \tag{4.2.12}$$

for $k = 1, \dots, n$, where

$$\varphi(t) = 1 - \sum_{j=1}^n \mu_{(j)}(t) D_j \log S(\mu_{(\cdot)}(t)), \quad t \in [0, T].$$

In this case, a.s., for $t \in [0, T]$,

$$\begin{aligned}
\sum_{i=1}^n \frac{\pi_{p_t(i)}(t)}{\mu_{(i)}(t)} d\mu_{(i)}(t) &= \sum_{i=1}^n D_i \log S(\mu_{(\cdot)}(t)) d\mu_{(i)}(t) + \varphi(t) \sum_{i=1}^n d\mu_{(i)}(t) \\
&= \sum_{i=1}^n D_i \log S(\mu_{(\cdot)}(t)) d\mu_{(i)}(t),
\end{aligned} \tag{4.2.13}$$

since $\sum_{i=1}^n d\mu_{(i)}(t) = 0$.

Now consider the last summation in (4.2.11). It follows from (4.2.12) that a.s., for $t \in [0, T]$,

$$\begin{aligned}
& \sum_{i,j=1}^n \pi_{p_t(i)}(t) \pi_{p_t(j)}(t) \tau_{(ij)}(t) \\
&= \sum_{i,j=1}^n D_i \log S(\mu_{(\cdot)}(t)) D_j \log S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) \\
&\quad + 2\varphi(t) \sum_{i,j=1}^n D_i \log S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) \\
&\quad + \varphi^2(t) \sum_{i,j=1}^n \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) \\
&= \sum_{i,j=1}^n D_i \log S(\mu_{(\cdot)}(t)) D_j \log S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t), \quad (4.2.14)
\end{aligned}$$

since Lemma 1.2.2 implies that $\mu_{(\cdot)}(t)$ is in the null space of $(\tau_{(ij)}(t))$. Equations (4.2.11), (4.2.13), and (4.2.14), imply that a.s., for $t \in [0, T]$,

$$\begin{aligned}
d \log(Z_\pi(t)/Z_\mu(t)) &= \sum_{i=1}^n D_i \log S(\mu_{(\cdot)}(t)) d\mu_{(i)}(t) \\
&\quad - \frac{1}{2} \sum_{i,j=1}^n D_i \log S(\mu_{(\cdot)}(t)) D_j \log S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) dt \\
&\quad + \frac{1}{2} \sum_{k=1}^{n-1} (\pi_{p_t(k+1)}(t) - \pi_{p_t(k)}(t)) d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t).
\end{aligned}$$

This equation and (4.2.9) imply (4.2.5). $\square$

This theorem extends Theorem 3.1.5 to cover functions of the ranked market weights, but Theorem 3.1.5 can also be extended in other directions. Here is an extension to the Gini coefficient, Example 3.4.7.

4.2.2 Example. Consider the Gini function

$$\mathbf{G}(x) = 1 - \frac{1}{2} \sum_{i=1}^n |x_i - n^{-1}|,$$

defined in Example 3.4.7. From Definition 4.1.1, we have a.s.,

$$d|\mu_i(t) - n^{-1}| = \operatorname{sgn}(\mu_i(t) - n^{-1}) d\mu_i(t) + 2 d\Lambda_{\mu_i - n^{-1}}(t), \quad t \in [0, T],$$

for $i = 1, \dots, n$. Hence, a.s., for $t \in [0, T]$,

$$\begin{aligned}
d\mathbf{G}(\mu(t)) &= -\frac{1}{2} \sum_{i=1}^n d|\mu_i(t) - n^{-1}| \\
&= -\frac{1}{2} \sum_{i=1}^n \operatorname{sgn}(\mu_i(t) - n^{-1}) d\mu_i(t) - \sum_{i=1}^n d\Lambda_{\mu_i - n^{-1}}(t)
\end{aligned}$$

$$= \frac{1}{2} \sum_{i=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) d\mu_i(t) - \sum_{i=1}^n d\Lambda_{\mu_i - n^{-1}}(t).$$

By Itô's rule, a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log \mathbf{G}(\mu(t)) &= \frac{d\mathbf{G}(\mu(t))}{\mathbf{G}(\mu(t))} - \frac{d\langle \mathbf{G}(\mu) \rangle_t}{2 \mathbf{G}^2(\mu(t))} \\ &= \frac{1}{2 \mathbf{G}(\mu(t))} \left(\sum_{i=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) d\mu_i(t) - 2 \sum_{i=1}^n d\Lambda_{\mu_i - n^{-1}}(t) \right) \\ &\quad - \frac{1}{8 \mathbf{G}^2(\mu(t))} \sum_{i,j=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) \operatorname{sgn}(n^{-1} - \mu_j(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) dt, \end{aligned}$$

since $d\langle \mu_i, \mu_j \rangle_t = \mu_i(t) \mu_j(t) dt$ by (1.2.14).

Define the portfolio π by

$$\pi_i(t) = \left(\frac{\operatorname{sgn}(n^{-1} - \mu_i(t))}{2 \mathbf{G}(\mu(t))} + \varphi(t) \right) \mu_i(t), \quad t \in [0, T],$$

for $i = 1, \dots, n$, where

$$\varphi(t) = 1 - \frac{1}{2 \mathbf{G}(\mu(t))} \sum_{j=1}^n \mu_j(t) \operatorname{sgn}(n^{-1} - \mu_j(t)), \quad t \in [0, T].$$

By Proposition 1.2.5, the value process Z_π satisfies, a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\pi(t)/Z_\mu(t)) &= \sum_{i=1}^n \pi_i(t) d \log \mu_i(t) + \gamma_\pi^*(t) dt \\ &= \sum_{i=1}^n \frac{\pi_i(t)}{\mu_i(t)} d\mu_i(t) - \frac{1}{2} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}(t) dt, \end{aligned}$$

as in (3.1.11). The first term on the right-hand side is

$$\begin{aligned} \sum_{i=1}^n \frac{\pi_i(t)}{\mu_i(t)} d\mu_i(t) &= \sum_{i=1}^n \left(\frac{\operatorname{sgn}(n^{-1} - \mu_i(t))}{2 \mathbf{G}(\mu(t))} + \varphi(t) \right) d\mu_i(t) \\ &= \frac{1}{2 \mathbf{G}(\mu(t))} \sum_{i=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) d\mu_i(t), \end{aligned}$$

since $\sum_{i=1}^n d\mu_i(t) = 0$, and the second term involves

$$\begin{aligned} \sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}(t) &= \sum_{i,j=1}^n \left(\frac{\operatorname{sgn}(n^{-1} - \mu_i(t))}{2 \mathbf{G}(\mu(t))} + \varphi(t) \right) \\ &\quad \times \left(\frac{\operatorname{sgn}(n^{-1} - \mu_j(t))}{2 \mathbf{G}(\mu(t))} + \varphi(t) \right) \mu_i(t) \mu_j(t) \tau_{ij}(t) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4 \mathbf{G}^2(\mu(t))} \sum_{i,j=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) \operatorname{sgn}(n^{-1} - \mu_j(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) \\
&\quad + \frac{\varphi(t)}{\mathbf{G}(\mu(t))} \sum_{i,j=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t) \\
&\quad + \varphi^2(t) \sum_{i,j=1}^n \mu_i(t) \mu_j(t) \tau_{ij}(t),
\end{aligned}$$

a.s., for $t \in [0, T]$. Since μ is in the null space of τ by Lemma 1.2.2, the last two summations vanish, so a.s., for $t \in [0, T]$,

$$\begin{aligned}
d \log(Z_\pi(t)/Z_\mu(t)) &= \frac{1}{2 \mathbf{G}(\mu(t))} \sum_{i=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) d\mu_i(t) \\
&\quad - \frac{1}{8 \mathbf{G}^2(\mu(t))} \sum_{i,j=1}^n \operatorname{sgn}(n^{-1} - \mu_i(t)) \operatorname{sgn}(n^{-1} - \mu_j(t)) \mu_i(t) \mu_j(t) \tau_{ij}(t).
\end{aligned}$$

Hence, a.s., for $t \in [0, T]$,

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{G}(\mu(t)) + \frac{1}{\mathbf{G}(\mu(t))} \sum_{i=1}^n d\Lambda_{\mu_i - n^{-1}}(t).$$

To summarize all this, we have shown that $\mathbf{G}$ generates the portfolio π with weights

$$\pi_i(t) = \left(\frac{\operatorname{sgn}(n^{-1} - \mu_i(t))}{2 \mathbf{G}(\mu(t))} + 1 - \sum_{j=1}^n \frac{\mu_j(t) \operatorname{sgn}(n^{-1} - \mu_j(t))}{2 \mathbf{G}(\mu(t))} \right) \mu_i(t),$$

for $i = 1, \dots, n$, and a drift process that satisfies

$$d\Theta(t) = \frac{1}{\mathbf{G}(\mu(t))} \sum_{i=1}^n d\Lambda_{\mu_i - n^{-1}}(t), \quad t \in [0, T], \quad \text{a.s.}$$

Compare this to the C^2 version of the Gini function in Example 3.4.7. $\square$

4.2.3 Problem. (!) Prove a general theorem for portfolio generating functions. It must include at least Theorem 3.1.5, Theorem 4.2.1, and Example 4.2.2.

4.3 Examples of Rank-Dependent Portfolios

Portfolios generated by ranked market weights are useful both for theoretical and practical applications. The theoretical applications we present here provide insight into the behavior of the biggest stock in the market as well

as into the cause of the size effect. As practical applications, we analyze *leakage* in diversity-weighted equity indices, and we present a method for measuring the effect that changes in the capital distribution of the market have on portfolio performance.

4.3.1 Example. (The biggest stock) Let $\mathbf{S}(x) = x_{(1)}$. Then Theorem 4.2.1 implies that $\mathbf{S}$ generates a portfolio π with weights $\pi_{p_t(1)}(t) = 1$ and $\pi_{p_t(k)}(t) = 0$ for $k = 2, \dots, n$. Here the smooth component of the drift process is 0, so

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mu_{(1)}(t) - \frac{1}{2} d\Lambda_{\log \mu_{(1)} - \log \mu_{(2)}}(t), \quad (4.3.1)$$

$t \in [0, T]$, a.s. Since $\mu_{(1)}(t) < 1$ and the local time component of the drift process is decreasing, the long-term relative performance of π will suffer if there are many changes of leadership in the market. Accordingly, for the biggest stock to perform well relative to the market, it must either pay higher than market-average dividends (ATT before the breakup), or else it must crush all competitors to allow no changes in market leadership (Microsoft). $\square$

4.3.2 Example. (The size effect) The *size effect* is the observed tendency of small stocks to have higher long-term returns than large stocks. Here is an explanation of the size effect based on Theorem 4.2.1.

Let $1 < m < n$, and suppose

$$\mathbf{S}_L(x) = x_{(1)} + \dots + x_{(m)}.$$

Then Theorem 4.2.1 implies that $\mathbf{S}_L$ generates a portfolio ξ with weights

$$\xi_{p_t(k)}(t) = \begin{cases} \frac{\mu_{(k)}(t)}{\mathbf{S}_L(\mu(t))}, & k \leq m, \\ 0, & k > m, \end{cases}$$

for $t \in [0, T]$. In this case, (3.1.2) becomes

$$d \log(Z_\xi(t)/Z_\mu(t)) = d \log \mathbf{S}_L(\mu(t)) - \frac{1}{2} \xi_{(m)}(t) d\Lambda_{\log \mu_{(m)} - \log \mu_{(m+1)}}(t), \quad (4.3.2)$$

$t \in [0, T]$, a.s. Here (4.1.2) implies that all but the last of the coefficients of the local times in (4.2.5) drop out. The value of the generating function

$$\mathbf{S}_L(\mu(t)) = \mu_{(1)}(t) + \dots + \mu_{(m)}(t)$$

represents the relative capitalization of a large-stock index composed of the m largest stocks in the market. The weights ξ_i , $i = 1, \dots, n$, represent the capitalization weights of the stocks as members of the large-stock index. Similarly,

$$\mathbf{S}_S(x) = x_{(m+1)} + \dots + x_{(n)}$$

generates a portfolio η that satisfies

$$d \log(Z_\eta(t)/Z_\mu(t)) = d \log \mathbf{S}_S(\mu(t)) + \frac{1}{2} \eta_{(m+1)}(t) d\Lambda_{\log \mu_{(m)} - \log \mu_{(m+1)}}(t),$$

$t \in [0, T]$, a.s., and $\mathbf{S}_S(\mu(t))$ represents the relative capitalization of a small-stock index composed of the $n - m$ smallest stocks in the market. It follows that a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\eta(t)/Z_\xi(t)) &= d \log(\mathbf{S}_S(\mu(t))/\mathbf{S}_L(\mu(t))) \\ &\quad + \frac{1}{2} (\xi_{(m)}(t) + \eta_{(m+1)}(t)) d\Lambda_{\log \mu_{(m)} - \log \mu_{(m+1)}}(t). \end{aligned} \quad (4.3.3)$$

Equation (4.3.3) decomposes the relative return of a small-stock index versus a large-stock index into the sum of two parts: the change in relative capitalizations of the two indices and the drift process. If the ratio of the relative capitalization of small stocks to that of the large stocks remains stable over time, then $\mathbf{S}_S(\mu(t))/\mathbf{S}_L(\mu(t))$ does not change much. Since the drift process in (4.3.3) is monotonically increasing, it is likely that the return on the small-stock index will eventually be greater than that of the large-stock index. Hence, the higher long-term return of the small-stock index is due to the increasing drift process, and the relative level of small-stock risk is irrelevant. $\square$

4.3.3 Problem. (!) Show that the singular dividend processes

$$\Delta_{p_t(k)}(t) = \frac{1}{2} (\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) - \Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}(t)),$$

$t \in [0, T]$, for $k = 1, \dots, n$, will nullify the drift process in (4.3.3). (Singular dividend processes are defined in Problem 1.1.12.) Is it possible to find smooth dividend payments such that, for all m ,

$$\lim_{t \rightarrow \infty} t^{-1} \log(\widehat{Z}_\eta(t)/\widehat{Z}_\xi(t)) = 0, \quad \text{a.s.}$$

4.3.4 Definition. Suppose $c_1, \dots, c_n$ are constants, each equal to 0 or 1. A portfolio π with weights that satisfy

$$\pi_i(t) = \frac{c_i \mu_i(t)}{c_1 \mu_1(t) + \dots + c_n \mu_n(t)}, \quad t \in [0, T],$$

$i = 1, \dots, n$, is called a *capitalization-weighted* portfolio.

The portfolios ξ and η generated by $\mathbf{S}_L$ and $\mathbf{S}_S$, respectively, in Example 4.3.2 are both capitalization-weighted portfolios, as is the market portfolio.

4.3.5 Example. (Leakage in a diversity-weighted large-stock index) Let ξ be the large-stock index generated by $\mathbf{S}_L$ in Example 4.3.2.

Consider a portfolio π of stocks held in ξ generated by the function $\mathbf{D}_p$ of Example 3.4.4. We wish to calculate the performance of the portfolio π relative to the index ξ .

For $1 < m < n$, let

$$\mathbf{S}(x) = \left(\sum_{i=1}^m x_{(i)}^p \right)^{1/p},$$

for $0 < p < 1$. Since π is generated by $\mathbf{S}$, Theorem 4.2.1 implies that, a.s., for $t \in [0, T]$,

$$\pi_{p_{t(k)}}(t) = \begin{cases} \frac{\mu_{(k)}^p(t)}{(\mathbf{S}(\mu(t)))^p}, & k \leq m, \\ 0, & k > m, \end{cases}$$

and

$$\begin{aligned} d \log(Z_\pi(t)/Z_\mu(t)) &= d \log \mathbf{S}(\mu(t)) + (1-p)\gamma_\pi^*(t) dt \\ &\quad - \frac{1}{2} \pi_{(m)}(t) d\Lambda_{\log \mu_{(m)} - \log \mu_{(m+1)}}(t). \end{aligned} \quad (4.3.4)$$

Since $\mathbf{S}$ is symmetric, (4.1.2) implies that all but the last of the coefficients of the local times in (4.2.5) drop out.

We are interested in the performance of π relative to ξ of Example 4.3.2 rather than relative to the whole market. Hence, we can subtract the left-hand side of (4.3.2) from the left-hand side of (4.3.4), and the right-hand side from the right-hand side, and we have a.s., for $t \in [0, T]$,

$$\begin{aligned} d \log(Z_\pi(t)/Z_\xi(t)) &= d \log \mathbf{D}_p(\xi_{(1)}(t), \dots, \xi_{(m)}(t)) + (1-p)\gamma_\pi^*(t) dt \\ &\quad + \frac{1}{2} (\xi_{(m)}(t) - \pi_{(m)}(t)) d\Lambda_{\log \mu_{(m)} - \log \mu_{(m+1)}}(t), \end{aligned} \quad (4.3.5)$$

since (3.4.2) implies that $\mathbf{S}(\mu(t))/\mathbf{S}_L(\mu(t)) = \mathbf{D}_p(\xi_{(1)}(t), \dots, \xi_{(m)}(t))$. The local time component of the drift process in (4.3.5) is called *leakage*, and since $\pi_{(m)}(t) \geq \xi_{(m)}(t)$, the leakage is monotonically decreasing. Heuristically, when a capitalization-weighted portfolio is contained in a larger market portfolio, leakage refers to the effect caused by stocks that cross over from the capitalization-weighted portfolio to the rest of the market portfolio: They “leak” out of the portfolio. $\square$

4.3.6 Problem. Derive an equation of the form (4.3.5) for general measures of diversity.

4.3.7 Problem. (!) Find dividend rates such that $\widehat{Z}_\pi$ and $\widehat{Z}_\xi$ in (4.3.5) grow at the same long-term growth rate,

$$\lim_{t \rightarrow \infty} t^{-1} \log(\widehat{Z}_\pi(t)/\widehat{Z}_\xi(t)) = 0, \quad \text{a.s.}$$

Is it possible to find (nontrivial) conditions under which

$$\lim_{t \rightarrow \infty} t^{-1} \log(Z_\pi(t)/Z_\xi(t)) = 0, \quad \text{a.s.}?$$

4.3.8 Example. (A portfolio with fixed weight ratios) The *weight ratios* for the portfolio π are defined by $\pi_{p_t(i)}(t)/\mu_{(i)}(t)$, for $i = 1, \dots, n$, where p_t is the permutation defined by (4.1.26). We say that the weight ratios of π are *increasing* if for $k = 1, \dots, n-1$,

$$\pi_{p_t(k)}(t)/\mu_{(k)}(t) \leq \pi_{p_t(k+1)}(t)/\mu_{(k+1)}(t), \quad t \in [0, T], \quad \text{a.s.}$$

We have seen that the buy-and-hold portfolio in Example 3.1.6 holds a fixed number of shares of each stock. In a similar manner we can fix the weight ratios of a portfolio, at least up to a common multiple. Consider constants $w_1, \dots, w_n \geq 0$ such that $\sum_{i=1}^n w_i > 0$ and let

$$\mathbf{S}(x) = w_1 x_{(1)} + \dots + w_n x_{(n)}. \quad (4.3.6)$$

Then Theorem 4.2.1 implies that $\mathbf{S}$ generates a portfolio π with weights

$$\pi_{p_t(i)}(t) = \frac{w_i \mu_{(i)}(t)}{w_1 \mu_{(1)}(t) + \dots + w_n \mu_{(n)}(t)}, \quad (4.3.7)$$

for $i = 1, \dots, n$. Hence, the weight ratios are proportional to the constants w_i .

For this portfolio, Theorem 4.2.1 and (4.1.2) imply that

$$d \log(Z_\pi(t)/Z_\mu(t)) \quad (4.3.8)$$

$$= d \log \mathbf{S}(\mu(t)) + \frac{1}{2\mathbf{S}(\mu(t))} \sum_{k=1}^{n-1} (w_{k+1} - w_k) \mu_{(k)}(t) d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t).$$

From this we see that if the weight ratios of a portfolio are increasing, the drift process for the portfolio will be increasing.

As in Example 3.1.6, the weight ratios of a portfolio can be interpreted as the (fractional) number of shares of each corresponding stock held in the portfolio. Since the fractional number of shares of each stock can be calculated for any portfolio, the first term on the right-hand side of (4.3.8) can also be calculated for any portfolio. This term measures the component of relative portfolio return due to changes in the capital distribution, i.e., the *distributional component* of the relative portfolio return. We apply this methodology in Chapter 7. $\square$

4.3.9 Problem. (!) In Example 4.3.8 we see that any portfolio can be approximated in a piecewise fashion by portfolios with fixed weight ratios. Can some limit argument be devised so that a continuous version of the decomposition (4.3.8) holds for a wide class of portfolios? How will this be compatible with Example 3.1.12, a portfolio with differentiable weights that is not functionally generated?

4.4 Notes and Summary

The capital distribution, i.e., the family of ranked market weights, is of central importance in stochastic portfolio theory, and this chapter combines this distribution with the portfolio generating functions of the previous chapter. Portfolios generated by functions of the ranked market weights first appeared in Fernholz (2001a).

In Section 4.1 we introduce semimartingale local time, Definition 4.1.1, in order to represent rank processes. Semimartingale local time is a generalization of Brownian local time, which was introduced by P. Levy (1948). Equation (4.1.1), which we use as a definition, is one of the *Tanaka–Meyer formulas* (Tanaka (1963), Meyer (1976)). Meyer (1976) showed that processes generated by convex functions of continuous semimartingales are themselves semimartingales, but his results do not provide an explicit representation. Representations have been derived in certain cases, but none specifically fits our needs (see, e.g., Carlen and Protter (1992) and Protter and San Martin (1993)).

Example 4.2.2, the Gini function as a portfolio generating function, was communicated to the author by I. Karatzas (2001).

The size effect, discussed in Example 4.3.2, was first noticed by Banz (1981) and Reinganum (1981). It is interesting to note that papers on the size effect began to appear in academic journals in the early 1980s, just after a ten-year rise in market diversity (see, for example, Figure 6.2 on page 122). The size effect is sometimes justified by the higher risk attributed to the small stocks, but the explanation in Example 4.3.2 is closer to the alternative hypothesis proposed in Fernholz (1998) (see also Section 6.4).

Example 4.3.5, leakage in a diversity-weighted portfolio, was mentioned in Fernholz et al. (1998), but the rigorous treatment that we present here first appeared in Fernholz (2001a).

Example 4.3.8, portfolios with fixed weight ratios, were applied in Fernholz (2001b) to determine the effect of company size on portfolio return (see also Section 7.4).

Chapter Summary

Section 4.1: The *capital distribution* of the market is defined to be the family of ranked market weights,

$$\mu_{(1)}(t) \geq \mu_{(2)}(t) \geq \cdots \geq \mu_{(n)}(t), \quad t \in [0, T].$$

In order to represent the ranked market weights as continuous semimartingales, it is necessary to use semimartingale local times.

For a continuous semimartingale X , the *local time* for X is defined by

$$\Lambda_X(t) = \frac{1}{2} \left(|X(t)| - |X(0)| - \int_0^t \operatorname{sgn}(X(s)) dX(s) \right), \quad (4.1.1)$$

for $t \in [0, T]$. The local time Λ_X measures the amount of time X spends near the origin, so $d\Lambda_X$ defines a positive measure on $[0, T]$ that is concentrated on the set of points $\{t \in [0, T] : X(t) = 0\}$.

For a family of continuous semimartingales $X_1, \dots, X_n$ that satisfy certain nondegeneracy conditions, the rank processes $X_{(k)}$, $k = 1, \dots, n$, are continuous semimartingales such that a.s., for $t \in [0, T]$,

$$\begin{aligned} dX_{(k)}(t) &= \sum_{i=1}^n I_{\{i\}}(p_t(k)) dX_i(t) \\ &\quad + \frac{1}{2} d\Lambda_{X_{(k)} - X_{(k+1)}}(t) - \frac{1}{2} d\Lambda_{X_{(k-1)} - X_{(k)}}(t), \end{aligned} \quad (4.1.15)$$

where p_t is the random permutation of $\{1, \dots, n\}$ such that for $k = 1, \dots, n$ and $t \in [0, T]$,

$$\begin{aligned} X_{p_t(k)}(t) &= X_{(k)}(t), \\ p_t(k) &< p_t(k+1) \quad \text{if} \quad X_{(k)}(t) = X_{(k+1)}(t). \end{aligned}$$

Equation (4.1.15) provides a representation that expresses the rank processes in terms of the original processes plus local times. This representation can be applied to functions of the ranked market weights and the portfolios they generate.

Section 4.2: Suppose that $\mathbf{S}$ is a function defined on Δ^n such that there exists a positive C^2 function S with

$$\mathbf{S}(x_1, \dots, x_n) = S(x_{(1)}, \dots, x_{(n)}), \quad x \in \Delta^n,$$

with $x_i D_i \log S(x)$ bounded for $x \in \Delta^n$, $i = 1, \dots, n$. If the stocks $X_1, \dots, X_n$ satisfy certain nondegeneracy conditions, then $\mathbf{S}$ generates a portfolio π such that for $k = 1, \dots, n$,

$$\pi_{p_t(k)}(t) = \left(D_k \log S(\mu_{(\cdot)}(t)) + 1 - \sum_{j=1}^n \mu_{(j)}(t) D_j \log S(\mu_{(\cdot)}(t)) \right) \mu_{(k)}(t), \quad (4.2.4)$$

for all $t \in [0, T]$, a.s., where p_t is the permutation defined above. The drift process Θ corresponding to π satisfies

$$\begin{aligned} d\Theta(t) &= \frac{-1}{2\mathbf{S}(\mu(t))} \sum_{i,j=1}^n D_{ij} S(\mu_{(\cdot)}(t)) \mu_{(i)}(t) \mu_{(j)}(t) \tau_{(ij)}(t) dt \\ &\quad + \frac{1}{2} \sum_{k=1}^{n-1} (\pi_{p_t(k+1)}(t) - \pi_{p_t(k)}(t)) d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t), \end{aligned} \quad (4.2.5)$$

for all $t \in [0, T]$, a.s., where

$$\tau_{(ij)}(t) = \tau_{p_t(i)p_t(j)}(t), \quad t \in [0, T]. \quad (4.2.2)$$

With the results of this section we can analyze the behavior of portfolios that depend on the capital distribution of the market.

Section 4.3: Four examples of rank-dependent portfolios were presented in this section: the biggest stock, the size effect, leakage in a diversity-weighted portfolio, and a portfolio with fixed weight ratios.

5

Stable Models for the Distribution of Capital

In Chapter 2 we examined certain characteristics of the distribution of capital, but since the analytical tools at our disposal were limited, the results were incomplete. In this chapter we shall apply the methods introduced in Chapters 3 and 4 to achieve a deeper understanding of the structure of the capital distribution. In particular, local times related to the market weight processes permit us to develop a detailed model for a stable capital distribution.

The capital distribution is equivalent to the size distribution of firms, and the size distribution of firms has been studied extensively using a number of methodologies. Much of the research in this area has been devoted to finding plausible models for corporate behavior that would result in the observed distribution of firm size. In contrast to the historical literature, here we construct an equity market model that is compatible with the observed long-term structure of the capital distribution, without regard to the mechanism that might have produced that distribution. We test the validity of our model on the capital distribution of the the U.S. equity market over the period from 1990 to 1999.

5.1 The Capital Distribution Curve

The *capital distribution curve* refers to the log-log plot of the market weights arranged in descending order, i.e., the logarithms of the market weights versus the logarithms of their respective ranks. Under certain ideal conditions, in a market in which the smallest stocks are replaced at a con-

stant rate, it can be shown that the capital distribution curve will be a straight line. This linear log-log weight distribution is called a *Pareto distribution*. An infinite market is similar to a finite market in which stocks enter and exit, and the corresponding capital distribution curves have similar characteristics.

5.1.1 Example. Suppose that we have a market in which the logarithms of the stock capitalizations are distributed according to a Poisson point process (or Poisson random measure, see Karatzas and Shreve (1991), Definition 6.2.3) with intensity measure $e^{-ax} dx$ on $\mathbb{R}$, for some positive constant a . Heuristically, this means that the expected number of stocks with log-capitalization between x and $x + dx$ is $e^{-ax} dx$. The expected number of stocks in such a market is infinite, but the total capitalization of the market will almost surely be finite if $0 < a < 1$. Suppose that a period of time t passes and the log-capitalizations evolve independently according to a probability distribution P_t . For example, for log-normal return processes, P_t will be a normal distribution with mean and variance that are linear in t . With this model, the intensity measure for the log-capitalizations at time t will be $\psi_t(x) dx$, with

$$\psi_t(x) = \int_{-\infty}^{\infty} e^{-a(x-y)} dP_t(y) = e^{-ax} e^{az_t},$$

where

$$e^{az_t} = \int_{-\infty}^{\infty} e^{ay} dP_t(y).$$

The result of the time evolution is to translate the log-capitalization intensity measure by z_t , so the expected capital distribution of the market does not change over this time period. This translation corresponds to an average growth rate of z_t/t for the log-capitalization of the market over the period. For example, if the distribution P_t is normal with zero mean and variance $\sigma^2 t$, then the growth rate of the log-capitalization is constant at $a\sigma^2/2$.

With the log-capitalization distributed according to the intensity measure $e^{az_t} e^{-ax} dx$, the quantity

$$e^{az_t} \int_{\log X}^{\infty} e^{-ax} dx = \frac{1}{a} e^{az_t} X^{-a}$$

represents the expected number of stocks with log-capitalization greater than or equal to $\log X$ at time t , which is the same as the expected rank of the stock with capitalization X . Hence, for $k = 1, 2, \dots$, if $X_{(k)}(t)$ represents the capitalization of the k th-largest stock at time t , then

$$X_{(k)}(t) \approx c e^{z_t} k^{-1/a}, \quad (5.1.1)$$

where $c = a^{-1/a}$. It follows from (5.1.1) that if $0 < a < 1$, then the sum of the capitalizations almost surely converges, so the total capitalization of the

market is almost surely finite. It also follows from (5.1.1) that a log-log plot of the capital distribution of this market will be approximately a straight line with slope $-1/a$, so the weights follow a Pareto distribution. $\square$

How closely does the Pareto distribution conform to actual capital distributions? To answer this question, let us consider some actual capital distribution curves and see what they look like. In Figure 5.1 we see eight capital distribution curves for the U.S. stock market for the dates December 31, 1929, 1939, 1949, 1959, 1969, 1979, 1989, 1999. These curves are generated using capitalization data from the monthly stock database of the Center for Research in Securities Prices (CRSP) at the University of Chicago. The market we consider, which we shall call the *CRSP universe*, consists of the stocks traded on the New York Stock Exchange (NYSE), the American Stock Exchange (AMEX), and the NASDAQ Stock Market, after the removal of all REITs, all closed-end funds, and those ADRs not included in the S&P 500 Index. Until 1962, the CRSP data include only NYSE stocks. The AMEX stocks are included after July 1962, and the NASDAQ stocks are included at the beginning of 1973. The number of stocks in the distribution curves varies from about 700 in 1929 to about 7500 in 1999. Since each successive curve contains a larger number of stocks than the previous one, each curve can be identified by this characteristic.

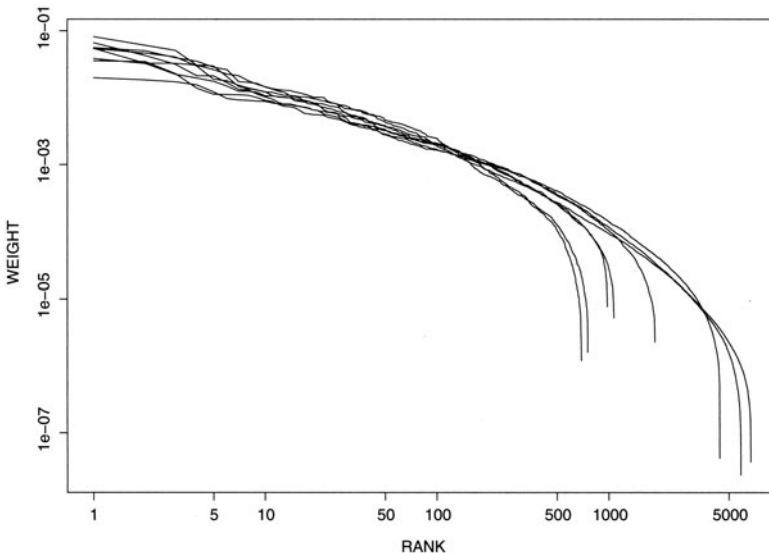


FIGURE 5.1. Capital distribution curves: 1929–1999.

The curves in Figure 5.1 are evidently not straight lines, but instead are generally concave. The tails of the distributions, containing the smallest weights, may be distorted somewhat, since companies in decline do not immediately leave an exchange. Whether or not the tails of the distributions

are removed, the curves have a similar shape, and this indicates that some type of stability is likely to be present. We shall investigate this stability in the following sections.

$$5.2 \quad dX(t) = -\alpha \operatorname{sgn}(X(t)) dt + \sigma dW(t)$$

In this section we present some technical results that will be needed for the analysis of stable capital distributions in the following sections. Consider the equation

$$dX(t) = -\alpha \operatorname{sgn}(X(t)) dt + \sigma dW(t), \quad t \in [0, \infty), \quad (5.2.1)$$

where α and σ are positive constants, and W is a Brownian motion. Here α represents the rate at which $X(t)$ is drawn back to the origin, and σ represents the amplitude of the random perturbation caused by W . From (5.2.1) it is clear that a solution X to this equation will be an absolutely continuous semimartingale. Equation (5.2.1) can be transformed into the simplified version

$$dZ(t) = -\operatorname{sgn}(Z(t)) dt + dB(t), \quad t \in [0, \infty), \quad (5.2.2)$$

by the change of variables

$$Z(t) = \frac{\alpha}{\sigma^2} X \left(\frac{\sigma^2 t}{\alpha^2} \right),$$

with B the Brownian motion process defined by

$$B(t) = \frac{\alpha}{\sigma} W \left(\frac{\sigma^2 t}{\alpha^2} \right).$$

A weak solution for (5.2.2) is given in Karatzas and Shreve (1991), Exercise 5.3.12, along with the probability density of the solution (Karatzas and Shreve (1991), Exercise 6.3.5) from which it follows that asymptotically the solution X of (5.2.1) has the double exponential density

$$\frac{\alpha}{\sigma^2} \exp \left(-\frac{2\alpha|u|}{\sigma^2} \right).$$

With this asymptotic density, it can be shown that for any constant $p > 0$,

$$\lim_{T \rightarrow \infty} \frac{X^p(T)}{T} = 0, \quad \text{a.s.} \quad (5.2.3)$$

Let us now calculate the asymptotic time averages for certain expressions related to (5.2.1).

5.2.1 Lemma. *Let X be an absolutely continuous semimartingale such that $\{t : X(t) = 0\}$ has Lebesgue measure 0, a.s. Suppose that X satisfies*

$$d|X(t)| = -\alpha dt + d\Lambda_{|X|}(t) + \sigma dB(t), \quad t \in [0, \infty), \quad \text{a.s.}, \quad (5.2.4)$$

where α and σ are positive constants, and B is a Brownian motion. Then

$$\lim_{t \rightarrow \infty} t^{-1} \Lambda_{|X|}(t) = \alpha, \quad \text{a.s.}, \quad (5.2.5)$$

and

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |X(t)| dt = \frac{\sigma^2}{2\alpha}, \quad \text{a.s.} \quad (5.2.6)$$

Proof. By Definition 4.1.1 and Lemma 4.1.9,

$$d\Lambda_{|X|}(t) = d|X(t)| - \operatorname{sgn}(X(t)) dX(t), \quad t \in [0, \infty), \quad \text{a.s.}$$

This and (5.2.4) imply that

$$\operatorname{sgn}(X(t)) dX(t) = -\alpha dt + \sigma dB(t), \quad t \in [0, \infty), \quad \text{a.s.},$$

and therefore, a.s., for $t \in [0, \infty)$,

$$\begin{aligned} dX(t) &= -\alpha \operatorname{sgn}(X(t)) dt + \sigma \operatorname{sgn}(X(t)) dB(t) \\ &= -\alpha \operatorname{sgn}(X(t)) dt + \sigma dW(t), \end{aligned}$$

where

$$W(t) = \int_0^t \operatorname{sgn}(X(s)) dB(s), \quad t \in [0, \infty),$$

is a Brownian motion. Hence X is a weak solution to (5.2.1), and (5.2.3) holds.

From (5.2.4) we have

$$\lim_{t \rightarrow \infty} t^{-1} |X(t)| = -\alpha + \lim_{t \rightarrow \infty} t^{-1} \Lambda_{|X|}(t) + \sigma \lim_{t \rightarrow \infty} t^{-1} B(t), \quad \text{a.s.} \quad (5.2.7)$$

The limit on the left-hand side of (5.2.7) is zero by (5.2.3), and the last limit on the right-hand side is zero by the strong law of large numbers for Brownian motion. This proves (5.2.5).

Itô's rule applied to (5.2.1) implies that a.s., for all $t \in [0, \infty)$,

$$dX^2(t) = 2X(t) dX(t) + \sigma^2 dt = -2\alpha |X(t)| dt + \sigma^2 dt + 2\sigma X(t) dW(t).$$

Hence, a.s.,

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{X^2(T)}{T} &= -2\alpha \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |X(t)| dt + \sigma^2 \\ &\quad + 2\sigma \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T X(t) dW(t). \end{aligned} \quad (5.2.8)$$

The limit on the left-hand side of (5.2.8) is zero by (5.2.3). We wish to show that last limit on the right-hand side of (5.2.3) also vanishes, i.e.,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T X(t) dW(t) = 0, \quad \text{a.s.}, \quad (5.2.9)$$

and this will suffice to prove (5.2.6).

Consider the process Y defined by

$$Y(t) = \int_0^t X(s) dW(s), \quad t \in [0, \infty).$$

Then, a.s.,

$$\langle Y \rangle_t = \int_0^t X^2(s) ds, \quad t \in [0, \infty),$$

so, a.s.,

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{-2} \langle Y \rangle_t \log \log t &= \lim_{t \rightarrow \infty} t^{-2} \log \log t \int_0^t X^2(s) ds \\ &\leq \lim_{t \rightarrow \infty} t^{-3/2} \int_0^t X^2(s) ds \\ &= \frac{2}{3} \lim_{t \rightarrow \infty} t^{-1/2} X^2(t) \\ &= 0, \end{aligned}$$

by (5.2.3). Hence Lemma 1.3.2 implies (5.2.9), and (5.2.6) follows. $\square$

Equation (5.2.4) shows that the movement of $|X(t)|$ in (5.2.4) is determined by three components: a reflective impulse $d\Lambda_{|X|}(t)$ at the origin, a compressive drift $-\alpha dt$, and a random vibration $\sigma dB(t)$. Equation (5.2.5) shows that equilibrium exists between the reflective impulse and the compressive drift. Equation (5.2.6) shows that the time average of $|X(t)|$ is proportional to the variance of the random vibration, and inversely proportional to the compressive drift.

With these results, we can now return to the analysis of stable capital distributions.

5.3 The Structure of Stable Capital Distributions

In this section we construct a model for a market with a stable capital distribution, and we show how it relates to the classical models that result in the Pareto distribution. We shall consider a coherent market $\mathcal{M}$ of pairwise mutually nondegenerate stocks $X_1, \dots, X_n$, along with the corresponding

market weights $\mu_1, \dots, \mu_n$. Recall that according to Definition 2.1.1, $\mathcal{M}$ is coherent if, for $i = 1, \dots, n$,

$$\lim_{t \rightarrow \infty} t^{-1} \log \mu_i(t) = 0, \quad \text{a.s.} \quad (5.3.1)$$

It follows from this that, a.s.,

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{-1} \log \mu_{(n)}(t) &= \lim_{t \rightarrow \infty} t^{-1} \log \left(\min_{1 \leq i \leq n} \mu_i(t) \right) \\ &= \lim_{t \rightarrow \infty} t^{-1} \left(\min_{1 \leq i \leq n} \log \mu_i(t) \right) \\ &= \min_{1 \leq i \leq n} \left(\lim_{t \rightarrow \infty} t^{-1} \log \mu_i(t) \right) \\ &= 0, \end{aligned} \quad (5.3.2)$$

as long as all the limits exist, which they do, and in fact are all 0, by (5.3.1). Since, for $k = 1, \dots, n$,

$$\mu_{(n)}(t) \leq \mu_{(k)}(t) \leq 1, \quad t \in [0, \infty),$$

it follows from (5.3.2) that

$$\lim_{t \rightarrow \infty} t^{-1} \log \mu_{(k)}(t) = 0, \quad \text{a.s.}, \quad (5.3.3)$$

for $k = 1, \dots, n$.

For $k = 1, \dots, n$, define the process g_k by

$$g_k(t) = \gamma_{p_t(k)}(t) - \gamma_\mu(t), \quad t \in [0, \infty), \quad (5.3.4)$$

where p_t is the permutation defined by (4.1.26), which means that $p_t(k)$ is the index of the k th-ranked stock. Corollary 4.1.12 states that, a.s., for $t \in [0, \infty)$,

$$\begin{aligned} d \log \mu_{(k)}(t) &= \sum_{i=1}^n I_{\{i\}}(p_t(k)) d \log \mu_i(t) \\ &\quad + \frac{1}{2} d \Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) - \frac{1}{2} d \Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}(t). \end{aligned}$$

But, a.s., for $t \in [0, \infty)$,

$$\begin{aligned} d \log \mu_i(t) &= d \log X_i(t) - d \log Z_\mu(t) \\ &= \gamma_i(t) dt + \sum_{\nu=1}^n \xi_{i\nu}(t) dW_\nu(t) \\ &\quad - \gamma_\mu(t) dt - \sum_{i, \nu=1}^n \mu_j(t) \xi_{j\nu}(t) dW_\nu(t), \end{aligned}$$

for $i = 1, \dots, n$. Hence, a.s., for all $t \in [0, \infty)$,

$$\begin{aligned}
d \log \mu_{(k)}(t) &= g_k(t) dt + \frac{1}{2} d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) \\
&\quad - \frac{1}{2} d\Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}(t) + \sum_{\nu=1}^n \xi_{p_t(k)\nu}(t) dW_\nu(t) \quad (5.3.5) \\
&\quad - \sum_{i,\nu=1}^n \mu_i(t) \xi_{i\nu}(t) dW_\nu(t),
\end{aligned}$$

for $k = 1, \dots, n$. With this equation we can analyze the asymptotic behavior of a stable capital distribution. We first need a formal definition.

5.3.1 Definition. The market $\mathcal{M}$ is *asymptotically stable* if it is coherent and, a.s.:

$$\begin{aligned}
(i) \text{ for } k = 1, \dots, n-1, \quad \lim_{t \rightarrow \infty} t^{-1} \Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) &= \lambda_{k,k+1}; \\
(ii) \text{ for } k = 1, \dots, n-1, \quad \lim_{t \rightarrow \infty} t^{-1} \langle \log \mu_{(k)} - \log \mu_{(k+1)} \rangle_t &= \sigma_{k:k+1}^2;
\end{aligned}$$

where $\lambda_{k,k+1}$ and $\sigma_{k:k+1}$ are positive constants. Let $\lambda_{0,1} = 0$ and $\lambda_{n,n+1} = 0$.

We shall also say that the capital distribution $\{\mu_{(1)}, \dots, \mu_{(n)}\}$ is *asymptotically stable* if Definition 5.3.1 holds. Conditions (i) and (ii) imply that asymptotic limits exist for the slopes of the processes $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}$ and $\langle \log \mu_{(k)} - \log \mu_{(k+1)} \rangle_t$, respectively, and both of these conditions are consistent with the heuristic idea of asymptotic stability. Condition (i) appears to be reasonably consistent with Figure 5.2 in Section 5.4, a plot of the values of $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t)$ for the U.S. equity market. An estimate of $\langle \log \mu_{(k)} - \log \mu_{(k+1)} \rangle_t$ in Section 5.4 indicates that condition (ii) also appears to be reasonable for the U.S. equity market.

Let us use the notation

$$g_k = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T g_k(t) dt \quad (5.3.6)$$

to represent the asymptotic relative growth rate of $X_{(k)}$, for $k = 1, \dots, n$.

5.3.2 Proposition. *If the market is asymptotically stable, then for $k = 1, \dots, n$,*

$$g_k = \frac{1}{2} \lambda_{k-1,k} - \frac{1}{2} \lambda_{k,k+1}, \quad \text{a.s.} \quad (5.3.7)$$

Proof. Let us take $\lim_{T \rightarrow \infty} T^{-1} \int_0^T$ of both sides of (5.3.5). The left-hand side of the equation vanishes by (5.3.3), and the last two terms on the right-hand side vanish by Lemma 1.3.2. Then the proposition follows from Definition 5.3.1(i). $\square$

It follows from (5.3.7) that for $k = 1, \dots, n$,

$$\lambda_{k,k+1} = -2(\mathbf{g}_1 + \dots + \mathbf{g}_k), \quad \text{a.s.} \quad (5.3.8)$$

From Definition 5.3.1(i) and (5.3.8) it follows that $\mathbf{g}_1 + \dots + \mathbf{g}_k < 0$ for $k = 1, \dots, n-1$, a.s., and that $\mathbf{g}_1 + \dots + \mathbf{g}_n = 0$. Hence, an asymptotically stable market is characterized by the $2n-2$ parameters $\mathbf{g}_1, \dots, \mathbf{g}_{n-1}$ and $\sigma_{1:2}^2, \dots, \sigma_{n-1:n}^2$, and these parameters will be called the *characteristic parameters* of the distribution.

We are interested in the behavior of the differences $\log \mu_{(k)} - \log \mu_{(k+1)}$ in an asymptotically stable market, and from (5.3.5) it follows that a.s., for $t \in [0, \infty)$,

$$\begin{aligned} d(\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)) &= (g_k(t) - g_{k+1}(t))dt \\ &\quad - \frac{1}{2}d\Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}(t) \\ &\quad - \frac{1}{2}d\Lambda_{\log \mu_{(k+1)} - \log \mu_{(k+2)}}(t) \\ &\quad + d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) \\ &\quad + \sum_{\nu=1}^n (\xi_{p_t(k)\nu}(t) - \xi_{p_t(k+1)\nu}(t))dW_\nu(t). \end{aligned} \quad (5.3.9)$$

The local times that appear in this equation participate in different manners. Both $\Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}$ and $\Lambda_{\log \mu_{(k+1)} - \log \mu_{(k+2)}}$ have negative coefficients and act as drift processes, much the same as the term containing g_k and g_{k+1} . However, the local time $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}$ participates in the vital role of reflecting $\log \mu_{(k)} - \log \mu_{(k+1)}$ at the origin in order to avoid negative values, just as $\Lambda_{|X|}$ does in (5.2.4).

Definition 5.3.1(i) implies that for large enough values of t ,

$$\Lambda_{\log \mu_{(k-1)} - \log \mu_{(k)}}(t) \approx \lambda_{k-1,k}t$$

and

$$\Lambda_{\log \mu_{(k+1)} - \log \mu_{(k+2)}}(t) \approx \lambda_{k+1,k+2}t,$$

and Definition 5.3.1(ii) implies that for large enough values of t ,

$$\langle \log \mu_{(k)} - \log \mu_{(k+1)} \rangle_t \approx \sigma_{k:k+1}^2 t.$$

These approximations can be used to define a *stable* version of (5.3.9) with

$$\begin{aligned} d(\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)) &= (\mathbf{g}_k - \mathbf{g}_{k+1})dt - \frac{1}{2}(\lambda_{k-1,k} + \lambda_{k+1,k+2})dt \\ &\quad + d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) + dM_{k:k+1}(t), \end{aligned} \quad (5.3.10)$$

for $t \in [0, \infty)$, where $M_{k:k+1}$ is a continuous martingale with

$$\langle M_{k:k+1} \rangle_t = \sigma_{k:k+1}^2 t, \quad t \in [0, \infty).$$

The local time $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}$ cannot be replaced in (5.3.10), since its role in (5.3.9) is essentially different from that of the other two local times. By (5.3.7),

$$\mathbf{g}_k - \mathbf{g}_{k+1} = \frac{1}{2} \boldsymbol{\lambda}_{k-1,k} - \boldsymbol{\lambda}_{k,k+1} + \frac{1}{2} \boldsymbol{\lambda}_{k+1,k+2},$$

so (5.3.10) is equivalent to

$$\begin{aligned} d(\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)) \\ = -\boldsymbol{\lambda}_{k,k+1} dt + d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) + dM_{k:k+1}(t), \end{aligned} \quad (5.3.11)$$

$t \in [0, \infty)$, for $k = 1, \dots, n-1$, and this is the *stable model* for the differences between consecutive ranked weights. Equation (5.3.11) is of the form (5.2.4) with $\alpha = \boldsymbol{\lambda}_{k,k+1}$ and $\sigma^2 = \sigma_{k:k+1}^2$. Hence, we can apply Lemma 5.2.1, which implies that (5.3.11) is consistent with Definition 5.3.1(i), and also implies that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)) dt = \frac{\sigma_{k:k+1}^2}{2\boldsymbol{\lambda}_{k,k+1}}, \quad \text{a.s.}, \quad (5.3.12)$$

for $k = 1, \dots, n-1$.

Let us consider again the capital distribution curve: the log-log plot of market weight versus rank. To estimate the slope of the capital distribution curve at the point above $\log k$, we can use (5.3.12) and the fact that for a large enough number k , $\log(k+1) - \log(k) \approx 1/k$. Hence, we have the estimate

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \frac{\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)}{\log(k) - \log(k+1)} dt \approx -\frac{k\sigma_{k:k+1}^2}{2\boldsymbol{\lambda}_{k,k+1}}, \quad \text{a.s.}, \quad (5.3.13)$$

for large enough k . From (5.3.8) and (5.3.13), we see that if the $\sigma_{k:k+1}^2$ and $\mathbf{g}_k$ change by a common factor, then there is no change in the capital distribution curve. If the $\sigma_{k:k+1}^2$ stay fixed, but the $\mathbf{g}_k$ are multiplied by a common factor that is greater than 1, then the slope of the capital distribution curve will decrease by that factor. This would seem to be reasonable, since (5.3.8) implies that for the higher-ranked stocks the $\mathbf{g}_k$ are likely to be negative, and for the lower-ranked stocks the $\mathbf{g}_k$ are likely to be positive, so multiplying by a factor greater than 1 would lower the growth rates for the larger stocks and raise the growth rates for the smaller stocks.

In the next section we shall test how closely the stable model (5.3.11) reproduces the capital distribution generated by actual data from the U.S. stock market, but here let us consider an example of a market similar in structure to that of Example 5.1.1.

5.3.3 Example. (The “Atlas” model) Let g and σ be positive constants, and suppose that we have stocks $X_1, \dots, X_n$ that satisfy

$$d \log X_i(t) = \gamma_i(t) dt + \sigma dW_i(t), \quad t \in [0, \infty), \quad (5.3.14)$$

where for $t \in [0, \infty)$,

$$\gamma_i(t) = \begin{cases} ng & \text{if } X_i(t) = X_{(n)}(t), \\ 0 & \text{otherwise.} \end{cases}$$

Hence, for $i = 1, \dots, n$,

$$\gamma_i(t) = ng I_{\{0\}}(X_i(t) - X_{(n)}(t)), \quad t \in [0, \infty), \quad \text{a.s.}$$

By symmetry, each of the stocks will asymptotically spend the same amount of time in the lowest rank, so it can be shown that, for $i = 1, \dots, n$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T I_{\{0\}}(X_i(t) - X_{(n)}(t)) dt = \frac{1}{n}, \quad \text{a.s.} \quad (5.3.15)$$

Therefore, by Lemma 1.3.2, for $i = 1, \dots, n$,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log X_i(t) = ng \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T I_{\{0\}}(X_i(t) - X_{(n)}(t)) dt = g, \quad \text{a.s.} \quad (5.3.16)$$

We can now apply Corollary 1.3.3 and Proposition 2.1.2(iii) to show that the market is coherent. Hence, for $k = 1, \dots, n$,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log X_{(k)}(t) = g, \quad \text{a.s.,} \quad (5.3.17)$$

and Proposition 2.1.2(ii) implies that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \gamma_\mu(t) dt = g, \quad \text{a.s.}$$

This is reasonable, since the smallest stock has growth rate ng and the other stocks have growth rate zero, and each stock is smallest about $1/n$ of the time, so the market should asymptotically grow at the rate g .

Proposition 4.1.11 implies that a.s., for all $t \in [0, \infty)$,

$$d \log X_{(n)}(t) = ng dt - \frac{1}{2} d\Lambda_{\log X_{(n-1)} - \log X_{(n)}}(t) + \sigma dB(t),$$

where B is a Brownian motion. Hence, (5.3.17) and Lemma 1.3.2 imply that

$$\lambda_{n-1,n} = 2(n-1)g, \quad \text{a.s.}$$

Since for $k = 1, \dots, n-1$, a.s., for all $t \in [0, \infty)$,

$$\begin{aligned} d \log X_{(k)}(t) &= \frac{1}{2} d\Lambda_{\log X_{(k)} - \log X_{(k+1)}}(t) \\ &\quad - \frac{1}{2} d\Lambda_{\log X_{(k-1)} - \log X_{(k)}}(t) + \sigma dB_k(t), \end{aligned} \quad (5.3.18)$$

where B_k is a Brownian motion, the same reasoning implies that

$$g = \frac{1}{2} \lambda_{k,k+1} - \frac{1}{2} \lambda_{k-1,k}, \quad \text{a.s.}, \quad (5.3.19)$$

and hence that

$$\lambda_{k,k+1} = 2kg, \quad \text{a.s.},$$

for $k = 1, \dots, n-1$. Equation (5.3.19) also implies that $g_k = -g$, for $k = 1, \dots, n-1$, and hence that $g_n = (n-1)g$.

For $k = 1, \dots, n-1$,

$$\sigma_{k:k+1}^2 = 2\sigma^2, \quad \text{a.s.},$$

so the characteristic parameters for this example are $g_1 = \dots = g_{n-1} = -g$ and $\sigma_{1:2}^2 = \dots = \sigma_{n-1:n}^2 = 2\sigma^2$.

It follows from (5.3.13) that for $k = 1, \dots, n-1$ the log-log slope between $\mu_{(k)}$ and $\mu_{(k+1)}$ is the constant $-\sigma^2/2g$, so the asymptotic capital distribution follows a Pareto distribution, just as in Example 5.1.1. Moreover, the relation between the slope of the log-log plot of the distribution and the growth rate of the market is the same as in that example. $\square$

In this example the entire market is supported by the smallest stock, whence the name “Atlas.” The example shows that a finite market in which the smallest stock has a positive growth rate that simulates the replacement of small stocks is similar to the infinite market of example Example 5.1.1, and both examples result in a Pareto distribution. Hence, there are three essentially equivalent modeling methodologies: a finite market with replacement, a finite market with the highest growth rate for the smallest stock, and an infinite market.

5.3.4 Example. (Equal growth rates) Suppose all the stocks in the market have the same growth rate process. Then Corollary 1.3.3 states that for $i = 1, \dots, n$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log X_i(t) - \int_0^T \gamma(t) dt \right) = 0, \quad \text{a.s.}$$

Therefore,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log X_{(k)}(T) - \int_0^T \gamma(t) dt \right) = 0, \quad \text{a.s.}, \quad (5.3.20)$$

for $k = 1, \dots, n$, so,

$$\lim_{t \rightarrow \infty} \frac{1}{t} (\log X_{(k)}(t) - \log X_{(k+1)}(t)) = 0, \quad \text{a.s.},$$

for $k = 1, \dots, n-1$. Now, Proposition 4.1.11 implies that, a.s., for $t \in [0, \infty)$,

$$\begin{aligned} d \log X_{(1)}(t) &= \sum_{i=1}^n I_{\{i\}}(p_t(k)) d \log X_i(t) + \frac{1}{2} d \Lambda_{\log X_{(1)} - \log X_{(2)}}(t) \\ &= \gamma(t) dt + \frac{1}{2} d \Lambda_{\log X_{(1)} - \log X_{(2)}}(t) + \sum_{\nu=1}^n \xi_{p_t(1)\nu}(t) dW_\nu(t), \end{aligned}$$

so by Lemma 1.3.2,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\log X_{(1)}(T) - \int_0^T \gamma(t) dt - \frac{1}{2} \Lambda_{\log X_{(1)} - \log X_{(2)}}(T) \right) = 0, \quad \text{a.s.}$$

This, along with (5.3.20), implies that, a.s.,

$$\lambda_{1,2} = \lim_{t \rightarrow \infty} \frac{1}{t} \Lambda_{\log X_{(1)} - \log X_{(2)}}(t) = 0.$$

By the same reasoning as in (5.3.18), it can be shown that

$$\lambda_{k,k+1} = 0, \quad \text{a.s.},$$

for $k = 1, \dots, n-1$. Therefore, this market is not asymptotically stable. $\square$

In Section 2.2 we saw that a (finite) market in which all the stocks share a common growth rate will not be diverse. Example 5.3.4 shows that in such a market, the stocks may change rank, but the asymptotic “intensity” of these changes, which is measured by the $\lambda_{k,k+1}$, is zero. Examples 5.3.3 and 5.3.4 indicate that the smaller stocks in the market must have higher growth rates than the larger stocks in order for the market to be asymptotically stable.

5.3.5 Problem. (!) In Example 5.3.3, show that (5.3.14) has a (weak) solution (see Karatzas and Shreve (1991)), and prove that (5.3.15) is valid.

5.3.6 Problem. (!) Note that the Atlas model in Example 5.3.3 admits an equivalent martingale measure (see Karatzas and Shreve (1991), Section 5.8 A, for a proof). This means that this market model is not weakly diverse (see Example 3.3.3). Determine what type of modification is needed to make the market weakly diverse in Example 5.3.3.

5.3.7 Problem. (!) Since the Atlas model in Example 5.3.3 is not weakly diverse, estimate the (nonzero) probability that over a given period of time, such a market exhibits behavior that violates weak diversity.

5.3.8 Problem. (!) For the Atlas model in Example 5.3.3, find

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mu_{(k)}(t) dt.$$

5.3.9 Problem. (!) Analyze a “top-down” market model, similar to Example 5.3.3, but with

$$\gamma_i(t) = \begin{cases} 0 & \text{if } X_i(t) = X_{(1)}(t), \\ g > 0 & \text{otherwise,} \end{cases}$$

for $t \in [0, \infty)$.

5.3.10 Problem. (!!) Develop the analysis in this section for countably infinite markets. What constraints are needed on the behavior of the structural parameters as $n \rightarrow \infty$?

5.4 Application to the U.S. Equity Market

We now have a model for a stable capital distribution, and we would like to see how well this model represents reality. To determine this, in this section we shall apply the methodology we have developed in this chapter to the U.S. equity market over the 10-year period from January 1, 1990, to December 31, 1999. Here we use monthly data from the CRSP universe, which was also used for the capital distribution curves in Figure 5.1. Of the 10 years considered, the smallest number of market weights in the database was 5758, in 1991, and the greatest was 7467, in 1998. As we noted in Section 5.1 above, the smallest weights in the capital distribution may be suspect, so here we shall consider only the largest 5120 weights.

To carry out the calculations of the previous section, we must estimate the parameters $\lambda_{k,k+1}$. To do so, we need to generate values for the local time processes $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}$, and to estimate these values we can use (4.3.2) of Example 4.3.2, which implies that for $k = 1, \dots, n$,

$$\begin{aligned} d \log(Z_\xi(t)/Z_\mu(t)) &= d \log(\mu_{(1)}(t) + \dots + \mu_{(k)}(t)) \\ &\quad - \frac{\mu_{(k)}(t)}{2(\mu_{(1)}(t) + \dots + \mu_{(k)}(t))} d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t), \end{aligned} \quad (5.4.1)$$

a.s., for $t \in [0, \infty)$, where ξ is the portfolio holding the k largest stocks. Since the relative return on the left-hand side of (5.4.1) and the first term on the right-hand side can both be estimated easily, this equation can be used to determine the value of $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t)$. With these estimates, we can proceed to estimate the parameters $\lambda_{k,k+1}$.

Figure 5.2 shows the local time processes $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}$ for the values of $k = 10, 20, 40, \dots, 5120$. Although the curves are not exactly straight

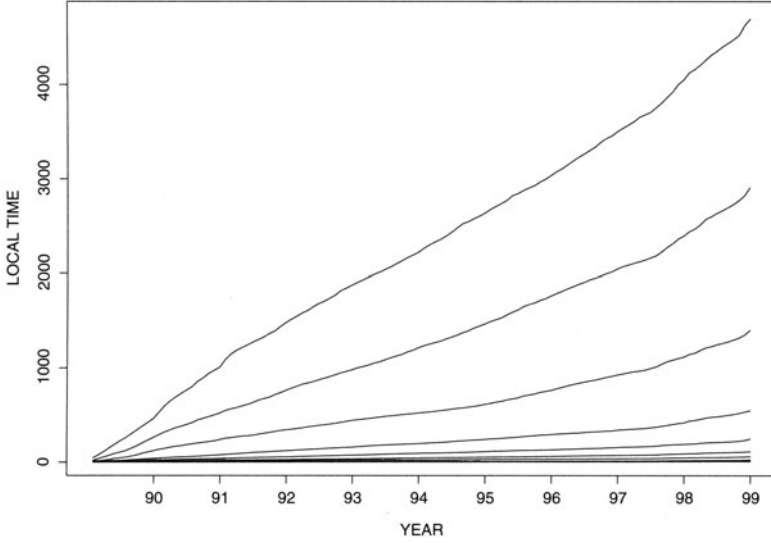


FIGURE 5.2. $\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t)$, $k = 10, 20, 40, \dots, 5120$.

lines, they are straight enough so that the parameters $\lambda_{k,k+1}$ can be reasonably estimated by the slopes of the lines connecting the endpoints of the curves, and this is the estimate that was used.

The values of the $\sigma_{k:k+1}^2$, for $k = 10, 20, 40, \dots, 5120$, are more difficult to estimate, since no equation of the form (5.4.1) is available. In the case of the $\lambda_{k,k+1}$ we were able to simplify the procedure by applying (5.4.1). Here we have to use more classical estimation techniques. The method we use here is chosen for its simplicity, and we make no claims that it is necessarily the most effective. Itô's rule implies that

$$d(\log \mu_{p_t(k)}(t) - \log \mu_{p_t(k+1)}(t))_t \approx (d(\log \mu_{p_t(k)}(t) - \log \mu_{p_t(k+1)}(t)))^2,$$

where we interpret d to signify the change in value over a short period of time. Accordingly, we can estimate the $\sigma_{k:k+1}^2$ by

$$\frac{1}{119(2k_1 + 1)} \sum_{j=k-k_1}^{k+k_1} \sum_{t=1}^{119} ((\log \mu_{p_t(k)}(t+1) - \log \mu_{p_t(k+1)}(t+1)) - (\log \mu_{p_t(k)}(t) - \log \mu_{p_t(k+1)}(t)))^2, \quad (5.4.2)$$

where the variable $t = 1, \dots, 120$ refers to the months from January 1990 to December 1999, and $k_1 = \max(2, [k/100])$, in which $[\cdot]$ denotes the integer part of a real number.

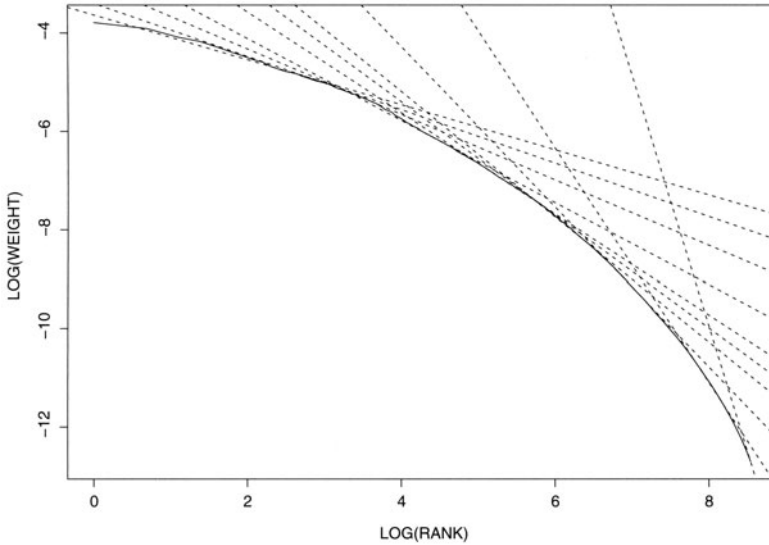


FIGURE 5.3. $\log \mu_{(k)}$ versus $\log k$, average value 1990 to 1999.
Tangent lines at $k = 10, 20, 40, \dots, 5120$.

From the $\lambda_{k,k+1}$ and the $\sigma_{k:k+1}^2$, the slopes of the tangents to the capital distribution curve were calculated using (5.3.13). The capital distribution curve and the tangent lines are displayed in Figure 5.3, and the accuracy of the slopes of the tangents in that chart confirm the accuracy of the estimates (5.4.2).

The solid line in Figure 5.3 is a log-log plot of the average capital distribution for the subset of the U.S. market comprising the largest 5120 stocks on December 31 of each year from 1990 to 1999. The horizontal coordinate is $\log k$, and the vertical coordinate is $\log \mu_{(k)}(t)$. The broken lines show the tangents at the points corresponding to the values of $k = 10, 20, 40, \dots, 5120$. There appears to be a local convexity in the curve near $k = 10$, and this is perhaps the cause of the tangent line being somewhat under the curve in that neighborhood. In general, the tangent lines appear to be quite accurate, and Figure 5.3 confirms that the methodology developed in Section 5.3 provides a valid model for the long-term structure of the capital distribution.

The characteristic parameters g_k and $\sigma_{k:k+1}^2$ are displayed in Figures 5.4 and 5.5, respectively. The $\lambda_{k,k+1}$ were estimated using (5.4.1), and from these values the g_k were calculated using (5.3.7):

$$g_k = \frac{1}{2}\lambda_{k-1,k} - \frac{1}{2}\lambda_{k,k+1}.$$

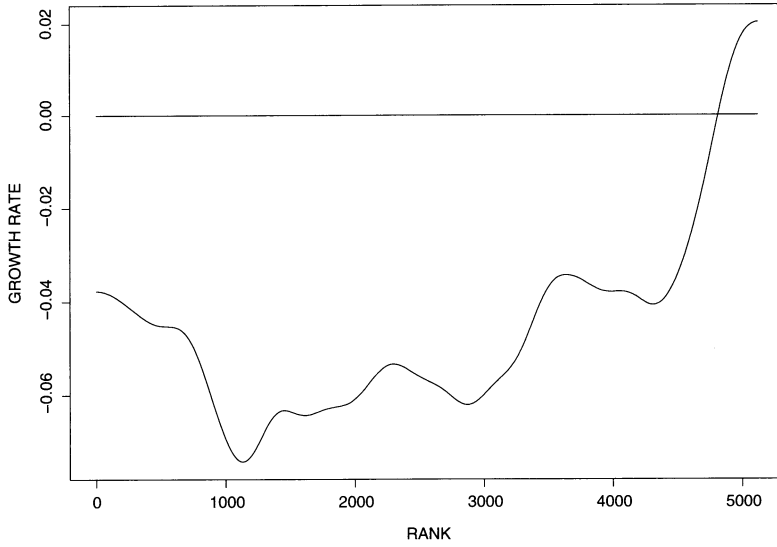


FIGURE 5.4. Smoothed annualized values of g_k for $k = 1, \dots, 5119$.
Calculated from 1990–1999 data.

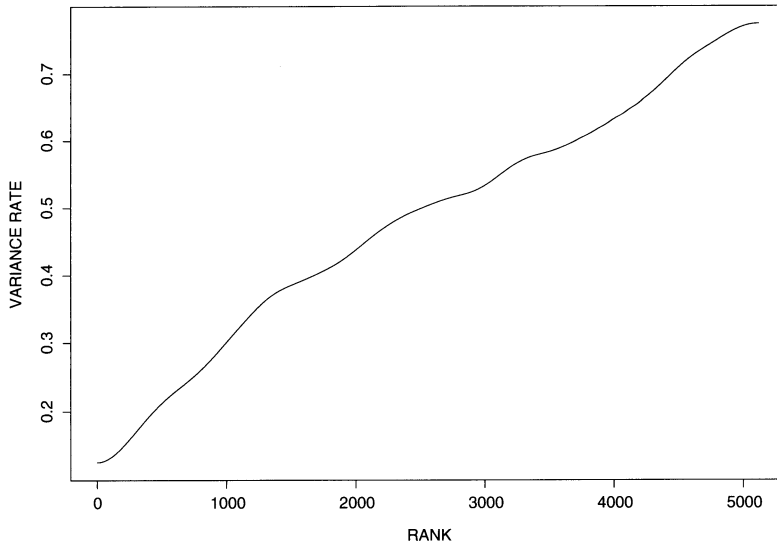


FIGURE 5.5. Smoothed annualized values of $\sigma^2_{k:k+1}$ for $k = 1, \dots, 5119$.
Calculated from 1990–1999 data.

These estimates of the $\mathbf{g}_k$ should be as accurate as the estimates of the $\lambda_{k,k+1}$. The values of the $\sigma_{k:k+1}^2$ were calculated from the $\lambda_{k,k+1}$ and the market weights using (5.3.12) in the form

$$\sigma_{k:k+1}^2 = 2\lambda_{k,k+1} \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)) dt.$$

This procedure is preferred to the estimates (5.4.2), since the time-averaged market weights are observable and the values of the $\lambda_{k,k+1}$ can be accurately determined by the methods we just discussed. The values of the characteristic parameters shown in Figures 5.4 and 5.5 are smoothed by convolution with a Gaussian kernel with $\pm 3.16\sigma$ spanning 1000 units on the horizontal axis, with reflection at the ends of the data.

The values of the $\mathbf{g}_k$, for $k = 1, \dots, 5119$, shown in Figure 5.4, go from negative for the larger stocks to positive for the smallest stocks. This curve shows considerable structure, which might not be stable over time, so this is an area for possible future investigation. The values of the $\sigma_{k:k+1}^2$ shown in Figure 5.5 form an increasing, approximately straight line. The $\sigma_{k:k+1}^2$ measure the relative variance of stocks that are adjacent in rank. Between the largest stocks, Figure 5.5 indicates that the relative annual standard deviation is about 35%, while between the smallest stocks it is about 85%.

5.4.1 Problem. (!) Can the parameters $\sigma_{k:k+1}$ be calculated using portfolio generating functions, in a manner similar to the calculation of the $\lambda_{k,k+1}$?

5.4.2 Problem. (!!) Determine whether the estimated values of the structural parameters displayed in Figures 5.4 and 5.5 are compatible with the Capital Asset Pricing Model. (Note that the structural parameters are compatible with the no-arbitrage hypothesis, since they are all bounded.) See Sharpe (1964) or Karatzas and Shreve (1998) for details.

5.5 A “First-Order” Market Model

In this section we construct a simple “first-order” approximation to a market model that is compatible with the $2n - 2$ structural parameters defined in the previous section. The goal here is to find the simplest model that is consistent with these parameters. The model we propose will have the same $\mathbf{g}_k$ and a slightly smoothed series of $\sigma_{k:k+1}^2$. We shall simulate the behavior of the first-order model to determine to what extent it reflects the behavior of the actual market.

Since the $\mathbf{g}_k$ represent growth rates, they can be used without modification as the growth rates of the stocks in the first-order model. In this case, $\mathbf{g}_k$ will be the growth rate of the k th-ranked stock $X_{(k)}$. The $\sigma_{k:k+1}^2$ represent variance parameters for the differences of the logarithmic returns

of consecutively ranked stocks, and hence cannot be assigned directly to individual stocks. Moreover, in the $\sigma_{k:k+1}^2$ there is little if any information about the cross-variation of the stock processes, so for the first-order model we shall assume that the stock processes are independent. This is clearly a strong assumption, and it will be interesting to observe its consequences.

Let us use the notation σ_k to represent the (constant) variance process of the k th-ranked stock. Under the assumption that the ranked stocks are independent, $\sigma_{k:k+1}^2$ will be equal to $\sigma_k^2 + \sigma_{k+1}^2$. Accordingly, we wish to find $\sigma_1^2, \dots, \sigma_n^2$ such that for each rank k , σ_k^2 depends only on $\sigma_{k-1:k}^2$ and $\sigma_{k:k+1}^2$, and such that $\sigma_k^2 + \sigma_{k+1}^2$ is equal to, or at least close to, $\sigma_{k:k+1}^2$. While an exact solution may be possible—although not assured: the $\sigma_{k:k+1}$ must be positive—an approximate solution is given by

$$\sigma_k^2 = \frac{1}{4}(\sigma_{k-1:k}^2 + \sigma_{k:k+1}^2).$$

We shall validate this claim below, but let us first present a precise formulation of the model.

5.5.1 Definition. Suppose that a market $\mathcal{M}$ has an asymptotically stable capital distribution, and that $\mathbf{g}_k$ and $\sigma_{k:k+1}^2$, for $k = 1, \dots, n-1$, are the characteristic parameters of the distribution. Let

$$\begin{aligned} \sigma_k^2 &= \frac{1}{4}(\sigma_{k-1:k}^2 + \sigma_{k:k+1}^2), \quad \text{for } k = 2, \dots, n-1, \\ \sigma_1^2 &= \frac{1}{2}\sigma_{1:2}^2, \quad \text{and} \quad \sigma_n^2 = \frac{1}{2}\sigma_{n-1:n}^2, \end{aligned} \tag{5.5.1}$$

and let q_t be the inverse of the permutation p_t in (4.1.26), for $t \in [0, \infty)$. Then the *first-order model* for the market is given by

$$d \log X_i(t) = \mathbf{g}_{q_t(i)} dt + \sigma_{q_t(i)} dV_i(t), \quad t \in [0, \infty), \tag{5.5.2}$$

for $i = 1, \dots, n$, where $(V_1, \dots, V_n)$ is an n -dimensional Brownian motion.

In (5.5.2), $q_t(i)$ represents the rank of the i th stock at time t . The equivalent forms of notation

$$j = q_t(i) \quad \text{and} \quad i = p_t(j)$$

both mean, roughly, that the i th stock, X_i , is in the j th rank at time t . In the original notation of (1.1.6) we have

$$\gamma_i(t) = \mathbf{g}_{q_t(i)} \quad \text{and} \quad \xi_{i\nu}(t) = \sigma_{q_t(i)} \delta_{i\nu}, \tag{5.5.3}$$

for $t \in [0, \infty)$, a.s., where $\delta_{ij} = 1$ if $i = j$, and $\delta_{ij} = 0$ otherwise. The covariance process for the first-order model is

$$\sigma_{ij}(t) = \delta_{ij} \sigma_{q_t(i)} \sigma_{q_t(j)}, \quad t \in [0, \infty), \quad \text{a.s.}$$

Obviously, the first-order market model defined by (5.5.2) is much simpler than the general market model given by (1.1.6), for it depends on $2n - 2$ constant parameters rather than $n^2 + n$ stochastic processes. With this model, the equation corresponding to (5.3.11) is

$$\begin{aligned} d(\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)) \\ = -\lambda_{k,k+1} dt + d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) + dN_{k:k+1}(t), \end{aligned} \quad (5.5.4)$$

a.s., for $t \in [0, \infty)$ and $k = 1, \dots, n - 1$, where the $N_{k:k+1}$ are continuous martingales such that a.s., for $t \in [0, \infty)$,

$$\langle N_{k:k+1} \rangle_t = \left(\frac{1}{4} \sigma_{k-1:k}^2 + \frac{1}{2} \sigma_{k:k+1}^2 + \frac{1}{4} \sigma_{k+1:k+2}^2 \right) t, \quad (5.5.5)$$

for $k = 2, \dots, n - 2$. Hence the variance parameters for (5.5.4) will be a smoothed version of those for (5.3.11), so we cannot expect the simulated capital distribution to be exactly the same as the actual capital distribution. However, if the series of variances $\sigma_{k:k+1}^2$ is reasonably smooth to begin with, the additional smoothing that occurs in the first-order model should have little effect. The values of the $\sigma_{k:k+1}^2$ shown in Figure 5.5 form a curve that is approximately a straight line, so in this case the smoothing effect is probably inconsequential. Perhaps a more serious weakness of the first-order model is that it approximates the martingales $M_{k:k+1}$ in (5.3.11) up to the quadratic variation processes $\langle M_{k:k+1} \rangle$, but ignores the cross-variation processes $\langle M_{j:j+1}, M_{k:k+1} \rangle$ for $j \neq k$. The effect of these and other shortcomings of our first-order model will be better understood after we consider a test of the model on actual data.

To test the first-order model, we generated simulated capital distributions using (5.5.2) where the characteristic parameters g_k and $\sigma_{k:k+1}^2$. We simulated 10 years of monthly data using (5.5.2), and compared the results of this simulation to the original 10 years of data from which the parameters were estimated. In Figure 5.6, the upper and lower envelopes of the actual 10 year-end capital distribution curves are shown as solid lines. These envelopes are not themselves capital distribution curves, but rather show at each point on the horizontal axis the maximum and minimum values of the 10 curves corresponding to that point. The upper and lower envelopes of the 10 year-end simulated capital distribution curves are shown in Figure 5.6 as broken lines. As can be seen, the actual curves showed a larger range of values than the simulated curves, particularly for the smaller weights. The average values of the actual curves and the simulated curves are not shown in Figure 5.6, but were quite close together.

Figure 5.6 suggests that the structural parameters g_k and $\sigma_{k:k+1}^2$ vary somewhat over time, and that this variation causes the spread in the actual curves that is not present in the simulated curves. In fact, the actual upper and lower envelopes in Figure 5.6 conform quite closely to the capital distribution curves generated by raising the average capital distribution over

the period to the 1.1 and 0.9 powers. These two curves cross at the narrow point between the upper and lower envelopes. Equation (5.3.12) implies that raising the weights to these powers has approximately the same effect as multiplying the parameters $\mathbf{g}_k$ by 0.9 and 1.1, respectively. This level of variation appears to be roughly consistent with the observed deviation from straight lines of the local time curves in Figure 5.2.

To measure the variation of capital distribution curve generated by the first-order model, we calculated the change in the entropy of the capital distribution over a simulated 60-year period. The cumulative values of the logarithmic change in market entropy appear in Figure 5.7. We can compare this chart to Figure 6.2 on page 122, which shows the change in the entropy for the U.S. equity market over an actual 60-year period. The scales of the two figures are about the same, so we see that the first-order model exhibits notably less variation in entropy than the actual market. Since entropy is a measure of the diversity of the capital distribution, this means that the variation of the diversity of the actual market is greater than the variation of the diversity of a simulated market generated by the first-order model. This observation is consistent with the behavior of the simulated envelopes displayed in Figure 5.6.

Let us consider the behavior of a portfolio π under the first-order model. It follows from (1.1.13) that the value process Z_π under the first-order model will satisfy

$$d \log Z_\pi(t) = \gamma_\pi(t) dt + \sum_{i=1}^n \pi_i(t) \sigma_{q_t(i)} dV_i(t), \quad (5.5.6)$$

for $t \in [0, \infty)$, a.s., where

$$\gamma_\pi(t) = \sum_{i=1}^n \pi_i(t) \mathbf{g}_{q_t(i)} + \frac{1}{2} \sum_{i=1}^n (\pi_i(t) - \pi_i^2(t)) \sigma_{q_t(i)}^2, \quad (5.5.7)$$

for $t \in [0, \infty)$. From (5.5.6), we see that the portfolio variance process for π is

$$\sigma_{\pi\pi}(t) = \sum_{i=1}^n \pi_i^2(t) \sigma_{q_t(i)}^2, \quad (5.5.8)$$

for $t \in [0, \infty)$.

Now, suppose we define

$$\tilde{\pi}_j(t) = \pi_{p_t(j)}(t), \quad t \in [0, \infty), \quad (5.5.9)$$

for $j = 1, \dots, n$, so $\tilde{\pi}_j(t)$ denotes the weight of the j th-ranked stock at time t . In order to avoid confusion with the usual portfolio weights $\pi_1, \dots, \pi_n$, let us refer to the weights $\tilde{\pi}_1, \dots, \tilde{\pi}_n$ as the *rank weights* of the portfolio. Then (5.5.3) implies that (5.5.7) becomes

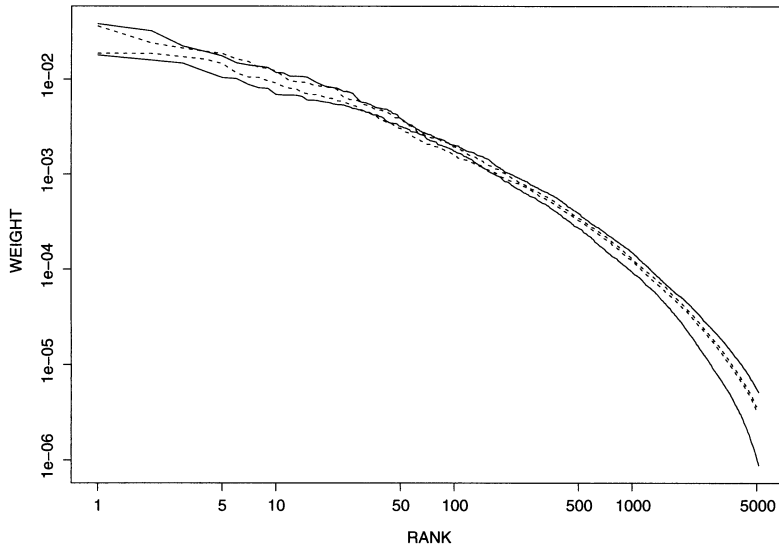


FIGURE 5.6. Upper and lower envelopes of the capital distribution curves. Actual data, solid lines; simulated data, broken lines.

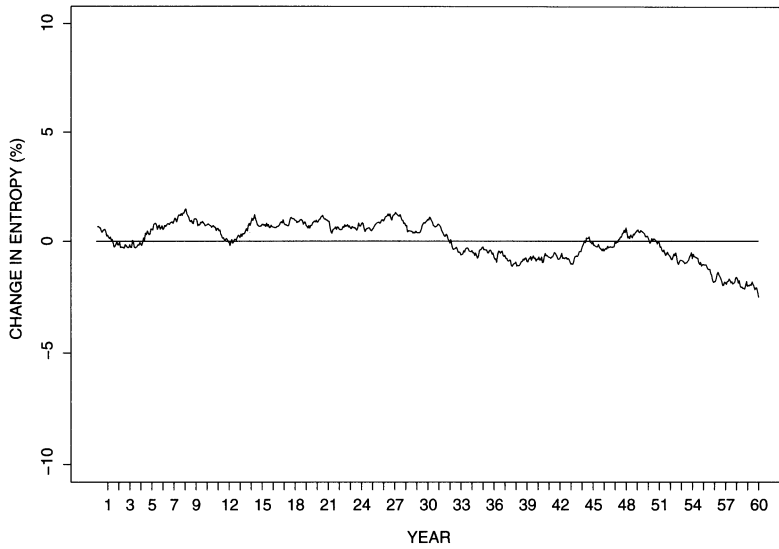


FIGURE 5.7. Change in market entropy for the first-order model. (Adjusted to have zero sample mean.)

$$\begin{aligned}
\gamma_\pi(t) &= \sum_{j=1}^n \pi_{p_t(j)}(t) \mathbf{g}_j + \frac{1}{2} \sum_{j=1}^n (\pi_{p_t(j)}(t) - \pi_{p_t(j)}^2(t)) \boldsymbol{\sigma}_j^2 \\
&= \sum_{j=1}^n \tilde{\pi}_j(t) \mathbf{g}_j + \frac{1}{2} \sum_{j=1}^n (\tilde{\pi}_j(t) - \tilde{\pi}_j^2(t)) \boldsymbol{\sigma}_j^2,
\end{aligned} \tag{5.5.10}$$

for $t \in [0, \infty)$, a.s., and similarly, (5.5.8) becomes

$$\sigma_{\pi\pi}(t) = \sum_{j=1}^n \tilde{\pi}_j^2(t) \boldsymbol{\sigma}_j^2, \tag{5.5.11}$$

for $t \in [0, \infty)$, a.s. From these equations we can develop a portfolio theory similar to the conventional theory, but where the portfolio weights pertain to the ranked stocks rather than to the original stocks in the market.

5.5.2 Problem. (!) Develop and test a market model that uses the first-order model's growth rate estimates, but uses a conventional estimate for the covariances among the stocks.

5.5.3 Problem. (!) Develop and test a market model that uses the first-order model's growth rate estimates, but, in the rank covariances, includes a factor that captures the variability of diversity displayed in Figure 5.7, perhaps based on the variation of

$$\log(\mu_{(1)}(t)/\mu_{(n)}(t)).$$

This might reintroduce positive correlation among stocks that are close to each other in rank, correlation that may have been lost in the $\boldsymbol{\sigma}_{k:k+1}^2$.

5.5.4 Problem. (!) Develop and test a market model that uses the first-order model's growth rate estimates, but includes an estimate of the whole *rank-covariance matrix* $\boldsymbol{\sigma} = (\boldsymbol{\sigma}_{jk})_{1 \leq j, k \leq n}$, where

$$\boldsymbol{\sigma}_{jk} = \lim_{t \rightarrow \infty} \frac{1}{t} \langle \log \mu_{(j)}, \log \mu_{(k)} \rangle_t,$$

$j, k = 1, \dots, n$. The $\boldsymbol{\sigma}_{jk}$ could, perhaps, be estimated by a procedure similar to (5.4.2).

5.6 Notes and Summary

The distribution of capital is equivalent to the size distribution of firms, and this has been studied extensively (see, for example, Gibrat (1931), Simon (1955), Simon and Bonini (1958), Steindl (1965), Ijiri and Simon

(1977)). Simon (1955) showed that under certain simple conditions, for a market in which the smallest stocks are replaced at a constant rate, the capital distribution would be a *Pareto distribution*. The use of the Pareto distribution with respect to firm size was developed in detail in Steindl (1965). Bak (1996) argued that the Pareto distribution applies to so many classes of size distributions that its ubiquity is the result of a fundamental law of nature.

The model in Example 5.1.1 was communicated to the author by M. Aizenman (2000). The model is related to work in mathematical physics by Bolthausen and Sznitman (1998), and is developed in more general form by Banner and Maguire (2001).

Ijiri and Simon (1974) observed that the log-log plots of actual capital distributions are usually concave, and hence are not accurately represented by a simple Pareto model. Since then, much of the literature has been devoted to explaining this observed discrepancy. Among the more current contributions, Jovanovic (1982) and Hopenhayn (1992) proposed dynamic models for corporate growth, Axtell (1999) used game-theoretic methods, and Hashemi (2000) applied the theory of evolutionary economics to explain the observed distribution of firm size. In this chapter we follow Fernholz (2001c), and show how the capital distribution curve depends on the mean and covariance structure of the ranked market weights.

Chapter Summary

Section 5.1: The *capital distribution curve* refers to the log-log plot of the market weights arranged in order of descending rank. For a market with an infinite number of stocks, but finite total capital, in which the logarithmic returns are independent and normally distributed with the same mean and variance, the capital distribution curve is approximately a straight line. This linear capital distribution curve is known as a *Pareto distribution*. Data from the U.S. equity market indicate that the capital distribution curves do not follow a Pareto distribution, but instead are somewhat concave in shape.

Section 5.2: The equation

$$dX(t) = -\alpha \operatorname{sgn}(X(t)) dt + \sigma dW(t), \quad t \in [0, \infty), \quad (5.2.1)$$

where α and σ are positive constants, and W is a Brownian motion, is useful in modeling the distance between adjacent points on the capital distribution curve, $\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)$. The solution to this equation is a continuous semimartingale with a double exponential asymptotic distribution. This fact can be used to show that

$$\lim_{t \rightarrow \infty} t^{-1} \Lambda_{|X|}(t) = \alpha, \quad \text{a.s.}, \quad (5.2.5)$$

and

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |X(t)| dt = \frac{\sigma^2}{2\alpha}, \quad \text{a.s.} \quad (5.2.6)$$

These expressions can be applied to the capital distribution curve.

Section 5.3: The market $\mathcal{M}$ is said to be *asymptotically stable* if it is coherent and, a.s.:

- (i) for $k = 1, \dots, n-1$, $\lim_{t \rightarrow \infty} t^{-1} \Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t) = \lambda_{k,k+1}$;
- (ii) for $k = 1, \dots, n-1$, $\lim_{t \rightarrow \infty} t^{-1} \langle \log \mu_{(k)} - \log \mu_{(k+1)} \rangle_t = \sigma_{k:k+1}^2$;

where $\lambda_{k,k+1}$ and $\sigma_{k:k+1}$ are positive constants.

With the help of (5.2.6), we can determine the asymptotic time-average structure of the capital distribution curve, with

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (\log \mu_{(k)}(t) - \log \mu_{(k+1)}(t)) dt = \frac{\sigma_{k:k+1}^2}{2\lambda_{k,k+1}}, \quad \text{a.s.}, \quad (5.3.12)$$

for $k = 1, \dots, n-1$.

Let

$$g_k = \frac{1}{2} \lambda_{k-1,k} - \frac{1}{2} \lambda_{k,k+1}, \quad (5.3.7)$$

for $k = 1, \dots, n$, where $\lambda_{0,1} = 0$ and $\lambda_{n,n+1} = 0$. Then the $2n-2$ parameters $g_1, \dots, g_{n-1}$ and $\sigma_{1:2}^2, \dots, \sigma_{n-1:n}^2$ are called the *characteristic parameters* of the capital distribution.

The “Atlas” model is a market model with stocks $X_1, \dots, X_n$ that satisfy

$$d \log X_i(t) = \gamma_i(t) dt + \sigma dW_i(t), \quad t \in [0, \infty), \quad (5.3.14)$$

where for $t \in [0, \infty)$,

$$\gamma_i(t) = \begin{cases} ng & \text{if } X_i(t) = X_{(n)}(t), \\ 0 & \text{otherwise,} \end{cases}$$

where g and σ are positive constants. In this case, the characteristic parameters are $g_1 = \dots = g_{n-1} = -g$ and $\sigma_{1:2}^2 = \dots = \sigma_{n-1:n}^2 = 2\sigma^2$, and the asymptotic capital distribution follows a Pareto distribution.

Section 5.4: In an analysis of the U.S. equity market over the ten-year period from 1990 to 1999, the characteristic parameters were calculated and are displayed in Figures 5.4 and 5.5.

Section 5.5: It is possible to construct a simple model that will have almost the same asymptotic time-average capital distribution as the market. Suppose that a market $\mathcal{M}$ has an asymptotically stable capital distribution, and that g_k and $\sigma_{k:k+1}^2$, for $k = 1, \dots, n-1$, are the characteristic parameters of the distribution. Let

$$\begin{aligned} \sigma_k^2 &= \frac{1}{4} (\sigma_{k-1:k}^2 + \sigma_{k:k+1}^2), \quad \text{for } k = 2, \dots, n-1, \\ \sigma_1^2 &= \frac{1}{2} \sigma_{1:2}^2, \quad \text{and} \quad \sigma_n^2 = \frac{1}{2} \sigma_{n-1:n}^2, \end{aligned} \quad (5.5.1)$$

and let q_t represent the rank of the i th stock at time t , for $t \in [0, \infty)$. Then the *first-order model* for the market is given by

$$d \log X_i(t) = \mathbf{g}_{q_t(i)} dt + \boldsymbol{\sigma}_{q_t(i)} dV_i(t), \quad t \in [0, \infty), \quad (5.5.2)$$

for $i = 1, \dots, n$, where $(V_1, \dots, V_n)$ is an n -dimensional Brownian motion.

For the U.S. market from 1990 to 1999, the first-order model accurately replicates the time-average capital distribution, but fails to capture the variation that occurs over time. Of course, since the first-order model is a time-average model, it cannot be expected to capture all of the variation over time.

6

Performance of Functionally Generated Portfolios

In this chapter we analyze the simulated behavior of several functionally generated portfolios in the U.S. stock market. The portfolios we consider are the entropy-weighted portfolio of Definition 2.3.3, the $\mathbf{D}_p$ -weighted portfolio of Examples 3.4.4 and 4.3.5, and the large-stock and small-stock portfolios from Example 4.3.2. We also look at the relative performance of the biggest stock in the market, discussed in Example 4.3.1.

Over the period we consider, the entropy-weighted and $\mathbf{D}_p$ -weighted portfolios that we simulate have higher return than the capitalization-weighted portfolios of the same stocks, while maintaining about the same level of risk. Moreover, absent a significant change in market structure, it would appear that the same type of outperformance can be expected in the future. The existence of such portfolios brings into doubt the concept of market “efficiency,” and even suggests that the no-arbitrage hypothesis discussed in Section 3.3 is questionable in actual markets, at least over the long term.

The data we use for all the simulations are from the CRSP universe, defined in Section 5.1, over the time period from 1939 to 1998. The number of stocks in this market varies from about 750, in 1939, to about 7500, in 1998. The CRSP universe is a subset of a more comprehensive, but essentially unobservable, universe of stocks that includes public stocks that are not traded on an exchange as well as privately held companies. Since the CRSP universe is a subset of a larger universe, leakage must be included in all the simulations.

Because the composition of actual equity markets is continually changing, certain accommodations must be made in order that Theorem 4.2.1 be

applicable. At the beginning of each month we observe the current composition of the market, and we use this composition, along with the capital gains on each stock over that month, to calculate the changes in the values of all capitalization-related time series during the month. For each of the time series, the cumulative value of these changes is presented in the results. This procedure excludes changes in the time series that would be caused by changes in the composition of the market or by corporate actions such as takeovers and breakups. This has the same effect as adjustments made to the “divisor” of a stock index.

Let us recall that by Definition 1.2.4, relative return always refers to relative logarithmic return. Accordingly, whenever arithmetic relative return is introduced, it is duly noted in the discussion. In the examples that follow, no transaction costs are subtracted from the returns of any of the portfolios considered. However, there is a discussion of portfolio turnover in Section 6.3, so the impact of transaction costs can be estimated. In each of the portfolio simulations we present, leakage will be included in the drift process.

6.1 Market Entropy

In this section we shall investigate the behavior of the market entropy function introduced in Section 2.3, and we shall simulate the performance of the entropy-weighted portfolio that it generates. The market we consider is the CRSP universe described in the introduction to this chapter, and the portfolio is simulated over the 60-year period from 1939 to 1998.

Recall that the entropy function is defined by

$$\mathbf{S}(x) = - \sum_{i=1}^n x_i \log x_i, \quad x \in \Delta^n.$$

This function generates the entropy-weighted portfolio with weights

$$\pi_i(t) = \frac{-\mu_i(t) \log \mu_i(t)}{\mathbf{S}(\mu(t))}, \quad t \in [0, T],$$

$i = 1, \dots, n$. The entropy-weighted portfolio satisfies

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{S}(\mu(t)) + \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} dt, \quad t \in [0, T], \quad \text{a.s.}$$

For simplicity, this equation does not include leakage, but leakage must be included in the simulation in order to produce realistic performance results. Accordingly, leakage has been calculated as described in Appendix A.

The results of the simulation are presented in Figures 6.1 through 6.4 on pages 122 and 123, which show the cumulative values of the components of

the relative logarithmic return of the entropy-weighted portfolio. The scales in these four charts are the same to facilitate comparison. Figure 6.1 shows that the entropy-weighted portfolio generated about 9% higher logarithmic return than the market over the 60-year period, and this resulted in a 0.15% average annual relative return.

Figure 6.2 shows the cumulative change in entropy over the period, centered to have zero sample mean. The cumulative change in entropy was about -8% , or -0.14% a year, which is small compared to total variation over the period, and the time series appears to be mean-reverting. Entropy is a measure of diversity, and it would seem reasonable for the diversity of a large equity market like the CRSP universe to remain fairly stable over the long term. It may also be worth noting that over the entire 60-year simulation, 1998 was the minimum for the cumulative change in entropy.

The drift process in Figure 6.3 is close to a pure trend process, and it contributed about 20%, or 0.34% a year, to the relative return of the entropy-weighted portfolio over the period studied. Since the drift process satisfies $d\Theta(t) = \gamma^*(t)/\mathbf{S}(\mu(t)) dt$, the slope of this process depends on the relative variances of the stocks in the market. From the looks of Figure 6.3, these relative variances appear to have been fairly stable most of the time, although the first half of the chart has slightly less slope than the second half.

The contribution of the differential dividend rate in Figure 6.4 was significantly less than that of the drift process. Although the differential dividend rate was flat for most of the period, after 1975 it averaged about 0.20% a year in favor of the market portfolio. Over the whole 60 years, dividend payments averaged about 0.05% a year higher for the market than for the entropy-weighted portfolio. To neutralize the drift process, it would have to have been more than three times greater even after 1975, so the net effect of drift and dividends clearly favors the entropy-weighted portfolio.

Leakage measures the effect of stocks crossing over between the CRSP universe and the nonobservable ambient universe. The leakage presented in Figure 6.4 could not be calculated directly, since the ambient universe is nonobservable. Instead, the leakage was estimated for the entropy-weighted portfolio consisting of the largest 0.7*n* of the stocks in the market. This method of using a partial market to estimate leakage seems to give satisfactory results when compared to direct measurement of leakage in cases where direct measurement is possible.

As noted at the beginning of this chapter, the number of stocks in the CRSP universe increased from about 750, in 1939, to about 7500, in 1998, and this increase probably would have caused an upward drift in the market entropy, except for the fact that the change in entropy in Figure 6.2 is corrected for changes in market composition. Since the normalized measure



FIGURE 6.1. $\log(\hat{Z}_\pi(t)/\hat{Z}_\mu(t))$ for the entropy-weighted portfolio.

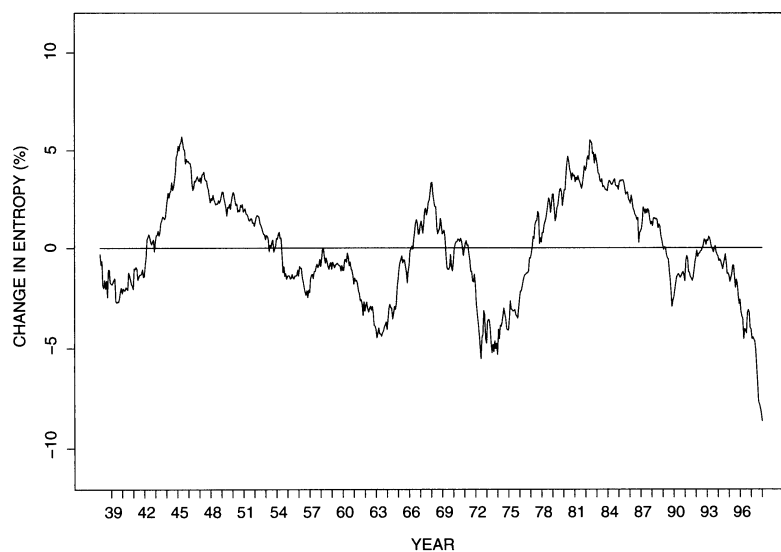


FIGURE 6.2. Change in market entropy.
(Adjusted to have zero sample mean.)

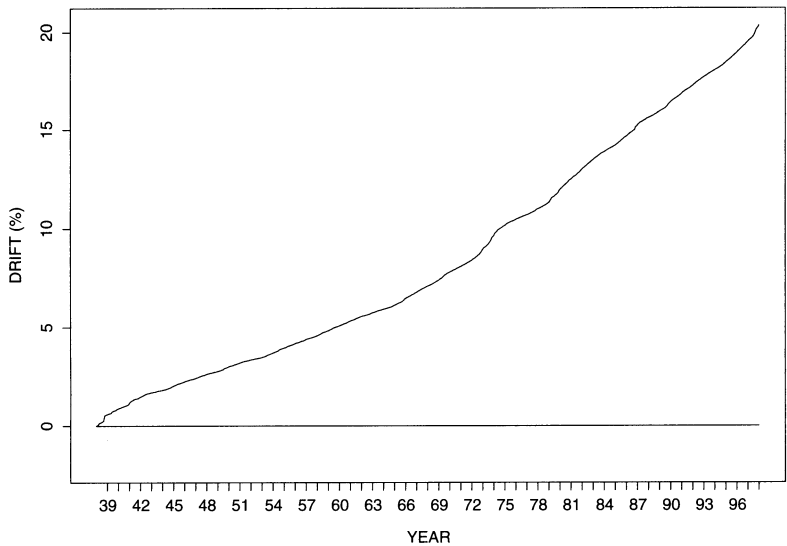


FIGURE 6.3. Drift process for the entropy-weighted portfolio.
Leakage is included in the drift process.

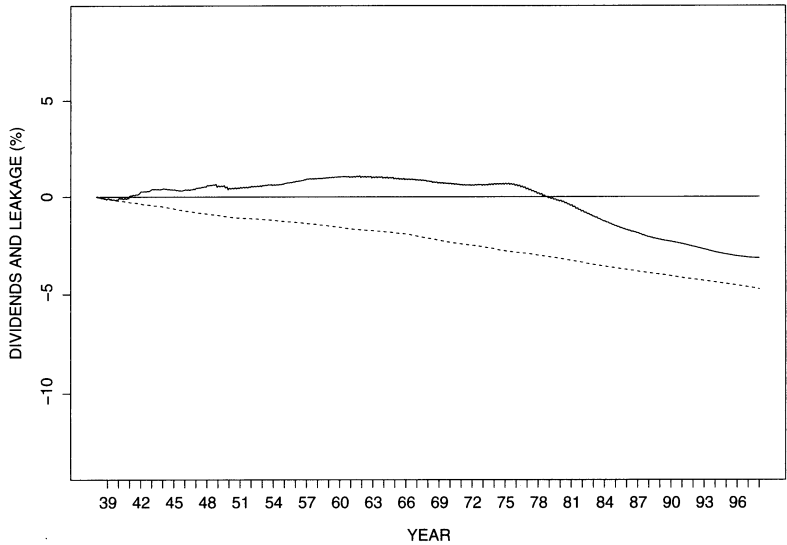
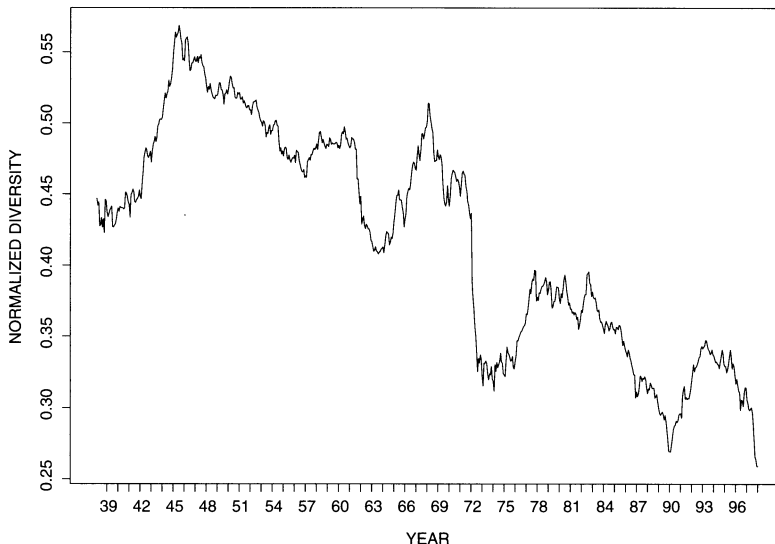


FIGURE 6.4. Dividends and leakage for the entropy-weighted portfolio.
Dividends (solid line) and leakage (broken line).

FIGURE 6.5. Diversity of CRSP universe using $\tilde{\mathbf{D}}_p$, $p = 0.5$.

of diversity $\tilde{\mathbf{D}}_p$, defined in (3.4.3) by

$$\tilde{\mathbf{D}}_p(x) = \left(n^{p-1} \sum_{i=1}^n x_i^p \right)^{1/p}, \quad x \in \Delta^n,$$

depends only on the shape of the capital distribution and not on the number of stocks in the market, it can provide a measure of the absolute diversity of the market even when the number of constituent stocks changes. Figure 6.5 shows the value of $\tilde{\mathbf{D}}_p(\mu(t))$ over the same period from 1939 to 1998. The most prominent difference between the two charts is the abrupt drop in diversity in Figure 6.5 when the NASDAQ stocks were added to the CRSP universe at the beginning of 1973. At that time, the number of stocks in the market roughly doubled from about 2000 to about 4000, and the drop in the chart indicates a change in the shape of the capital distribution. There is also a smaller drop in the middle of 1962 that coincides with the addition of the AMEX data to the CRSP universe. If these two drops are deleted, the structure of the two charts will be about the same.

Although the entropy-weighted portfolio had a higher return than the market portfolio during the period we considered, financial theory would claim that this improved return must be offset by higher risk. However, the time series in Figure 6.2 appears to be mean-reverting, and the relaxation time of the series appears to be on the order of about 10 years. This means

that if we sample the return series for the entropy-weighted portfolio and for the market portfolio at intervals on the order of the relaxation time, or greater, the sample variance will probably be about the same. Therefore, at such a sampling interval, the higher return is not offset by higher risk.

Financial theory implies that a portfolio that has higher expected return than the market portfolio must have a “beta” greater than 1. Nevertheless, the beta of the entropy-weighted portfolio with respect to the market is about 0.95 when the series are sampled at 10-year intervals. This means that the entropy-weighted portfolio appears not to be behaving in a manner consistent with current financial theory. We shall see in the following sections that other functionally generated portfolios also appear to be in violation of financial theory as it is currently accepted.

6.1.1 Problem. (!!) Determine the structure of the time series for the change in market entropy. Develop a predictor for this time series. The difficulty here is that there are probably too few data here for conventional time series analysis, and instead, some structural theory may be necessary.

6.2 A Diversity-Weighted Portfolio

In this section we investigate the simulated behavior of a portfolio generated by $\mathbf{D}_p$ from Examples 3.4.4 and 4.3.5, with parameter $p = 0.5$. Recall that $\mathbf{D}_p$ was defined in Example 3.4.4 by

$$\mathbf{D}_p(x) = \left(\sum_{i=1}^n x_i^p \right)^{1/p}, \quad x \in \Delta^n, \quad (6.2.1)$$

where $0 < p < 1$. The function $\mathbf{D}_p$ is a measure of diversity, and the portfolio it generates is called the $\mathbf{D}_p$ -weighted portfolio with weights

$$\pi_i(t) = \frac{\mu_i^p(t)}{(\mathbf{D}_p(\mu(t)))^p}, \quad t \in [0, T], \quad (6.2.2)$$

$i = 1, \dots, n$. The $\mathbf{D}_p$ -weighted portfolio satisfies

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{D}_p(\mu(t)) + (1 - p) \gamma_\pi^*(t) dt, \quad (6.2.3)$$

a.s., for $t \in [0, T]$. For simplicity, this equation does not include leakage, but leakage for the $\mathbf{D}_p$ -weighted portfolio is discussed in detail in Example 4.3.5 and Appendix A, and is included in the simulation.

In the simulation we use the largest 1000 stocks, or largest 80% of all the stocks, whichever is smaller, from the CRSP universe. Hence, in the early part of the simulation, when there are fewer than 1250 stocks in the CRSP universe, the 80% criterion is used, and in the later part, the largest

1000 stocks are used. This ensures that the market is somewhat larger than the portion we consider, and hence we can obtain a reasonable estimate of the leakage that occurs. Since this is precisely the setting considered in Example 4.3.5, this application should provide a good example of leakage in a diversity-weighted portfolio.

The results of the simulation are presented in Figures 6.6 through 6.9 on pages 127 and 128, which show the cumulative values of the components of the relative logarithmic return. The scales in these four charts are the same to facilitate comparison. Figure 6.6 shows that the $\mathbf{D}_p$ -weighted portfolio generated about 29% higher logarithmic return than the capitalization-weighted portfolio over the 60-year period, and this resulted in a 0.48% average annual relative return.

Figure 6.7 shows the cumulative change in $\mathbf{D}_p$, centered to have zero sample mean over the period. The change in $\mathbf{D}_p$ was responsible for almost all of the volatility of the relative return, but over the 60 years total change was only about -16% , or -0.27% a year. This negative change in $\mathbf{D}_p$ is small compared to total variation over the period, and the time series appears to be mean-reverting, just as the market entropy appeared to be in Figure 6.2.

The drift process in Figure 6.8 is close to a pure trend process, and it contributed about 46% to the relative return of the $\mathbf{D}_p$ -weighted portfolio over the 60 years, or about 0.77% a year. The drift process had a smooth component contribution of 1.46% a year, and leakage contribution of -0.69% a year.

The cumulative contribution of the differential dividend rate in Figure 6.9 was almost nil over the 60-year period studied. Although the differential dividend rate was flat for most of the period, after 1975 it averaged about 0.25% a year in favor of the capitalization-weighted portfolio. But to neutralize the drift process, the differential dividend rate would have to have been more than three times greater even after 1975.

It is interesting to note that the leakage shown in Figure 6.9 is about the same as the drift in Figure 6.8. Since leakage is included in the drift process, this means that on average, leakage was about half as great as the smooth component of the drift. After about 1965, the slope of the leakage process becomes more negative, so leakage subtracts more than half the drift generated by the smooth component of Θ . Despite this, the slope of the drift process appears to be quite constant, even after 1965.

As usual, accepted financial theory would claim that the higher return of the $\mathbf{D}_p$ -weighted portfolio is due to higher risk. However, the same argument holds here as in Section 6.1: if the $\mathbf{D}_p$ process is mean reverting, as suggested by Figure 6.7 as well as market stability arguments, then when measured over time periods greater than or equal to the relaxation time of this process, the estimated variance rates of $\log \hat{Z}_\pi(t)$ and $\log \hat{Z}_\xi(t)$ will be about the same. As with the market entropy, this relaxation time appears to be on the order of 10 years. Measured with a 10-year frequency, the

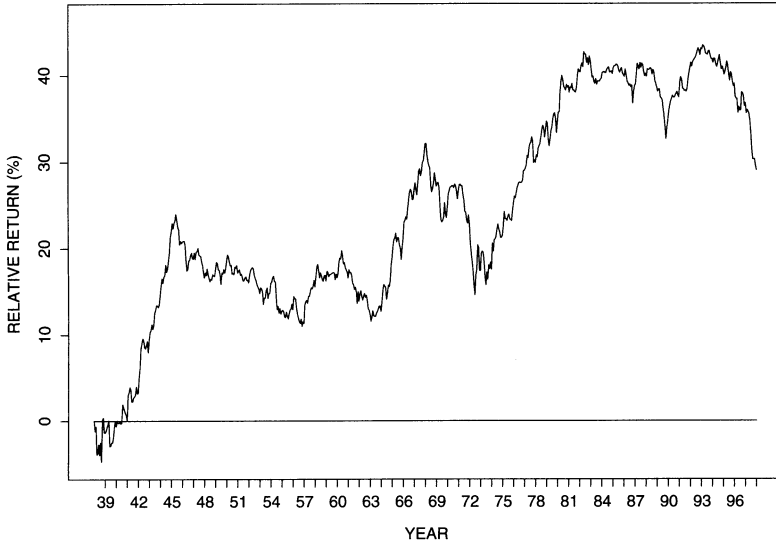


FIGURE 6.6. $\log(\hat{Z}_\pi(t)/\hat{Z}_\xi(t))$ for the $\mathbf{D}_p$ -weighted portfolio.

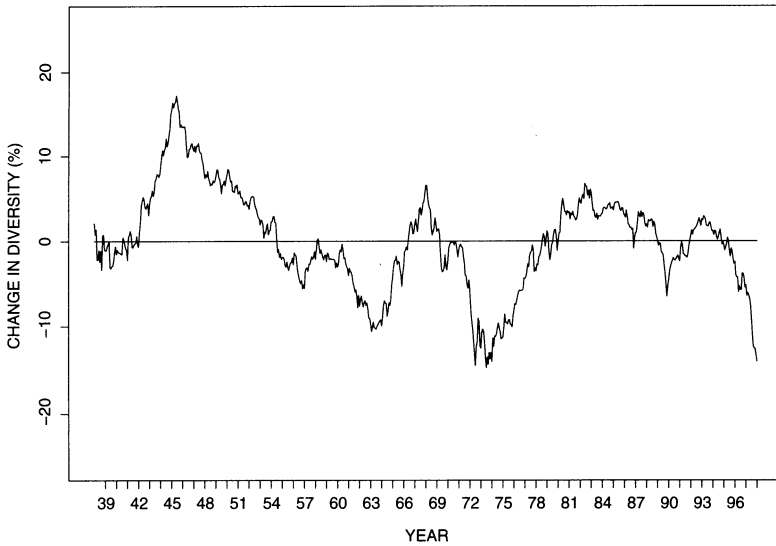


FIGURE 6.7. Change in $\mathbf{D}_p$ for the largest 1000 stocks, $p = 0.5$.
(Adjusted to have zero sample mean.)

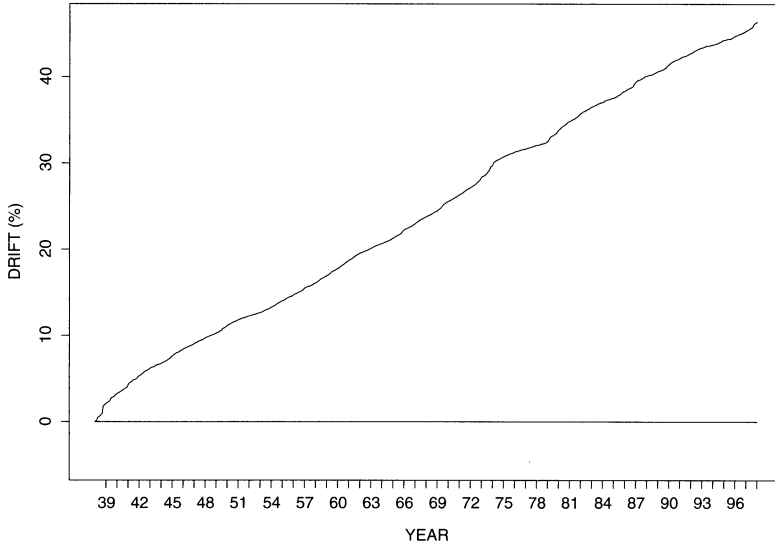


FIGURE 6.8. Drift process for the $\mathbf{D}_p$ -weighted portfolio.

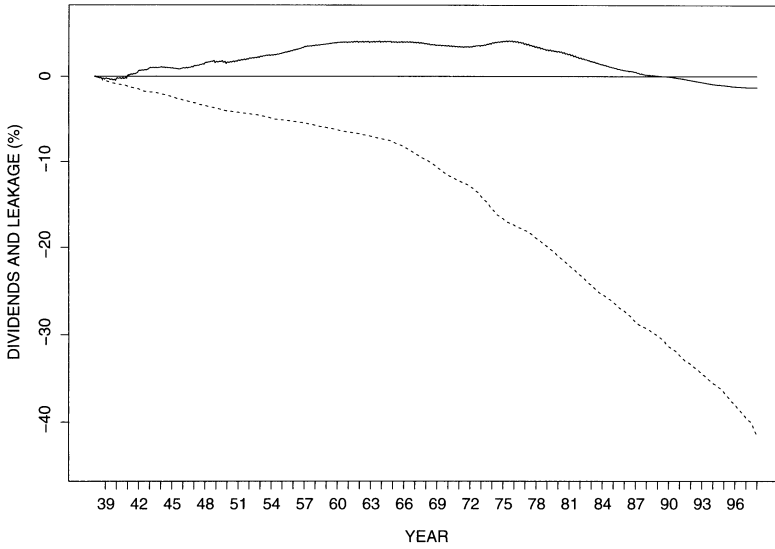


FIGURE 6.9. Dividends and leakage for the $\mathbf{D}_p$ -weighted portfolio.
Dividends (solid line) and leakage (broken line).

Sharpe ratio of the $\mathbf{D}_p$ -weighted portfolio was 1.29 versus 1.11 for the capitalization-weighted portfolio of the largest 1000 stocks. Hence, for long-term investors, the $\mathbf{D}_p$ -weighting would seem to be preferable to capitalization-weighting in this case.

After the appropriate change in scale, Figures 6.6 and 6.7 are strikingly similar to Figures 6.1 and 6.2, respectively. This suggests that different measures of diversity provide similar results, at least up to a scale factor, even when one measurement is for the whole market and the other is for the top 1000 stocks. The similarities in these charts notwithstanding, the drift process in Figure 6.8 is somewhat straighter than that in Figure 6.3, and the leakage process in Figure 6.9 is significantly more pronounced than that in Figure 6.4. This difference in the leakage processes is understandable: leakage from the top 1000 stocks is likely to be much greater than leakage from the whole CRSP universe.

6.2.1 Problem. (!!) Determine whether the behavior of the $\mathbf{D}_p$ -weighted portfolio discussed in this section is consistent with the Capital Asset Pricing Model (see Sharpe (1964) or Karatzas and Shreve (1998)).

6.3 Turnover in a Diversity-Weighted Portfolio

Since diversity-weighted portfolios have dynamically changing weights, an actual equity portfolio following such a strategy must be rebalanced from time to time in order to maintain the actual portfolio weights close enough to the theoretical weights. Hence there is a certain amount of portfolio turnover, and portfolio turnover causes trading costs. In any application to a portfolio of actual stocks, control of trading costs is essential, for trading costs directly affect portfolio returns. Here we estimate how much turnover a diversity-weighted portfolio generates. We use the $\mathbf{D}_p$ measure of diversity as the generating function in our estimate, and we shall assume in this section that the market has bounded variance.

The turnover of a diversity-weighted portfolio can be decomposed into three parts. First is the part of the turnover generated by the rebalancing trades; second is the turnover caused by leakage, manifested here by the continual replacement of the smallest stocks in the portfolio; and third is the turnover from dividend reinvestment. Here we estimate only the turnover generated by rebalancing such a portfolio. The turnover from leakage is a multiple of the leakage-induced turnover in the capitalization-weighted portfolio, and this multiple depends only on the weight ratios of the smallest stocks in the portfolio. Turnover from dividend reinvestment is equal to half the dividend rate (half because turnover pairs buy and sell trades), so this turnover is similar to that of the benchmark index.

Since we are interested only in turnover due to rebalancing, we can ignore leakage, and use the generating function $\mathbf{D}_p$ in (6.2.1) with weights given by

(6.2.2), rather than the generating function based on rank in Example 4.3.5. From (6.2.3) we see that the drift process satisfies

$$d\Theta(t) = (1 - p)\gamma_\pi^*(t) dt. \quad (6.3.1)$$

We consider the case in which the portfolio is traded when the actual portfolio weights differ from the desired weights in (6.2.2) by a fixed multiple δ of the desired weights. We use the notation $O(x)$ to represent a generic function that satisfies

$$\limsup_{x \rightarrow 0} \left| \frac{O(x)}{x} \right| < \infty,$$

and $o(x)$ to represent a generic function for which

$$\lim_{x \rightarrow 0} \frac{o(x)}{x} = 0.$$

Suppose that a period of time dt passes, along with a change in the i th desired weight to $\pi_i(t) + d\pi_i(t)$, for $i = 1, \dots, n$. Itô's rule implies that over this period

$$\begin{aligned} dZ_\pi^2(t) &= 2Z_\pi(t) dZ_\pi(t) + (dZ_\pi(t))^2 + o(dt) \\ &= 2Z_\pi(t) dZ_\pi(t) + d\langle Z_\pi \rangle_t + o(dt). \end{aligned}$$

Hence,

$$(dZ_\pi(t))^2 = d\langle Z_\pi \rangle_t + o(dt) = O(dt),$$

since the market is assumed to have finite variance. Similarly,

$$dX_i(t) dZ_\pi(t) = d\langle X_i, Z_\pi \rangle_t + o(dt) = O(dt).$$

Therefore, the i th portfolio weight becomes

$$\begin{aligned} \pi_i(t) \frac{X_i(t) + dX_i(t)}{X_i(t)} \frac{Z_\pi(t)}{Z_\pi(t) + dZ_\pi(t)} &= \pi_i(t) \left(1 + \frac{dX_i(t)}{X_i(t)} \right) \left(1 - \frac{dZ_\pi(t)}{Z_\pi(t)} + O(dt) \right) \\ &= \pi_i(t) \left(1 + \frac{dX_i(t)}{X_i(t)} - \frac{dZ_\pi(t)}{Z_\pi(t)} + O(dt) \right). \end{aligned}$$

When

$$\left| \pi_i(t) \left(1 + \frac{dX_i(t)}{X_i(t)} - \frac{dZ_\pi(t)}{Z_\pi(t)} + O(dt) \right) - \pi_i(t) - d\pi_i(t) \right| = \delta \pi_i(t), \quad (6.3.2)$$

a trade is made. We estimate the expected time between such trades, and use this to estimate the expected turnover.

Now, (6.3.2) is equivalent to

$$\left(\frac{dX_i(t)}{X_i(t)} - \frac{dZ_\pi(t)}{Z_\pi(t)} - \frac{d\pi_i(t)}{\pi_i(t)} + O(dt) \right)^2 = \delta^2.$$

We also have

$$\begin{aligned}
& \left(\frac{dX_i(t)}{X_i(t)} - \frac{dZ_\pi(t)}{Z_\pi(t)} - \frac{d\pi_i(t)}{\pi_i(t)} + O(dt) \right)^2 \\
&= (d \log X_i(t) - d \log Z_\pi(t) - d \log \pi_i(t) + O(dt))^2 \\
&= (d \log \mu_i(t) - d \log(Z_\pi(t)/Z_\mu(t)) - d \log \pi_i(t) + O(dt))^2 \\
&= d \langle \log \mu_i - \log(Z_\pi/Z_\mu) - \log \pi_i \rangle_t + o(dt),
\end{aligned}$$

by Itô's rule. Equation (6.2.2) implies that

$$d \log \pi_i(t) = p d \log \mu_i(t) - p d \log \mathbf{D}_p(\mu(t)),$$

so,

$$\begin{aligned}
d \log \mu_i(t) - d \log(Z_\pi(t)/Z_\mu(t)) - d \log \pi_i(t) \\
&= (1-p) d \log \mu_i(t) - d \log(Z_\pi(t)/Z_\mu(t)) + p d \log \mathbf{D}_p(\mu(t)) \\
&= (1-p) d \log \mu_i(t) - (1-p) d \log(Z_\pi(t)/Z_\mu(t)) + p d \Theta(t).
\end{aligned}$$

Hence,

$$\begin{aligned}
d \langle \log \mu_i - \log(Z_\pi/Z_\mu) - \log \pi_i \rangle_t &= d \langle (1-p) \log \mu_i - (1-p) \log(Z_\pi/Z_\mu) \rangle_t \\
&= (1-p)^2 (\tau_{ii}(t) - 2\tau_{i\pi}(t) + \tau_{\pi\pi}(t)) dt.
\end{aligned} \tag{6.3.3}$$

Suppose that the time scale is short enough so that the differences among the growth rates of the stocks will have a negligible contribution. Then we can make the approximation

$$\begin{aligned}
d \log \mu_i(t) - d \log(Z_\pi(t)/Z_\mu(t)) - d \log \pi_i(t) \\
\approx (1-p) (\tau_{ii}(t_0) - 2\tau_{i\pi}(t_0) + \tau_{\pi\pi}(t_0))^{1/2} dB(t),
\end{aligned} \tag{6.3.4}$$

where B is a Brownian motion, and $t_0 \geq 0$ is a starting time such that

$$\log \mu_i(t_0) - \log(Z_\pi(t_0)/Z_\mu(t_0)) - \log \pi_i(t_0) = 0.$$

If we let T_1 represent the stopping time for the first occurrence of

$$\log \mu_i(t) - \log(Z_\pi(t)/Z_\mu(t)) - \log \pi_i(t) = \pm \delta,$$

then it follows from (6.3.4) that

$$ET_1 \approx \frac{\delta^2}{(1-p)^2 (\tau_{ii}(t_0) - 2\tau_{i\pi}(t_0) + \tau_{\pi\pi}(t_0))} \tag{6.3.5}$$

(see Karatzas and Shreve (1991) for stopping times).

Each time that a trade occurs, the process in (6.3.4) is reset to the origin, so there is a sequence of stopping times $T_1 < T_2 < \dots$ corresponding to

the the sequence of trades. Let us assume that these stopping times are i.i.d. random variables, and let $k(t)$ denote the the number of trades up to time t . Then

$$\begin{aligned} E(T_1 + \cdots + T_{k(t)}) &= E(E(T_1 + \cdots + T_{k(t)} | k(t))) \\ &= E(k(t) ET_1) = Ek(t) ET_1. \end{aligned}$$

Since

$$T_1 + \cdots + T_{k(t)} \leq t < T_1 + \cdots + T_{k(t)+1},$$

we have

$$Ek(t) ET_1 \leq t < E(k(t) + 1) ET_1.$$

Hence,

$$\frac{t}{ET_1} - 1 < Ek(t) \leq \frac{t}{ET_1},$$

so for large enough t ,

$$Ek(t) \approx \frac{t}{ET_1} \approx \frac{t}{\delta^2} (1-p)^2 (\tau_{ii}(t_0) - 2\tau_{i\pi}(t_0) + \tau_{\pi\pi}(t_0)). \quad (6.3.6)$$

Suppose that the unit of time is one year, and all parameters are measured accordingly. Since the proportion of the portfolio value traded in each such trade is $\pi_i(t)\delta$, (6.3.6) implies that the expected annual trading generated by the i th stock is approximately

$$\frac{\pi_i(t)}{\delta} (1-p)^2 (\tau_{ii}(t) - 2\tau_{i\pi}(t) + \tau_{\pi\pi}(t)).$$

Hence, the expected annual trading of the whole portfolio is

$$\frac{1}{\delta} (1-p)^2 \sum_{i=1}^n \pi_i(t) (\tau_{ii}(t) - 2\tau_{i\pi}(t) + \tau_{\pi\pi}(t)) = \frac{2}{\delta} (1-p)^2 \gamma_{\pi}^*(t),$$

and (6.3.1) implies that this is $2(1-p)/\delta$ multiplied by the rate of change of the smooth component of the drift process.

In Section 6.2 we had $p = 0.5$, so the expected annual trading would have been equal to the rate of change of the smooth component of the drift process, divided by δ . The rate of change of the smooth component of the drift process was 1.46% a year, so

$$\text{expected annual trading} \approx 0.0146/\delta.$$

With $\delta = 0.1$, this implies that the expected annual trading due to re-balancing is about 14.6% a year, which is 7.3% a year expected annual turnover (since turnover pairs buy and sell trades). If trading were to be

done as a market maker at the bid–ask for each stock, this would mean that for a spread of 10 cents on a \$40 stock,

$$\delta \approx \frac{0.05}{40} = 0.00125.$$

If we assume this uniformly for all the stocks in the portfolio, it would result in expected turnover of about 584% a year due to rebalancing.

6.3.1 Problem. Calculate the turnover if all portfolio holdings are traded to their theoretical weights at times $t_0, 2t_0, 3t_0, \dots$, where $t_0 > 0$ is fixed.

6.3.2 Problem. (!) Generalize the analysis of this section to a wider class of generating functions, e.g., all measures of diversity.

6.3.3 Problem. (!!) Find a measure of diversity that maximizes the ratio of drift to turnover for the fixed trading time model of Problem 6.3.1.

6.4 The Size Effect

In Example 4.3.2 we considered a model to explain the *size effect*, the tendency for small firms to generate higher long-term returns than large firms. Here we shall test this model on actual data.

As in Section 6.2, we consider the subset of the CRSP universe composed of the largest 1000 stocks, or 80% of the stocks in the market, whichever is smaller. Of this subset, we define a large-stock index consisting of the largest 100 stocks, and a small-stock portfolio consisting of the remainder of the stocks. We decompose the relative return of the small-stock index versus the large-stock index into three components: the change in relative capitalization of small stocks versus the large stocks, the drift process, and the differential dividend rate. We show that over the period studied, the drift process was the dominant component.

The results of the simulation are presented in Figures 6.10 through 6.13 on pages 134 and 135, and these figures show the cumulative values of the components of the relative return. Figure 6.10 shows that the small-stock index generated about 64% higher logarithmic return than the large-stock index over the 60-year period, which averages to about 1.06% a year relative return. Over the same 60-year period, a small-stock index composed of the ninth and tenth size deciles of the U.S. market averaged about 2.5% a year higher logarithmic return than S&P 500 Index. The stocks in this small-stock index were considerably smaller than the stocks we have considered here, so the higher return on those stocks would be consistent with the model proposed in Example 4.3.2.

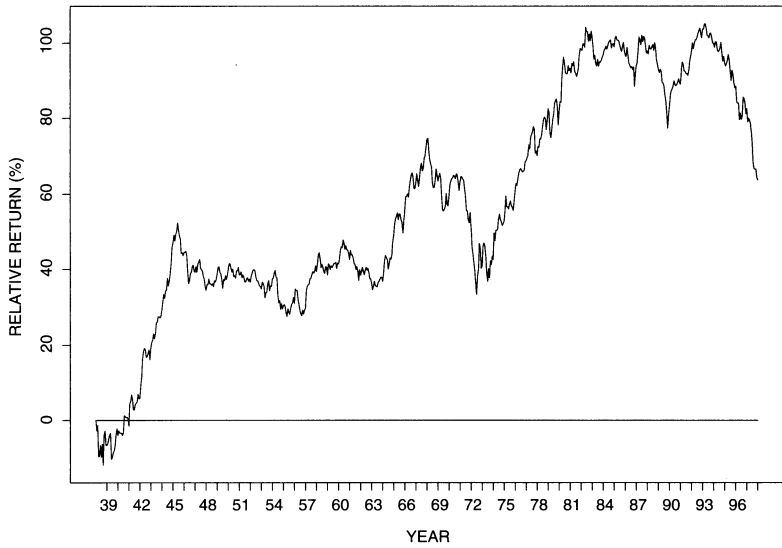


FIGURE 6.10. $\log(\hat{Z}_\eta(t)/\hat{Z}_\xi(t))$ for the small/large portfolios.



FIGURE 6.11. Change in relative capitalization of the small/large portfolios.
(Adjusted to have zero sample mean.)

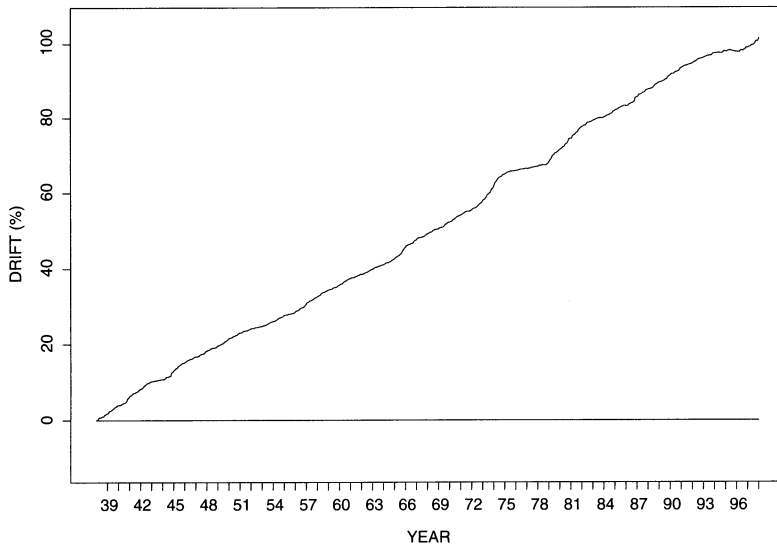


FIGURE 6.12. Drift process for the small/large portfolios.

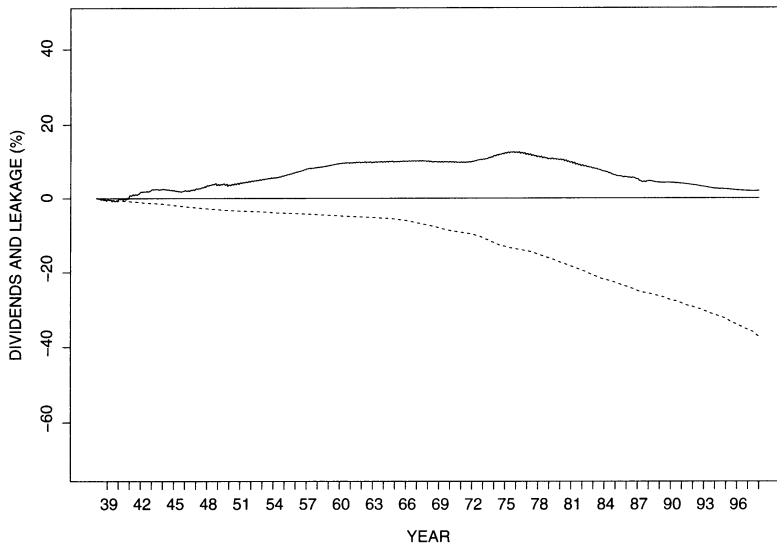
FIGURE 6.13. Dividends and leakage for the small/large portfolios.
Dividends (solid line) and leakage (broken line).

Figure 6.11 shows the cumulative change in relative capitalization, shifted on the vertical axis to have zero sample mean over the period. The change in relative capitalization was responsible for almost all of the volatility of the relative return, but it appears to be mean-reverting with a total change of about -40% over the 60-year period, or -0.66% a year. Note that the periods of concentration of capital in Figure 6.11 occur at the same time as those in Figures 6.2 and 6.7.

The drift process in Figure 6.12 is close to a pure trend process, and it contributed about 101% , or 1.69% a year, to the relative return of the small-stock index over the period studied. In Figure 6.13 we see that the differential dividend rate was essentially negligible over the 60 years studied, and had little effect on the relative return. From Figures 6.12 and 6.13 we see that although leakage increased after about 1966, the slope of the drift process, which includes leakage, remained essentially unchanged.

If corporations repurchase their own shares, this can have an effect similar to the payment of dividends. However, companies that repurchase their shares often do so in order to issue call options to employees. Since the issuance of call options has an effect opposite to share repurchase, the net effect on the differential dividend rate in our simulation is not clear. In any case, the change in relative capitalization in Figure 6.11 is adjusted for corporate actions that change the number of shares of stock outstanding, and with this adjustment, the repurchase or issuance of shares need not be considered in the differential dividend rate.

It appears that the results of this simulation are consistent with the explanation of the size effect proposed in Example 4.3.2. In this case, the smaller stocks will have superior performance whether or not they have higher volatility. The size effect is due to the structure of the large-stock and small-stock indices as portfolios generated by functions of the ranked market weights, not to differences in risk. Hence, the superior performance of smaller stocks is a natural characteristic of a stable equity market in which the larger stocks do not pay adequate dividends.

6.5 The Biggest Stock

In Example 4.3.1 the relative return of the biggest stock with respect to the market was decomposed into its change in market weight plus a decreasing drift process. It was remarked that in order to offset the decreasing drift process, the biggest stock would have to pay higher dividends than the market average. Here we determine how much higher the dividends should be, and compare this to the actual dividend stream. As in the previous sections of this chapter, here we again consider the period from 1939 to 1998 for the CRSP universe of stocks.

The relative return of the biggest stock in the market is decomposed in

(4.3.1) as

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mu_{(1)}(t) - \frac{1}{2} d \Lambda_{\log \mu_{(1)} - \log \mu_{(2)}}(t), \quad (6.5.1)$$

for $t \in [0, T]$, where Z_π is the value of the portfolio consisting of the biggest stock. The differential dividend rate, which is not included in this equation, will be compared to the drift process, the last term in the equation. The results of the simulation are presented in Figures 6.14 through 6.17 on pages 138 and 139, and include the three terms in (6.5.1), as well as the differential dividend rate.

Figure 6.14 shows that the return of the biggest stock was about 11% lower than that of the market over the 60-year period. While this difference is not great, it is negative, and hence would be a disincentive to investment in the biggest stock. The change in market weight in Figure 6.15 contributed about -4% to the relative return over the period. This change in market weight is corrected for additions and deletions of stocks in the market, so the decline evident in Figure 6.15 cannot be attributed to the fact that the number of stocks in the market has increased. Figure 6.16 shows that the drift process contributed about -8% over the period, and Figure 6.17 indicates that the dividend process was able to offset this negative drift by only about 0.5% . The contribution of leakage was positive in this case, but was essentially negligible compared to the other components.

Over the period considered, the dividend rate of the biggest stock was only slightly greater than the market rate, and not enough to offset the negative drift. To offset the negative drift, the biggest stock would have had to pay about 0.13% a year more dividends than the market average. Although this does not seem like a very significant difference, Figure 6.17 indicates that the dividend rate of the biggest stock reached this modest premium at best rarely.

It may be of interest to consider for a moment the case in which all the stocks have the same growth rate. From Example 5.3.4 we see that if the growth rates of all the stocks are the same, then

$$\lim_{t \rightarrow \infty} \frac{1}{t} \Lambda_{\log X_{(1)} - \log X_{(2)}}(t) = 0, \quad \text{a.s.}$$

Therefore, if the growth rates of all the stocks are the same, the biggest stock is not noticeably penalized over the long term by the negative drift process in (6.5.1). Hence,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log(Z_\pi(t)/Z_\mu(t)) = 0, \quad \text{a.s.},$$

so the long-term return of the biggest stock equals that of the market. With equal growth rates, the crossovers between the first- and second-ranked stocks become so rare that the contribution of the negative drift in (6.5.1) becomes negligible. The crossovers become rare because the biggest stock dominates the market, as we have seen in Chapter 2.



FIGURE 6.14. Relative logarithmic return for the biggest stock.

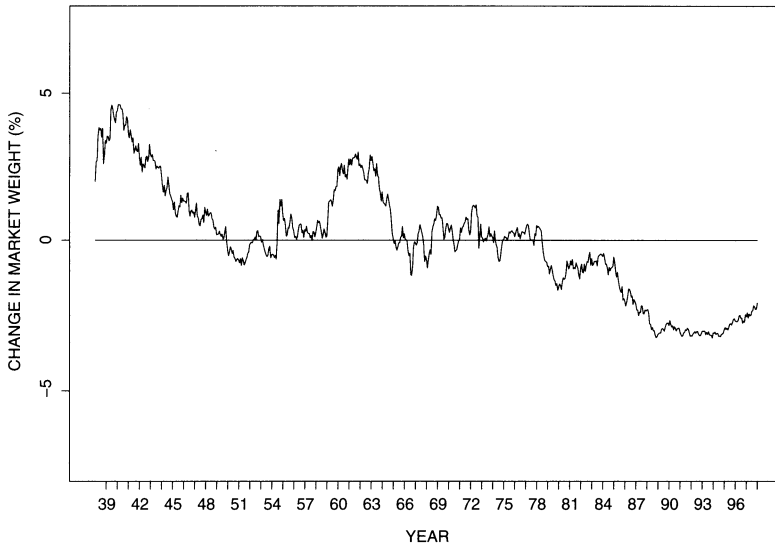


FIGURE 6.15. Change in the market weight of the biggest stock.
(Adjusted to have zero sample mean.)

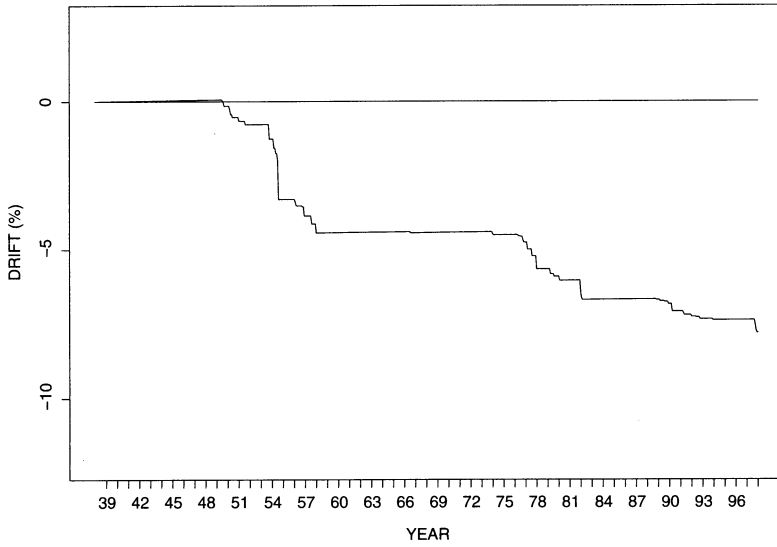


FIGURE 6.16. Drift process for the biggest stock.

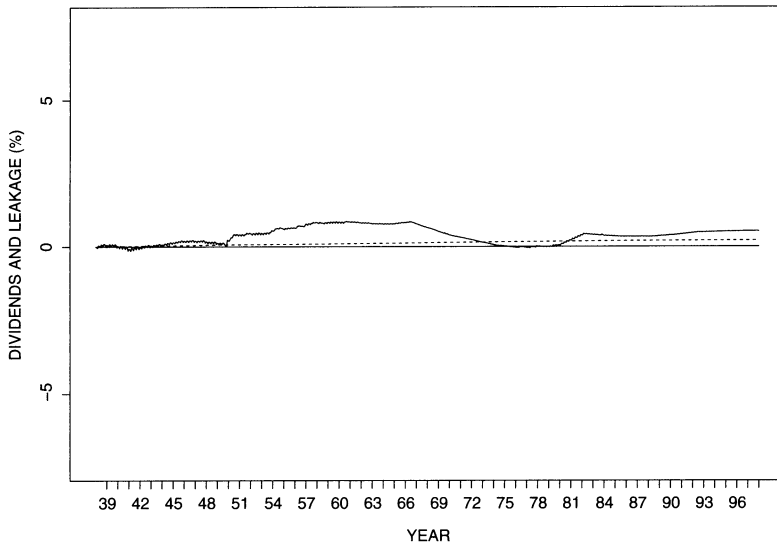


FIGURE 6.17. Dividends and leakage for the biggest stock.
Dividends (solid line) and leakage (broken line).

6.5.1 Problem. (!) It would seem that for some k , $1 \leq k \leq n$, the k th-ranked stock, $X_{(k)}$, would have the same total return as the market. Find this k , and interpret the result.

6.6 Notes and Summary

Markowitz (1952), Sharpe (1964), and Merton (1969) developed the concept of “efficient” equity markets. The original CAPM of Sharpe (1964), a more modern version of which can be found in Karatzas and Shreve (1998), implies that if the market is efficient, then a portfolio that has higher expected return than the market portfolio must have a “beta” greater than 1. Moreover, a portfolio’s Sharpe ratio, discussed in Sharpe (1975), should not exceed 1 in an efficient market. The portfolios studied in Sections 6.1 and 6.2 suggest that all of these conditions fail for the U.S. equity market.

The size effect studied in Section 6.4 was first identified by Banz (1981) and Reinganum (1981) as a tendency for smaller stocks to have higher return than larger stocks. There has been a difference of opinion as to whether or not this higher return can be explained by the higher risk that is usually attributed to small stocks (see Roll (1981), Handa et al. (1989), and Jegadeesh (1992)). Our results indicate that the higher return of the small stocks can be explained by portfolio generating functions, and is independent of risk.

In Section 6.4, the data on the small-stock index came from Ibbotson Associates (1997). The small-stock data consist of the monthly return of Dimensional Fund Advisors (DFA) Small Company Fund, a capitalization-weighted index of the ninth and tenth (size) deciles of the NYSE, plus stocks listed on the AMEX and NASDAQ with capitalization less than or equal to the upper bound of the NYSE ninth decile.

Regarding the effect of stock repurchasing mentioned in Section 6.4, Cole, Helwege, and Laster (1996) studied this practice over the period from 1975 to 1995, and found that this could increase the effective dividend rate of the S&P 500 Index by as much as 80 basis points a year in some of the later years. In another study, Shulman, Brown, and Narayanan (1997) showed that if the issuance of options and new shares is also considered, the effective dividend rates were actually reduced.

Chapter Summary

Table 6.1 summarizes the results of the simulations in this chapter. All of the values in the table are logarithmic and are in the form of annual averages. The relative return in the first column is the sum of the components in the other three columns. In each simulation, the contribution of the drift process is substantially greater than the contribution of either the generating function or the dividends.

	Relative return	Generating function	Drift process	Differential dividend rate
Entropy	0.15%	−0.14%	0.34%	−0.05%
$\mathbf{D}_p$	0.48%	−0.27%	0.77%	−0.02%
Small/Large	1.06%	−0.66%	1.69%	0.03%
Biggest Stock	−0.19%	−0.06%	−0.13%	0.01%

TABLE 6.1. Results of simulations, 1939–1998.
Annual logarithmic means.

Section 6.1: Recall that the entropy function generates the entropy-weighted portfolio with weights

$$\pi_i(t) = \frac{-\mu_i(t) \log \mu_i(t)}{\mathbf{S}(\mu(t))}, \quad t \in [0, T],$$

$i = 1, \dots, n$, and the entropy-weighted portfolio satisfies

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{S}(\mu(t)) + \frac{\gamma_\mu^*(t)}{\mathbf{S}(\mu(t))} dt, \quad t \in [0, T], \quad \text{a.s.}$$

The results of the simulated performance of the entropy-weighted portfolio relative to the U.S. equity market are presented in Figures 6.1, 6.2, 6.3, and 6.4, as well as in Table 6.1. From the results of the simulation, it appears that the market entropy is stable and mean-reverting over the long term, and therefore, with an increasing drift process, the entropy-weighted portfolio outperforms the market.

Section 6.2: The measure of diversity $\mathbf{D}_p$ generates the $\mathbf{D}_p$ -weighted portfolio with weights

$$\pi_i(t) = \frac{\mu_i^p(t)}{(\mathbf{D}_p(\mu(t)))^p}, \quad t \in [0, T], \quad (6.2.2)$$

$i = 1, \dots, n$, and the portfolio satisfies

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{D}_p(\mu(t)) + (1 - p) \gamma_\pi^*(t) dt, \quad (6.2.3)$$

a.s., for $t \in [0, T]$. (This equation does not include leakage.) We simulated a $\mathbf{D}_p$ -weighted portfolio, with $p = 0.5$, of the 1000 largest stocks in the U.S. market, and calculated the relative return of this portfolio versus the capitalization-weighted portfolio of the same stocks. The results of the

simulation are presented in Figures 6.6, 6.7, 6.8, and 6.9, with the average values given in Table 6.1 above. Note that these four charts are strikingly similar to the corresponding charts for the entropy-weighted portfolio in the previous section.

Section 6.3: Portfolios incur trading expenses, so it is important in practice to know the amount of turnover that is necessary to maintain the dynamically changing weights of a functionally generated portfolio. For a $\mathbf{D}_p$ -weighted portfolio, if the portfolio is traded whenever a portfolio weight differs from its theoretical value by a factor of $1 \pm \delta$ or more, where $\delta > 0$ is a (small) constant, then the estimated portfolio turnover is

$$(1 - p)^2 \gamma_{\pi}^*(t) / \delta.$$

For the $\mathbf{D}_p$ -weighted portfolio considered in the previous section, with $p = 0.5$, this turnover would be about 7.3% a year if $\delta = 0.1$.

Section 6.4: In Example 4.3.2 we presented a model to explain the size effect, the observed tendency for smaller stocks to outperform larger ones, in terms of portfolios generated by functions of the ranked market weights. To test the size effect, in this section we simulated a large-stock portfolio holding the largest 100 stocks in the U.S. market, and a small-stock portfolio holding the next 900 by rank. The decomposition of the relative return followed Example 4.3.2, and is presented in Figures 6.10, 6.11, 6.12, and 6.13, with the average values given in Table 6.1. The charts for this simulation are quite similar to those of Sections 6.1 and 6.2, and indicate that the size effect is a structural phenomenon, similar in nature to the behavior of diversity-weighted portfolios.

Section 6.5: The biggest stock is an extreme case of the size effect, and in this case, the size effect will act against the biggest stock, so we should expect negative return relative to the market. This is indeed verified by the simulation, as can be seen from Figures 6.14, 6.15, 6.16, and 6.17, and from the average values given in Table 6.1.

7

Applications of Stochastic Portfolio Theory

In the previous chapters we have seen a number of theoretical applications of stochastic portfolio theory; in this chapter we shall consider some practical applications. As a first application, we show how the first-order model can be used in portfolio optimization. Next, we discuss a passive strategy based on a $\mathbf{D}_p$ -weighted version of the S&P 500 Index that has been used for institutional accounts since 1996. Manager performance is related to the change in market diversity, and we analyze this relationship and consider its implications. We propose a direct method to measure the effect that changes in the distribution of capital have on portfolio return, and use this method to analyze the poor performance of value stocks during the 1990s. Our analysis indicates that the principal cause of this disappointing performance was a shift in the capital distribution that favored the larger stocks over the period considered.

7.1 Optimization and the First-Order Model

A critical problem in portfolio optimization is the estimation and prediction of the growth rate parameters that are needed to carry out the optimization. While the variance parameters can frequently be estimated adequately from past data, it is well known that the past return on a stock is not a good predictor of its future return. In Examples 1.1.7, 1.3.5, and 1.3.8 this problem was addressed, and in Example 1.3.8 the growth rate parameters were all assumed to be equal, for lack of any better estimator.

The first-order model we introduced in Section 5.5 provides estimates of

the growth rates of the stocks based on rank. These estimates should be more accurate than estimates based on the assumption that all the growth rates are equal, for the first-order estimates are compatible with the structure of the capital distribution, while equal growth rates are not. Hence, we would expect that portfolio optimization based on the first-order estimates should provide better investment results than portfolio optimization based on the assumption of equal growth rates.

7.1.1 Example. (Portfolio optimization IV) Suppose we consider a hybrid model that uses the $\mathbf{g}_k$ of the first-order model for growth rates, but retains the usual covariances $\sigma_{ij}(t)$ and relative covariances $\tau_{ij}(t)$ among the stocks. This gives us a combination of the first-order model defined in (5.5.2) with the standard model in (1.1.6), so we have

$$d \log X_i(t) = \mathbf{g}_{q_t(i)} dt + \sum_{\nu=1}^n \xi_{i\nu}(t) dW_\nu(t), \quad t \in [0, T], \quad (7.1.1)$$

for $i = 1, \dots, n$, where $q_t(i)$ represents the rank of the i th stock at time t .

With the growth rate estimates in (7.1.1), we can replace the growth rates $\gamma_i(t)$ in (1.3.13) of Example 1.3.5 with the estimates $\mathbf{g}_{q_t(i)}$. Here we shall minimize the variance relative to the market, but the generalization to an arbitrary benchmark index is straightforward. In this case, the optimization corresponding to Example 1.3.5 will minimize

$$\sum_{i,j=1}^n \pi_i(t) \pi_j(t) \tau_{ij}(t),$$

under the constraints

$$\sum_{i=1}^n \pi_i(t) \mathbf{g}_{q_t(i)} + \frac{1}{2} \sum_{i=1}^n \pi_i(t) \tau_{ii}(t) \geq \gamma_0, \quad (7.1.2)$$

where $\gamma_0 > 0$ is constant, and

$$\pi_1(t) + \dots + \pi_n(t) = 1 \quad \text{with} \quad \pi_1(t), \dots, \pi_n(t) \geq 0.$$

The constraint in (7.1.2) is linear, so conventional quadratic programming can be used. $\square$

In the portfolio optimization problems we have seen, the optimal weights depend on the growth rates and covariances of the stocks. Since the growth rate and variance parameters of the first-order model, $\mathbf{g}_1, \dots, \mathbf{g}_n$ and $\sigma_1, \dots, \sigma_n$, are constant, optimal weights that depend on these parameters will also be constant. Hence, these optimizations will result in portfolios with constant rank weights, so for a portfolio π , we can denote the rank weights by $\tilde{\pi}_j(t) = \tilde{\pi}_j$, $j = 1, \dots, n$, for all $t > 0$. With these constant rank

weights, the portfolio growth rate in (5.5.10) is the constant

$$\gamma_\pi = \sum_{j=1}^n \tilde{\pi}_j \mathbf{g}_j + \frac{1}{2} \sum_{j=1}^n (\tilde{\pi}_j - \tilde{\pi}_j^2) \sigma_j^2, \quad t \in [0, T],$$

and the portfolio variance in (5.5.11) is the constant

$$\sigma_{\pi\pi} = \sum_{j=1}^n \tilde{\pi}_j^2 \sigma_j^2, \quad t \in [0, T].$$

There are a number specific portfolio optimization problems related to reaching a particular performance goal that can be solved in the setting of the first-order model.

7.1.2 Example. (Maximize the probability of reaching a goal) Suppose that we start a portfolio π with initial value $Z_\pi(0) = 1$ and try to maximize the probability of reaching a goal $b > 1$ before descending to a level $a < 1$. This can be accomplished by maximizing the “growth-to-variance” ratio

$$\frac{\gamma_\pi}{\sigma_{\pi\pi}} = \frac{\sum_{j=1}^n \tilde{\pi}_j (\mathbf{g}_j + \sigma_j^2/2)}{\sum_{j=1}^n \tilde{\pi}_j^2 \sigma_j^2} - \frac{1}{2}, \quad (7.1.3)$$

over all $(\tilde{\pi}_1, \dots, \tilde{\pi}_n) \in \Delta^n$. □

7.1.3 Example. (Minimize the expected time to reach a goal) Suppose that we start a portfolio π with initial value $Z_\pi(0) = 1$ and try to minimize the expected time required to reach a goal $b > 1$. This can be accomplished by maximizing the portfolio growth rate

$$\gamma_\pi = \sum_{j=1}^n \tilde{\pi}_j \mathbf{g}_j + \frac{1}{2} \sum_{j=1}^n (\tilde{\pi}_j - \tilde{\pi}_j^2) \sigma_j^2,$$

over all $(\tilde{\pi}_1, \dots, \tilde{\pi}_n) \in \Delta^n$. □

7.1.4 Example. (Maximize the probability of reaching a goal before a deadline) Suppose that we start a portfolio π with initial value $Z_\pi(0) = 1$ and try to maximize the probability of reaching a goal $b > 1$ during the time interval $[0, T]$. In this case, if there is a portfolio for which the maximum in (7.1.3) can be attained simultaneously with a maximum of the variance

$$\sigma_{\pi\pi} = \sum_{j=1}^n \tilde{\pi}_j^2 \sigma_j^2,$$

over all $(\tilde{\pi}_1, \dots, \tilde{\pi}_n) \in \Delta^n$, then this portfolio solves the problem. This simultaneous maximization is somewhat problematical, and without the simultaneous maximization, the solution is apparently unknown. □

7.2 Diversity-Weighted Indexing

*Diversity-weighted indexing*¹ is a passive index strategy based on portfolios generated by the $\mathbf{D}_p$ generating function defined in Example 3.4.4. The strategy was introduced as an institutional investment product in 1996 in the form of a diversity-weighted version of the S&P 500 Index, but it could be applied to any large-stock index. Diversity-weighted indexing can legitimately be considered passive, since it involves neither optimization nor estimation, and the portfolio weights follow a simple, publicly disclosed mathematical algorithm.

The choice of the generating function $\mathbf{D}_p$ allows a certain level of flexibility, since the parameter p can be chosen anywhere between 0 and 1. The value $p = .76$ was selected, since with this value of p , the diversity-weighted version of the S&P 500 Index retains characteristics common to other well-known large-stock indices such as the (capitalization-weighted) S&P 500 Index or Russell 1000. These characteristics are, first, that the index holds a representative selection of large companies; second, that the selection and weighting of the securities in the index are objectively established; and third, that the portfolio turnover is minimal. Clearly, the diversity-weighted version of the S&P 500 Index shares the first two characteristics. Regarding the third characteristic, the index is rebalanced to exact diversity weights at the end of each month, and we shall see that with this convention, it has low turnover. Besides conforming to these three characteristics, it was shown in Fernholz et al. (1998) that the covariance characteristics of the diversity-weighted version of the S&P 500 are similar to the covariance characteristics of the S&P 500 and the Russell 1000.

The capital distributions for both the capitalization-weighted S&P 500 and the diversity-weighted S&P 500 on December 29, 1999, can be seen in Figure 7.1. The chart shows that the diversity weights are lower than the capitalization weights for the largest stocks, and higher for the smallest. If we compare Figure 7.1 with Figure 4.1 on page 70, we see that the diversity weights for the S&P 500 at the end of 1999 were remarkably similar to the capitalization weights for the S&P 500 at the end of 1997. Hence, diversity weights are not a radical departure from common experience, but rather provide a measure of moderation in periods when there is an extreme concentration of capital into the largest stocks. To move from capitalization weighting to diversity weighting in Figure 7.1 would require about 14% turnover of the portfolio. Since the diversity weights are not far from the corresponding capitalization weights, the diversity-weighted S&P 500 Index retains the general performance characteristics of a large-stock index.

¹U.S. patent 5,819,238 was granted to the author in 1998 for the process used in managing a functionally generated portfolio.

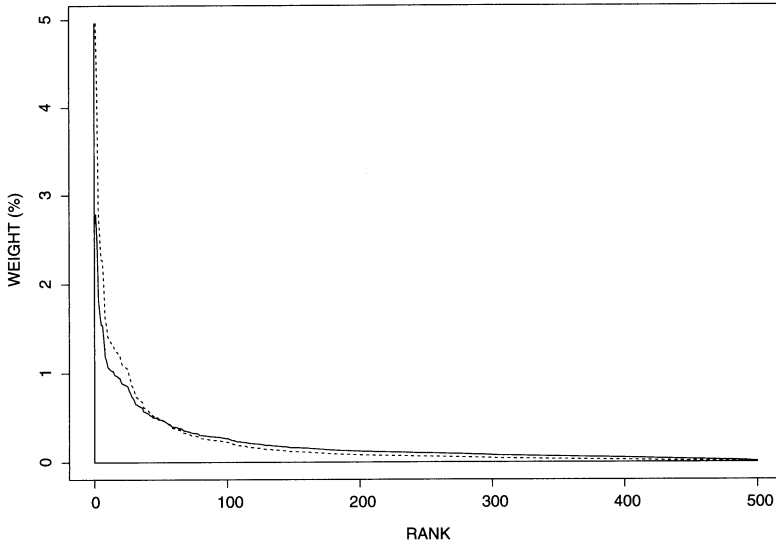


FIGURE 7.1. Diversity weights and capitalization weights.
Diversity (solid line), capitalization (broken line).

We present here a simulation of this portfolio over the period from 1966 to 1998, using the monthly data from the CRSP universe. The starting date of the simulation is January 1966, since data on the composition of the S&P 500 Index are available only from that date on.

The results of the simulation are presented in Figures 7.2 through 7.5 on pages 148 and 149, which show the cumulative values of the relative return, the change in $\mathbf{D}_p$, the differential dividend rate, and the drift process over the 33-year period from 1966 to 1998. Note that these results represent an index and not an actual portfolio, so there are no transaction costs. The charts are quite similar to Figures 6.6 through 6.9 for the $\mathbf{D}_p$ -weighted portfolio in Section 6.2, and again they show that almost all the short-term volatility in the relative return is caused by changes in $\mathbf{D}_p$, while the long-term outperformance is generated by the drift process. The value of the parameter p was 0.5 for the simulation exhibited in Figures 6.6 through 6.9, and the smaller value of this parameter causes the scale in these figures to be greater than the scale in Figures 7.2 through 7.5, where $p = 0.76$.

Over the whole period, the logarithmic return of the diversity-weighted S&P 500 Index averaged about 0.26% a year higher than that of the capitalization-weighted S&P 500. The last few years in Figure 6.2 witnessed the greatest concentration of capital in the whole 60-year period, and Figure 7.3 indicates that a similar concentration occurred within the S&P 500. Hence, the relative performance of the 33 years we are considering is unlikely to be representative of the long-term average. If diversity does not

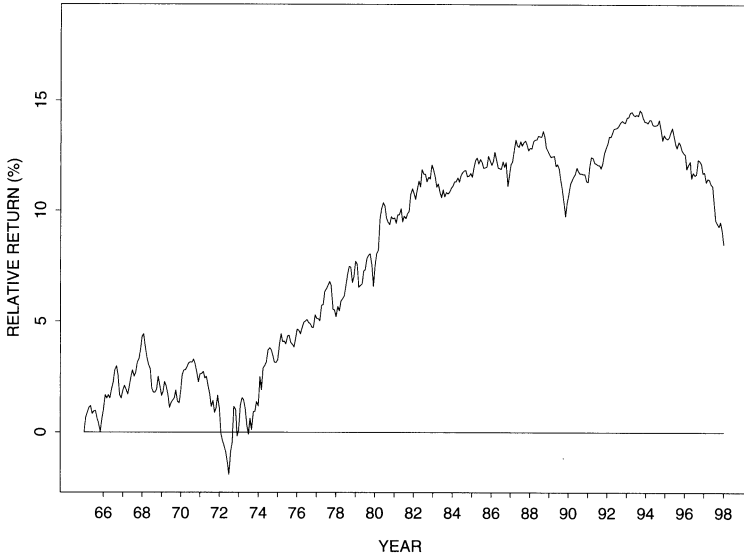


FIGURE 7.2. Relative return of the diversity-weighted S&P 500 Index.

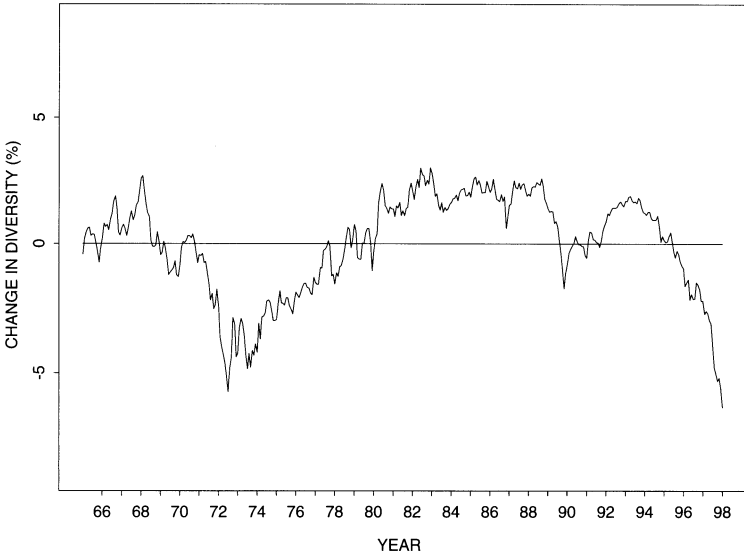


FIGURE 7.3. Change in D_p for the S&P 500 Index.
(Adjusted to have zero sample mean.)

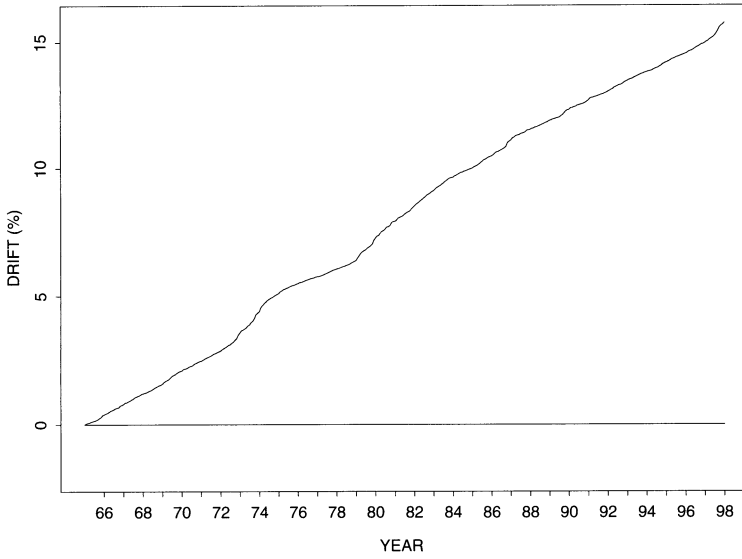
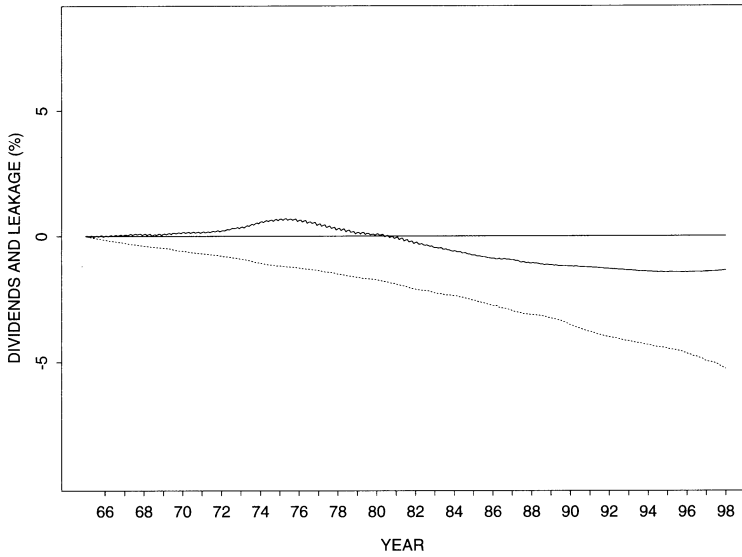


FIGURE 7.4. Drift process for the diversity-weighted index.

FIGURE 7.5. Dividends and leakage for the diversity-weighted index.
Dividends (solid line) and leakage (broken line).

change over a given period, the diversity-weighted index can probably be expected to generate about 0.47% a year higher logarithmic return than the capitalization-weighted index. It is also important to note that the corresponding arithmetic value would be increased by the average appreciation of the index over the period. For example, average index appreciation of 10% a year would increase this 0.47% to a mean annual arithmetic relative return of about 0.52%.

The change in D_p in Figure 7.4 represents the cumulative logarithmic change in the diversity of the S&P 500 Index. It has behaved as a mean-reverting process with an annual standard deviation of about 1.34% and annual mean of about -0.19% , statistically insignificant compared to the standard deviation. The stability of D_p implies that the diversity-weighted S&P 500 Index has about the same long-term risk as the S&P 500 Index.

Figure 7.5 shows the differential dividend rate between the two indices. (The sawtooth effect in the chart is caused by the seasonal nature of dividend payments.) It is clear from Figure 7.5 that dividend payments by the larger stocks do not nearly offset the drift process. At the beginning of the period studied, the diversity-weighted S&P 500 Index actually had a higher dividend rate than the capitalization-weighted S&P 500. This reversed around 1975, and the differential dividend rate has been against the diversity-weighted S&P 500 Index since then, although over the last few years it has been flat. The average annual contribution of the differential dividend rate over the whole period was about -0.04% .

The superior performance of the diversity-weighted index over the 33-year period studied gave it a Sharpe ratio about 4% higher than that of the S&P 500 Index, even though there was a significant contraction in diversity. Theoretically, we know that a diversity-weighted index has a significant performance advantage over a capitalization-weighted index under conditions of stable or increasing diversity, unless the larger companies pay enough dividends to offset the drift process. Over the period studied, the dividends of the larger companies have not been much higher than those of smaller companies, and not nearly high enough to eliminate this differential.

The annual standard deviation of the relative return of the diversity-weighted S&P 500 versus the capitalization-weighted S&P 500 was about 1.33%. For the sake of comparison, the annual standard deviation of the relative return of the Russell 1000 Index, another well-known large-stock index, versus the S&P 500 Index is about 1.55%. The total volatility of the diversity-weighted S&P 500 was about the same as the total volatility of the capitalization-weighted S&P 500, about 15.3% over the period for each. Over some periods one version of the index had a slightly higher volatility, and over other periods the other one did, but when averaged over the whole simulation, the volatility was about the same for the two strategies.

The annual portfolio turnover for the diversity-weighted S&P 500 averages about 12%. In comparison, the capitalization-weighted S&P 500 Index

averages about 6% annual turnover, and the Russell 1000 Index about 8%. For all three indices, most of the turnover is generated by stocks entering or leaving the index. None of these turnover numbers represents a practical impediment to fund management.

Diversity weighting also offers an opportunity for asset allocation between large-stock indices. In times of low diversity, the capitalization-weighted S&P 500 can become quite concentrated into the largest stocks. Diversity weighting has lower weights for the largest stocks, so it offers investors some protection against this concentration of capital. If the change in $\mathbf{D}_p$ of the index continues to be mean-reverting, allocating more funds to diversity weighting when diversity is low could significantly improve returns, as well as more adequately diversifying risk. Hence, on a practical level, diversity-weighted indexing should prove to be an attractive passive strategy.

7.3 Manager Performance and Change in Diversity

The S&P 500 Index is frequently used as a benchmark for active equity managers' performance. During some periods, managers do well against this benchmark, and over other periods they do not do well. Along with the managers' performance, the popularity of active management versus passive management ebbs and flows. However, after a period of poor performance by active managers, it might be unwise to move to passive strategies.

In Figure 7.6, annual manager performance relative to this index is plotted versus the annual changes in diversity over the period from 1971 to 1998, with each of the data points represented by a number denoting the corresponding year. The horizontal axis is the annual change in $\mathbf{D}_p$, $p = 0.5$, for the largest 1000 stocks in the CRSP universe, as in Figure 6.7 on page 127. The vertical axis is the difference between the annual logarithmic return of the median equity manager and the annual logarithmic return of the S&P 500, calculated using data from the Domestic Equity Database of Callan Associates. The diagonal line is the least-squares regression line for the data. Analysis of the regression indicates that slightly more than half of the annual variation in relative manager performance can be explained by the change in diversity ($R^2 = 0.55$). This means that there can be no other variable independent of the change in diversity that explains as much of the annual variation in relative manager performance.

The fact that managers' relative performance is positively correlated with change in diversity is an indication that managers' portfolios on average are likely to be less concentrated in the larger stocks than the S&P 500 benchmark. This suggests that a diversity-weighted S&P 500 Index would probably be a more appropriate benchmark for active managers, because its stock holdings more closely reflect those of the managers. In fact, if

we use ordinary least-squares regression, we find that when the annual logarithmic returns of the S&P 500 Index are used as the explanatory variable for the managers' annual logarithmic return, the residual R^2 is 5.75%. If the annual logarithmic return of the diversity-weighted S&P 500 is used as the explanatory variable in the regression, the residual R^2 is 3.68%. Hence, the evidence indicates that the diversity-weighted version of the S&P 500 would be a more appropriate manager benchmark than the standard, capitalization-weighted version.

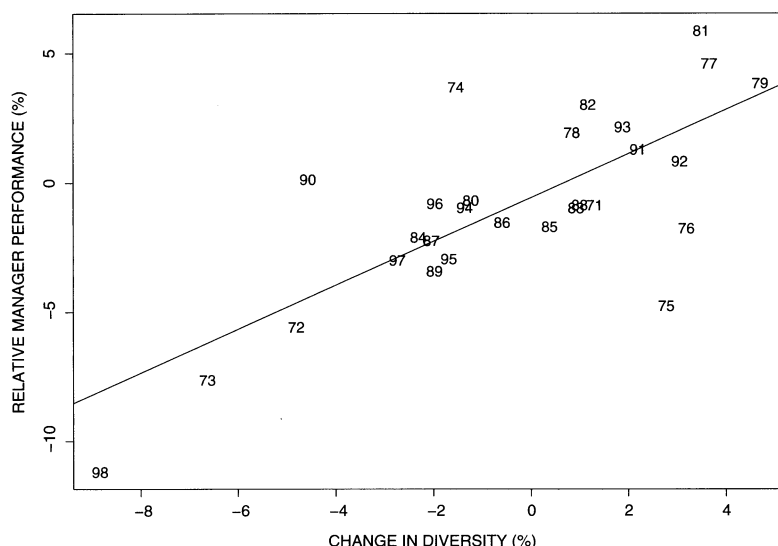


FIGURE 7.6. Manager performance relative to S&P 500 vs. change in D_p .
Manager performance data from Callan Associates: 1971–1998.

While the diversity-weighted S&P 500 is a more accurate benchmark in terms of the annual variation in managers' returns, it will also be a more rigorous benchmark in terms of managers' long-term performance. As we have seen in Section 7.2, the diversity-weighted S&P 500 is likely to outperform the capitalization-weighted S&P 500 by about a half percent a year over periods in which diversity remains stable. If managers can generate a half percent a year over the capitalization-weighted S&P 500 merely by using diversity weights, one would hope that they could provide at least as much return with active management. Although, on average, managers historically have underperformed passive benchmark indices, nevertheless there may be some evidence that managers can choose superior stocks. Data on active managers suggest that, on average, the stocks the managers select outperform their benchmarks, but that the gains are lost in the managers' high transaction costs. So perhaps managers could outperform the indices if they were able to control their trading costs effectively.

The popularity of passive managers versus active managers varies with their relative performance, and this varies with the change in market diversity. Over periods of increasing diversity, the median manager outperformed the S&P 500: by about 1.2% a year from 1975 to 1983 and by about 1.4% a year from 1991 to 1993. From Figure 7.6 we see that 1998 was not a good year for active managers. However, the 1998 data point is not far from the regression line, so most of the manager underperformance in 1998 can be explained by the change in diversity that year. Figure 6.7 on page 127 shows that diversity appears to be a mean-reverting process, so if the relation between managers' relative performance and diversity continues to hold, then active managers' performance should also be mean-reverting. Hence, it could be unwise to abandon active managers in favor of capitalization-weighted indexing after periods in which the managers' relative performance is weak.

7.3.1 Problem. (!) Determine the value of the parameter p that maximizes the part of the variation in manager logarithmic return explained by the logarithmic return of a $\mathbf{D}_p$ -weighted index. Determine how this parameter p changes over time, and interpret the results.

7.4 The Distributional Component of Equity Return

It has been understood for some time that the size of companies is an important factor affecting the return of stocks, and hence affecting portfolio performance. Moreover, in Section 7.3 we saw that change in market diversity is related to manager performance, and change in market diversity depends on the relative behavior of large and small stocks. Hence, it is important to be able to accurately measure the component of portfolio return that is related to company size. In this section we consider a method of direct calculation for the size component of portfolio return. The method we propose is derived from the fixed-weight-ratio portfolio constructed in Example 4.3.8.

Traditionally the size component of equity return has been estimated by statistical methods related to regression analysis. Unfortunately, regression analysis can be complicated by instability, nonlinearity of response, and correlation among multiple factors. Moreover, there is no a priori reason why regression techniques must be used, and in some cases they are definitely inappropriate. For example, since stock return can be measured directly, regression analysis is inappropriate for measuring the component of portfolio return due to a particular stock. Regression minimizes the value of the mean-squared residuals, but the rationale for this criterion is questionable, especially when direct measurement is possible.

To accurately measure the total effect of size on the return of a portfolio, it is necessary to consider the capital distribution of the market and analyze

the effect that changes in the capital distribution have on the portfolio. In Section 4.3 we noted that for a portfolio generated by a function of the ranked market weights, the effect on the relative return of the portfolio caused by a change in the capital distribution is measured by the change in value of the portfolio generating function. Here, we use this observation to construct a direct method to measure the effect that changes in the capital distribution have on an arbitrary portfolio.

In Example 4.3.8 we showed how portfolios with fixed weight ratios are generated by functions of the ranked market weights. Since weight ratios correspond to the number of (fractional) shares of each stock held in the portfolio, we can represent an arbitrary portfolio with no short sales as a portfolio with fixed weight ratios, at least at a given moment in time. Since a portfolio with fixed weight ratios is generated by a function of the ranked market weights, we can measure the effect that changes in the capital distribution have on it. This gives us a general method to measure the effect that changes in the capital distribution have on an arbitrary portfolio.

Suppose we have a portfolio π with weight ratios $w_1(t), \dots, w_n(t) \geq 0$ at time $t > 0$, defined by

$$w_i(t) = \pi_{p_t(i)}(t) / \mu_{(i)}(t),$$

for $i = 1, \dots, n$, where p_t is the permutation defined by (4.1.26), so that $p_t(i)$ is the index of the i th ranked stock at time t . For a fixed moment in time t_0 , the fixed-weight-ratio portfolio with the same weight ratios as π is generated by

$$\mathbf{S}(x) = w_1(t_0)x_{(1)} + \dots + w_n(t_0)x_{(n)}, \quad x \in \Delta^n, \quad (7.4.1)$$

as in (4.3.6). It follows from (4.3.8) that the effect on the relative return of the portfolio caused by change in the capital distribution at time t is

$$d \log \mathbf{S}(\mu(t)) = d \log(w_1(t_0)\mu_{(1)}(t) + \dots + w_n(t_0)\mu_{(n)}(t)),$$

so over the (short) period of time from t_0 to $t_1 > t_0$, the effect on the relative return will be approximately

$$\begin{aligned} & \log(w_1(t_0)\mu_{(1)}(t_1) + \dots + w_n(t_0)\mu_{(n)}(t_1)) \\ & \quad - \log(w_1(t_0)\mu_{(1)}(t_0) + \dots + w_n(t_0)\mu_{(n)}(t_0)) \\ & = \log(w_1(t_0)\mu_{(1)}(t_1) + \dots + w_n(t_0)\mu_{(n)}(t_1)), \end{aligned} \quad (7.4.2)$$

since $\mathbf{S}(\mu(t_0)) = 1$. We define this to be the *distributional component* of the relative return of π over the period from t_0 to t_1 . This distributional component measures the contribution to the relative return of the portfolio due to change in the capital distribution. If there is no change in the capital distribution, then there is no distributional component in the portfolio's return, or, for that matter, in the return of any other portfolio.

It is possible to generalize this methodology to measure certain other components of portfolio return. Let $c = (c_1, \dots, c_n)$, $c_i > 0$, for $i = 1, \dots, n$, and then define $(\cdot)^c$ to be the ranking operation on $1, \dots, n$ such that

$$c_1\mu_{(1)^c}(t) \geq c_2\mu_{(2)^c}(t) \geq \dots \geq c_n\mu_{(n)^c}(t), \quad t \in [0, T].$$

Let r_t be a permutation defined similarly to p_t in (4.1.26) such that

$$\mu_{r_t(k)}(t) = \mu_{(k)^c}(t), \quad t \in [0, T],$$

for $k = 1, \dots, n$. For a portfolio π , define weight ratios $v_1(t), \dots, v_n(t)$, where

$$v_i(t) = \pi_{r_t(i)}(t) / \mu_{(i)^c}(t), \quad t \in [0, T],$$

for $i = 1, \dots, n$. Then, as in (7.4.2), we define the c -component of the return of π over the period from t_0 to $t_1 > t_0$ to be

$$\log(v_1(t_0)\mu_{(1)^c}(t_1) + \dots + v_n(t_0)\mu_{(n)^c}(t_1)). \quad (7.4.3)$$

As a possible application of this generalized procedure, consider the book values of the companies in the market. If we assume that the book values vary slowly over time (which may or may not be true), then they can be approximated by constants over the short term. Suppose that b_i is the book value of the i th company, and let $c_i = 1/b_i$. Then the c -component of the return in this case can be considered a measure of the *price-to-book* component of the relative return of π . The price-to-book component is one of the factors that is claimed to have a significant effect on portfolio return. This generalized procedure offers a means to measure this component directly.

7.4.1 Problem. (!!) Determine whether the book value of companies changes slowly enough to make the proposed procedure effective. Develop a method to simultaneously measure the distributional component and the price-to-book component of portfolio return. Calculate the decomposition of portfolio return using the simultaneous procedure, and compare the results to those using the Sharpe/Fama/French methodology (Sharpe (1988) and Fama and French (1995)). Find a method to determine the exposure of a portfolio to these two factors.

7.5 Measurement of the Distributional Component

In this section we measure the distributional component of the return on a simulated “active-core” portfolio over the 10-year period from 1989 to 1998, using the method of direct measurement proposed in the previous section. Over this period, the diversity of the largest 1000 stocks in the CRSP universe declined significantly, as we saw in Figure 6.7. We saw in Section 7.3 that managers tend to have more exposure to smaller stocks than the S&P 500 Index, and our simulated portfolio is not an exception.

Since any portfolio with size exposure is likely to have been affected in some way by the decline in diversity over the period, we should expect this to be the case for our portfolio.

The portfolio we simulated was composed of stocks selected from the S&P 500 Index, and the performance of the portfolio was measured relative to that benchmark. The methodology of the previous section is valid for a broad market such as the market of all exchange-traded U.S. stocks. However, the model must be modified if the market is replaced by a subset of the market such as the S&P 500 Index, in which smaller stocks are systematically dropped and replaced by larger ones. In this case we must consider the S&P 500 itself to be a portfolio of stocks within the broad market. To calculate the contribution of the distributional component for a portfolio relative to the S&P 500, we must first calculate the distributional component of the portfolio relative to the market, and then subtract from this the distributional component of the S&P 500 relative to the market. This procedure corrects for leakage in a manner similar to that of Example 4.3.5.

The results of the analysis we conducted are presented in Figures 7.7 through 7.10 on pages 158 and 159. The goal of the active-core management style is to generate annual return about one or two percentage points higher than a benchmark large-stock index such as the S&P 500, while at the same time maintaining control over the standard deviation of the return relative to the benchmark. Active-core portfolios can be quite large, sometimes holding several hundred stocks selected from the benchmark index, and this is the case for our simulated portfolio.

The distributional component of the portfolio return is presented in Figure 7.8. The similarity of Figure 7.8 to the period from 1989 to 1998 in Figure 6.7 indicates that the distributional component of the portfolio return is likely to be highly correlated with change in market diversity. This correlation with change in market diversity is consistent with our observations regarding manager performance in Section 7.3.

The cumulative monthly relative logarithmic return of the simulated portfolio versus the S&P 500 Index over the period from January 1, 1989, to December 31, 1998, is presented in Figure 7.7. As we can see, the portfolio outperformed the benchmark by about 2 percentage points a year for the five years from 1989 to 1993, went into a slump for four years, and then came back in 1998. We see from Figure 7.8 that from 1989 to 1993 there was very little cumulative effect of the distributional component. However, from 1994 to 1998 it declined about 2 percentage points a year. This decline over the last five years had a significant effect on the relative return.

The residual component of the relative return is the difference of the relative return of the portfolio minus the distributional component. Conventionally, the residual component in portfolio factor analysis is interpreted to represent the return a portfolio manager generates from stock selection, and hence is referred to as the *stock selection* component. With this con-

vention, we see from Figure 7.9 that the stock selection component of the return of our simulated portfolio was about 2 percentage points a year for the first five years, and then flattened out from 1994 to 1997. However, in 1998 the return from stock selection came back strongly. In any case, from 1994 to 1997 the flat stock selection component, combined with a significantly negative distributional component, gave the portfolio four years of poor performance relative to its benchmark.

It is also of interest to determine how the size component looked using conventional regression analysis. In order to use regression, appropriate explanatory variables must be found. Unfortunately, there is no natural variable to represent the size factor in regression: it could be represented, for example, by the relative return of the largest 100 stocks in the S&P 500 Index versus the remaining 400, or of the Russell 1000 Index versus the Russell 2000 Index, or by the change in market diversity, as in Section 7.3. Moreover, no single regression variable can accurately represent a nonlinear, multidimensional relationship. In fact, Barra, a well-known investment consulting firm, announced that they would represent size as a two-dimensional variable in order to deal with the nonlinearity of its effect. The arbitrary nature of the size variable used in the regression-based methods casts doubt on the efficacy of these techniques in portfolio analysis.

Although there appears to be no natural explanatory variable available for conventional regression analysis, let us proceed anyway by using two possible variables, and then compare the results. Figure 7.10 presents two regression estimates of the cumulative size component in the portfolio return. The first estimate is given by the solid line, in which the explanatory variable is the relative logarithmic return of the largest 25 stocks in the S&P 500 Index versus the Index itself. The broken line is the corresponding estimate using the largest 100 stocks versus the S&P 500. Since both of the curves in Figure 7.10 are of about the same magnitude, the choice of explanatory variable does not appear to be significant, at least between these two variables. From the look of these charts, the size component estimates in Figure 7.10 have roughly the same shape as that in Figure 7.8, but the magnitude is only about one-fourth as great (the scale is the same in Figures 7.8, 7.7, 7.9, and 7.10).

There is one more regression variable that may be of interest: our calculated values of the distributional component of the portfolio return. With this explanatory variable, we find that the regression coefficient is approximately 0.51, and that this explains about 16% of the monthly variation of the relative return. Regression is a linear model, so perhaps this 51% represents the linear contribution to the distributional component.

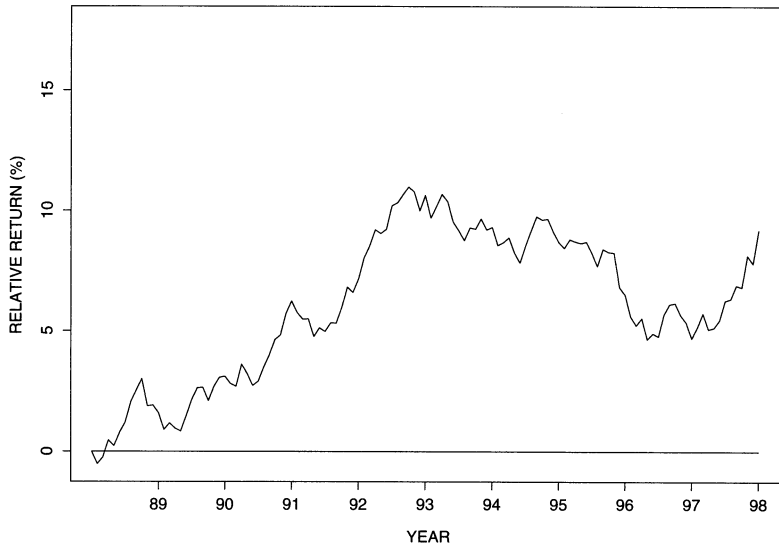


FIGURE 7.7. Relative return of the simulated portfolio.

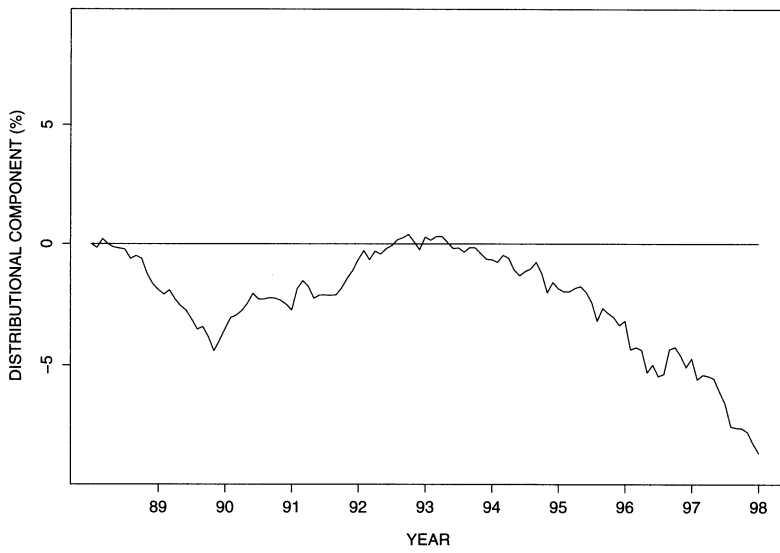


FIGURE 7.8. Distributional component for the simulated portfolio.

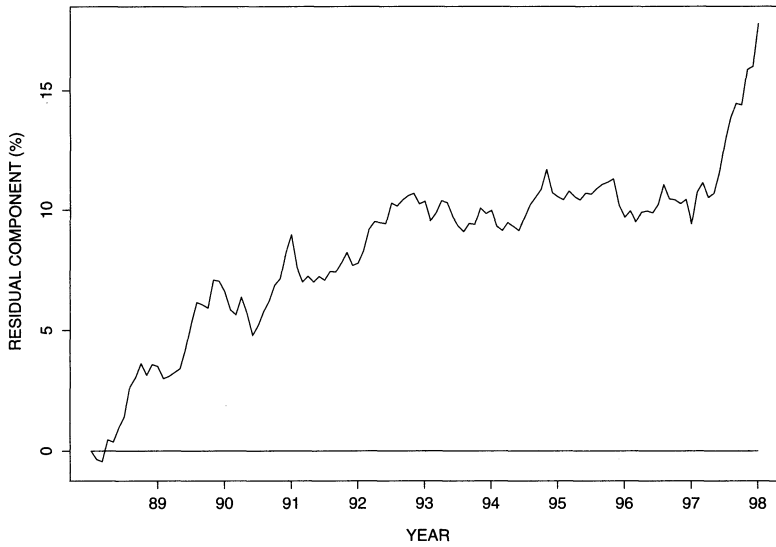


FIGURE 7.9. Residual component for the simulated portfolio.

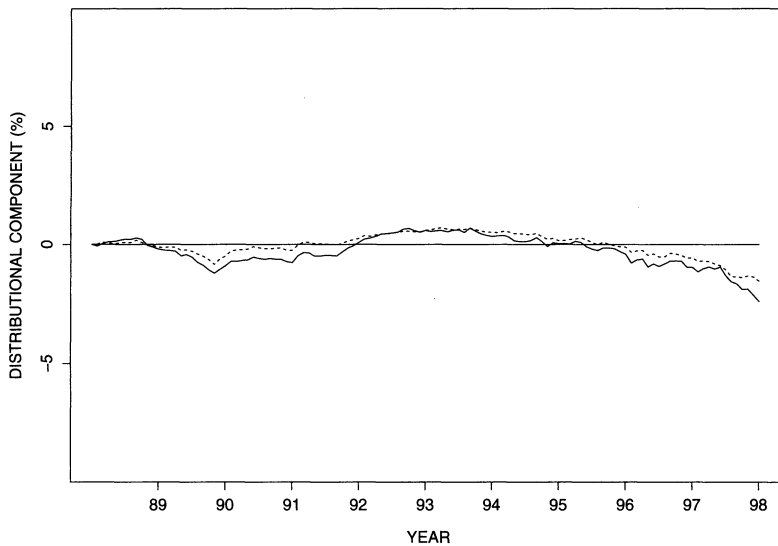


FIGURE 7.10. Regression estimates of size component.

Solid line: vs. relative return of largest 25 S&P 500 stocks.

Broken line: vs. relative return of largest 100 S&P 500 stocks.

7.6 An Analysis of Value Stocks

Value stocks are sometimes defined as those stocks with low price-to-book ratios. It has been claimed by many influential investors that value stocks tend by their very nature to be underpriced, and hence are likely to have above-average return in the future. For this reason, some investors favor value stocks for their equity investments.

The S&P/Barra Value Index is the subindex comprising about half the capitalization of the S&P 500 Index and consisting of those stocks with lower price-to-book ratios. The S&P/Barra Value Index significantly underperformed the S&P 500 Index over the 10-year period from 1989 through 1998, so much so that it has led to doubts about the desirability of investing in value stocks in general. However, the S&P/Barra Value Index usually tends to have lower average capitalization than the whole S&P 500, and if this size bias can explain some of the underperformance of the value stocks, then perhaps these stocks are not such a bad bet after all. We now use the methodology of Section 7.4 to determine how much of the underperformance of the S&P/Barra Value Index was due to the distributional component of the relative return.

The cumulative values of the relative logarithmic return of the S&P/Barra Value Index versus the S&P 500 Index, as well as the distributional component and the residual component, are displayed in Figures 7.11 through 7.13. Note that the values in these charts could differ slightly from the actual historical values for these indices, since monthly data were used in the simulation, while daily data are used for the actual values.

In Figure 7.11 we see that over the 10-year period studied, the S&P/Barra Value Index underperformed the S&P 500 by over 20%. However, Figure 7.12 shows that more than half of this underperformance was due to the distributional component. This is reasonable, since value stocks usually have relatively low average capitalization, and large stocks generally outperformed small stocks from 1989 to 1998, as seen in Figure 6.7. Figure 7.12 is quite similar in shape to the change in diversity over the corresponding time period in Figure 6.7, although the similarity is not as great as between Figure 7.8 and Figure 6.7. Nevertheless, the similarity between Figures 6.7 and 7.12 indicates that change in diversity is likely to be a significant factor affecting the relative return of value stocks.

The residual component in Figure 7.13 contributed about -10% over the period, but since 1991 it has been flat. Hence, although this 10-year period has certainly not been favorable for value stocks, much of the underperformance was due to the distributional factor, so value investment may not be as dismal as Figure 7.11 suggests. Moreover, if diversity reverts to the mean, as it has in the past, this should give a boost to the relative performance of value stocks.

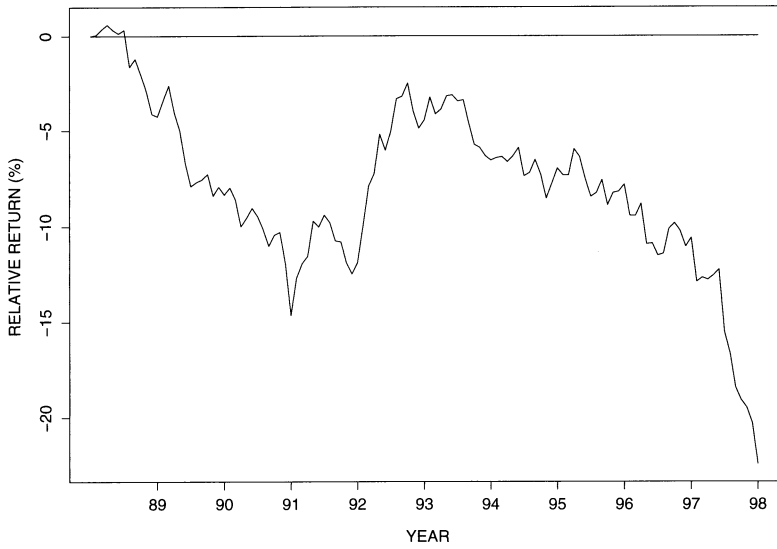


FIGURE 7.11. Relative return of the S&P/Barra Value Index.

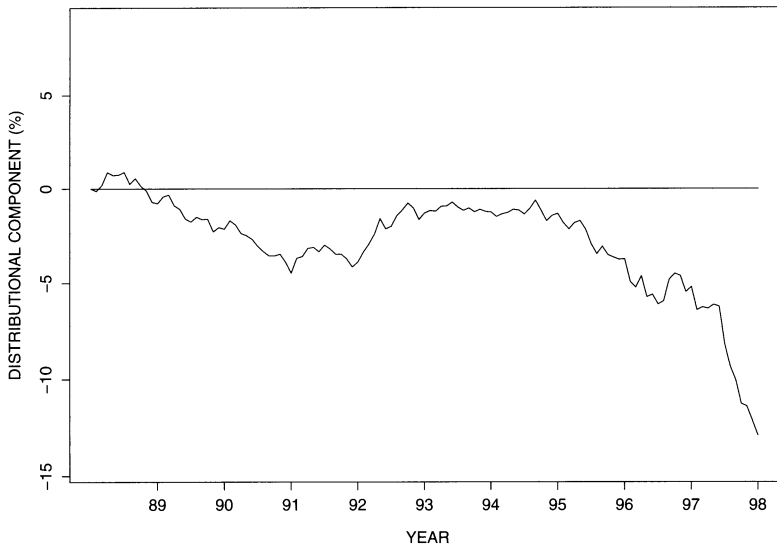


FIGURE 7.12. Distributional component for S&P/Barra Value Index.

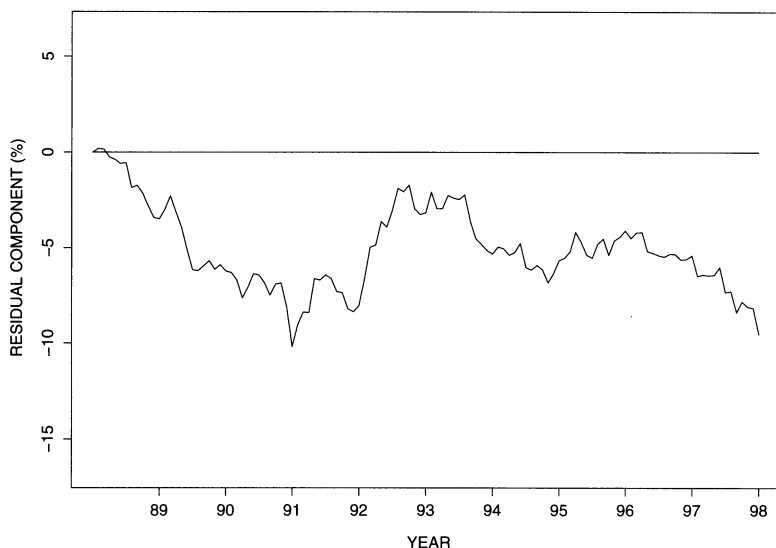


FIGURE 7.13. Residual component for S&P/Barra Value Index.

7.7 Notes and Summary

In Section 7.1, the growth rates of the first-order model are used as an alternative to analysts' predicted returns in portfolio optimization, since according to Malkiel (1990), a well-known source for efficient market doctrine, analysts' predicted returns appear to be no better than random. Of the examples in Section 7.1, Example 7.1.2, maximizing the probability of reaching a goal, is from Pestien and Sudderth (1985); Example 7.1.3, minimizing the expected time to reach a goal, is from Heath and Sudderth (1984) and Heath et al. (1987); and Example 7.1.4, maximizing the probability of reaching a goal before a deadline, is from Sudderth and Weerasinghe (1989).

Diversity-weighted indexing, described in Section 7.2, was first proposed in Fernholz et al. (1998). The Sharpe ratio, one of the measures used to determine the efficacy of diversity-weighted indexing, is discussed in Sharpe (1975).

The relation between manager performance and change in diversity presented in Section 7.3 was observed by Fernholz and Garvy (1999). Malkiel (1990) provides evidence that managers historically have underperformed passive benchmark indices, but Wagner and Ryavec (1997) found that the

stocks selected by managers outperform their benchmarks on average. In Plexus Group (1995) it is argued that managers lose their advantage because of high transaction costs.

The method developed in Sections 7.4, 7.5, and 7.6 to measure the distributional component of equity return comes from Fernholz (2001b). Ross (1976) and Fama and French (1993, 1995, 1996) presented evidence that the size of companies is an important factor affecting the return of stocks, and hence affecting portfolio performance. At least since Sharpe (1988), the size component of equity return has been estimated by statistical methods related to regression analysis. Sharpe (1988) and Fama and French (1995) suggested that the price-to-book component, mentioned with regard to (7.4.3), is one of the factors that has a significant effect on portfolio return. See Barra (1997) for the announcement that they would represent size as a two-dimensional variable.

Value stocks, discussed in Section 7.6, have been acclaimed by Graham and Dodd (1934) among others, but Loughran (1997) casts doubt on their putative superiority. The rules for the composition of the S&P/Barra Value Index can be found in Standard & Poor's (1999).

Chapter Summary

Section 7.1: The parameters $\mathbf{g}_k$, for $k = 1, \dots, n$, of the first-order model of Section 5.5 can be used as growth rates in a hybrid model for the market. In this case

$$d \log X_i(t) = \mathbf{g}_{q_t(i)} dt + \sum_{\nu=1}^n \xi_{i\nu}(t) dW_\nu(t), \quad t \in [0, T], \quad (7.1.1)$$

for $i = 1, \dots, n$, where $q_t(i)$ represents the rank of the i th stock at time t . The use of this model in portfolio optimization avoids the intractable problem of forecasting the growth rates of the individual stocks.

Section 7.2: Diversity-weighted indexing is an institutional equity management product based on a $\mathbf{D}_p$ -weighted version of the S&P 500 Index. The simulated relative return of this portfolio versus the S&P 500 Index is shown in Figure 7.2, and the components of the relative return are shown in Figures 7.3, 7.4, and 7.5. The results of the simulation are summarized in Table 7.1. The simulation indicates that diversity-weighted indexing should generate about a half a percent higher annual return than conventional capitalization-weighted indexing, with about the same volatility.

Section 7.3: Manager performance relative to the S&P 500 Index is correlated with the change in market diversity ($R^2 = 0.55$). This observation means that since market diversity appears to be mean-reverting and stable over the long term, the same is likely to be true for manager performance. It also means that a diversity-weighted index is likely to be a more appropriate benchmark than the conventional capitalization-weighted S&P 500 Index. In fact, the residual R^2 is 3.68% for the diversity-weighted S&P 500

Relative return	Change in $\mathbf{D}_p$	Drift process	Differential dividend rate
0.26%	-0.19%	0.47%	-0.04%

TABLE 7.1. Diversity-weighted S&P 500 Index, 1966–1998.
Annual logarithmic means.

as an explanatory variable for median manager return, while the residual R^2 is 5.75% for the capitalization-weighted S&P 500 as an explanatory variable.

Section 7.4: Let π be a portfolio with weight ratios $w_1(t), \dots, w_n(t) \geq 0$ at time $t > 0$, defined by

$$w_i(t) = \pi_{p_t(i)}(t) / \mu_{(i)}(t),$$

for $i = 1, \dots, n$, where $p_t(i)$ is the index of the i th ranked stock at time t . Then the *distributional component* of the relative return of π over the (short) period of time from t_0 to $t_1 > t_0$ will be approximately

$$\log(w_1(t_0)\mu_{(1)}(t_1) + \dots + w_n(t_0)\mu_{(n)}(t_1)). \quad (7.4.2)$$

This expression directly measures the effect that changes in the capital distribution of the market have on the relative return of the portfolio.

Section 7.5: The method of the previous section is applied to a simulated “active-core” portfolio. The method produces a decomposition of the relative return that is similar to the decomposition for functionally generated portfolios. The relative return of the simulated portfolio is presented in Figure 7.7, and the components of relative return are presented in Figures 7.8 and 7.9. The distributional component of the relative return measured by this method is about four times greater than the size component measured by conventional means based on linear regression, shown in Figure 7.10.

Section 7.6: The method of the previous two sections is applied to the S&P/Barra Value Index over the period from 1990 to 1999. The results of the decomposition are presented in Figures 7.11, 7.12, and 7.13. The results show that although the Value Index underperformed the S&P 500 Index as a whole over the period analyzed, most of the underperformance was due to the distributional component.

Appendix A

Evaluation of Local Times

Summary: Let F be a C^2 function defined on Δ^n , and suppose that

$$\mathbf{S}(x) = F(x_{(1)}, \dots, x_{(n)}), \quad x \in \Delta^n.$$

The function $\mathbf{S}$ generates a portfolio, and we wish to estimate the contribution to the relative return of this portfolio that is due to the local time component of the drift function over the (short) period of time from t_0 to $t_1 > t_0$. If we represent this quantity by $dL_\pi(t_0)$, then

$$dL_\pi(t_0) \approx \log F(\mu_{i_1}(t_1), \dots, \mu_{i_n}(t_1)) - \log F(\mu_{(1)}(t_1), \dots, \mu_{(n)}(t_1)),$$

where for $k = 1, \dots, n$, i_k is the index of the stock that is ranked k th at time t_0 .

In any simulation of functionally generated portfolios that involves actual stock data, the evaluation of the local time component of the drift process will be involved, even if only for the calculation of leakage. As we have seen in Example 4.3.2, Theorem 4.2.1 provides a method for the theoretical calculation of individual local times. This method was used in (5.4.1) to calculate the local times in Figure 5.2, but in simulations it can be tedious to calculate individually the value of each of the local times that may be involved in the local time component of the drift process. Fortunately, in most simulations it is possible to use a direct calculation of local time component that is based on Theorem 4.2.1, but does not involve the evaluation of the individual local times.

Suppose that F is a C^2 function defined on Δ^n , and suppose that

$$\mathbf{S}(x) = F(x_{(1)}, \dots, x_{(n)}), \quad x \in \Delta^n.$$

In this case, Theorem 4.2.1 states that $\mathbf{S}$ generates a portfolio π with weights

$$\pi_{p_t(k)}(t) = \left(D_k \log F(\mu_{(1)}(t), \dots, \mu_{(n)}(t)) + 1 - \sum_{j=1}^n \mu_{(j)}(t) D_j \log F(\mu_{(1)}(t), \dots, \mu_{(n)}(t)) \right) \mu_{(k)}(t), \quad (\text{A.1.1})$$

for $k = 1, \dots, n$ and $t \in [0, T]$, and a drift process Θ_π that satisfies

$$d\Theta_\pi(t) = \theta_\pi(t) dt + dL_\pi(t), \quad t \in [0, T], \quad \text{a.s.}, \quad (\text{A.1.2})$$

where, a.s.,

$$\theta_\pi(t) = \frac{-1}{2\mathbf{S}(\mu(t))} \sum_{j,k=1}^n D_{jk} F(\mu_{(1)}(t), \dots, \mu_{(n)}(t)) \mu_{(j)}(t) \mu_{(k)}(t) \tau_{(jk)}(t)$$

for $t \in [0, T]$, and

$$dL_\pi(t) = \frac{1}{2} \sum_{k=1}^{n-1} (\pi_{p_t(k+1)}(t) - \pi_{p_t(k)}(t)) d\Lambda_{\log \mu_{(k)} - \log \mu_{(k+1)}}(t)$$

for $t \in [0, T]$. Equation (A.1.2) is precisely the decomposition of the drift process Θ into its smooth and local time components, θ_π and L_π . We wish to estimate the value of $L_\pi(t)$ for $t \in [0, T]$.

Now suppose that $(i_1, \dots, i_n)$ is a permutation of $(1, \dots, n)$, and that

$$\mathbf{R}(x) = F(x_{i_1}, \dots, x_{i_n}), \quad x \in \Delta^n.$$

Theorem 3.1.5 implies that $\mathbf{R}$ generates a portfolio η with weights

$$\eta_{i_k}(t) = \left(D_k \log F(\mu_{i_1}(t), \dots, \mu_{i_n}(t)) + 1 - \sum_{j=1}^n \mu_{i_j}(t) D_j \log F(\mu_{i_1}(t), \dots, \mu_{i_n}(t)) \right) \mu_{i_k}(t), \quad (\text{A.1.3})$$

for $t \in [0, T]$ and $k = 1, \dots, n$, and a drift process Θ_η with

$$d\Theta_\eta(t) = \theta_\eta(t) dt,$$

a.s., for $t \in [0, T]$, where

$$\theta_\eta(t) = \frac{-1}{2\mathbf{R}(\mu(t))} \sum_{j,k=1}^n D_{jk} F(\mu_{i_1}(t), \dots, \mu_{i_n}(t)) \mu_{i_j}(t) \mu_{i_k}(t) \tau_{i_j i_k}(t).$$

Suppose that at a particular time $t_0 \in [0, T]$,

$$i_k = p_{t_0}(k), \quad k = 1, \dots, n.$$

By the definition of the permutation p_t , this means that

$$\mu_{i_k}(t_0) = \mu_{(k)}(t_0), \quad k = 1, \dots, n,$$

and it follows immediately from (A.1.1) and (A.1.3) that, for $i = 1, \dots, n$,

$$\pi_i(t_0) = \eta_i(t_0), \quad (\text{A.1.4})$$

and, from the definitions of θ_π and θ_η , that

$$\theta_\pi(t_0) = \theta_\eta(t_0). \quad (\text{A.1.5})$$

For the purposes of simulation, differentials are interpreted as small changes in the values of the corresponding functions. Hence, it follows from (A.1.4) that in a simulation, since π and η have identical weights at time t_0 ,

$$d \log(Z_\pi(t_0)/Z_\mu(t_0)) = d \log(Z_\eta(t_0)/Z_\mu(t_0)). \quad (\text{A.1.6})$$

Now, a.s., for $t \in [0, T]$,

$$d \log(Z_\pi(t)/Z_\mu(t)) = d \log \mathbf{S}(\mu(t)) + \theta_\pi(t) dt + dL_\pi(t),$$

and

$$d \log(Z_\eta(t)/Z_\mu(t)) = d \log \mathbf{R}(\mu(t)) + \theta_\eta(t) dt.$$

These two equations combined with (A.1.5) and (A.1.6) imply that for the purposes of simulation,

$$\begin{aligned} dL_\pi(t_0) &= d \log \mathbf{R}(\mu(t_0)) - d \log \mathbf{S}(\mu(t_0)) \\ &= d \log F(\mu_{i_1}(t_0), \dots, \mu_{i_n}(t_0)) - d \log F(\mu_{(1)}(t_0), \dots, \mu_{(n)}(t_0)). \end{aligned} \quad (\text{A.1.7})$$

This means that for the (short) period of time from t_0 to $t_1 > t_0$, we have

$$\begin{aligned} dL_\pi(t_0) &\approx (\log F(\mu_{i_1}(t_1), \dots, \mu_{i_n}(t_1)) - \log F(\mu_{i_1}(t_0), \dots, \mu_{i_n}(t_0))) \\ &\quad - (\log F(\mu_{(1)}(t_1), \dots, \mu_{(n)}(t_1)) - \log F(\mu_{(1)}(t_0), \dots, \mu_{(n)}(t_0))) \\ &= \log F(\mu_{i_1}(t_1), \dots, \mu_{i_n}(t_1)) - \log F(\mu_{(1)}(t_1), \dots, \mu_{(n)}(t_1)). \end{aligned}$$

For any $t \in [0, T]$, we can partition $[0, t]$ into subintervals of the form $[t_0, t_1]$, and evaluate $L_\pi(t)$ as the sum the estimates $dL_\pi(t_0)$ corresponding to the partition.

A.1.1 Example. (Leakage in a D_p -weighted portfolio) Let us apply (A.1.7) to the evaluation of leakage considered in Example 4.3.5. Recall that in that example p and m are constants with $0 < p < 1$ and $1 < m < n$. Here we wish to evaluate the local time component in (4.3.5), which, for our purposes, can be written

$$\begin{aligned} d \log(Z_\pi(t)/Z_\xi(t)) &= d \log(Z_\pi(t)/Z_\mu(t)) - d \log(Z_\xi(t)/Z_\mu(t)) \\ &= d \log \mathbf{D}_p(\xi_{(1)}(t), \dots, \xi_{(m)}(t)) + \theta_\pi(t) dt + dL(t), \end{aligned}$$

a.s., for $t \in [0, T]$. The local time component L is the difference between the leakage terms for the portfolios π and ξ , and appears as the last term in (4.3.5). The generating function here is defined by

$$\mathbf{D}_p(\xi_{(1)}(t), \dots, \xi_{(m)}(t)) = \mathbf{S}(\mu(t))/\mathbf{S}_L(\mu(t)),$$

where $\mathbf{S}$ and $\mathbf{S}_L$ are defined in Examples 4.3.5 and 4.3.2, respectively. In our current context, this generating function corresponds to the function F defined by

$$F(x) = \left(\sum_{i=1}^m x_i^p \right)^{1/p} / \sum_{i=1}^m x_i, \quad x \in \Delta^n.$$

In terms of this function, $dL(t_0)$ is estimated by

$$\log F(\mu_{i_1}(t_1), \dots, \mu_{i_m}(t_1)) - \log F(\mu_{(1)}(t_1), \dots, \mu_{(m)}(t_1)).$$

In terms of $\mathbf{D}_p$, this estimate can be expressed as

$$\log \mathbf{D}_p(\xi_{i_1}(t_1), \dots, \xi_{i_m}(t_1)) - \log \mathbf{D}_p(\xi_{(1)}(t_1), \dots, \xi_{(m)}(t_1)), \quad (\text{A.1.8})$$

where for $k = 1, \dots, m$, $i_k = p_{t_0}(k)$,

$$\xi_{i_k}(t) = \frac{\mu_{i_k}(t)}{\mu_{i_1}(t) + \dots + \mu_{i_m}(t)},$$

and

$$\xi_{(k)}(t) = \frac{\mu_{(k)}(t)}{\mu_{(1)}(t) + \dots + \mu_{(m)}(t)}.$$

Hence, $L(t)$ will be evaluated as the sum of expressions of the form (A.1.8) corresponding to an appropriate partition of the interval $[0, t]$. $\square$

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